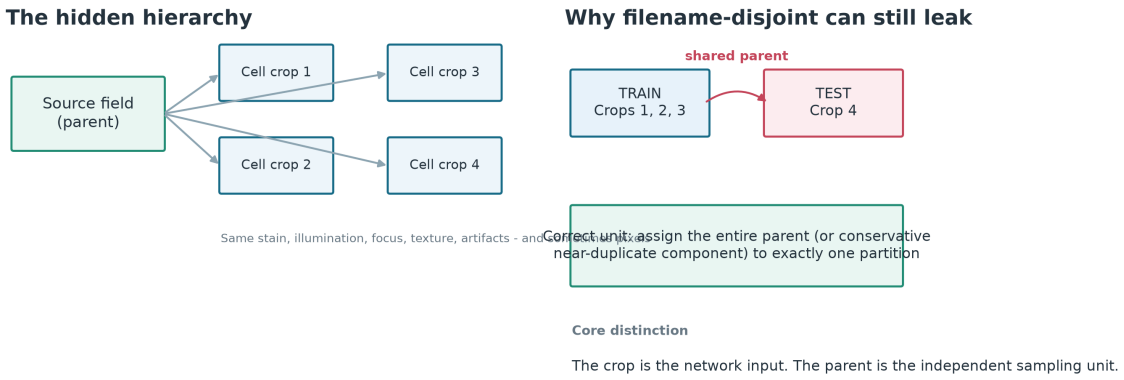
Do We Train on Test Cells?

The Complete CytoFair Lifecycle Record

Study 1, the scaling roadmap, the preregistered Study 2, Amendments 6–9, and the follow-up experiments:
motivations, designs, engineering, every result, what worked, what did not, and what we got out of it



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Evidence through 20 October 2026 (Asia/Kolkata)

2,606 trained models, 640 evaluation-only runs, 82 commits, two registered studies

About this record

This is the complete record of the CytoFair project from its first registered commit on 22 August 2026 to the end of the follow-up experiments on 20 October 2026. It is deliberately exhaustive. It keeps the original Study 1 record intact as Part II, and adds everything that happened afterwards: the review that turned Study 1's weaknesses into a scaling roadmap, the separately preregistered Study 2, the engineering needed to run 2,921GPU tasks on a shared host with a four-core CPU allowance, every registered result, the eight follow-up hypotheses of Amendment 8, the bugs and setbacks, and what the project established, refuted, and left open.

Evidence boundary

Every number in Parts I and III–VIII is generated from committed outputs: the registered analysis files in `results/study2/analysis` and `results/study2/followup`, the exported run records in `results/study2/run_records`, the power simulation, the provenance maps, and the git history. The generators are `scripts/study2/paper_numbers.py`, `record/scripts/make_record_tables.py`, and `record/scripts/make_record_figures.py`; no result is typed by hand. Part II is the Study 1 record of 13 September 2026, carried over verbatim (labels prefixed so they cannot collide), and its numbers come from the Study 1 result files as they stood then.

What "preregistered" means here

Study 2 and Amendment 8 were registered as commits in version control before any model whose result they govern was trained, and the registration commit's hash is recorded inside every run record. That is strong internal evidence, but the commits have not yet been pushed or deposited with a third party, so an outside reader cannot yet verify their dates independently. Amendment 8 was written after Study 2's results were known and is reported as a registered follow-up, not as part of Study 2.

How the record is organised

Part I, the lifecycle at a glance
: One chapter with the whole story: timeline, every registered test and its verdict, the hypothesis map, and the answers to "what did we get out of it?".

Part II, Study 1
: The frozen Study 1 record: motivation, provenance recovery, structural leakage, literal overlap, the parent-identity probe, the SIPaKMeD and Cx22 rescoring, statistics, execution failures, verification, and release.

Part III, from Study 1 to Study 2
: The roadmap and our response to each item, Phase 0 checks, the Study 2 preregistration,

power analysis, every design (splits, factorial, transplants, adversary, CRIC), the structural closed forms, and the post hoc reanalysis of Study 1.

Part IV, engineering and execution
The host, its constraints, the GPU-resident input path, determinism, the trainer's artifact schema, the scheduler, failure handling, smoke tests, the execution timeline, compute, archiving, and the complete log of errors and fixes.

Part V, Study 2 results
H1 to H5 with every sensitivity analysis and secondary analysis, training curves, calibration, confusion matrices, and per-architecture detail.

Part VI, Amendment 8
The follow-up question, the eight registered hypotheses, their designs, the preparer bug (Amendment 9), every result, and the mechanism by elimination.

Part VII, synthesis
What was established, refuted, and left open; what worked and what did not; lessons; threats to validity; recommendations; publication options; decisions only the authors can make.

Part VIII, appendices
The Study 1 appendices, complete run ledgers for every Study 2 and follow-up run, variance components, power grid, risk report, the full commit chronology, registration hashes, and a reproduction cookbook.

Reading paths

- *Ten minutes:* Part I.
- *The science:* Part I, Chapter 21, Part V, Part VI, Chapter 41.
- *Reproducing or extending the work:* Part IV, Appendix T, and the ledgers.
- *Writing a paper from it:* Chapter 41 (claims and their strength) and Section 44.3.1.

Conventions

P1 is a stratified crop-level 80:20 split; P1-prime (P1′) a size-matched control; P2 a parent-disjoint split (field, source image, or slide); P3 a parent-disjoint split after merging near-duplicate-linked parents. "pp" means percentage points. Unless stated otherwise, intervals are 95%, times are Asia/Kolkata (IST), and "supported" means a registered test passed its registered criterion (adjusted $p < 0.05$ with the predicted sign). In figures, crop-level (leaky) series are orange, parent-level blue, and component-level aqua; leak models are orange and clean models blue.

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PART I

The Lifecycle at a Glance

1

The whole project on a few pages

1.1 The question

Public cervical cytology benchmarks are released as cell crops, and most crops are cut from far fewer microscope fields or slide images. When a benchmark is split at the crop level, crops from one field land on both sides of the train–test boundary. Files are disjoint, but the test set shares stain, illumination, focus, background, and sometimes literal pixels with training. The project asked three questions in order: *is the structure there*, *is the shared parent recognisable*, and *does it inflate reported performance*; and then a fourth: *how do models use it*.

1.2 What happened, in order

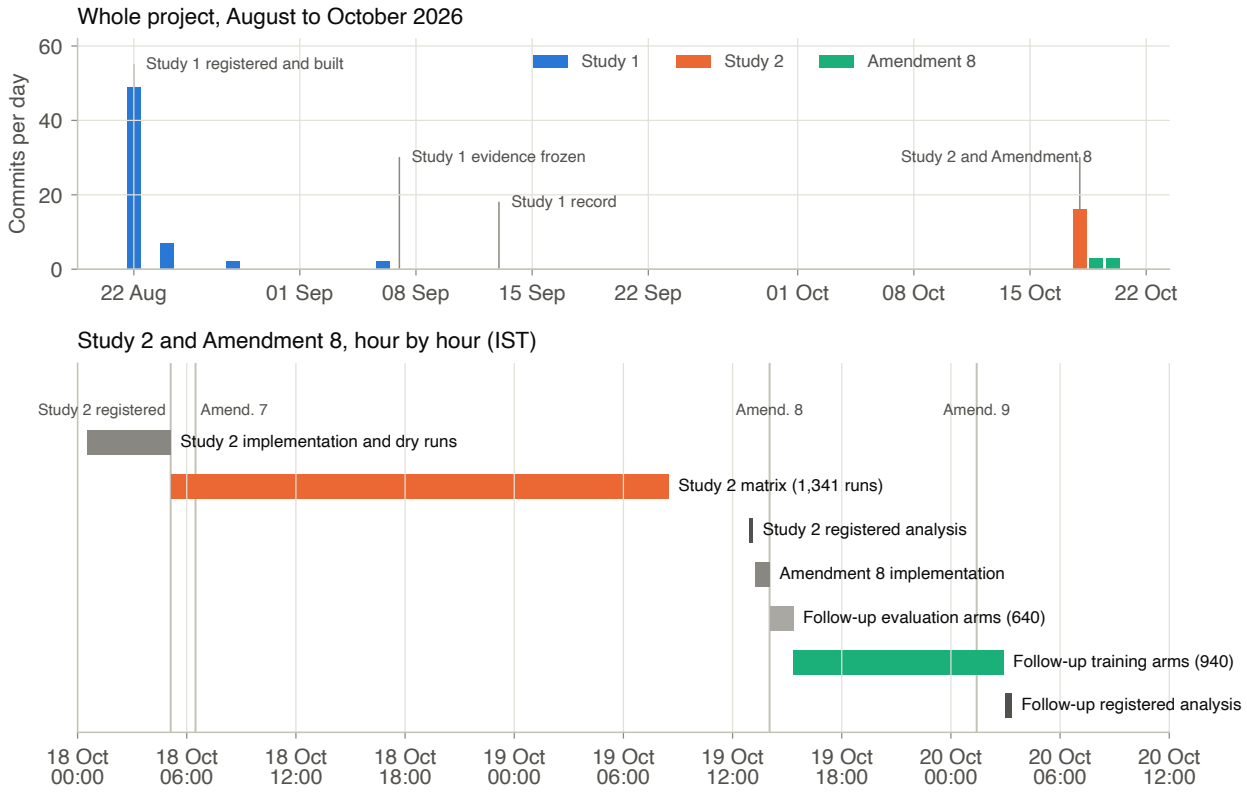


Figure 1.1: The lifecycle. Top: commits per day across the project. Bottom: Study 2 and Amendment 8 hour by hour; every span is measured from run records or commits.

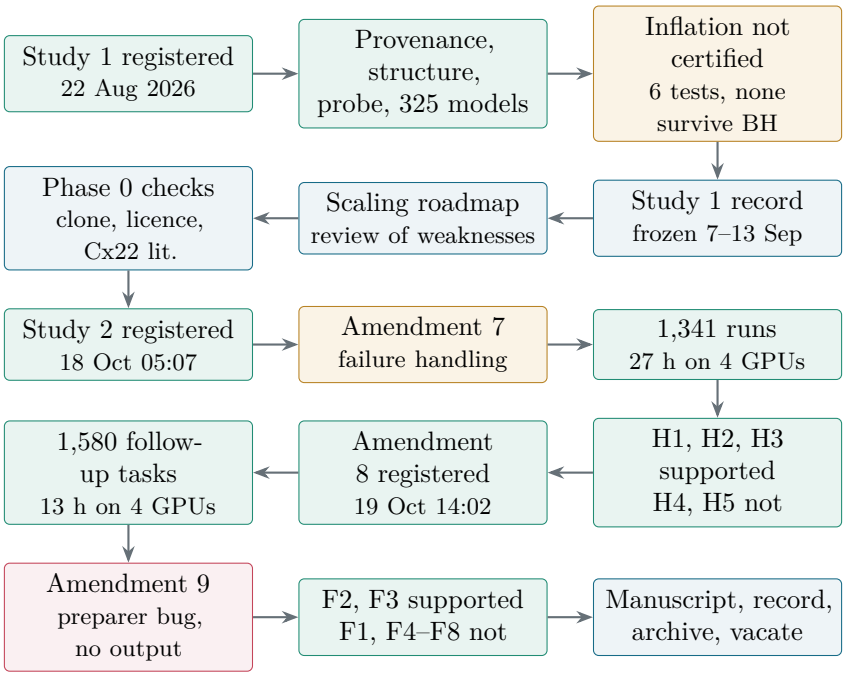


Figure 1.2: The whole lifecycle as a sequence of registered decisions (green), amendments and setbacks (gold, red), and reviews (blue). Every arrow is a commit in the repository; Appendix O lists all of them.

1. **Study 1 (22 August – 13 September 2026).** The crop-to-parent hierarchy of four public datasets was recovered with independent, fail-closed branches; random crop splitting placed a training parent behind 92.6% of SIPaKMeD and 99.97% of CRIC test cells; more than half of SIPaKMeD test crops shared source pixels with a training crop; a 424-way parent probe reached 109 times chance. Classifiers scored higher under crop-level splits for all six architectures and all seven Cx22 segmentation models, but none of the per-architecture tests survived the registered Benjamini–Hochberg correction. Part II is that record.

2. **The roadmap (October 2026).** A review identified four weaknesses: one question asked six times, availability not distinguished from use, leakage confounded with training diversity, and only one rescored benchmark. It proposed a single mixed-model test sized by simulation, causal arms, more architectures and seeds, preregistration, and keeping every per-sample artifact (Chapter 21).

3. **Study 2 (18–19 October).** Registered at commit `a62e8e7` (2026-10-18 05:07 IST), before any Study 2 model was trained: one primary test (H1) and a Holm family of four mechanism tests (H2–H5), 1,341 runs over 27.4 hours on four GPUs. H1, H2, and H3 were supported; H4 and H5 were not.

4. **Amendment 8 (19–20 October).** Eight new hypotheses about *how* the leak is used, registered at `80171fe` (2026-10-19 14:02 IST) after the Study 2 analysis and before any follow-up run: 1,580 tasks over 12.8 hours. F2 and F3 were supported; F1 went the opposite way; F4–F8 were null.

5. **Amendment 9 (20 October, 01:25).** A labelling bug in the follow-up preparer halted the analysis before it produced output; fixed and recorded before any affected estimate was seen.

6. **Close-out.** Manuscript (`paper/v2`), this record, 121 GB of outputs archived and verified on the network home, and the node-local disk vacated.

1.3 Every registered test

Table 1.1: Every registered confirmatory test of Study 2 and Amendment 8. H1 is tested alone; H2–H5 and F1–F8 are separate Holm families. Units as in the text (pp; slopes in pp per pp).

	Question	Estimate	95% CI	p	Holm p	Verdict
H1	SIPaKMeD P1 − P2 accuracy (primary)	0.97	0.59 to 1.39	<0.001	alone	**supported**
H2	CRIC P1 − P2 accuracy	12.08	11.16 to 12.99	<0.001	<0.001	**supported**
H3	2C leakage at matched diversity	1.73	0.95 to 2.51	0.006	0.02	**supported**
H4	2B own-context route	0.42	−0.35 to 1.18	0.23	0.23	**not supported**
H5	2A gap vs recoverability	0.059	−0.007 to 0.125	0.08	0.16	**not supported**
F1	CRIC slide-context pull	−4.29	−8.60 to −1.37	<0.001	0.001	**not supported**
F2	test-time stain transfer	1.27	0.38 to 2.16	0.005	0.03	**supported**
F3	leak with backgrounds masked	1.86	1.42 to 2.30	<0.001	<0.001	**supported**
F4	colour augmentation	−0.23	−0.81 to 0.35	0.43	1.00	**not supported**
F5	LEACE erasure	−0.01	−0.10 to 0.07	0.72	1.00	**not supported**
F6	stain normalisation	0.02	−0.41 to 0.44	0.94	1.00	**not supported**
F7	probe-matched adversary	0.000	−0.038 to 0.039	0.99	1.00	**not supported**
F8	pooled dose-response	0.000	−0.005 to 0.005	0.93	1.00	**not supported**

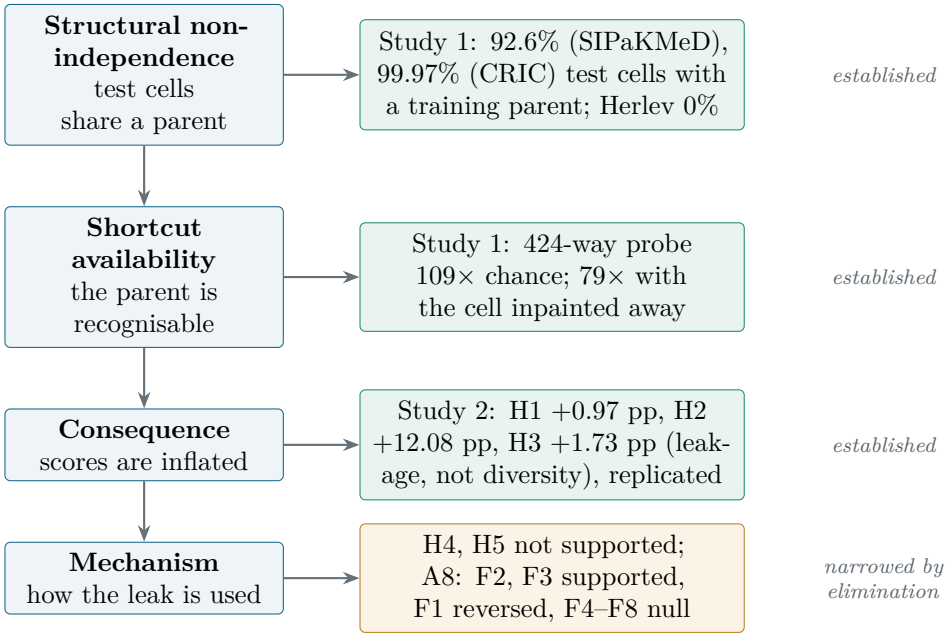


Figure 1.3: The four claims the project separates, and the evidence each received. Study 1 established the first two; Study 2 the third; Amendment 8 narrowed the fourth.

1.4 What we got out of it

Established

- Crop-level splitting inflates accuracy on SIPaKMeD by 0.97 pp (95% CI 0.59–1.39; all twelve architectures positive) and on CRIC by 12.08 pp; the parent-level error rate is 1.34 and 2.27 times the crop-level one.
- The inflation is leakage, not the extra training diversity crop-level splits bring: at matched crop and field counts the leak effect is 1.73 pp and the diversity effect 0.00 pp; the leak effect replicated on fresh seeds (1.47 pp).
- The leak needs no background (F3: 1.86 pp with every background masked) and is tied to the field's colour rendering at test time (F2: stain transfer cuts it from 1.72 to 0.46 pp).
- Neither a linearly decodable field code nor Grad-CAM-visible attention carries it: erasing or adversarially suppressing field identity cut probe recoverability by up to 91% without moving the gap, and Grad-CAM maps of leak and clean models were indistinguishable.

Not supported, reversed, or open

- H4 (a background context route) and H5 (the gap tracks gradient-reversal suppression) were not supported; the ConvNeXt-T adversary was evaded.
- F1 reversed: CRIC models were *not* pulled toward the slide's majority label.
- F4: strong colour augmentation did not prevent leakage, which qualifies F2.
- Parents are not patients; one training recipe was used throughout; Study 2's registrations still need an external timestamp.

The practical message

Probes, saliency maps, background removal, and colour augmentation cannot certify that an evaluation is free of hierarchical leakage. Only a split at the deepest released parent unit removes it, and the closed forms of Chapter 25 predict which benchmarks are exposed before any model is trained.

1.5 Numbers at a glance

Quantity	Value
Trained models (Study 1, Study 2, Amendment 8)	2,606 (325 + 1,341 + 940)
Evaluation-only runs on saved models (Amendment 8)	640
Training epochs logged (Study 2 and Amendment 8)	45,600
GPU-hours, Study 2 / Amendment 8	102.7 / 49.8
Study 2 matrix wall clock	27.4 h (18 Oct 05:07 to 19 Oct 08:28)
Amendment 8 matrix wall clock	12.8 h (19 Oct 14:03 to 20 Oct 02:52)
Failed tasks, retries, diverged runs	0, 0, 0
Smoke tests before registration	30
Commits	82
Registered confirmatory tests	1 primary + 4 (Holm) + 8 (Holm)

PART II

Study 1: The Original Audit (August–September 2026)

Reading Part II in context

Part II is the Study 1 record exactly as it was compiled on 13 September 2026 from evidence frozen on 7 September. It is reproduced verbatim so that the original claims, their strength, and their stated limitations remain on the record unchanged. Three things happened afterwards that a reader of Part II should keep in mind.

What changed after Study 1

- **The inflation question was re-asked and answered.** Study 1's six per-architecture tests could not certify an accuracy inflation. Study 2 asked the question once, with a crossed mixed model sized by simulation and twenty new seeds, and found 0.97 pp on SIPaKMeD (all twelve architectures positive) and 12.08 pp on CRIC (Part V). The six Study 1 architectures, re-run with fresh seeds, all moved in the same direction (Figure 30.5).
- **The ordering failure stands.** Study 1 registered that P3 $\leq$ P2 $\leq$ P1 and found it failed for four of six architectures. Study 2's P2 $-$ P3 contrast was 0.03 pp ($p = 0.90$): merging near-duplicate-linked parents on top of parent-level splitting changes nothing, so the ordering failure reflects noise between two equivalent protocols, not a defect of P3.
- **Availability is not use.** Study 1 showed the parent fingerprint is learnable. Amendment 8 showed that the fingerprint a probe can decode can be almost entirely removed without touching the inflation (Part VI). Study 1's warning that availability does not prove use was correct, and the distinction turned out to matter.

Where Part II says that per-sample predictions and checkpoints were not retained, that applied to Study 1; every Study 2 and Amendment 8 run kept per-crop logits, final weights, and probe-split features.

2

Study 1 record: status, scope, and abstract

This is the long-form technical record of the CytoFair project. It is intentionally broader than the six-page conference paper. It explains why the project was started, how every analysis was designed, how the code represents the data, what was measured, what worked, what failed, what was changed during execution, what remains unverified, and what artifacts can be reused. The report is written from the repository state and frozen result files; it does not add a new experiment.

Evidence boundary

The quantitative claims in this document come from committed CSV, JSON, Parquet, source, and report artifacts in the repository. The experiment inventory contains 120 SIPaKMeD classification runs, 30 parent-identity probe runs, and 175 Cx22 segmentation runs: 325 completed neural-network runs in total. Structural analyses are additional deterministic or Monte Carlo computations, not neural runs. No post-hoc classifier confusion matrix or qualitative model prediction is invented, because the frozen classifier JSON files retain aggregate metrics and epoch histories but not per-sample predictions or checkpoints.

How to interpret "leakage" in this report

Structural non-independence, literal pixel overlap, and a learnable parent fingerprint are demonstrated. These facts establish that a shortcut is available. They do *not* prove that every cervical-classification model causally uses that shortcut, and the multiplicity-corrected classifier contrasts do not establish a universal accuracy-inflation effect. This distinction is maintained throughout the report.

Abstract of the Study 1 record (13 September 2026)

Cervical-cell datasets are frequently released as crops, even when several crops came from one microscopic field or source image. A random crop split can therefore place sibling crops in both training and test sets, making file-level disjointness look like biological or acquisition-level independence. CytoFair recovers the hidden crop-to-parent hierarchy, validates it with fail-closed checks, measures structural and literal overlap, constructs parent- and component-disjoint splits, probes whether source identity remains learnable after transformations, and rescores classification and segmentation models under matched protocols. The audit recovered all 4,049 SIPaKMeD crops, 1,320 Cx22 regions, 11,534 CRIC cells, and 917 Herlev cells. Under random splitting, 92.62% of SIPaKMeD test crops, 85.23% of Cx22 test regions, and 99.97% of CRIC test cells had a parent in training, whereas Herlev correctly remained at zero. More than half of SIPaKMeD test crops shared literal source pixels with a training crop. A 424-way parent probe reached 25.73% accuracy against 0.2358% chance. All six SIPaKMeD classifier P1–P2

point estimates and all seven Cx22 segmentation estimates were positive, but no confirmatory contrast survived its preregistered Benjamini–Hochberg family. The project therefore supports a strong benchmark-construction conclusion—evaluate at the independent parent unit—while retaining a narrower and appropriately uncertain model-performance conclusion.

3

Executive overview

3.1 The entire project in one page

The initial problem was simple to state: *when a benchmark calls every image file an independent example, is that file really the unit that was independently sampled?* In cervical cytology, the answer can be no. One microscope field can emit several cell crops. If those crops are split independently, their source field can appear on both sides of evaluation.

The project followed a staged answer:

1. define the independent parent unit for four public datasets;
2. recover exact crop-to-parent mappings, using independent recovery branches where the release permits them;
3. refuse downstream analysis unless agreement, geometry, counts, and uniqueness pass;
4. calculate how often random crop splitting separates siblings;
5. detect literal pixel overlap and conservative cross-parent near-duplicate components;
6. create four fixed protocols that isolate independence from training-set size;
7. ask whether parent identity is visually learnable after removing candidate cues;
8. re-run multiple classifier and segmentation families over five fixed seeds;
9. use paired inference, effect sizes, bootstrap intervals, and declared BH families;
10. preserve failed attempts and execution amendments rather than erasing them.

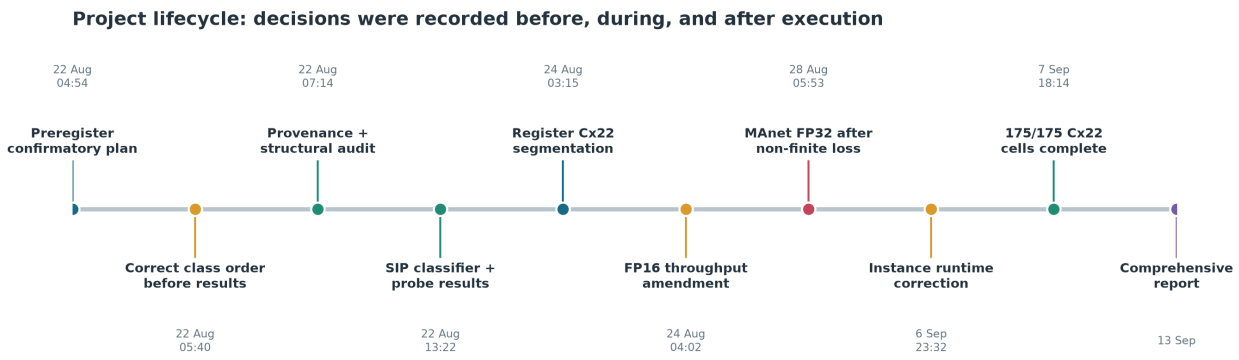


Figure 3.1: The project lifecycle. Blue events register decisions, green events record completed evidence, gold events document execution-driven amendments, red records a numerical failure, and purple marks this report.

3.2 Headline findings, without spin

Table 3.1: Headline conclusions and their correct strength.

Question	Answer	Evidence and boundary
Can provenance be recovered?	Yes	SIPaKMeD branches agree on 4,049/4,049 crops; all boxes are contained; all dataset and class totals match.
Does random crop splitting share parents?	Yes	Mean P1 test-parent overlap is 92.62% on SIPaKMeD, 85.23% on Cx22, and 99.97% on CRIC.
Does the negative control remain negative?	Yes	Herlev has one released cell per parent and returns exactly 0.00% structural leakage for all seeds.
Are literal pixels shared?	Yes, in SIPaKMeD P1	52.62% of test crops share at least one source pixel with a training crop; P2/P3 return zero.
Is source identity learnable?	Yes	A 424-way ResNet-18 reaches 25.73% versus 0.2358% chance; shuffled labels return 0.17% and cover chance.
Is P1 accuracy universally inflated?	Not established	All six SIPaKMeD and all seven Cx22 P1–P2 means are positive, but no contrast survives BH correction.
Does stronger grouping always lower score?	No	SIPaKMeD's registered P3 $\leq$ P2 $\leq$ P1 model ordering fails for four of six architectures. The failure is retained.
Is Cx22's official split defective?	No such claim	The authors' Multi-Train/Multi-Test parent sets are disjoint. The pooled stress test asks what happens under a different downstream reuse pattern.
Is the release fully ready?	Not yet	The clean-clone/virgin-host evidence, lab-paper crosswalk, and independent Figure-1 reader gate remain pending.

4

Motivation and problem formulation

4.1 Why cervical cytology makes independence important

Cervical cancer caused an estimated 660,000 new cases and 350,000 deaths worldwide in 2022 [37]. Cytology screening is a high-volume recognition problem: a small number of clinically important abnormal cells must be found among many plausible normal cells. Automation is attractive because it can help direct expert attention, but only if reported generalization represents genuinely unseen biological or acquisition units.

Modern cervical-cell classifiers report very high crop-level scores. DeepPap reported 98.3% on Herlev [38]; DeepCervix reported 99.14% on SIPaKMeD [24]; and a knowledge-distillation system reported 98.52% on SIPaKMeD [10]. These achievements motivated a basic audit question that is logically prior to choosing a stronger network: *what population unit does the test set represent?*

4.2 Observation, parent, and evaluation unit

Let x_i be a released image observation and let $g(x_i) = p_j$ map it to the acquisition unit that generated it. In this project:

- x_i is a SIPaKMeD cell crop and p_j is a cluster field;
- x_i is a Cx22 generated region and p_j is a CCEDD source image;
- x_i is a CRIC annotated cell and p_j is its source smear image;
- x_i is a Herlev single-cell image and is also its own parent.

The crucial idea is that the *model input* and the *independent sampling unit* need not be the same object. A network consumes x_i, but the statistical independence boundary may be p_j.

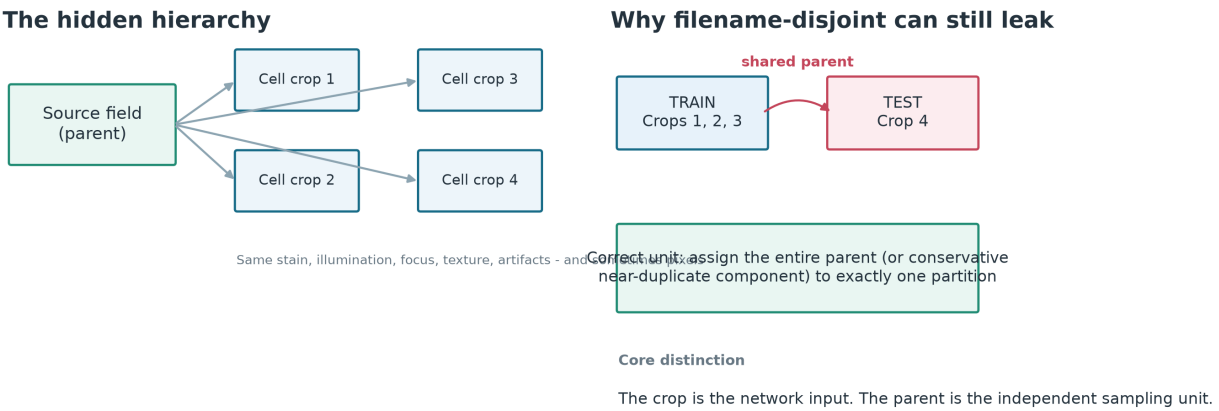


Figure 4.1: The hidden hierarchy and the central failure mode. Several file-disjoint crops can share a parent field, source-specific nuisance factors, or literal source pixels.

4.3 Three different claims that must not be conflated

Claim 1: structural non-independence

Some parents contribute observations to both train and test. This can be established exactly from provenance and split assignments. It requires no model.

Claim 2: shortcut availability

A model can identify the source parent from held-out sibling crops at greater-than-chance accuracy. This shows that parent information survives in the released input. It does not yet show that a disease or morphology classifier uses it.

Claim 3: downstream score inflation

A crop-level protocol produces a larger score than a parent-disjoint protocol under matched models, seeds, and sample sizes. This is an empirical paired comparison. It needs uncertainty, multiplicity correction, and careful interpretation.

The project obtains strong evidence for Claims 1 and 2. For Claim 3, point estimates are directionally consistent across every tested architecture, but the confirmatory families do not cross the multiplicity-adjusted threshold. The report therefore never upgrades "available shortcut" into "proven universal causal inflation."

4.4 Relationship to leakage and duplicate audits

Duplicate-aware evaluation has been studied in general vision: ciFAIR removed near-duplicate CIFAR test counterparts [2]; license-plate work re-evaluated models under duplicate-free splits [17]; and medical-image re-identification demonstrated that patient identity can be highly learnable [22]. Kapoor and Narayanan place these problems in a broader taxonomy of data leakage and reproducibility failures [15]. CytoFair's distinctive concern is an exact cervical-cytology source hierarchy: sibling crops can be non-independent without being duplicate files.

5

Preregistration and analysis governance

5.1 Why the plan was committed first

The original preregistration was committed at 04:54:10 IST on 22 August 2026 (commit `67e8ded`). The first result commit was recorded at 13:22:42 IST that day (commit `1c2da7f`). The plan therefore predates the committed results. This matters because a five-seed matrix contains many possible comparisons; fixing endpoints, seeds, exclusions, and multiplicity families before seeing scores reduces analyst degrees of freedom.

The fixed confirmatory seeds were $17, 29, 43, 71, 101$. Five additional robustness seeds were named but were not needed for the reported confirmatory summaries and were not substituted for any unfavorable seed. Every protocol-architecture cell had to contain at least five valid paired runs before inferential output could be emitted.

5.2 Registered endpoints

The structural endpoints were (i) fraction of parents straddling train and test and (ii) fraction of test observations whose parent appears in training. The mechanistic endpoint was 424-way parent-identification accuracy. The SIPaKMeD model endpoint was test accuracy; macro-F1 and balanced accuracy were concordance metrics. For Cx22 semantic segmentation, the primary endpoint was mean per-image cytoplasm/nucleus Dice. For Cx22 instance segmentation, the primary endpoint was per-image mean ground-truth-instance Dice after maximum-weight matching, with unmatched targets assigned zero.

5.3 Amendments: what changed and why

Table 5.1: Registered amendments and their evidential status.

Date	Amendment	Trigger	Why it did not select a favorable result
22 Aug	SIPaKMeD class order	The integrity gate exposed a transcription swap between Metaplastic and Dyskeratotic.	Corrected before any SIPaKMeD result; numeric target vector 126/108/238/271/223 did not change.
24 Aug	Cx22 segmentation extension	Structural Cx22 results motivated a separately scoped model rescore.	Registered before any Cx22 model was trained; official split retained as descriptive control.
24 Aug	FP16 execution	Full FP32 preflight left three A6000s under-used and made one U-Net epoch exceed 4.5 minutes.	Changed numerical execution before an epoch emitted a result; models, splits, seeds, endpoints, and epoch count stayed fixed.
28 Aug	MAnet in FP32	The first MAnet P1 seed-17 attempts produced non-finite losses at epochs 23 and 24.	No completed MAnet result existed. All MAnet cells restarted from their fixed seeds rather than selecting a pre-divergence checkpoint.
28 Aug	Deterministic CUDA workspace	Startup warnings showed deterministic matrix multiplication required `CUBLAS_WORKSPACE_CONFIG=:4096:8`.	Frozen before a corrected MAnet or instance epoch existed; applied to all remaining cells.
6 Sep	Instance runtime correction	Evaluation every epoch, CPU mask matching, and frequent network checkpoint writes caused severe under-utilization.	The registered endpoint was final epoch. Intermediate test inference was removed, exact binary intersections moved to GPU, and recovery checkpoints became five-epoch; final computation stayed identical.

A useful demotivation

The project began with the possibility of a large, architecture-general accuracy correction. The statistics did not support that broad claim. Instead of hiding this, the project narrowed its claim: the benchmark hierarchy is demonstrably non-independent, the source signal is demonstrably learnable, and parent grouping removes the measured sharing. The size of downstream score inflation remains model- and task-dependent.

6

Datasets and the hidden hierarchy

6.1 Overview

Table 6.1: Audited datasets and independent units.

Dataset	Observations	Parents	Mean/parent	Released task	Parent definition
SIPaKMeD	4,049	966	4.19	5-class classification	cluster field
Cx22	1,320	514	2.57	semantic/instance segmentation	CCEDD source image
CRIC	11,534	400	28.84	cytology classification resource	source smear image
Herlev	917	917	1.00	7-class single-cell classification	released cell image

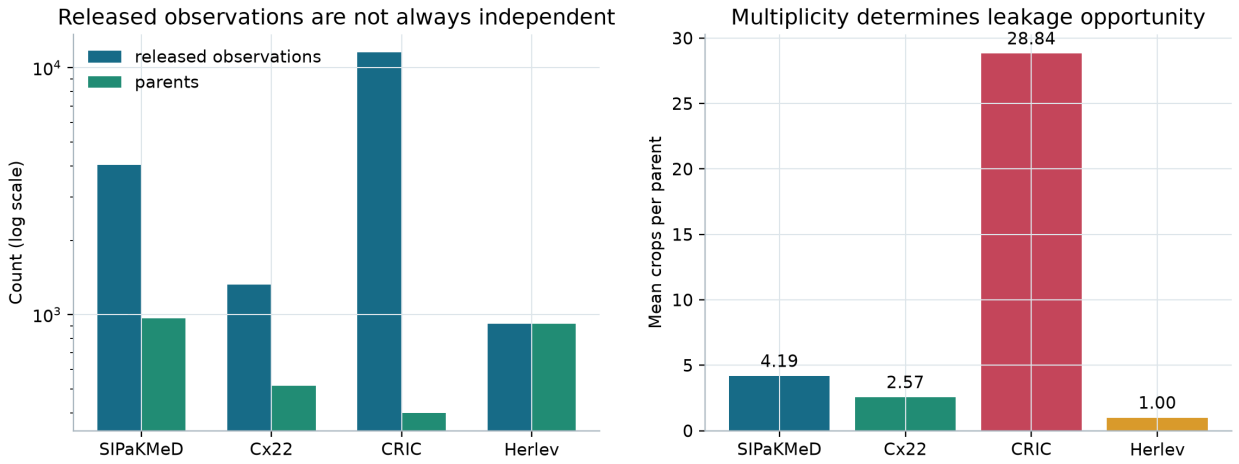


Figure 6.1: Dataset scale and mean parent multiplicity. The log-scale left panel prevents CRIC's 11,534 observations from visually erasing the smaller datasets.

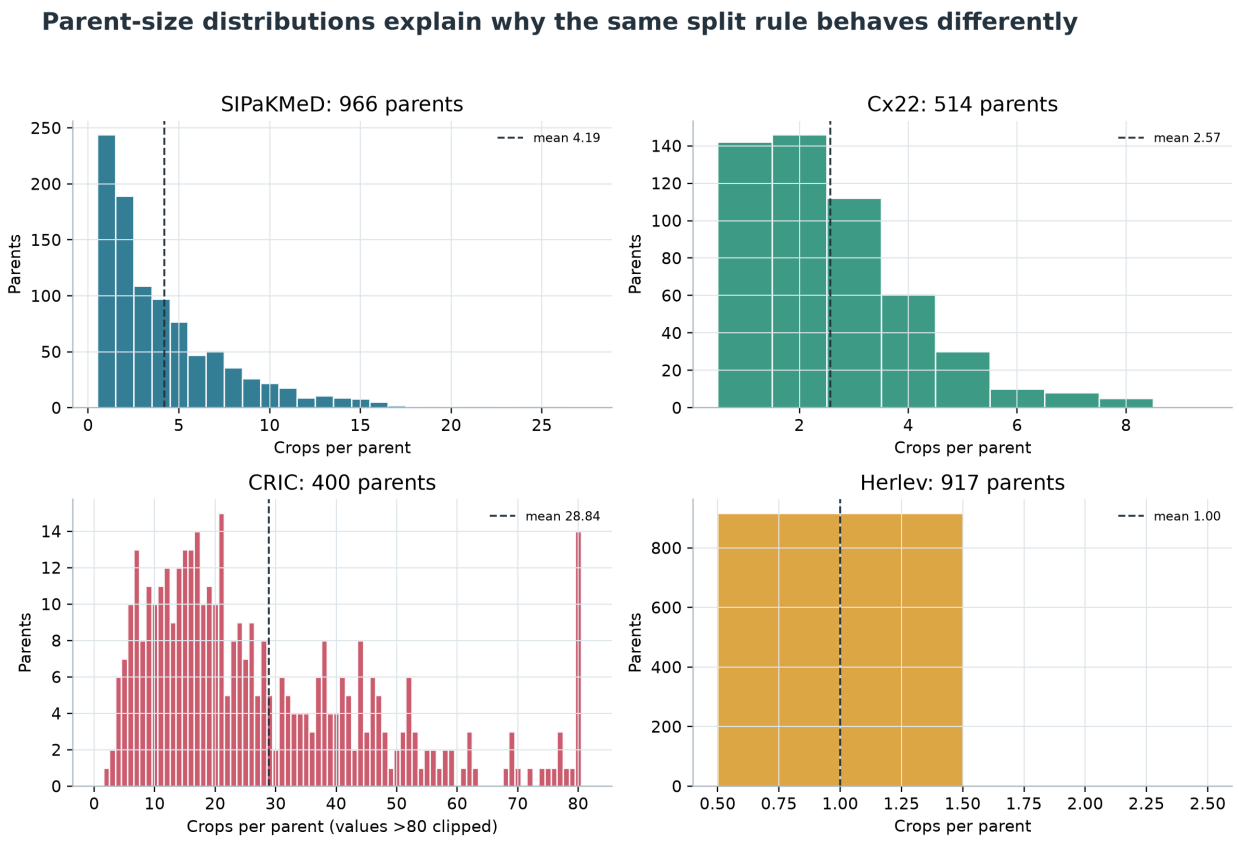


Figure 6.2: Complete parent-size distributions recovered from the Parquet maps. High multiplicity drives the probability that random crop splitting places a parent in both arms.

6.2 SIPaKMeD

SIPaKMeD contains 4,049 isolated cell images derived from 966 cluster fields [23]. The released hierarchy is unusually valuable for an audit: full fields and a `CROPPED` directory coexist, and crop filenames encode the field identifier. The project uses the cluster field—not the crop file—as the parent.

Table 6.2: Independently reconstructed SIPaKMeD class totals in release order.

Class	Crops	Parents	Crops/parent
Superficial–Intermediate	831	126	6.60
Parabasal	787	108	7.29
Koilocytotic	825	238	3.47
Metaplastic	793	271	2.93
Dyskeratotic	813	223	3.65
Total	4,049	966	4.19

The class order deserves explicit mention because the original planning text swapped the final two names even though its numerical vector was right. The integrity gate exposed that inconsistency, and Amendment 1 corrected it before results existed. This is an example of a check preventing a quiet metadata error.

6.3 Cx22 and CCEDD

Cx22 provides 1,320 generated regions and instance labels, drawing source images from CCEDD [19]. Three released subsets are involved: Multi-Train, Multi-Test, and Pair. The parent is the named CCEDD source image. Generator metadata provide source name and requested ROI coordinates; the authors' MATLAB generator defines how a source rectangle is clipped, resized, and pasted into a 512-by-512 white canvas.

The official Multi-Train and Multi-Test arms have zero shared parents. Pair is separately released and shares 55 parents with Multi-Test (Jaccard 0.126) and 233 with Multi-Train (Jaccard 0.475). The later model matrix pools all 1,320 observations only as a preregistered stress test of downstream reuse. It is not an accusation that the official authors used a leaking split.

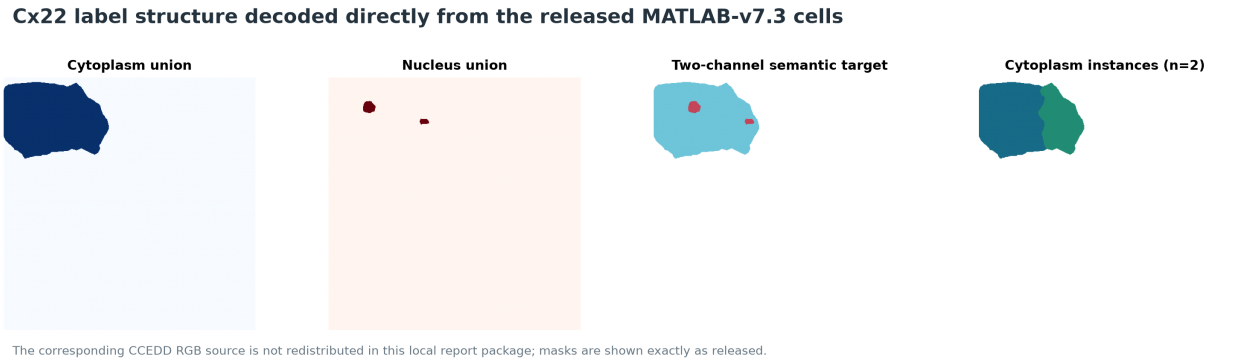


Figure 6.3: One released Cx22 Pair target decoded from the MATLAB-v7.3 structures. Cytoplasm and nucleus unions define two non-exclusive semantic channels; individual cytoplasm masks define the instance task. The CCEDD RGB source is not redistributed in this report.

6.4 CRIC

CRIC contains 11,534 classified cells associated with 400 source images [27]. The released classification table supplies source identity and cell coordinates, which makes source-level grouping deterministic. A reliable patient identifier is not present in the released table, so the project does not claim patient-disjoint evaluation. This matters: source-image grouping is stronger than crop grouping but can still be weaker than patient grouping.

6.5 Herlev as a structural negative control

Herlev contains 917 released single-cell images [14]. Under the project's declared data unit, each image is one observation and one parent. It therefore has no multi-observation parent structure and must return exactly zero parent straddling. If Herlev showed a nonzero structural effect, the implementation would be wrong. The control passed for all five seeds.

6.6 Data-use constraints

The repository publishes identifiers, coordinates, hashes, split assignments, code, and derived statistics. It does not redistribute raw images. Cx22/CCEDD material is restricted to non-commercial research/education in the declared source manifest, and CCEDD redistribution is prohibited. CRIC and the local Herlev repackaging are declared CC BY 4.0. SIPaKMeD's source

notes require citation and experimental use. The complete URL/checksum/terms ledger appears in table D.1.

7

System architecture and fail-closed philosophy

7.1 Pipeline

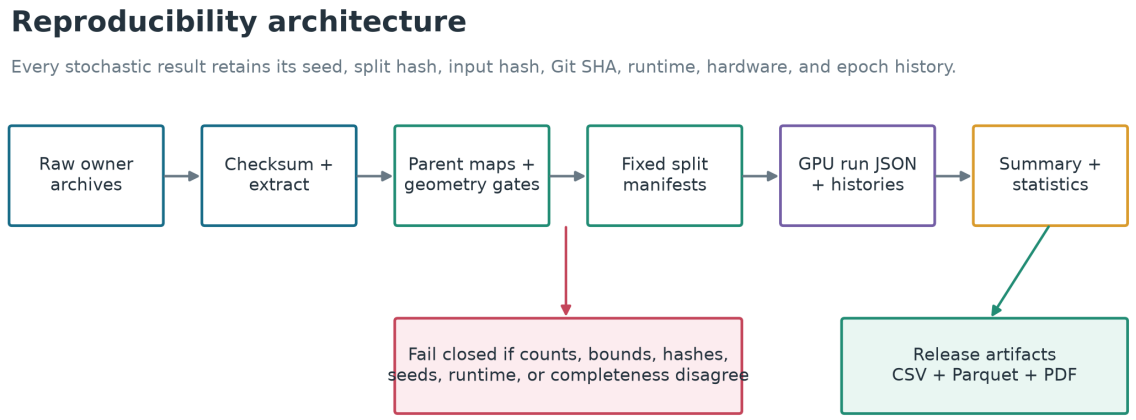


Figure 7.1: Reproducibility architecture. Deterministic provenance and split artifacts gate the stochastic GPU experiments, and every final result carries the hashes needed to reconnect it to its inputs.

The system deliberately separates three artifact layers:

1. **Canonical identity:** exact or released parent IDs used to construct splits.
2. **Independent audit:** image-matched parent IDs, boxes, confidence, runner-up margin, and an auxiliary diagnostic flag.
3. **Sensitivity grouping:** P3 connected-component IDs built from thresholded cross-parent near-duplicate candidates.

Keeping these layers separate prevents a lower-confidence image match from overwriting an exact filename identity. It also prevents a conservative pHash/dHash candidate from being narrated as a proven duplicate.

7.2 Core artifact schemas

The canonical SIPaKMeD map records `crop_id`, `parent_id`, `class_name`, paths, recovery method, and confidence. Branch B adds x, y, w, h, crop and parent dimensions, coarse and refined NCC, second-best confidence, margin, method, and diagnostic `accepted`. P3 adds `p3_group_id`. Split CSVs record `crop_id`, `parent_id`, `partition`, `protocol`, and `seed`.

Every neural result JSON records task, architecture, protocol/mode, seed, primary and secondary metrics, train/test counts, hyperparameters, exact split/input hashes, Git SHA, timestamps, elapsed time, visible GPU, platform and library versions, and full epoch history. This is more useful than a final number alone: it permits matrix completeness checks, runtime analysis, and learning-curve reconstruction without checkpoints.

7.3 Fail-closed algorithm

Listing 7.1: Conceptual fail-closed workflow.

```
for each released dataset:
    recover exact/released parent identity
    if an independent recovery branch exists:
        recover parent and bounding box without consulting branch A
        require >= 99% A/B agreement
    require one parent per observation
    require all boxes contained and all totals exact

construct P1, P1-prime, P2, and P3 for every fixed seed
measure structural and literal overlap
run probe/classifier/segmenter matrix with bounded GPUs

before summarizing:
    require every registered cell and seed exactly once
    require split hashes, runtime versions, and sample counts
    pair comparisons by seed
    attach an effect size and interval to every p-value
    apply BH over the entire declared family once

if any required evidence is absent:
    emit FAIL or PENDING; never infer PASS
```

8

Exact provenance recovery

8.1 SIPaKMeD Branch A: filename hierarchy

Branch A parses the released directory and filename convention. For example, `im_Dyskeratotic/CROPPED/001_03.bmp` maps to parent `Dyskeratotic/001`. This route is deterministic, fast, and used for the canonical split map. Its weakness is that a parsing bug could produce internally tidy but wrong identifiers. That is why Branch B was required.

8.2 SIPaKMeD Branch B: independent content matching

For crop x and candidate parent region r, normalized cross-correlation is

$$\mathrm{NCC}(x, r) = \frac{\sum_u (x_u - \bar{x})(r_u - \bar{r})}{\sqrt{\sum_u (x_u - \bar{x})^2} \sqrt{\sum_u (r_u - \bar{r})^2}}. \tag{8.1}$$

Branch B never uses Branch A's parent assignment in the query. It ranks every same-class parent at quarter resolution, refines the five best candidates at full resolution, and records the winning location as a bounding box. A multiscale fallback exists only for unresolved cases. Matching is bounded by class working sets so the complete parent image collection does not have to remain in memory.

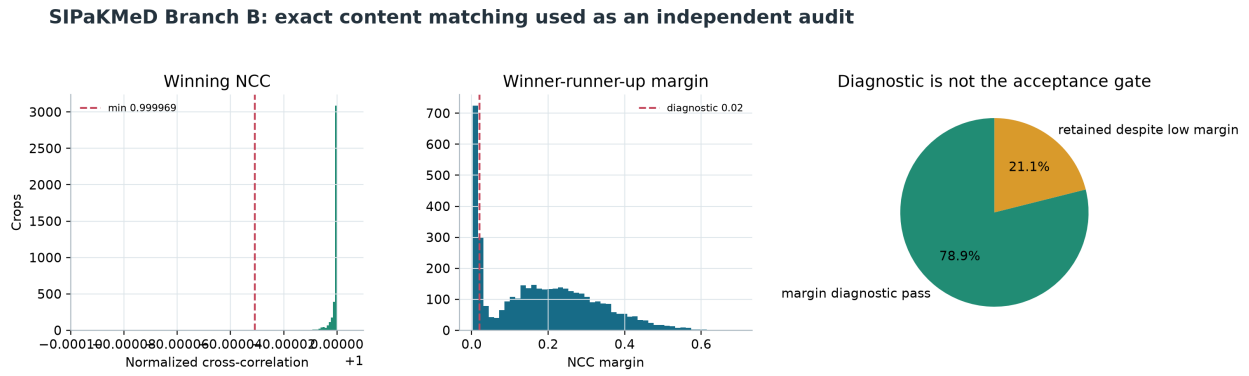


Figure 8.1: Branch-B diagnostics. The minimum winning NCC is 0.999969. The 0.02 runner-up-margin rule is intentionally only a diagnostic: 3,195 crops pass it and 854 do not, yet all rows agree with Branch A and all boxes remain contained.

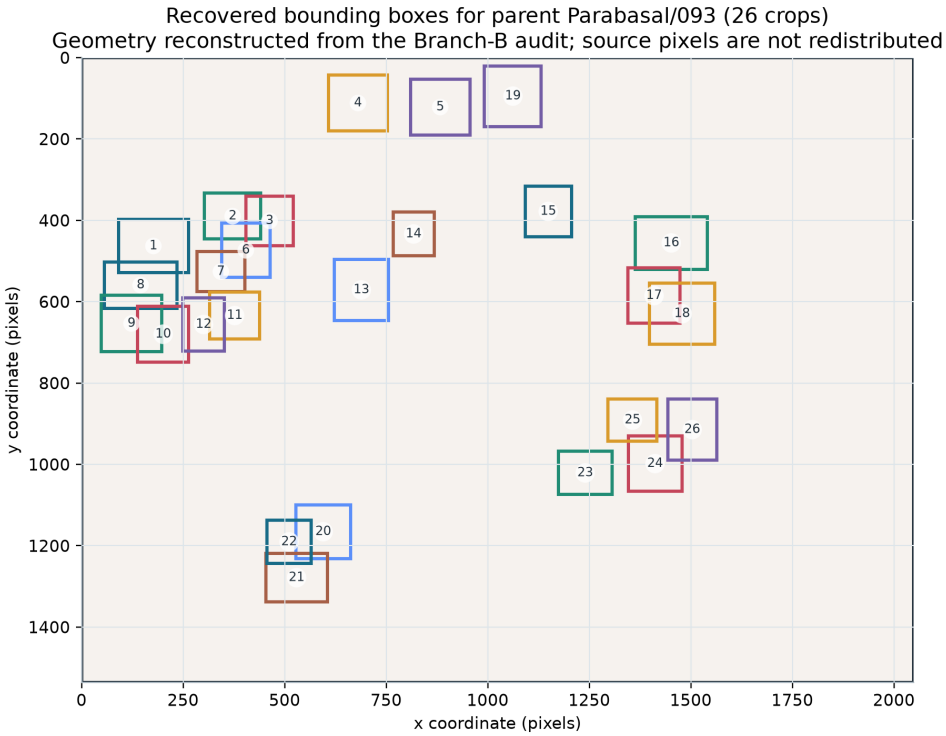


Figure 8.2: A real recovered-coordinate example for the parent with the largest crop count. Only rectangle geometry is displayed; source pixels are not redistributed. Overlapping rectangles make literal source-pixel sharing possible.

8.3 Why the auxiliary acceptance flag is false for many correct rows

The `accepted` field in Branch B encodes a conservative confidence-plus-margin diagnostic. It is not the registered provenance acceptance outcome. Many microscope fields contain visually repetitive background, so a correct near-perfect winner can have another near-perfect candidate. Substituting this diagnostic for the registered 99% cross-branch agreement rule would discard 854 correctly mapped crops after looking at an arbitrary margin. The project instead retains every row and exposes the ambiguity measure.

8.4 Cx22, CRIC, and Herlev recovery

Cx22 uses source names and ROI coordinates decoded from the authors' three MATLAB generator releases. Every source resolves against the declared 686-image CCEDD set, and rare edge-touching requested boxes are clipped exactly as the generator code specifies. The released Cx22 archives contain generator metadata and masks rather than a second independent RGB crop array, so an A/B image-content branch cannot be constructed from the public package alone. Applicable checks are source existence, exact metadata decoding, generator geometry, bounds, and subset overlap.

CRIC uses the released classifications table: source-image identity and coordinates directly define the map. Herlev assigns each released cell to itself. Neither route pretends to recover unavailable patient identity.

8.5 Verification outcome

Table 8.1: Provenance gates.

Dataset	Observations	Parents	Applied evidence	Outcome
SIPaKMeD	4,049	966	4,049/4,049 A/B agreement; 4,049 contained boxes; exact class totals	**PASS**
Cx22	1,320	514	generator names/ROIs; 686 sources resolve; 1,320 contained clipped boxes	**PASS**
CRIC	11,534	400	released source identities and coordinates; no patient-ID claim	**PASS**
Herlev	917	917	one released cell per parent	**PASS**

No observation was excluded. The five SIPaKMeD parent counts are exactly 126/108/238/271/223 in corrected release order. All crop counts sum to 4,049.

9

Split protocols and structural leakage

9.1 Why four protocols are needed

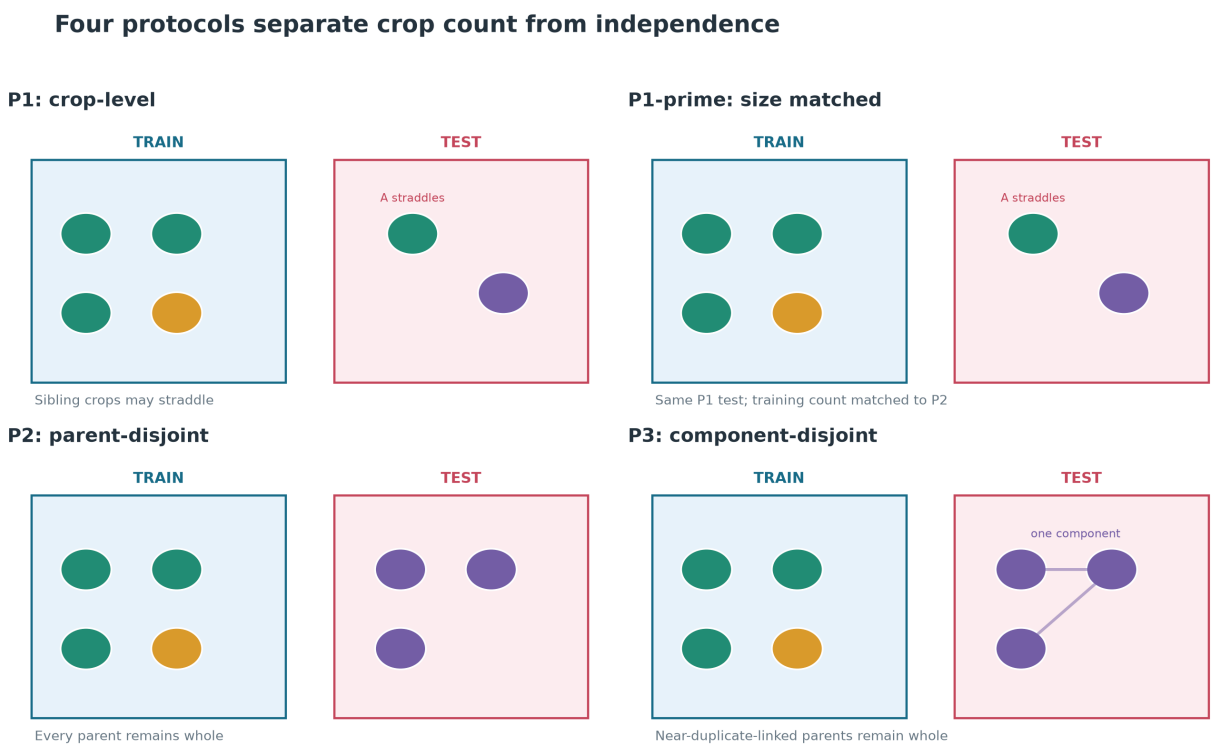


Figure 9.1: The four protocols. P1 asks what common crop splitting shares. P1-prime holds its test set but matches the P2 training count. P2 groups exact parents. P3 groups whole near-duplicate-linked parent components.

All splits use an 80:20 target and the five fixed seeds. P1 is class-stratified at crop level. P1-prime retains P1's test observations and downsamples only its training observations to exactly the P2 training count, as closely class-balanced as integer counts permit. P2 uses group-aware allocation by exact parent. P3 repeats parent-disjoint allocation after cross-parent candidate links are collapsed into connected components.

On the frozen SIPaKMeD environment, P2 contains 3,239 train and 810 test crops for every seed. P1 already has exactly those counts, so P1-prime is identical to P1. This is not a redundant

hidden comparison: it shows that the primary P1–P2 difference cannot be attributed to a larger P1 training set in this implementation.

9.2 Analytic straddle probability

For a parent with k crops independently assigned to training with probability 0.8 and test with probability 0.2, the only non-straddling outcomes are "all train" and "all test." Hence

$$P(\text{no straddle} \mid k) = 0.8^k + 0.2^k, \tag{9.1}$$

$$P(\text{straddle} \mid k) = 1 - 0.8^k - 0.2^k. \tag{9.2}$$

For a dataset with parent crop counts $k_1, \ldots, k_m$, the analytic expectation is

$$\bar{P}_{\text{straddle}} = \frac{1}{m} \sum_{j=1}^{m} \left(1 - 0.8^{k_j} - 0.2^{k_j}\right). \tag{9.3}$$

The exact class-stratified fixed-size splitter is not perfectly independent Bernoulli assignment. Therefore the analytic quantity is checked against a 10,000-allocation Monte Carlo interval generated with the actual count vector and allocation rule.

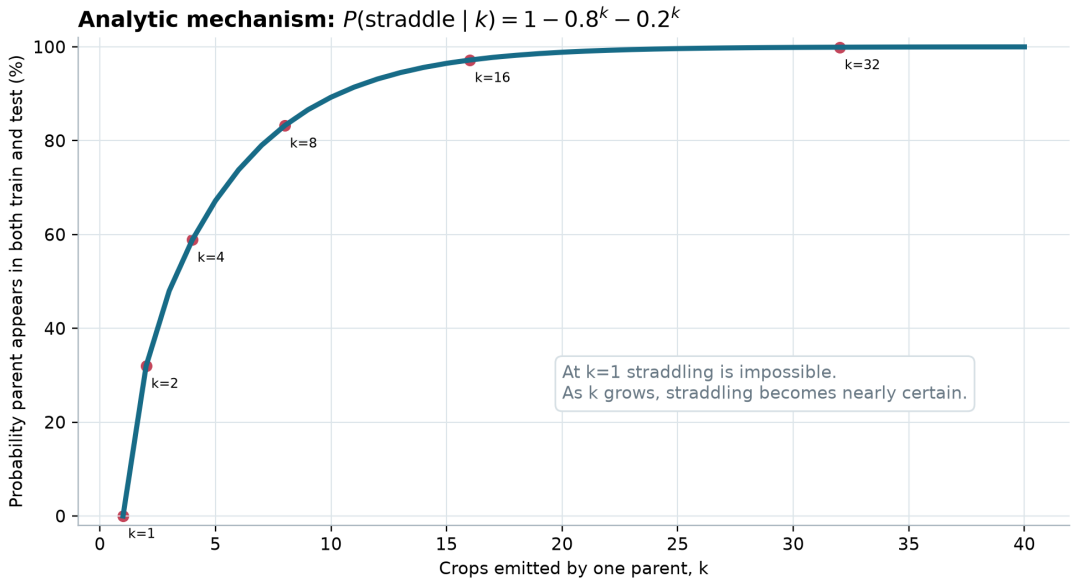


Figure 9.2: The analytic mechanism. A one-crop parent cannot straddle; a high-multiplicity parent almost certainly straddles under independent 80:20 crop allocation.

9.3 Structural endpoints

Let G_{tr} and G_{te} be training and test parent sets, and let T be the test observations. The parent-straddle fraction and leaked-test-observation fraction are

$$L_p = \frac{|G_{\text{tr}} \cap G_{\text{te}}|}{|G|}, \qquad L_c = \frac{1}{|T|} \sum_{x \in T} \mathbf{1}\{g(x) \in G_{\text{tr}}\}. \tag{9.4}$$

Table 9.1: Structural leakage. Stochastic values are mean ± sample SD over five seeds.

Dataset	Crops	Parents	Analytic P1 straddle	Observed P1 straddle	P1 test with train parent	P2/P3 straddle
SIPaKMeD	4,049	966	44.83%	45.63±1.04%	92.62±0.40%	0.00±0.00%
Cx22	1,320	514	33.93%	34.01±1.10%	85.23±3.51%	0.00±0.00%
CRIC	11,534	400	94.88%	94.35±1.10%	99.97±0.08%	0.00±0.00%
Herlev	917	917	0.00%	0.00±0.00%	0.00±0.00%	0.00±0.00%

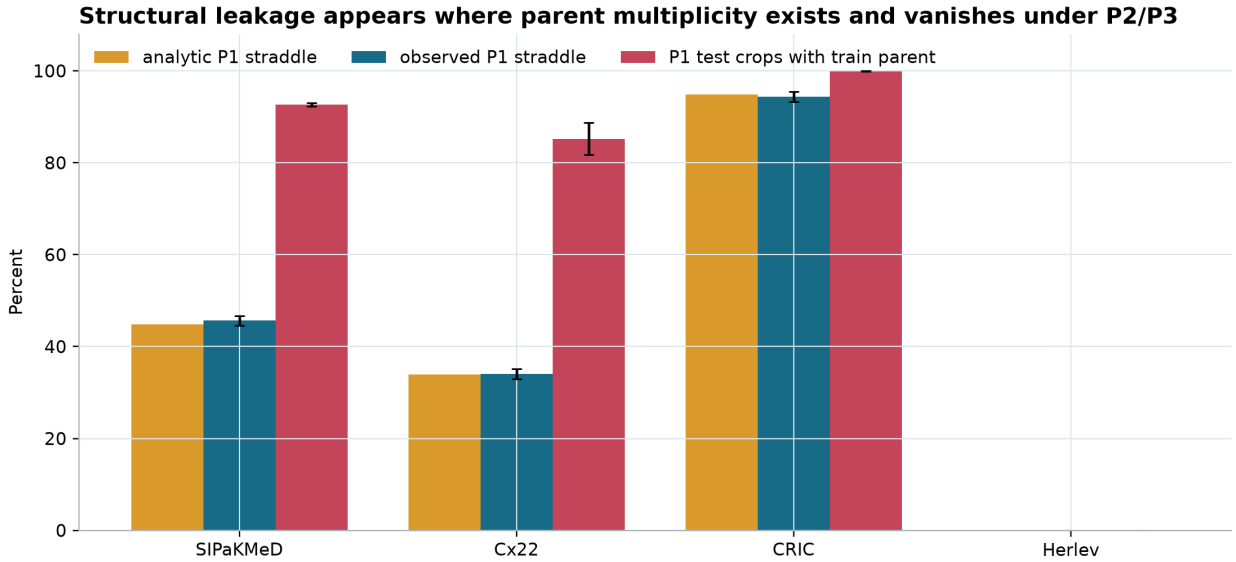


Figure 9.3: Analytic and observed leakage. P2 and P3 are omitted from the bars because both are exactly zero on all four datasets.

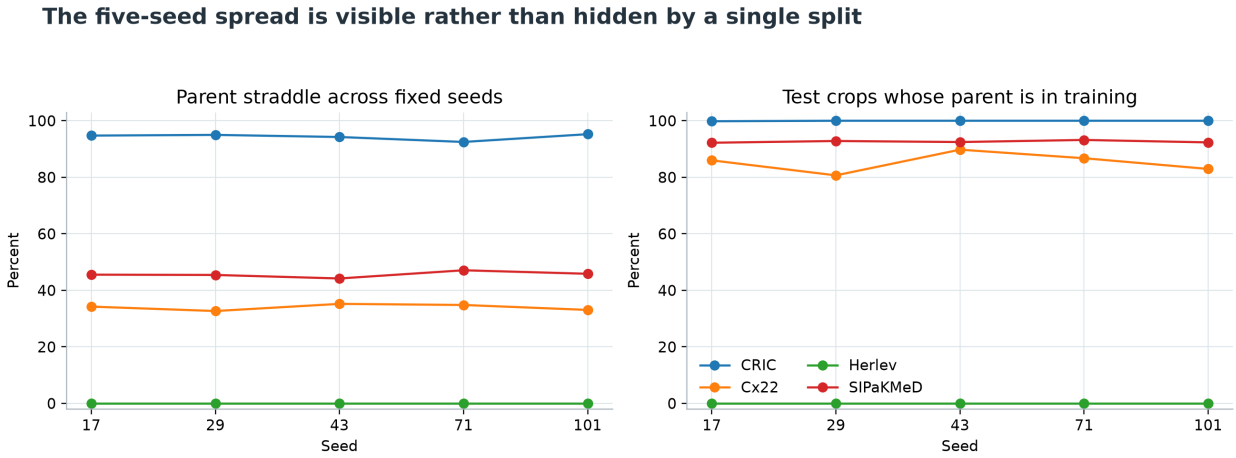


Figure 9.4: Seed-level structural results. Reporting the spread prevents one convenient split from defining the claim.

9.4 Interpretation

CRIC is the most extreme because its parents emit 28.84 crops on average: essentially every random-split test cell has a source image in training. SIPaKMeD and Cx22 have lower parent multiplicity, yet 92.62% and 85.23% of their test observations respectively still have a training sibling. Herlev establishes implementation specificity: the same code returns zero when the data

genuinely have one observation per parent.

The structural ordering P3 $\leq$ P2 $\leq$ P1 passes for all datasets. This statement concerns shared parents, not model scores. Later, the analogous model-score ordering fails for four of six SIPaKMeD classifiers; the two outcomes are logically compatible.

10

Literal overlap and conservative duplicate grouping

10.1 Literal source-pixel overlap

SIPaKMeD's recovered boxes permit a stronger question than shared parent identity. Let boxes B_i and B_j come from the same field. Their intersection is non-empty when

$$\max(x_i, x_j) < \min(x_i + w_i, x_j + w_j) \quad \text{and} \quad \max(y_i, y_j) < \min(y_i + h_i, y_j + h_j). \tag{10.1}$$

The intersection-over-union is

$$\mathrm{IoU}(B_i, B_j) = \frac{|B_i \cap B_j|}{|B_i \cup B_j|}. \tag{10.2}$$

Across 13,064 within-parent crop pairs, 2,185 share at least one pixel and the maximum IoU is 0.546. Under P1, 16.80±0.77% of cross-split sibling pairs overlap and 52.62±1.97% of test crops overlap at least one source pixel with a training crop. Both quantities are 0.00±0.00% under P2 and P3.

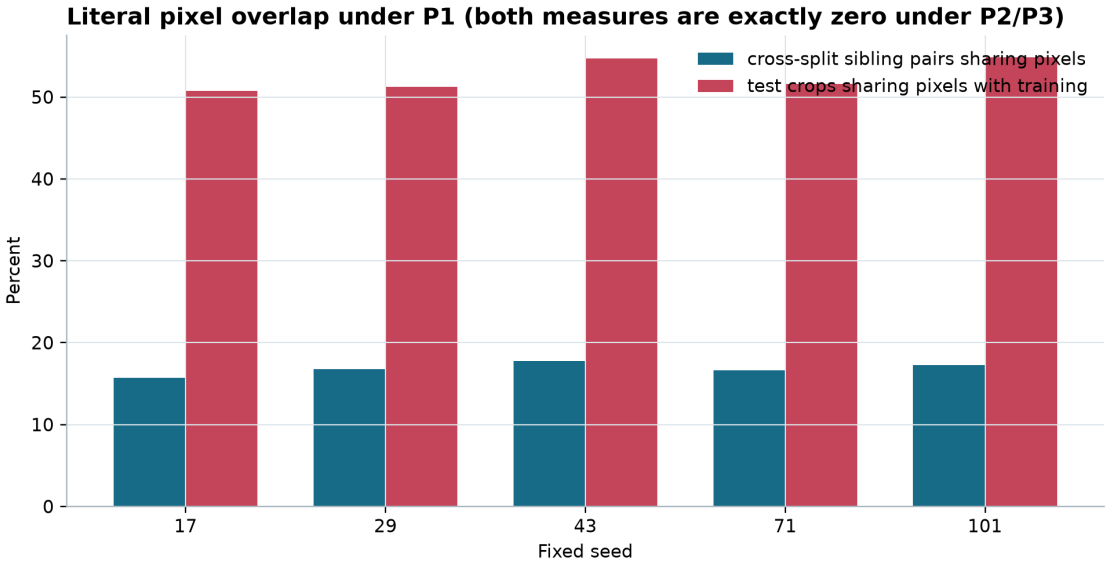


Figure 10.1: Literal overlap for every P1 seed. The larger test-crop fraction answers a different question from the pair fraction: whether each test crop has at least one overlapping training sibling.

10.2 Perceptual hashes and P3

For each crop the implementation computes a 64-bit perceptual hash from the median-thresholded low-frequency 8×8 DCT block and a 64-bit difference hash from adjacent comparisons in an 8×9 downsample. Two crops become a candidate pair when they are pixel-exact or when both Hamming distances are at most six. Cross-parent candidate pairs are joined by union-find; each connected component becomes one P3 group.

At the frozen thresholds, the audit found 201 cross-parent candidate pairs. They connect 170 parent IDs into 74 components, the largest containing eight parents, and reduce 966 parents to 870 P3 groups. No pair is pixel-exact. A synthetic calibration detected all three injected transformations and rejected all 100 deterministic unrelated-parent negatives, yielding precision 1.00 and recall 1.00 on that calibration set.

P3 is a conservative sensitivity grouping, not a claim that every candidate is a duplicate

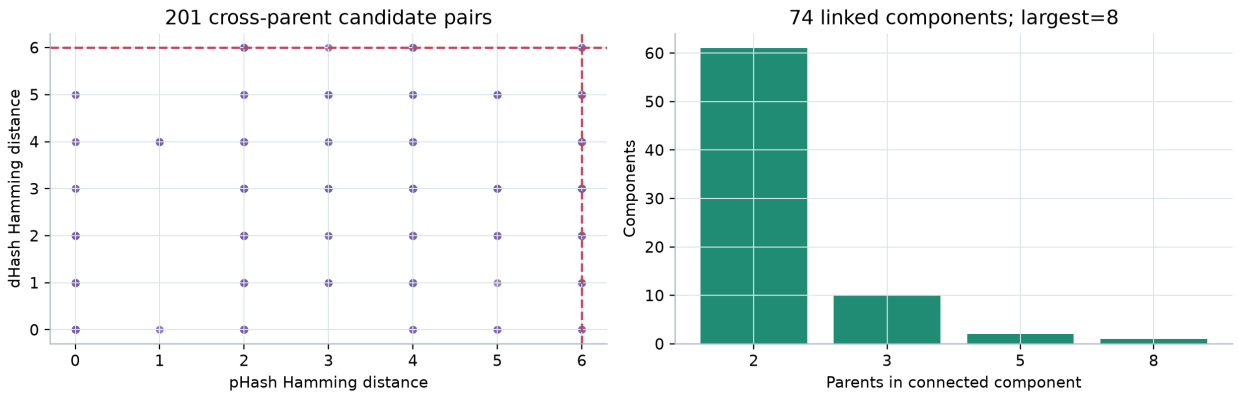


Figure 10.2: Near-duplicate sensitivity audit. The thresholded candidates are used conservatively for grouping; they are not treated as ground-truth duplicate annotations.

What the duplicate experiment did not do

The broader planning idea mentioned a three-detector majority vote involving learned features. The frozen release implements only the pHash+dHash consensus audit and its synthetic calibration. Therefore the report does not claim a CNN/domain-encoder duplicate ensemble, human-confirmed duplicate truth, or general detector precision outside the calibration set.

11

The parent-identity mechanism probe

11.1 Motivation

Structural overlap says that siblings are present. The parent probe asks whether their shared source is visually recoverable after cellular content and color cues are altered. A positive answer makes the shortcut concrete: a learner can recognize the parent field from a different crop. It still does not prove that a morphology classifier uses that signal.

11.2 Balanced task construction

Parents with at least four crops are eligible. The minimum eligible size allows two training and two test crops per parent. To make chance exact, each eligible parent contributes the same number of training and test crops. The final task contains 424 parents, 848 training crops, and 848 test crops, so chance is

$$P_{\text{chance}} = \frac{1}{424} = 0.00235849 = 0.235849\%. \tag{11.1}$$

A ResNet-18 [11] initialized from ImageNet replaces its final layer with a 424-way linear classifier. It trains for 15 epochs with AdamW [20], learning rate 10^{-4}, weight decay 10^{-4}, batch size 64, and fixed seeds. Training augmentation is a random resized crop with scale 0.85–1.0 plus horizontal and vertical flips; testing resizes to 224×224. ImageNet normalization follows every image-mode transform.

11.3 Image transformations and their motivations

11.3.1 RGB

RGB is the maximal-information condition. It permits morphology, stain, illumination, background, focus, acquisition artifacts, and any other visible cue.

11.3.2 Grayscale

Grayscale removes hue and saturation while retaining luminance, edges, morphology, focus, and texture. Persistence here means identity is not solely a stain-color phenomenon.

11.3.3 LAB stain normalization

RGB is converted to CIE LAB. For channel vector z, source mean μ_s, source standard deviation σ_s, and fixed target statistics $\mu_t = (190, 140, 135)$ and $\sigma_t = (35, 14, 14)$, the implemented transform is

$$z' = \text{clip}\left((z - \mu_s) \odot \frac{\sigma_t}{\max(\sigma_s, 1)} + \mu_t, 0, 255\right). \tag{11.2}$$

This is a fixed-statistics color-transfer normalization based on the logic of Reinhard et al. [26]. Persistence means simple first- and second-moment color normalization does not erase the source fingerprint.

11.3.4 Background-only: Otsu, morphology, and Telea inpainting

This condition deliberately removes the likely cell foreground. The implementation converts RGB to HSV and grayscale, then independently obtains Otsu thresholds [21] on saturation S and luminance Y. The preliminary foreground mask is

$$M_0(u) = \mathbf{1}\{S(u) > t_S \ \vee \ Y(u) < t_Y\}. \tag{11.3}$$

A 5×5 morphological closing fills small holes and connects narrow gaps:

$$M = (M_0 \oplus K) \ominus K. \tag{11.4}$$

Finally, the marked region is filled from its boundary with Telea's fast-marching inpainting method using radius 5 [34]. The result keeps background statistics while suppressing the cell body and nucleus. A successful classifier in this condition shows that background alone carries considerable parent information.

11.3.5 High-pass

For Gaussian blur $G_{\sigma=3} * I$, the residual is

$$I_{\text{hp}} = \text{clip}\left(I - G_{\sigma=3} * I + 128, 0, 255\right). \tag{11.5}$$

This suppresses low-frequency illumination and emphasizes local texture, edges, compression, and focus structure.

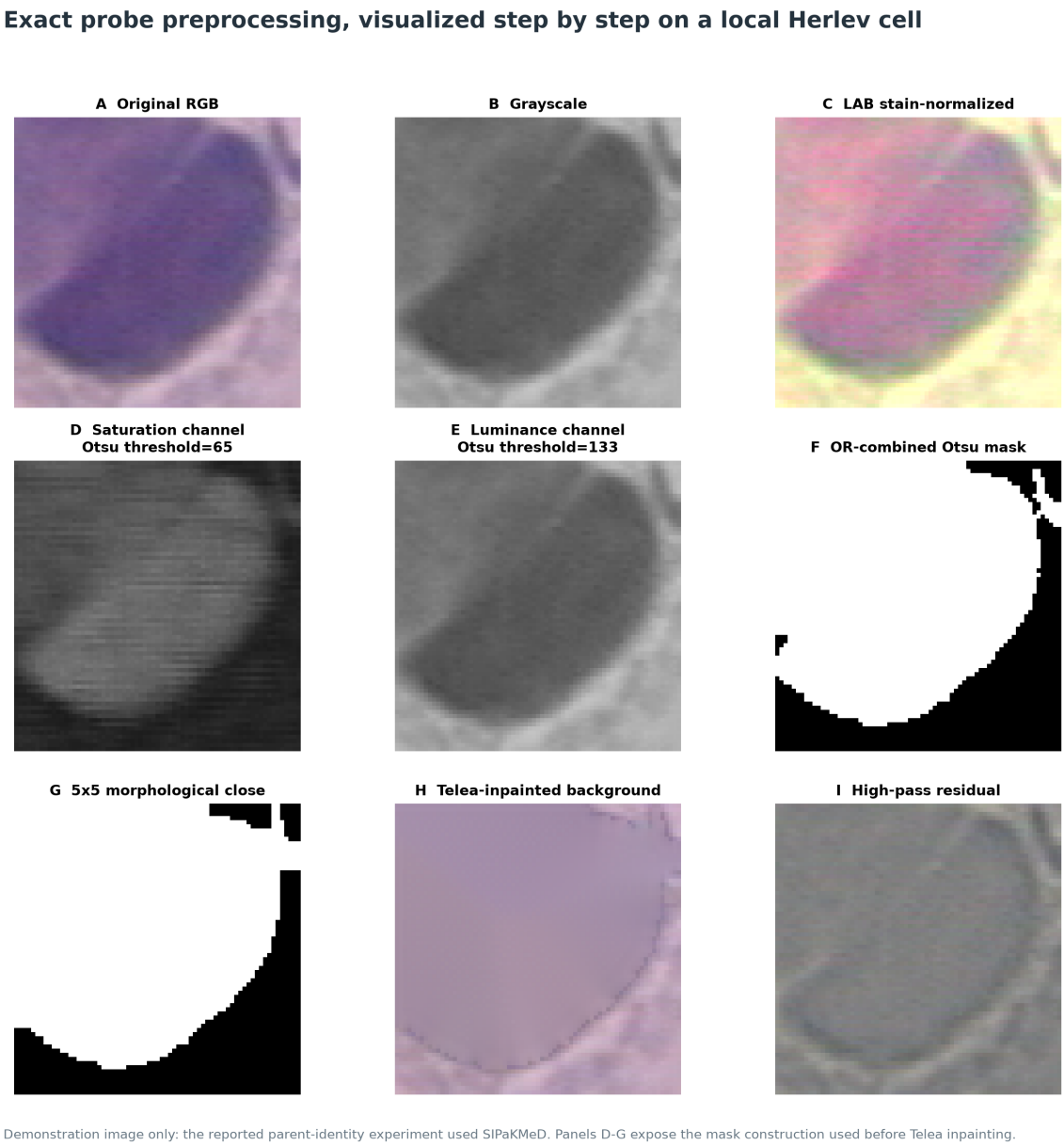


Figure 11.1: Every preprocessing stage shown on an actual local Herlev cell. This is an implementation demonstration, not a probe result image: the reported parent-identification experiment used SIPaKMeD crops. Panels D–G make Otsu thresholding, mask combination, and morphological closing visible before Telea inpainting.

11.4 Shuffled-label negative control

Only the training labels are permuted, using the same images, optimizer, epochs, seeds, and task size. The five-seed 95% interval must cover the exact chance baseline. Otherwise the probe would be rejected because a bookkeeping or imbalance artifact could explain apparent identity learning.

11.5 Results

Table 11.1: Parent-identity probe. Mean ± sample SD over five seeds.

Input condition	Accuracy	Multiple of chance
RGB	25.73±0.66%	109.1×
Grayscale	13.80±1.12%	58.5×
Stain-normalized	17.31±0.64%	73.4×
Background-only	18.68±1.25%	79.2×
High-pass	16.27±0.60%	69.0×
Shuffled RGB labels	0.17±0.24%	0.7×

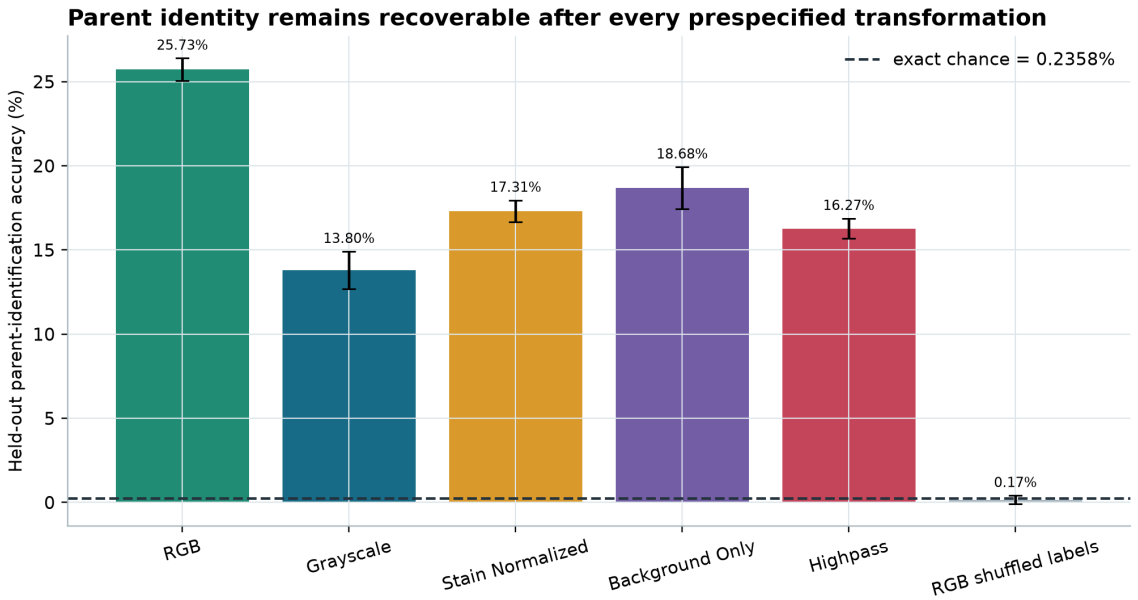


Figure 11.2: All prespecified image conditions remain far above chance. Error bars are sample SD across five seeds.

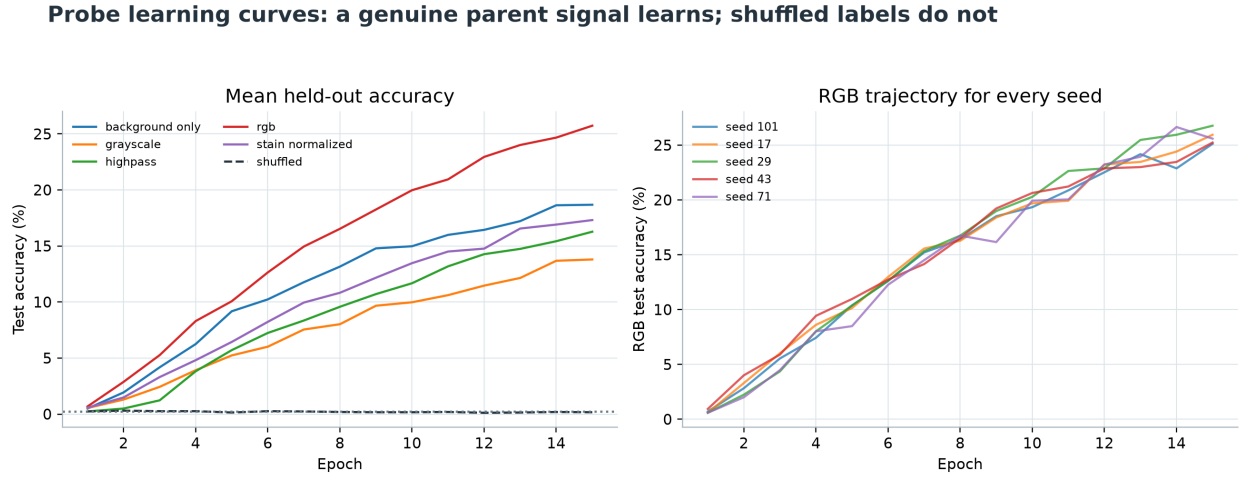


Figure 11.3: Probe learning curves reconstructed from every retained epoch history. The RGB signal rises smoothly across seeds; shuffled labels remain near chance.

The shuffled mean is 0.1651%, SD 0.2445 percentage points, with a five-seed interval from −0.1385% to 0.4687%; the exact 0.2358% chance value lies inside. A negative lower confidence limit is an ordinary symmetric small-sample interval artifact and does not imply negative accuracy.

Background-only accuracy is higher than grayscale and stain-normalized accuracy. That does not mean background is the only cue: RGB remains highest, and transformations alter multiple image properties at once. The careful conclusion is that parent information is distributed across color, low- and high-frequency structure, and background.

12

SIPaKMeD classification rescore

12.1 Why re-run models rather than quote published scores?

Published numbers are entangled with architectures, preprocessing, augmentations, and—most importantly—different splits. Quoting them would not isolate the quantity of interest. The rescore therefore trains six representative image classifiers under the same code path, preprocessing, optimizer family, epoch budget, seeds, and four fixed split protocols. The only intended change across protocols is the independence constraint and, for the explicitly size-matched control, the number of retained training crops.

The six families cover different inductive biases and capacities: ResNet-18 and ResNet-50 [11], DenseNet-121 [13], MobileNet-V2 [30], EfficientNet-B0 [33], and ViT-B/16 [6]. Each ImageNet-pretrained head is replaced with a five-way linear classifier. Re-running all six also prevents the conclusion from depending on one especially favorable architecture.

12.2 Inputs, augmentation, and optimization

Every crop is decoded as RGB, converted to a tensor, and normalized by the ImageNet channel statistics. Training augmentation consists of a random resized crop to 224×224 with scale in $[0.8, 1.0]$, a horizontal flip, and a vertical flip. Test preprocessing is a deterministic resize to 224×224. The code records the fully resolved split hash and configuration beside every result.

All classifiers use AdamW [20], cross-entropy loss, and a fixed learning rate. The retained endpoint is the final epoch; there is no separate validation split or post-hoc best-epoch selection. A run is valid only when the stored identifiers match its split, the final file is complete, and all reported quantities are finite. Table 12.1 gives the execution contract.

Table 12.1: SIPaKMeD classifier execution settings.

Item	Setting
Architectures	ResNet-18/50, DenseNet-121, MobileNet-V2, EfficientNet-B0, ViT-B/16
Initialization	ImageNet pretrained
Input	RGB, 224×224
Loss	Five-class cross entropy
Optimizer	AdamW
Learning rate	10^{-4}, fixed
Weight decay	10^{-4}
Batch size	64
Epochs	20, final-epoch endpoint
Seeds	17, 29, 43, 71, 101
Primary metric	Test accuracy
Hardware	NVIDIA RTX A6000; at most three concurrent GPU jobs

12.3 Metric definitions

For true class y_i and prediction $\hat{y}_i$, accuracy is

$$\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\}. \tag{12.1}$$

For class c, with precision P_c and recall R_c,

$$F_{1,c} = \frac{2P_cR_c}{P_c + R_c}, \qquad F_{1,\mathrm{macro}} = \frac{1}{C}\sum_{c=1}^{C} F_{1,c}. \tag{12.2}$$

Balanced accuracy is the unweighted mean class recall,

$$\mathrm{BAcc} = \frac{1}{C}\sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c}. \tag{12.3}$$

The preregistered primary endpoint is accuracy. Macro-F1 and balanced accuracy check that the direction is not a class-frequency artifact; they are not used to shop for a smaller p-value.

12.4 Aggregate scores

Table 12.2: SIPaKMeD test accuracy (%), mean ± sample SD over five seeds. P1′ is numerically identical to P1 because the frozen stratified crop split already contained the registered 3,239 training and 810 test observations.

Architecture	P1	P1′	P2	P3
DenseNet-121	98.12±0.28	98.12±0.28	96.72±0.69	97.38±0.50
EfficientNet-B0	97.78±0.49	97.78±0.49	96.69±1.49	97.41±0.71
MobileNet-V2	97.41±0.45	97.41±0.45	95.16±1.50	96.20±0.93
ResNet-18	97.19±0.59	97.19±0.59	96.00±0.89	96.35±1.28
ResNet-50	97.51±0.47	97.51±0.47	96.79±0.42	96.27±0.53
ViT-B/16	97.14±0.95	97.14±0.95	95.56±0.84	95.53±1.68

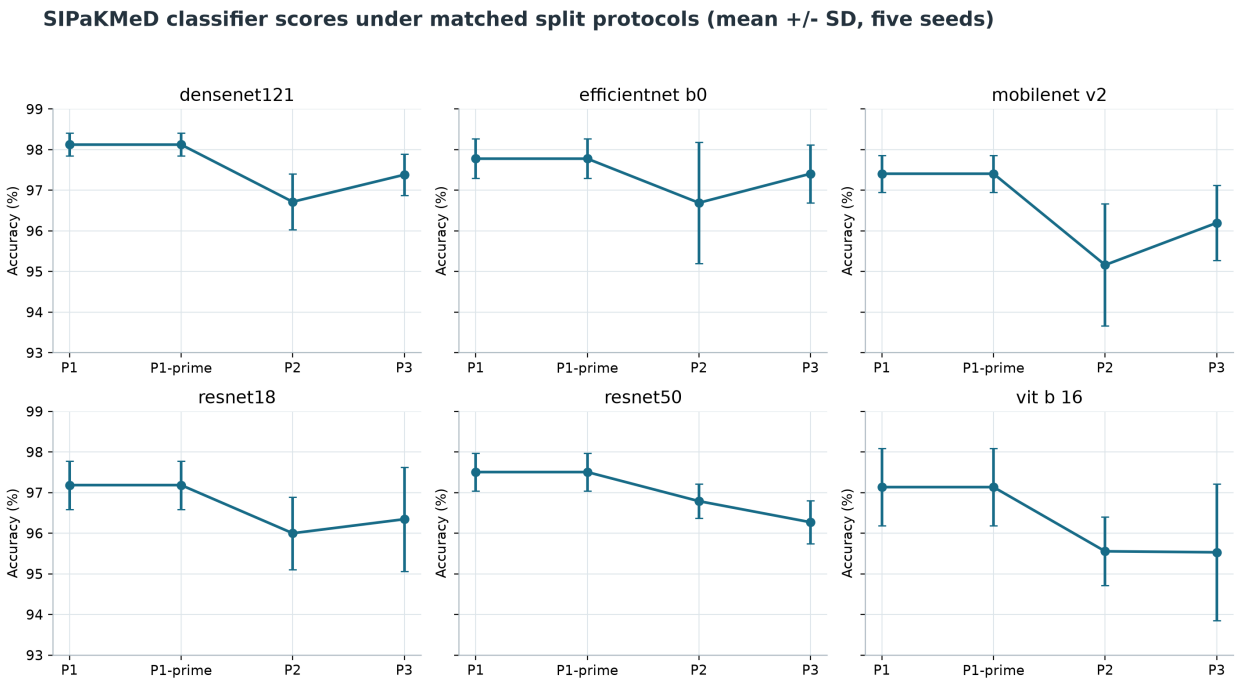


Figure 12.1: All frozen SIPaKMeD accuracy observations. Lines connect the same architecture across protocols; points show individual seeds and bars show the mean.

The top score under P1 is DenseNet-121 at 98.12±0.28%. Under parent-disjoint P2, DenseNet-121 remains highest at 96.72±0.69%, narrowly followed by ResNet-50. The key scientific observation is not which backbone wins; it is that all six paired P1−P2 mean differences are positive.

12.5 Matched contrasts, uncertainty, and the ordering failure

For architecture a and seed s, the primary paired difference is

$$d_{a,s} = 100\,(\mathrm{Acc}_{a,s,\mathrm{P1}} - \mathrm{Acc}_{a,s,\mathrm{P2}}) \tag{12.4}$$

in percentage points. The paired standardized effect is Cohen's

$$d_z = \frac{\bar{d}}{s_d}, \tag{12.5}$$

where s_d is the sample SD of the five seed-paired differences. Reported effect-size confidence intervals are seed-paired bootstrap intervals, so a bootstrap resample always keeps the two protocol results from a seed together. The test is a paired t-test because every registered five-difference vector passed its Shapiro–Wilk screen; the switch to Wilcoxon was reserved for a failed screen.

Table 12.3: Primary SIPaKMeD P1−P2 contrasts. The six paired t-tests form one preregistered BH family. CI is the paired-bootstrap 95% interval for Cohen's d_z, not for the raw percentage-point mean.

Architecture	Mean±SD (pp)	95% CI for d_z	p	BH q	d_z
DenseNet-121	1.41±0.90	[0.985, 4.747]	0.0251	0.1109	1.561
EfficientNet-B0	1.09±1.85	[−0.316, 4.668]	0.2591	0.2591	0.588
MobileNet-V2	2.25±1.63	[0.809, 6.686]	0.0370	0.1109	1.377
ResNet-18	1.19±1.28	[0.442, 2.365]	0.1081	0.1511	0.922
ResNet-50	0.72±0.83	[0.085, 4.025]	0.1260	0.1511	0.863
ViT-B/16	1.58±1.70	[0.057, 4.950]	0.1059	0.1511	0.931

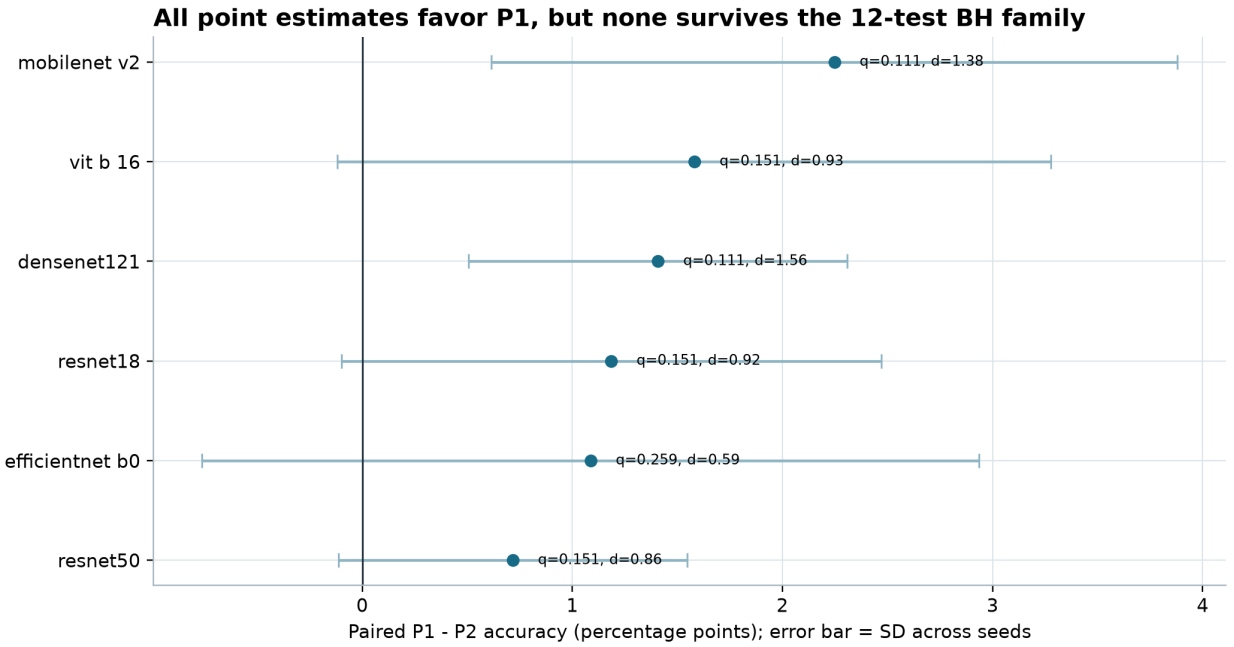


Figure 12.2: Paired P1−P2 effects. Every mean favors the crop-level split, but none survives its preregistered multiplicity correction. The vertical line at zero marks no protocol effect.

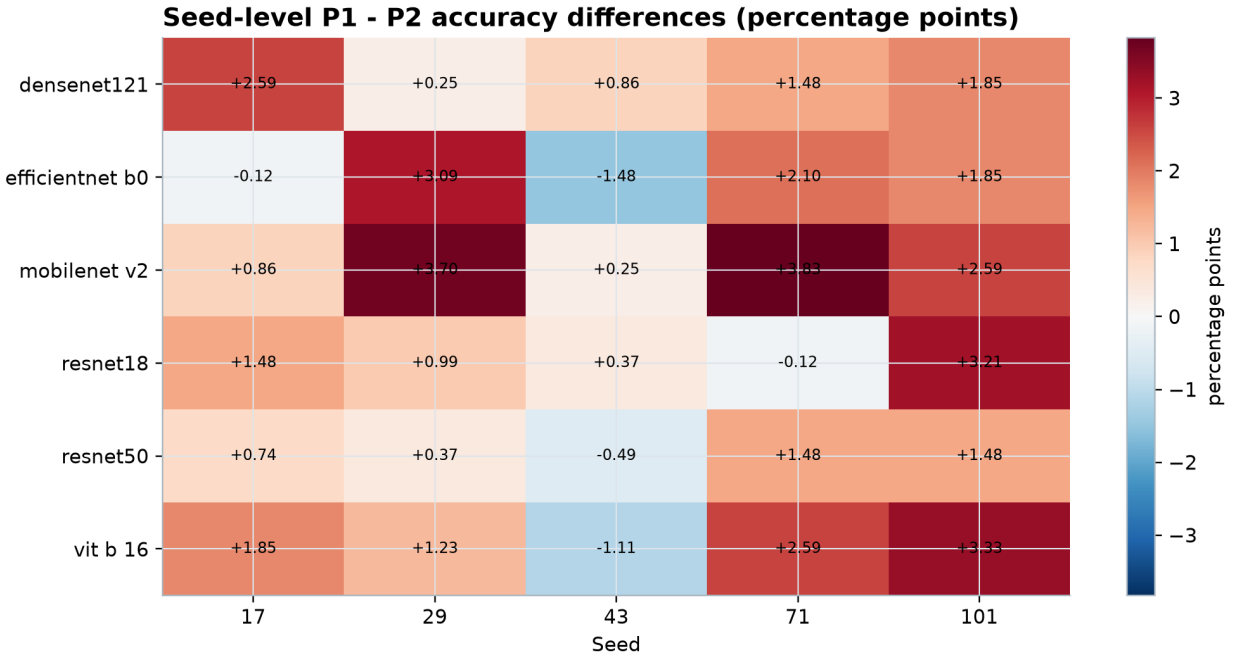


Figure 12.3: Seed-level paired differences. This figure exposes the small effective sample size and the architecture-specific variability that an aggregate table can hide.

The registered monotonicity expectation was P3≤P2≤P1. Only ResNet-50 and ViT-B/16 satisfy it. The other four improve from P2 to P3 by 0.35–1.04 points. Those reversals are not deleted or relabeled: the corresponding P2−P3 differences are −0.67, −0.72, −1.04, −0.35, +0.52, and +0.02 points in table order, and none has an adjusted confirmatory claim.

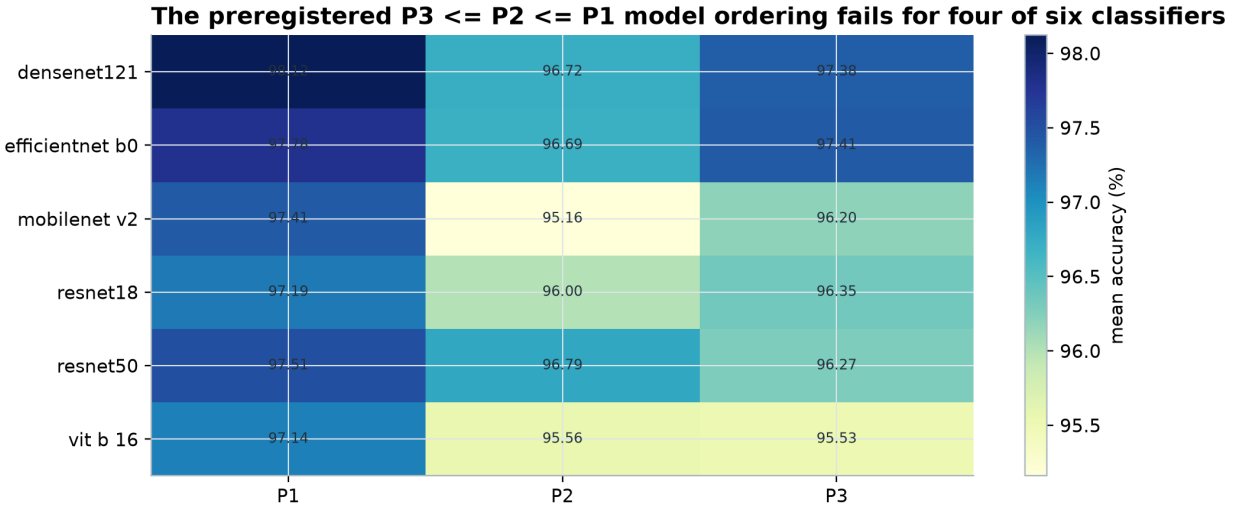


Figure 12.4: The registered ordering check. Cell values are the protocol means; the title records the gate outcome. Four of six architectures fail P3≤P2≤P1, so the acceptance contract correctly records one failed gate.

Architecture ranks are also unstable across P1 and P2: Kendall's $\tau = 0.467$ with bootstrap interval [0.071, 0.733] and $p = 0.2722$; four of the 15 architecture pairs reverse. This is a descriptive warning that a benchmark split can alter conclusions about model selection even when absolute score differences look modest.

12.6 What the classification rescore does and does not show

The same-sign result across six architectures is scientifically suggestive and consistent with the structural mechanism. The five-seed paired variability is nevertheless too large for any architecture to pass the six-test BH family. The correct report is therefore: *crop-disjoint scores were higher on average for every tested classifier, by 0.72–2.25 percentage points, but the preregistered architecture-wise confirmatory family did not establish individual effects after multiplicity correction.*

No classifier confusion matrix appears because the frozen result schema retains aggregate test metrics and epoch histories, but not per-sample predictions or model checkpoints. Generating such a matrix now would require a new experiment after the declared research stop. Omitting it is more faithful than reconstructing or simulating unobserved cells.

12.7 Training behavior and compute

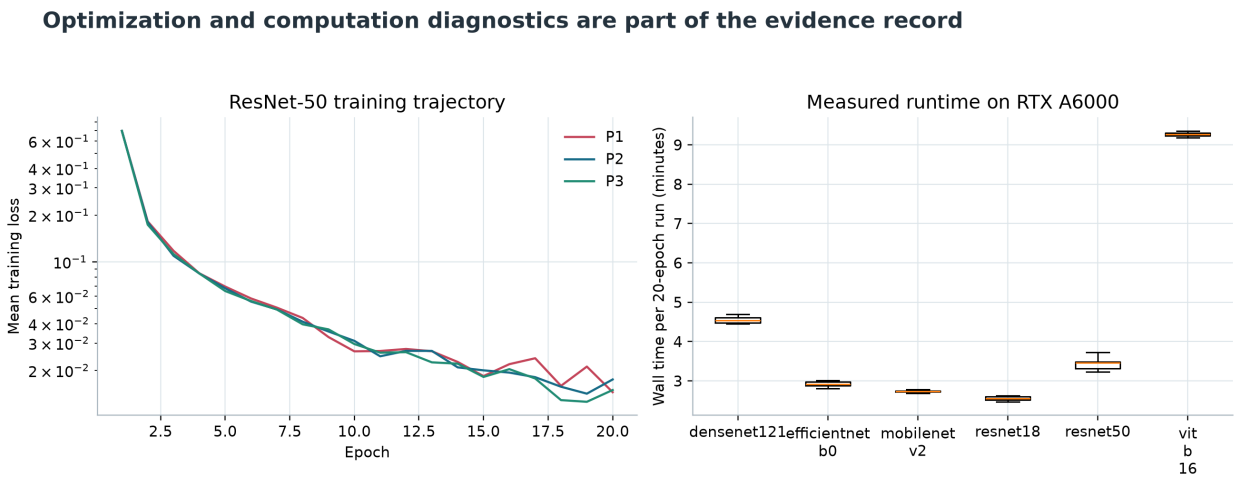


Figure 12.5: Classifier wall-clock summaries from the retained run JSON files. These are same-system measurements and should not be compared to timings reported on unrelated hardware.

The 120 retained classification runs consumed 8.48 summed run-hours. Because up to three jobs ran concurrently, this is computational work, not elapsed calendar duration. Shorter wall time does not imply a more accurate or better-calibrated model; it is included so the execution cost is auditable.

13

Cx22 segmentation rescore

13.1 Why add a second task and dataset?

Classification alone could leave two alternative explanations: perhaps the phenomenon is specific to SIPaKMeD, or perhaps it is specific to assigning a single label to a crop. Cx22 offers a different hierarchy and two pixel-level tasks. It therefore tests whether the split logic transfers to semantic and instance segmentation without pretending that its official split is the same estimand as the pooled stress test.

The pooled analysis treats all 1,320 released generated regions as the observation pool and the 514 CCEDD sources as parents. The protocolOFFICIAL arm uses only Cx22-Multi-Train (400 observations) and Cx22-Multi-Test (100 observations). These official train and test source-parent sets are disjoint. Pair is not added to the official test comparison: 55 of its source parents overlap Multi-Test and 233 overlap Multi-Train, with Jaccard similarities 0.126 and 0.475 respectively. Pooling and re-splitting is thus a downstream-reuse stress test, not an accusation that the authors' official Multi split is flawed.

13.2 Targets and models

The semantic target has two independent binary channels: cytoplasm and nucleus. Five encoder–decoder families are re-run: U-Net [28], U-Net++ [39], DeepLabV3+ [5], FPN [18], and MAnet [9]. ResNet-34 is the encoder except for DeepLabV3+, which uses ResNet-50. All encoders start from ImageNet weights. The output activation is not embedded in the network; logits are passed through sigmoid in the loss and evaluation code.

For image i, channel c, prediction probability p_{icu}, and binary target y_{icu} at pixel u, the training objective is

$$\mathcal{L}_{\text{semantic}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{softDice}}, \tag{13.1}$$

$$\mathcal{L}_{\text{softDice}} = 1 - \frac{1}{2N} \sum_{i,c} \frac{2 \sum_u p_{icu} y_{icu} + 1}{\sum_u p_{icu} + \sum_u y_{icu} + 1}. \tag{13.2}$$

The instance task uses Mask R-CNN with a ResNet-50 FPN backbone and its torchvision V2 implementation [12]. The classification and mask heads are replaced for one foreground category plus background. Ground-truth cytoplasm instances are reconstructed from the released MATLAB label structure before resizing. Empty instances after resizing are rejected rather than silently retained.

13.3 Deterministic input construction

All images and masks are resized to 256×256; RGB images use antialiased interpolation, while masks use nearest-neighbor interpolation. Training augmentation is determined by a SHA-256 digest of seed, epoch, and crop ID. Digest bits choose horizontal flip, vertical flip, and one of four right-angle rotations. This produces per-epoch augmentation without relying on worker scheduling order. Images are ImageNet-normalized; masks remain binary.

The original JPEGs and generated masks were packed into memory-mappable arrays, each named in a provenance manifest with SHA-256 hashes and a cache version. This was not a scientific transformation. It removed repeated JPEG/MATLAB/NFS decoding from the inner loop while making the exact input bytes auditable.

Table 13.1: Cx22 segmentation execution settings.

Item	Semantic	Instance
Image size	256×256	256×256
Epochs	30	30
Batch size	16	2 (test batch 1)
Optimizer	AdamW	AdamW
Learning rate	10^{-4} fixed	10^{-4} fixed
Weight decay	10^{-4}	10^{-4}
Precision	FP16 except MAnet FP32	FP16
Endpoint schedule	every epoch; final retained	final epoch only
Checkpointing	every epoch	every 5 epochs
Seeds	17, 29, 43, 71, 101	17, 29, 43, 71, 101
Hardware	NVIDIA RTX A6000; at most three concurrent GPU jobs	

13.4 Semantic metrics

Probabilities are thresholded at 0.5. For predicted set A and target set B,

$$\mathrm{Dice}(A,B) = \frac{2|A \cap B|}{|A| + |B|}, \qquad \mathrm{IoU}(A,B) = \frac{|A \cap B|}{|A \cup B|}. \tag{13.3}$$

The primary semantic value first averages cytoplasm and nucleus Dice within each image, then averages over test images. This prevents a large cytoplasm region from numerically drowning the smaller nucleus. Channel-specific Dice and IoU are retained as diagnostic endpoints.

13.5 Instance matching metric

Predictions with object score below 0.5 are discarded. Let D_{jk} be Dice between predicted instance j and ground-truth instance k. The Hungarian assignment [16] maximizes

$$\max_{\pi} \sum_{(j,k)\in\pi} D_{jk} \tag{13.4}$$

subject to one-to-one matching. Every unmatched ground-truth instance receives Dice and IoU zero. The primary image value is the mean over that image's ground-truth instances; images are then averaged. Recall@0.70 is the fraction of ground-truth instances whose matched Dice is at least 0.70. This endpoint penalizes missed cells and does not allow an image with many easy instances to dominate the result.

Listing 13.1: Exact logic of the matched-instance endpoint.

```
keep predictions with confidence >= 0.50
threshold predicted and target masks at 0.50
compute all predicted-by-target Dice values on the GPU
solve maximum-weight one-to-one assignment
write matched Dice into a zero vector of length n_targets
average the vector within the image
average image means over the test set
```

13.6 Semantic results

Table 13.2: Cx22 semantic macro-Dice (%), mean ± sample SD over five seeds. The official column uses a different 400/100 sample and is descriptive, not a paired protocol contrast.

Model	P1	P1′	P2	P3	OFFICIAL
DeepLabV3+	88.90±0.45	88.84±0.45	88.64±0.34	88.10±0.38	82.96±1.54
FPN	88.90±0.43	88.97±0.55	88.89±0.48	88.33±0.30	84.43±0.33
MAnet	90.03±0.60	90.07±0.63	89.97±0.45	89.63±0.21	85.50±1.41
U-Net	90.29±0.48	90.32±0.52	89.99±0.52	89.77±0.16	85.64±0.58
U-Net++	90.56±0.51	90.58±0.54	90.38±0.35	90.05±0.21	87.30±1.13

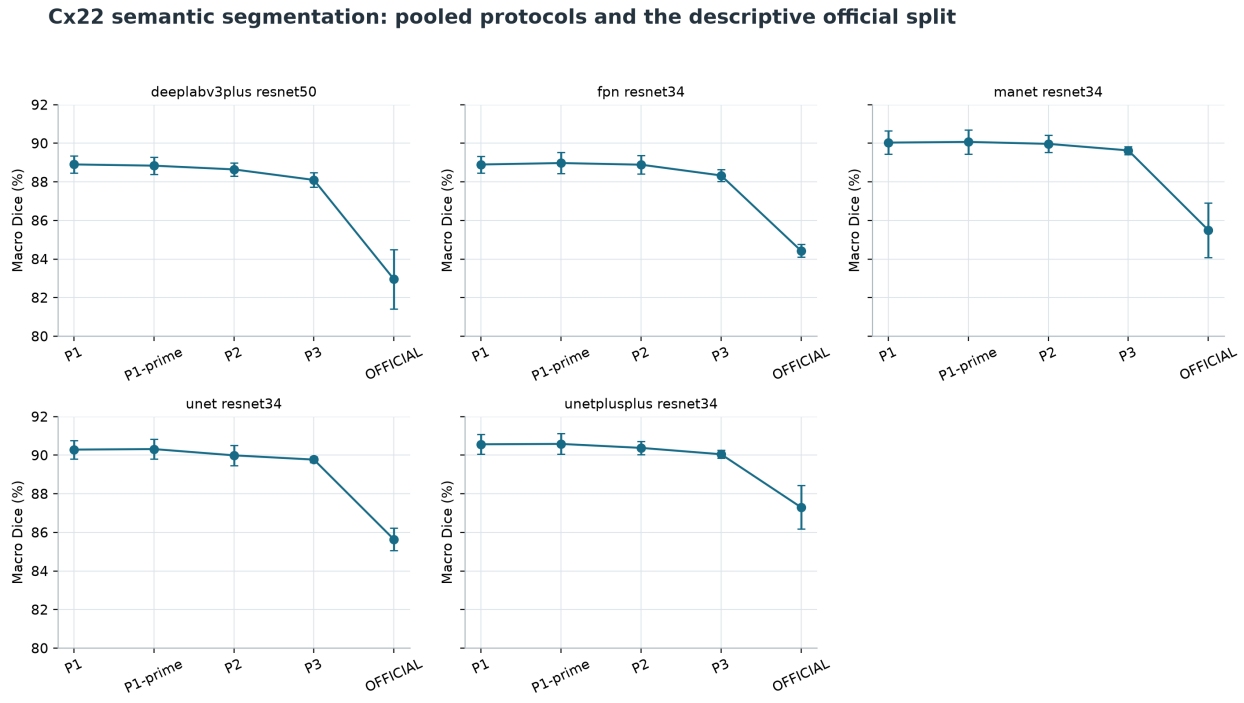


Figure 13.1: Cx22 semantic macro-Dice by architecture and protocol. The official arm is visually separated because it is not a same-observation contrast.

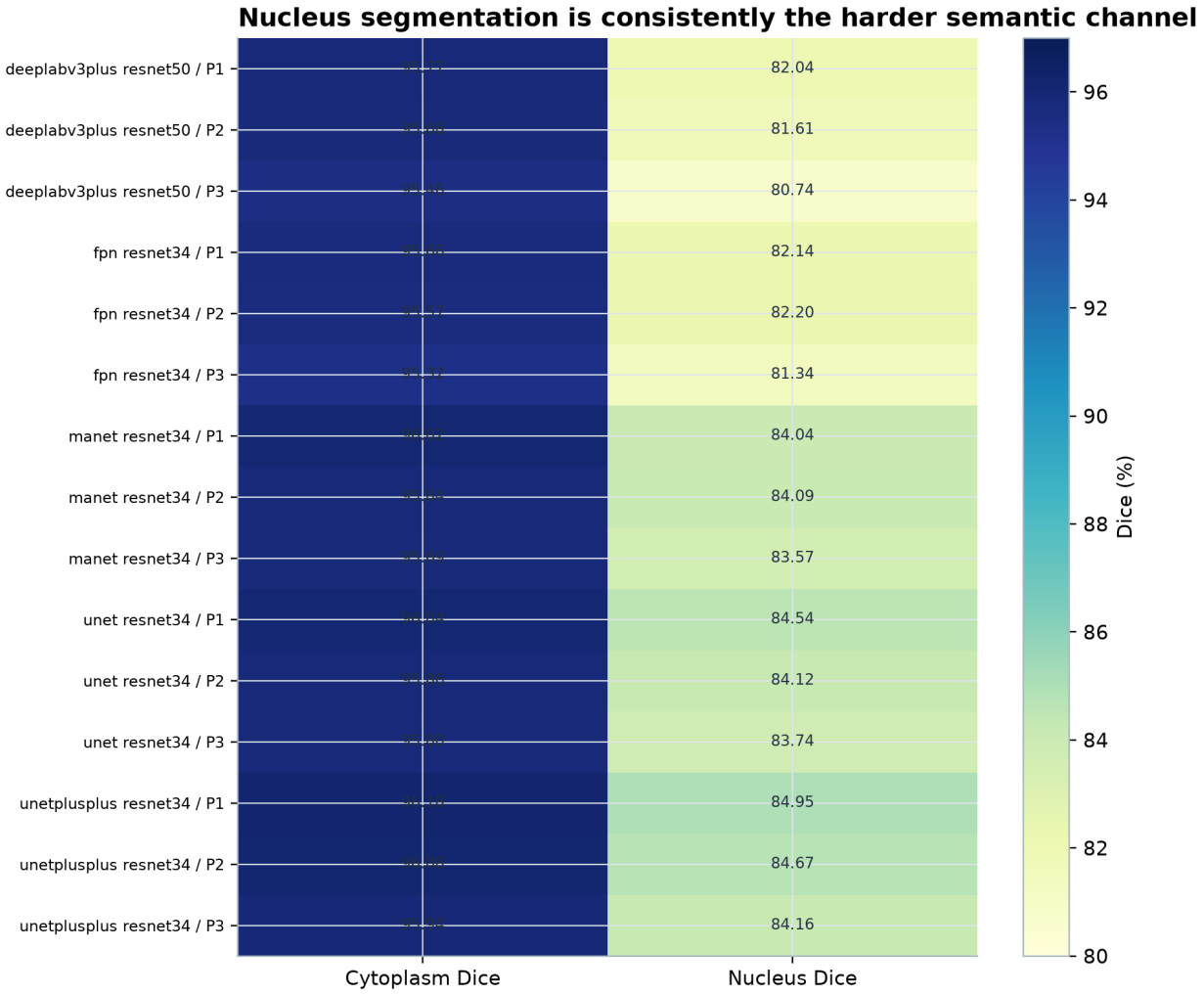


Figure 13.2: Channel-level Dice makes the central difficulty visible: nuclei are consistently harder than cytoplasm. This is the segmentation analogue of a confusion-matrix diagnostic and uses retained aggregate channel measurements, not invented pixel predictions.

U-Net++ is best in every pooled protocol and on the official Multi arm. The channel breakdown explains the lower official macro score: across models, cytoplasm Dice is roughly 94–95% while nucleus Dice is roughly 72–79%. Nuclei are smaller, so a fixed boundary displacement consumes a larger fraction of their area; faint staining and overlap also make the boundary less stable. This is a checkable data property visible in the released masks rather than a generic explanation added after the fact.

13.7 Instance results

Table 13.3: Cx22 matched-instance Dice (%), mean ± sample SD over five seeds.

Model	P1	P1′	P2	P3	OFFICIAL
Mask R-CNN	92.10±0.52	92.05±0.52	91.59±0.42	91.11±1.09	92.29±0.32
Mask R-CNN V2	91.99±0.57	91.95±0.58	91.36±0.82	90.60±0.52	92.18±0.71

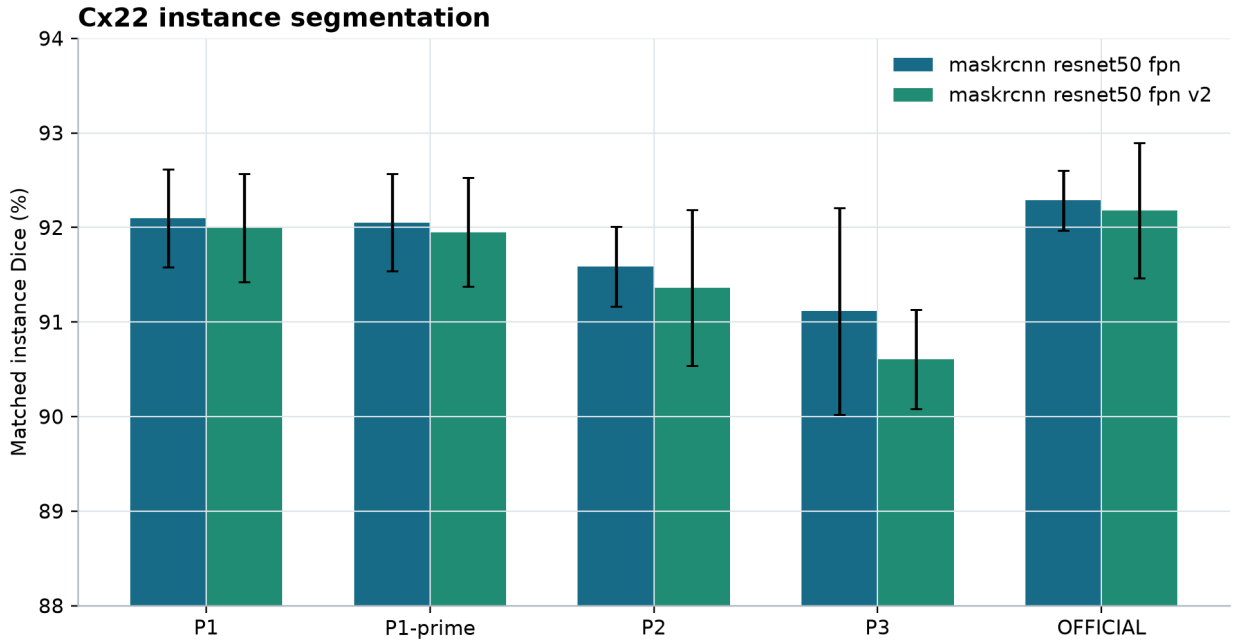


Figure 13.3: Cx22 matched-instance Dice. Both Mask R-CNN variants follow the complete registered P3≤P2≤P1 ordering.

Unlike SIPaKMeD classification, all seven Cx22 architecture/task rows satisfy the registered ordering. The differences are still small: P1−P2 is 0.01–0.30 percentage points for semantic models and 0.51–0.63 points for instance models.

13.8 Inference and rank stability

Table 13.4: Primary Cx22 P1−P2 contrasts. Values are mean ± SD percentage points; q is within the task-specific preregistered family. Effect is Cohen's d_z except DeepLabV3+, whose non-normal seed differences trigger rank-biserial effect size.

Task	Model	Mean±SD	Raw p	BH q	Effect
Semantic	DeepLabV3+	0.258±0.418	0.3125	0.8458	0.600†
Semantic	FPN	0.010±0.468	0.9625	0.9625	0.022
Semantic	MAnet	0.065±0.450	0.7612	0.8458	0.146
Semantic	U-Net	0.297±0.767	0.4353	0.8458	0.387
Semantic	U-Net++	0.188±0.406	0.3582	0.8458	0.464
Instance	Mask R-CNN	0.509±0.581	0.1220	0.2440	0.875
Instance	Mask R-CNN V2	0.630±1.204	0.3068	0.3116	0.523

†rank-biserial correlation; all other effects are Cohen's d_z.

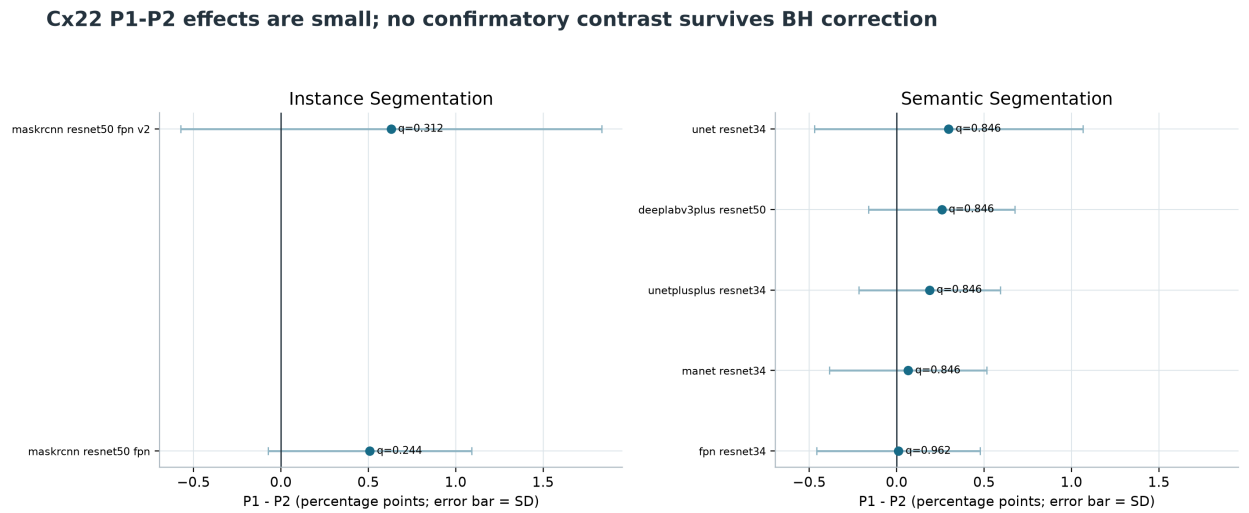


Figure 13.4: Cx22 paired contrast overview. Positive values favor the crop-level protocol. The plot separates point-estimate consistency from multiplicity-adjusted evidence.

No primary Cx22 test survives BH correction. The size-matched $\mathsf{P1}'-\mathsf{P2}$ Mask R-CNN contrast has raw $p = 0.0377$, $d_z = 1.368$, and $q = 0.1508$; it is not promoted to a significant claim. In the explicitly secondary $\mathsf{P2}-\mathsf{P3}$ family, FPN has raw $p = 0.0226$ and $d_z = 1.613$; it remains secondary and is not used to rescue the main conclusion.

Semantic model ranking is comparatively stable: Kendall's $\tau = 0.8$ with bootstrap interval [0.6, 1.0], $p = 0.0833$, and one of ten model pairs reordered. With only two instance models, $\tau = 1.0$ but its bootstrap interval spans $[-1,1]$, which is appropriately uninformative.

13.9 Training failures, recovery, and runtime

The first MAnet P1, seed-17 attempts produced non-finite losses at epochs 23 and 24 under FP16, despite dynamic loss scaling. Retaining an epoch-22 model or changing only that seed would have biased the matrix. The matrix stopped at 100 of 175 cells, the failure was amended in writing, and every MAnet cell was restarted in FP32 from its fixed seed. The corrected runs also set `CUBLAS_WORKSPACE_CONFIG=:4096:8` before any corrected MAnet or instance result existed.

Instance segmentation exposed a different failure: not numerical instability, but pipeline starvation. Per-epoch test inference, CPU pairwise mask matching, and network-filesystem checkpoint writes made GPUs wait. The final endpoint was already preregistered, so evaluation was restricted to the final epoch, exact binary intersections moved to the GPU, checkpoints became five-epoch recovery files, and immutable arrays were staged locally. These changes preserved the endpoint and substantially improved utilization; they did not alter splits, seeds, models, thresholds, or the final metric.

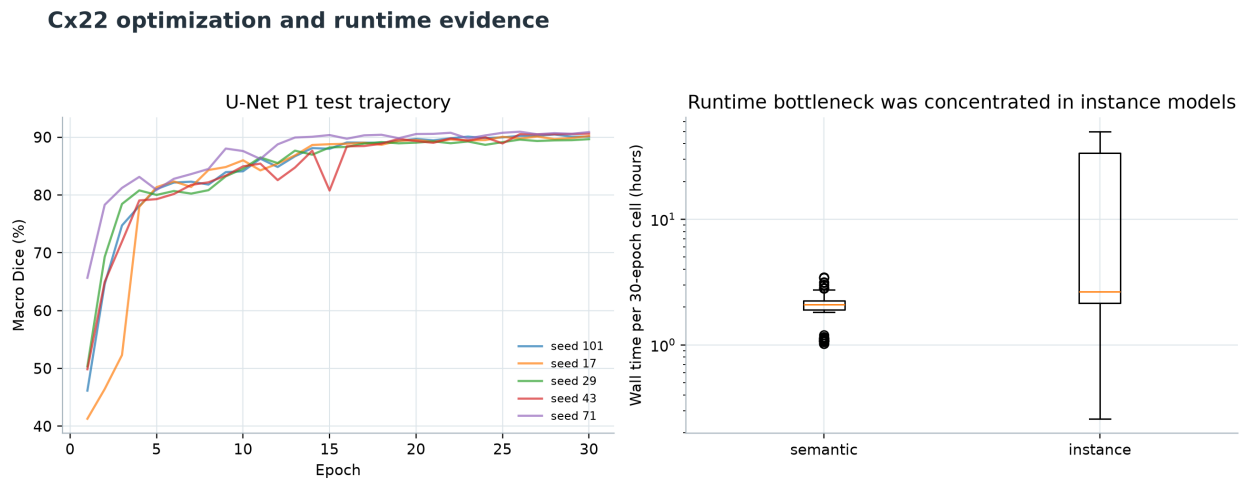


Figure 13.5: Cx22 run-time distribution. Instance segmentation dominates the computational cost and has a long tail; the plot makes the operational bottleneck explicit.

The 125 semantic cells consumed 253.33 summed run-hours, with median cell time 2.09 hours. The 50 instance cells consumed 646.76 summed run-hours, with median 2.64 hours and range 0.258–49.79 hours. The complete Cx22 matrix consumed 900.08 summed run-hours. Up to three jobs were concurrent, so the sum is not wall-clock project duration.

As in classification, final model weights and per-pixel predictions were not retained after the completed JSON was written. Figure 6.3 therefore visualizes real released ground truth, not a simulated qualitative prediction. A post-hoc prediction gallery would require fresh inference and is deliberately outside the frozen record.

14

Statistical analysis in full

14.1 The experimental unit is the seed-paired run

Every reported neural summary uses the five fixed seeds 17, 29, 43, 71, and 101 and gives the arithmetic mean with sample standard deviation ($n - 1$ denominator). The protocol contrast is paired by seed. This pairing removes seed-to-seed variation that affects both arms and prohibits a favorable run in one protocol from being compared with an unrelated run in another.

For values m_s over $S = 5$ seeds,

$$\bar{m} = \frac{1}{S}\sum_{s=1}^{S} m_s, \qquad s_m = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}(m_s - \bar{m})^2}. \tag{14.1}$$

A summary is suppressed if fewer than five seeds are present. This is enforced in code rather than remembered during table writing.

14.2 Test-selection rule

The preregistered rule first applies Shapiro–Wilk [32] to the five paired differences. If $p \geq 0.05$, a two-sided paired t-test is emitted; otherwise a two-sided Wilcoxon signed-rank test [36] is emitted. The effect is Cohen's d_z for a t-test and rank-biserial correlation for Wilcoxon. All-zero differences are descriptive; a constant nonzero difference has an infinite standardized effect by definition and is flagged explicitly.

With only five observations, normality screening has little power and neither test can create information that is not present. The rule is retained because it was specified before results, but conclusions emphasize effect direction, variability, and corrected evidence rather than treating the test label as a guarantee.

14.3 Bootstrap effect intervals

Ten thousand deterministic bootstrap replicates resample the vector of paired differences with replacement. Each replicate recomputes the relevant effect-size estimator. The interval is the 2.5th and 97.5th percentiles of finite estimates. A SHA-256-derived bootstrap seed from the dataset/-model/protocol key makes intervals repeatable without coupling them to the training seed.

These are intervals for the standardized effect, not raw mean-score intervals. Because $n = 5$, they can be wide or asymmetric and can contain values much larger than the point estimate. That apparent awkwardness is a correct display of limited information.

14.4 Multiplicity families

Benjamini–Hochberg [4] is applied once to the exact supplied family:

- SIPaKMeD classification: all six P1−P2 and six P1′−P2 tests in the registered joint family;
- Cx22 semantic: five P1−P2 and five P1′−P2 tests;
- Cx22 instance: two P1−P2 and two P1′−P2 tests.

Secondary P2−P3 tests and ranking stability are clearly labeled secondary/unadjusted and cannot substitute for a failed primary family. For ordered raw values $p_{(1)} \le \cdots \le p_{(M)}$, BH computes monotone adjusted values based on

$$q_{(i)} = \min_{j \ge i} \left(1, \frac{M}{j} p_{(j)}\right). \tag{14.2}$$

14.5 Structural sampling error

The expected parent-straddle probability for a parent of size k and 80/20 crop assignment is exact: $1 - 0.8^k - 0.2^k$. Dataset-level analytic expectations average this quantity over recovered parents. Monte Carlo means and SDs over the same five fixed seeds check the split implementation, while 10,000 additional simulations form the broad analytic envelope recorded in the structural table. Herlev is a stronger software test than a confidence interval: because every parent has $k = 1$, both the analytic and observed straddle fractions must be exactly zero.

14.6 Why a non-significant result is not “no effect”

A BH-adjusted $q > 0.05$ says that the current five-seed family does not meet its declared evidential threshold. It does not prove a zero effect. Conversely, six same-sign classifier means do not license six independent significance claims. The report keeps both facts visible: the direction is consistent, and the adjusted confirmatory evidence is insufficient.

15

Execution lifecycle, failures, and demotivations

15.1 Run inventory

Table 15.1: Frozen neural-run inventory and summed run time.

Experiment	Cells	Retained complete	Summed hours
SIPaKMeD classification	120	120	8.48
SIPaKMeD parent probe	30	30	5.75
Cx22 semantic segmentation	125	125	253.33
Cx22 instance segmentation	50	50	646.76
Total	325	325	914.31

The total is the sum of recorded per-run elapsed seconds. It is intentionally not converted into "GPU-days" without qualification: host I/O stalls and overlapping runs mean elapsed seconds include more than active kernel time, while three concurrent GPUs mean the sum exceeds calendar duration.

15.2 The first classification matrix was invalid

An early classifier attempt used scikit-learn 1.7.2. The resulting P2 split counts disagreed across cells, violating the requirement that every architecture see the same seed-specific observations. Twelve JSON files from that attempt were excluded in full. The environment was pinned to scikit-learn 1.9.0, the expected split hashes and counts were made acceptance checks, and the entire 120-cell matrix restarted. Three additional one-epoch files were smoke tests and were never mixed into the result family.

This failure changed the engineering standard: a result is not accepted merely because training finished. Dataset name, architecture, protocol, seed, train/test counts, library versions, provenance hash, split hash, Git SHA, GPU field, metric finiteness, and history length must all match the matrix contract.

15.3 MAnet numerical failure

MAnet attention matrix multiplications overflowed under FP16 late in training. Two attempts failed non-finitely at epochs 23 and 24 for the first MAnet cell. The tempting choices—use the last finite epoch, lower the learning rate only for that cell, or replace seed 17—would each change the estimand after seeing its behavior. The accepted response was broader: freeze an amendment, run every MAnet cell in FP32, restart from the same seeds, and keep the failures in the record.

15.4 The host bottleneck

The remote machine had three available RTX A6000 GPUs but a separate storage/host bottleneck. Heavy concurrent decoding and checkpoint writes could make all GPUs appear slow and could destabilize the SSH session. Launchers therefore queried memory usage and utilization, admitted no more than three idle devices, used atomic result writes, and kept resumable checkpoints. The instance-segmentation redesign reduced expensive network I/O and moved exact matching work to the GPU.

What did not work

The project did not produce a universal multiplicity-corrected classifier inflation result, did not validate the preregistered classification ordering, did not retain prediction-level classifier outputs for a confusion matrix, and has not yet demonstrated a raw-download-to-Table-1 run on a machine that has never seen the project. Each limitation is part of the result, not a cosmetic omission.

15.5 Research stop and evidence freeze

After the publication verdict, exploration stopped. No architecture was added to chase significance; no seed was replaced; no endpoint was changed; no new confusion matrix was generated. The long-form report reconstructs plots only from frozen aggregates and epoch histories. Images used to explain Otsu/Telea and Cx22 label decoding are implementation demonstrations from locally available released data and are labeled as such.

16

Verification against the acceptance contract

16.1 Gate-by-gate status

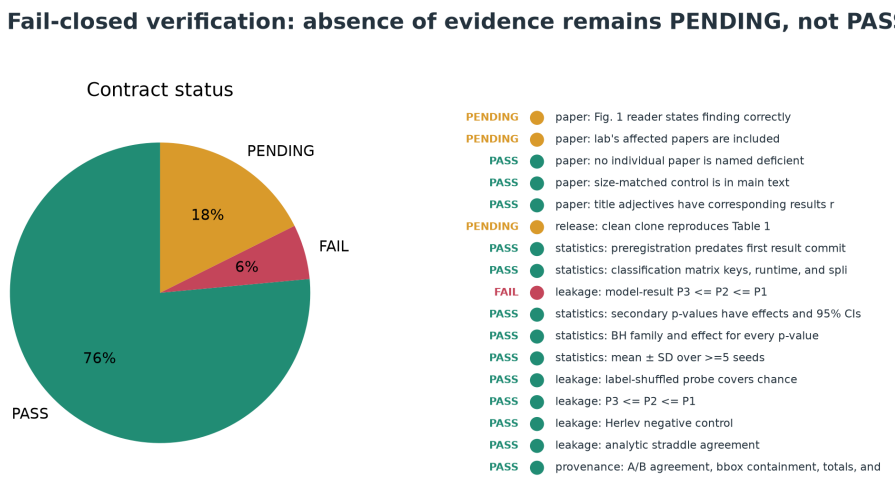


Figure 16.1: Machine-readable acceptance status: 13 gates pass, one fails, and three remain pending. A pending release gate is not painted green simply because the numerical experiments are complete.

Table 16.1: Acceptance contract with evidence.

Gate	Status	Evidence
SIPaKMeD branch agreement $\geq 99\%$	**PASS**	4,049/4,049 mappings agree (100%).
Every bbox lies in parent bounds	**PASS**	4,049/4,049 pass geometry checks.
Crop and per-class parent totals match	**PASS**	4,049 crops; parent vector 126/108/238/271/223.
Observed straddle matches analytic prediction	**PASS**	All audited multi-crop datasets fall inside the recorded simulation envelope.
Herlev has no structural effect	**PASS**	All five seeds return exactly zero in P1; P2/P3 also zero.

Gate	Status	Evidence
P3≤P2≤P1, structural	**PASS**	Grouped protocols eliminate measured parent sharing.
Shuffled probe confirms chance	**PASS**	95% t interval $[-0.1385, 0.4687]$% contains 0.2358%.
Every neural summary has ≥ 5 seeds	**PASS**	120 classifier, 30 probe, and 175 Cx22 files pass matrix completeness.
Plan predates first result commit	**PASS**	22 Aug 2026 04:54:10 IST versus 13:22:42 IST.
BH applied over stated families	**PASS**	Stored family labels and q values are reproduced from the exact supplied rows.
Effect accompanies every inferential p	**PASS**	d_z, rank-biserial, or Kendall τ is present.
Classifier P3≤P2≤P1	**FAIL**	Only ResNet-50 and ViT-B/16 pass; four of six fail.
Cx22 model ordering	**PASS**	All five semantic and two instance rows pass.
Official Cx22 parent disjointness	**PASS**	Multi-Train and Multi-Test have no shared recovered source parent.
Virgin-host clean-clone reproduction	**PENDING**	Script exists, but execution evidence from a never-used host is absent.
Lab's own affected papers in audit	**PENDING**	No verified crosswalk in the frozen repository.
Figure-1-only independent reader	**PENDING**	No documented colleague response in the frozen repository.

16.2 Why the classification ordering gate fails

The structural ordering and model-score ordering are different propositions. P3 has stricter grouping than P2, but it also creates a different balanced partition. Removing a small duplicate-connected component can alter class composition, parent difficulty, and which observations land in test. A finite test set can therefore score higher under the stricter split even though it contains less structural sharing. The failure warns against using monotonic score ordering as a definition of correctness; leakage absence should be checked directly from IDs and components.

16.3 Release-readiness boundary

The repository contains a one-script verification path, but the strongest release criterion requires a truly clean machine that has never held the repository or data. Running the script on the development host, even in a new directory, would not fully establish that environmental condition. This report therefore leaves the release gate pending. The same standard applies to the human communication gate and lab-paper crosswalk: absence of evidence is not converted into a pass.

17

Interpretation, limitations, and practical guidance

17.1 What the evidence supports

The source hierarchy is real and recoverable. Random crop splitting predictably separates siblings. In SIPaKMeD it can also put overlapping source pixels into training and test. Parent identity is learnable even after large parts of the apparent cell signal are removed. Parent- and component-disjoint protocols eliminate the directly measured sharing. These conclusions follow from deterministic identities, geometry, and controls, not from a single favorable neural score.

The downstream effect is subtler. Every tested $\mathsf{P1}-\mathsf{P2}$ mean is positive, including six classifiers and seven segmentation models, but the Cx22 semantic differences are often tiny and none of the task-specific confirmatory families survives BH correction. The evidence supports changing split construction because independence is a design requirement, not because a guaranteed score penalty has been estimated.

17.2 Two hard visual structures in Cx22

The most persistent semantic error source is the nucleus channel. Nuclei occupy fewer pixels, so boundary uncertainty has a larger Dice penalty; pale nuclei and overlaps further reduce contrast. The second hard structure is touching cytoplasm instances. A union mask can remain visually plausible while an instance model merges two touching cells; maximum matching then assigns a good overlap to one target and zero or low overlap to the other. These explanations correspond to the actual label geometry in Figure 6.3 and the channel results in Figure 13.2.

17.3 Real limitations

1. **Dataset hierarchy stops at the released identifier.** SIPaKMeD fields and Cx22/CRIC sources are not guaranteed to be patient IDs. Parent-disjoint is therefore necessary but may not be sufficient for clinical independence.

2. **Five seeds leave broad uncertainty.** The complete matrix is large in GPU hours but small in inferential replication. Shapiro screening and bootstrap effects are fragile at $n = 5$.

3. **No clinical validation.** The study audits public research benchmarks; it does not estimate

screening sensitivity, specificity, prevalence calibration, reader impact, or deployment safety.

4. **Hardware and host constraint.** Execution occurred on one login node with at most three A6000 GPUs and an unstable I/O bottleneck. Wall times are not portable performance claims.
5. **Probe transformations are not surgical interventions.** Grayscale, normalization, inpainting, and high-pass filtering each alter multiple cues. They localize signal classes but do not identify a unique causal artifact.
6. **Near-duplicate grouping is conservative and calibrated synthetically.** It found no exact cross-parent duplicates and validated candidate detection on 3 positive/100 negative synthetic examples, but it is not a proof that every biological near-duplicate was linked.
7. **Prediction-level artifacts were not retained.** Post-hoc classifier confusion matrices and qualitative prediction panels cannot be reconstructed without new inference.
8. **Reproduction remains pending on a virgin host.** Dependency or URL drift may still be exposed by that final test.

17.4 Recommended evaluation rule

For any crop-derived cytology dataset, publish at least four fields for every released observation: a stable observation ID, an acquisition/source parent ID, a patient/specimen ID when ethically and legally possible, and a duplicate-component ID if cross-parent duplication was audited. Construct folds at the highest available independent level and publish the exact fold manifest. Report sample counts at both observation and parent levels. Never compare methods under nominally identical dataset names unless the split identifiers also match.

17.5 What to say in a paper

The defensible headline is: “Random crop-level evaluation in public cervical cytology can place sibling and even pixel-overlapping crops across train and test; source-disjoint manifests remove this non-independence.” A stronger sentence—“all published scores are inflated”—is not supported. No individual prior paper needs to be named as deficient; the target is a reusable benchmark practice.

18

Reproduction and release engineering

18.1 From owner downloads to structural Table 1

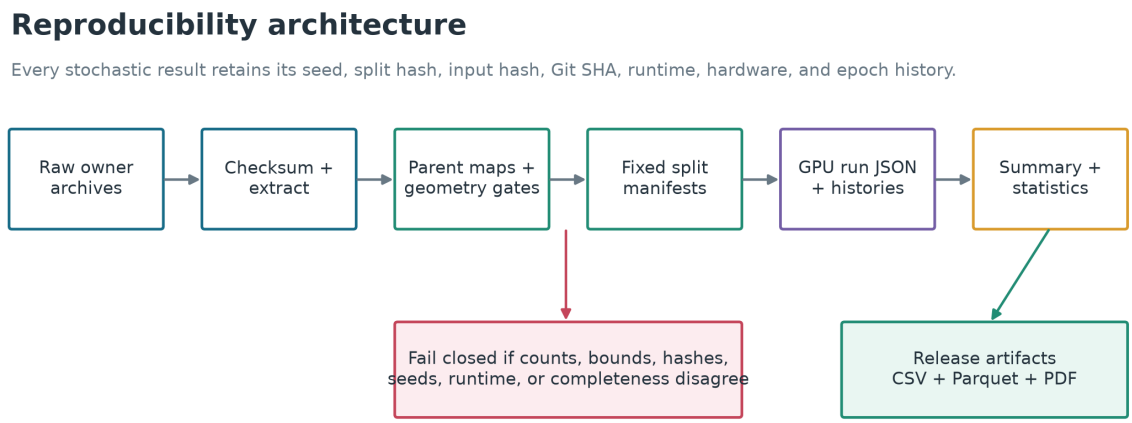


Figure 18.1: The intended clean-clone path. Each boundary writes an auditable artifact and performs validation before the next stage begins.

The GPU-free structural result has one public entry point:

Listing 18.1: Raw-download-to-Table-1 entry point.

```
./scripts/reproduce_table1.sh
```

The script performs a locked dependency sync; downloads all declared owner files with at most three connections; verifies exact sizes and SHA-256 values; extracts archives; recovers SIPaKMeD, Cx22, CRIC, and Herlev provenance; runs the SIPaKMeD duplicate and pixel-overlap audits; creates fixed splits; and emits the dataset-level and seed-level structural tables.

Raw images are intentionally excluded from version control. The repository distributes URLs, checksums, IDs, coordinates, derived maps, split assignments, and results, subject to each source's terms. In particular, Cx22/CCEDD assets are non-commercial research resources and are not repackaged in this report.

18.2 Clean-clone attestation

The wrapper requires a repository URL and a destination that does not yet exist:

Listing 18.2: Clean-clone verification wrapper.

```
./scripts/verify_clean_clone.sh \
  REPOSITORY_URL NEW_DESTINATION [GIT_REF]
```

It uses `gitclone--no-local`, optionally checks out an exact detached ref, runs the reproduction script, and writes an attestation. A human/operator must set `CYTOFAIR_VIRGIN_HOST_ATTESTED=YES` only on a host that genuinely has never seen the project. The current repository lacks that final attested evidence, so release remains pending rather than assumed.

18.3 Pinned software and result metadata

The package requires Python 3.11 or newer. The result contract records NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.1, scikit-learn 1.9.0, Pillow 12.3.0, OpenCV 5.0.0, PyTorch 2.9.0+cu128, torchvision 0.24.0+cu128, and CUDA 12.8 for the accepted classifier matrix. The lock file, not these prose values, is the install authority. The explicit versions remain useful diagnostics when a future lock update changes a hash.

18.4 Atomicity and resumability

Long GPU runs write checkpoints locally and copy them to a temporary destination name before atomic rename. A checkpoint is accepted only when its protocol, architecture, seed, and manifest hash match the requested cell. The launcher skips a result only after validating its completed JSON. A second launch can therefore resume interrupted work without accepting a mismatched or half-written file.

18.5 Human-readable and machine-readable layers

The project separates three layers:

- **canonical data**: Parquet provenance maps and CSV split assignments;
- **machine evidence**: JSON run records, CSV summaries, hashes, and acceptance states;
- **human interpretation**: Markdown reports, figures, the conference paper, and this lifecycle monograph.

This prevents a narrative edit from changing the measured evidence and makes it possible to regenerate prose tables from structured files.

19

Reusable outputs and future work

19.1 What exists now

The most durable asset is not a new image collection. It is a stable interface between released observations and independent units: canonical parent maps, duplicate-component identifiers, fixed split CSVs, adjacent hashes, a leak-checking API, a run JSON schema, and paired statistics. A downstream model joins on `crop_id`; it never infers a split from row order or regenerates folds with a library default.

19.2 Two publication units

The conference unit is the complete provenance/leakage argument, mechanism probe, fixed split release, and matched SIPaKMeD rescore. A journal extension can add a separately preregistered annotation study, expanded cross-dataset validation, a broader detector ensemble, and power-oriented learning curves. These are partitioned future studies, not unreported parts of the present evidence.

19.3 Projects unblocked

Table 19.1: Immediate technical reuse. These rows describe enabled work, not completed future experiments.

Project	Reused asset	Work no longer rebuilt
F3: Cx22 seed robustness	Parent map, fixed seeds, run schema, mean/SD and paired inference	Source grouping, seed ledger, rescore scaffold
F8: cross-dataset transfer	Canonical maps, split API, runtime metadata, label-crosswalk slot	Leakage-safe source folds and experiment provenance
F2: foundation encoders	P1/P1′/P2/P3 files and architecture-neutral summarizer	Fair split comparison, seed accounting, BH/effect reporting

19.4 Pre-specified V2 in fish imagery

Fish-Pak contains three views of the same specimen; Croatian Fish contains frames sampled at one frame per second from continuous BRUV/DOV video. The independent units become specimen and video episode. The planned V2 will: recover those units, compute analytic and observed frame straddling, construct specimen/episode-disjoint manifests, run the identity probe, and re-score fixed baselines under $\mathsf{P1}/\mathsf{P1}'/\mathsf{P2}/\mathsf{P3}$. No fish-domain result is claimed here.

19.5 Adoption goal

If versioned CytoFair manifests become the default evaluation interface, future papers can cite both the original dataset and the split asset while reporting exactly which parent-disjoint partition they used. This is an intended impact pathway, not a citation prediction or an empirical result.

20

Conclusion

This project asked whether a file-level train/test boundary corresponds to an independent acquisition boundary in public cervical cytology. It recovered and validated the hidden observation-to-parent hierarchy across four datasets, quantified the analytic and observed structural sharing, demonstrated literal source-pixel sharing in SIPaKMeD, and showed with a shuffled-label-controlled 424-way probe that parent identity remains strongly learnable after multiple cue-removal transformations. It then re-ran six classifiers and seven segmentation models over five fixed seeds and four matched protocols, preserving effect sizes, bootstrap intervals, multiplicity families, failures, and runtime metadata.

The strongest result is methodological: parent-disjoint evaluation is necessary whenever multiple released observations descend from one acquisition unit. The downstream neural results point in a consistent direction but do not justify a universal multiplicity-corrected accuracy-inflation claim. Four of six classifier architectures also violate the preregistered monotonic score ordering, a useful negative result showing that stricter grouping and lower finite-sample score are not synonymous.

CytoFair turns that lesson into reusable artifacts: provenance maps, duplicate components, fixed splits, hashes, fail-closed validators, run ledgers, and a raw-download structural pipeline. The next scientific step is a separately preregistered specimen/episode-disjoint fish-imagery V2; the next release step is a virgin-host clean-clone attestation. Until the latter, the numerical study is complete but the strongest release gate remains honestly pending.

PART III

From Study 1 to Study 2: Roadmap and Design

21

The roadmap and what we did with it

21.1 What Study 1 left unresolved

Study 1 (Part II) ended with a strong benchmark-construction conclusion and a deliberately weak model-performance conclusion. Five problems stood between them.

1. **One question, six tests.** "Does crop-level splitting inflate accuracy?" was tested once per architecture inside a twelve-test Benjamini–Hochberg family. Every point estimate was positive, every test was underpowered, and none survived.
2. **Availability is not use.** A 424-way probe recognised fields far above chance, but that shows the shortcut is *available*, not that classifiers route their predictions through it.
3. **Leakage confounded with diversity.** At equal crop counts a crop-level training set contains about 914 distinct fields and a field-level one about 773; P1 differs from P2 in training diversity as well as in leakage.
4. **One rescored benchmark.** Only SIPaKMeD was rescored for classification; CRIC, whose crop-level split puts a training parent behind essentially every test cell, was not.
5. **The host.** The original classification matrix had to be repeated (Chapter 15 is the record), and runs were slower than expected for reasons only diagnosed in Study 2 (Section 26).

21.2 The roadmap, item by item

The October review turned these into a phased roadmap. Table 21.1 records what was done with every item, including the ones deliberately not done.

Table 21.1: The scaling roadmap and the response to each item.

Roadmap item	What was done	Status
Phase 0: arXiv readiness		
Fresh-clone check	Public repository cloned into a clean directory; its 45 tests pass. The public `main` branch stops at 22 August and lacks the Cx22 commits and a LICENSE file.	done; push and licence are authors' decisions

Roadmap item	What was done	Status
Text extraction of PDFs	Ligatures broke copy-paste in the drafts; `glyphtounicode` and T1 encoding added to every preamble.	done
Draft claims	The ISBI draft's sentence implying P3 is strictly more conservative than P2 flagged as an overreach; the draft's conclusion now says replication is in progress.	done
Phase 1: fix the statistics		
Single omnibus test	One crossed mixed model, `acc ~ x + (1+x|architecture) + (1|seed) + (1|split)`, with the split random effect that the roadmap's formula omitted (twelve architectures share each test set).	done (H1)
Simulation-based power	Custom R simulation of the exact model, 1,000 simulations per design cell, from the Study 1 variance components (Chapter 23).	done
More architectures	Twelve: the six Study 1 models plus ConvNeXt-T, Swin-T, RegNetY-1.6GF, EfficientNetV2-S, MaxViT-T, and self-supervised DINOv2 ViT-S/14.	done
15–25 seeds	Twenty, chosen by a registered power rule.	done
Preregistration	`PREREGISTRATION_STUDY2.md`, registered as Amendment 6 before any Study 2 model was trained.	done
Keep predictions, logits, checkpoints	Every run keeps per-crop logits, float16 final weights, and probe-split features; this is what made Amendment 8's evaluation-only arms possible.	done
Phase 2: causal arms		
2C leakage vs diversity	Factorial over one common test set at matched crop and field counts.	done (H3)
2A fingerprint suppression	Class-conditional parent adversary behind gradient reversal, seven-level λ sweep.	done (H5)
2B background transplant	Counterfactual composites with a same-parent (sibling) control.	done (H4)
Phase 3: generalisation		
Other domains (fish imagery)	Gated on Phase 1 and 2 outcomes; not started.	deferred by design
Part 5: datasets		
CRIC second rescore	Rescored with six architectures and ten seeds.	done (H2)
Keep Cx22 un-rescored	Kept; the official Cx22 split is parent-disjoint.	done
Cx22 literature check	Four open-access studies examined (Section 21.4).	done
JUCYT ground truth	Not publicly available.	not possible
Part 6 and Part 8		
Single-column neutral draft	`paper/v2`, every number generated.	done
Avoid scope creep	Study 2 ran only registered arms; Amendment 8 was added after Study 2's results with an explicit justification and its own Holm family.	done

21.3 Phase 0 in detail

The repository was cloned from its public remote into an empty directory, its environment created from the lock file, and its test suite run: 45 tests passed. Three gaps were recorded in `reports/`

`study2/phase0_arxiv_checklist.md`. The public branch stops at 22 August 2026 and does not contain the Cx22 segmentation commits; there is no LICENSE file, so nobody may legally reuse the toolkit; and several drafts produced PDFs whose ligatures (fi, fl) did not survive copy-paste. The third was fixed in every preamble. The first two are decisions for the authors and remain open.

21.4 The Cx22 literature check

The Cx22 release ships an official parent-disjoint split (Multi-Train/Multi-Test). The question was whether downstream users keep it. Of four open-access studies that train on Cx22, three use the official split. One describes image-level five-fold cross-validation over all 1,320 generated regions, which pools regions from the same CCEDD source image across folds. Following the project's no-naming rule for third-party work, only the aggregate is reported (`reports/study2/cx22_literature_check.md`).

21.5 Post hoc reanalysis of Study 1

While validating the analysis code, the registered H1 model was fitted to the sixty Study 1 SIPaKMeD P1 and P2 runs. It gives 1.37 pp (SE 0.38, Satterthwaite $df = 9.1$, $p = 0.005$). The same model on the Cx22 semantic-segmentation runs gives 0.16 Dice points ($p = 0.43$). Both analyses were specified after the Study 1 data were seen; they motivated Study 2 and supplied its pilot variances, and they are not evidence for Study 2 (`results/study2/posthoc/README.md`).

22

The Study 2 preregistration

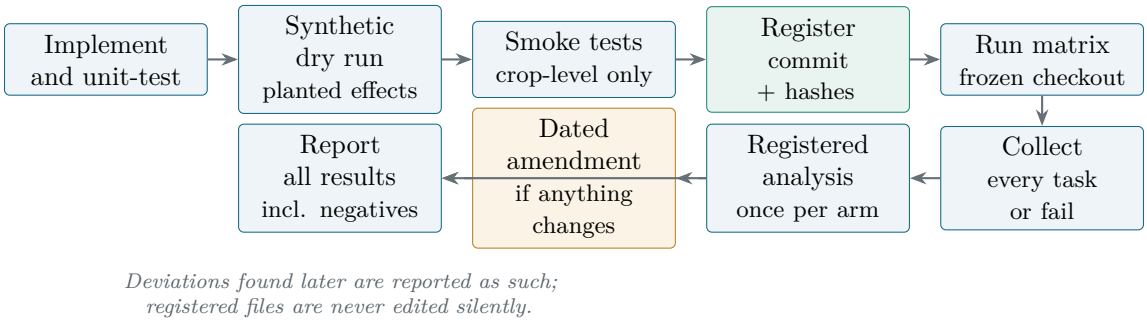


Figure 22.1: The governance loop used for Study 2 (Amendment 6), Amendment 7, Amendment 8, and Amendment 9. The registration commit fixes the analysis code, split manifests, task files, and cache hashes before any model whose result it governs is trained.

22.1 Status at registration

`PREREGISTRATION_STUDY2.md` was committed at `a62e8e7` (2026-10-18 05:07 IST) as Amendment 6 of the original plan. At that moment no Study 2 model had been trained for any registered cell. Before the commit: every split, design table, transplant plan, and input cache was generated and hashed; the full analysis pipeline was dry-run on a synthetic run matrix with planted effects (all recovered); and one-epoch smoke tests were run only on crop-level (P1) splits, so no protocol contrast could be observed (Table M.6).

22.2 What the plan fixes

Recipe
ImageNet initialisation, new linear head, AdamW (learning rate 10^{-4}, weight decay 10^{-4}), batch 64, 20 epochs, cross-entropy, final-epoch endpoint, no validation split, early stopping, or checkpoint selection; random resized crop (scale 0.8–1.0, ratio 3/4–4/3), horizontal and vertical flips, ImageNet normalisation.

Architectures
Twelve (Table 24.1).

Seeds
137, 173, 211, 257, 307, 349, 401, 457, 503, 557, 601, 653, 701, 751, 809, 853, 907, 953, 1009, 1051. Phase 1 uses all twenty, the mechanism arms and CRIC the first ten. The first five had been named as robustness seeds in the original plan and never run. No seed, architecture, or cell may be added or replaced after any result is seen.

Splits
Generated before registration under `splits/study2/` with SHA-256 hashes; trainers refuse any other.

H1 The crossed model on SIPaKMeD P1 and P2 accuracy; REML, bobyqa, Satterthwaite degrees of freedom, two-sided $\alpha = 0.05$; supported if $p < 0.05$ and positive. Singular fits are accepted; on convergence failure the model is simplified (uncorrelated slope, then no seed intercept) but the architecture slope and split effect are never dropped.

Assumption trigger
If no model converges, or residual |skewness| > 1, or excess kurtosis > 3, the primary result becomes a two-way cluster bootstrap (architectures and seeds resampled independently, 10,000 replicates).

Sensitivity
Kenward–Roger degrees of freedom, a binomial GLMM with an observation-level random effect, a seed-level exact sign-flip test, and the cluster bootstrap.

H2–H5
CRIC (H2), leakage at matched diversity (H3), the background context route (H4), and the gap's dependence on suppressed recoverability (H5), in one Holm family.

Exclusions
Runs are excluded only for logged infrastructure failure, non-finite loss, or a failed integrity check, never for magnitude or direction.

Gates
Gate A (H1) decides whether the inflation is reported as certified; Gate B decides whether the causal evidence is "clean" or "murky". Both decide framing only; every result is reported.

22.3 Amendment 7: failure handling

While Phase 1 ran, and before any 2A run had started, two cases the plan did not cover were registered (commit at 06:29 on 18 October). Infrastructure failures (another user's process exhausting GPU memory, loss of the host or controller, a filesystem error) do not count toward the one-retry limit. And because runs are deterministic, a gradient-reversal run whose loss becomes non-finite diverges again on retry; in that arm only, such a run is recorded as *diverged*, its λ leaves that architecture's regime, and it is excluded from H5. No Study 2 run failed or diverged, so neither provision was used.

23

Power analysis

23.1 Method

`analysis/study2/power_simulation.R` fits the H1 model to the sixty Study 1 runs and simulates it. The pilot gives a mean accuracy of 96.84%, an effect of 1.37 pp, architecture intercept and slope SDs of 0.44 and 0.36, split SDs of 0.31 (P1) and 0.59 (P2), a residual SD of 0.69, and a seed SD of zero. Six architectures put the intercept–slope correlation on its -1 boundary, so the data-generating process sets it to zero. Each design cell (seeds $\in \{10, 15, 20, 25, 30\}$, architectures $\in \{6, 8, 10, 12\}$, effect $\in \{0.5, 0.8, 1.0\}$ pp, variance scale $\in \{1, 1.5\}$) was simulated 1,000 times and fitted with the registered model.

23.2 The registered rule and the choice

The rule chose the smallest seed count, with twelve architectures, whose power for a 0.8-pp effect is at least 0.90 under pilot variances and at least 0.75 with every variance inflated by half. Twenty seeds give 0.977 and 0.782; fifteen fell short of the inflated criterion. A 0.8-pp effect exceeds the spread between published SIPaKMeD scores that motivates the question.

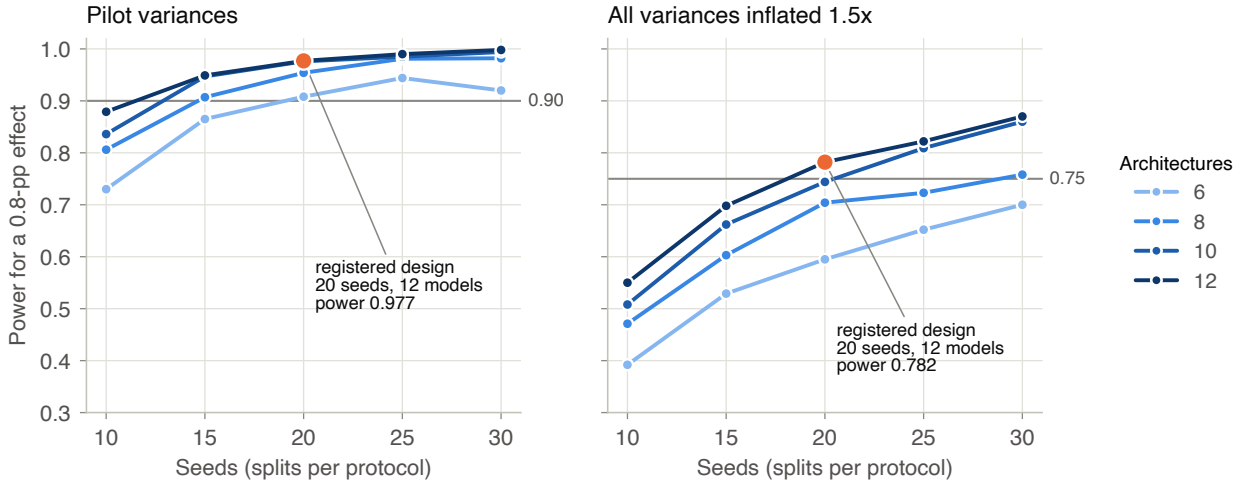


Figure 23.1: Simulated power of the H1 model for a 0.8-pp effect (1,000 simulations per design).

Table 23.1: Simulated power (1,000 simulations per cell) of the registered H1 model for a 0.8-pp effect, by variance scale, number of architectures, and number of seeds.

Variance	Archs	10 seeds	15 seeds	20 seeds	25 seeds	30 seeds
1×	6	0.730	0.865	0.908	0.944	0.920
1×	8	0.806	0.907	0.954	0.981	0.982
1×	10	0.836	0.947	0.976	0.985	0.994
1×	12	0.879	0.949	0.977	0.990	0.998
1.5×	6	0.392	0.529	0.595	0.652	0.700
1.5×	8	0.471	0.603	0.704	0.723	0.758
1.5×	10	0.508	0.662	0.744	0.809	0.860
1.5×	12	0.550	0.698	0.782	0.822	0.870

24

The designs in detail

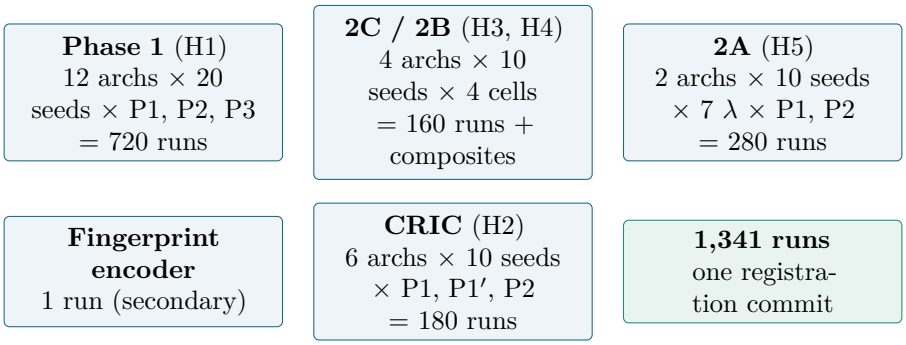


Figure 24.1: The registered Study 2 run matrix. P1-prime equals P1 byte for byte on every SIPaKMeD seed (P2 always has exactly 810 test crops), so it needed no separate SIPaKMeD runs.

24.1 Splits

P1 is a stratified 80:20 crop split; P2 a parent-disjoint split taken from `StratifiedGroupKFold` as the fold whose test size is closest to 810 crops; P3 the same on near-duplicate component groups. For every SIPaKMeD seed P2 has exactly 810 test crops, so P1-prime (a P1 split resampled to P2's size) is byte-identical to P1 and was not trained separately. For CRIC, P1-prime differs from P1 by 2–11 training cells and was trained.

24.2 Architectures

Table 24.1: The twelve Study 2 backbones: penultimate feature dimension, ImageNet weights, and mean Phase 1 training and total minutes per run.

Architecture	Dim	Weights	Train min	Total min
ResNet-18	512	ResNet18_Weights. IMAGENET1K_V1	0.85	0.87
ResNet-50	2048	ResNet50_Weights. IMAGENET1K_V2	2.60	2.64
DenseNet-121	1024	DenseNet121_Weights. IMAGENET1K_V1	3.25	3.29

Architecture	Dim	Weights	Train min	Total min
EfficientNet-B0	1280	EfficientNet_B0_Weights. IMAGENET1K_V1	1.78	1.80
MobileNet-V2	1280	MobileNet_V2_Weights. IMAGENET1K_V2	1.45	1.48
ViT-B/16	768	ViT_B_16_Weights. IMAGENET1K_V1	9.09	9.17
ConvNeXt-T	768	ConvNeXt_Tiny_Weights. IMAGENET1K_V1	5.15	5.21
Swin-T	768	Swin_T_Weights. IMAGENET1K_V1	4.61	4.66
RegNetY-1.6GF	888	RegNet_Y_1_6GF_Weights. IMAGENET1K_V2	2.47	2.50
EfficientNetV2-S	1280	EfficientNet_V2_S_Weights. IMAGENET1K_V1	3.46	3.50
MaxViT-T	512	MaxVit_T_Weights. IMAGENET1K_V1	7.76	7.86
DINOv2 ViT-S/14	384	timm:vit_small_patch14_ dinov2.lvd142m	4.08	4.13

24.3 2C: leakage at matched diversity

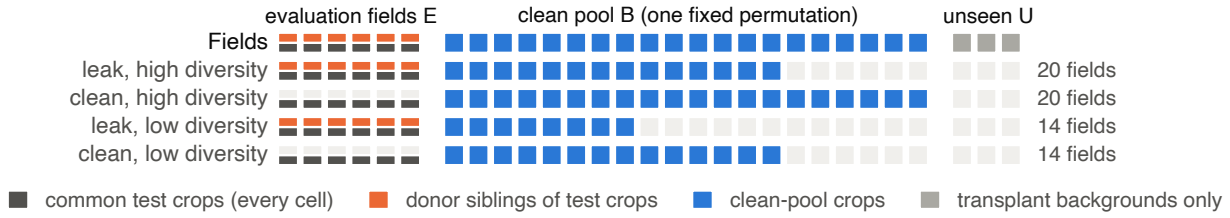


Figure 24.2: The 2C factorial for one seed: one common test set; leak cells add the evaluation fields' sibling crops, clean cells use other fields; within a diversity level crop and field counts are identical, and low-diversity field sets are nested in high-diversity ones.

For each seed and class, 30% of fields with at least two crops are evaluation fields; half of each one's crops (rounded down) form the common test set (about 527 crops) and the rest are its donor siblings. Six fields per class with at least three crops are reserved as unseen background donors for 2B. Each training set holds exactly 220 crops per class. The clean cells draw from non-evaluation fields along one fixed permutation; the leak cells replace the last $|E|$ fields of that permutation with the evaluation fields' donor siblings; the low-diversity cells use the first 70% of the permutation. High-diversity cells therefore contain 720 distinct fields and low-diversity cells 504. Every test crop has a training sibling in the leak cells (2.0 on average at high diversity, 3.0 at low) and none in the clean cells. A 240-crop budget was infeasible for seed 307 and for 10 of 100 unrelated seeds, so 220 was chosen before any training; it was feasible for 100 of 100.

24.4 2B: counterfactual background transplants

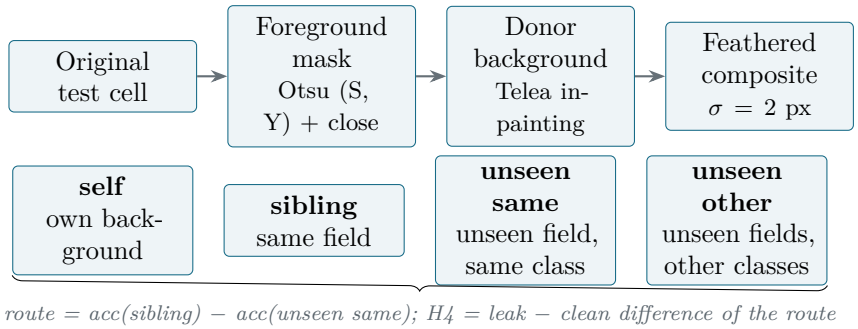


Figure 24.3: The 2B counterfactual background transplants. The *self* condition measures seam and inpainting artifacts; every contrast compares conditions that share them.

24.5 2A: adversarial fingerprint suppression

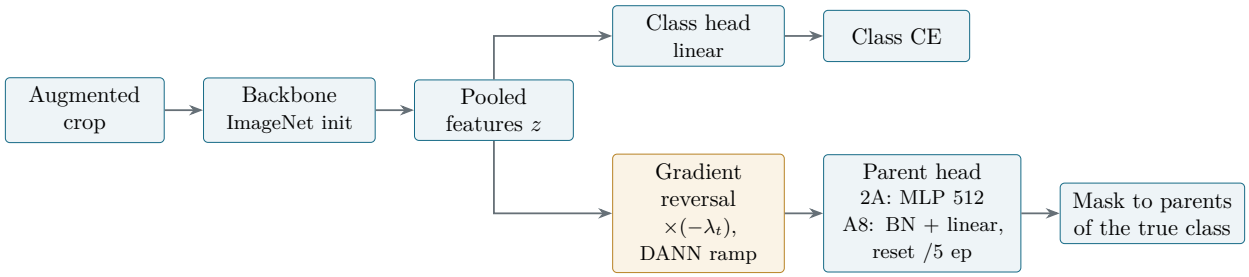


Figure 24.4: The adversarial suppression arms. The parent adversary only competes among parents of the sample's own class, so suppressing it never requires removing class information. Study 2 (2A) used a two-layer MLP; Amendment 8 (F7) used a probe-matched head (per-feature standardisation and one linear layer, its own optimiser at 10^{-3}, re-initialised every five epochs) so that features could not hide parent identity from it while leaving it readable by a linear probe.

The maximum reversal coefficient followed the DANN ramp $\lambda\,(2/(1+e^{-10p})-1)$ over training progress p, with $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3, 1, 3\}$. Recoverability R is the within-class field-identification accuracy of a multinomial logistic regression (C = 1, standardised features) on frozen features of the balanced 424-field probe split (two fitting and two evaluation crops per field). The *regime* is the set of λ whose mean P2 accuracy stays within 1 pp of $\lambda = 0$; the *manipulation check* requires R at the strongest in-regime λ to fall by at least a quarter of its above-floor baseline.

24.6 CRIC

CRIC cells are 100×100 windows centred on the published nucleus coordinates (shifted inward at image borders) of the 400 Figshare slide images, whose MD5 and SHA-256 were verified (`data_manifests/cric_images_verified.json`). Six Bethesda classes; ResNet-18, ResNet-50, DenseNet-121, EfficientNet-B0, ConvNeXt-T, ViT-B/16; P1, P1-prime, P2; ten seeds.

25

Structure before training

For a parent with k crops allocated independently with test fraction q, the parent straddles the split with probability $1-(1-q)^k-q^k$; a test crop has a training sibling with probability $1-q^{k-1}$ and $(1-q)(k-1)$ training siblings on average; and the ratio of distinct training parents between a crop-level and a parent-level split of equal size is $\sum_j\{1-q^{k_j}\}/\{(1-q)m\}$ for m parents. Class–parent dependence is the mutual information between parent and class divided by the class entropy. `cytofair risk-report` computes all of them from any crop-to-parent table (`src/cytofair/risk.py`).

Table 25.1: `cytofair risk-report` closed forms for the four datasets at a 20% test fraction: crops per parent, expected straddle and sibling rates (percent), siblings per test crop, training-diversity ratio, and class–parent dependence (fraction of class entropy explained by the parent).

Dataset	Obs.	Parents	Obs./parent	Straddle	Test w/ sibling	Siblings	Diversity	Class dep.
SIPaKMeD	4,049	966	4.19	44.8	91.7	5.16	1.176	1.000
Cx22	1,320	514	2.57	33.9	83.6	1.95	1.164	–
CRIC	11,534	400	28.84	94.9	100.0	35.56	1.250	0.676
Herlev	917	917	1.00	0.0	0.0	0.00	1.000	1.000

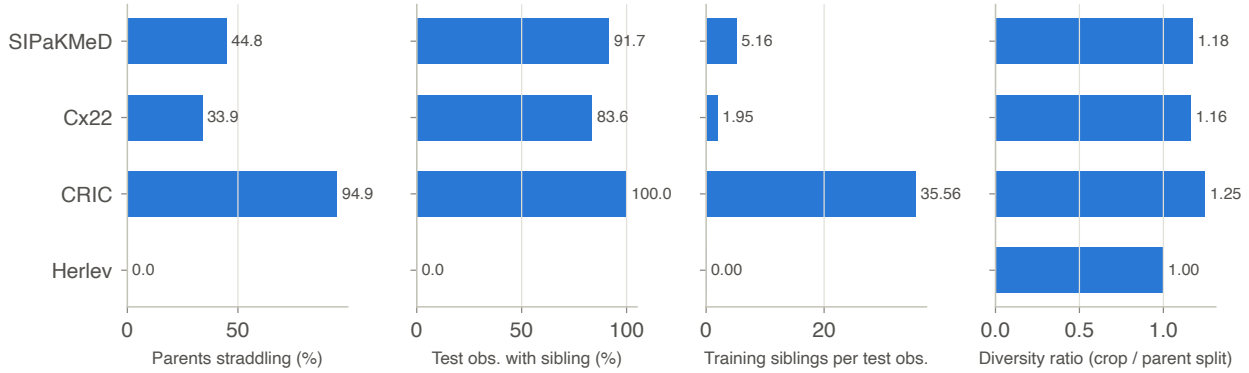


Figure 25.1: Closed-form structural exposure of the four datasets to a random 80:20 crop split.

Under five random P1 splits, the share of test observations sharing at least one source pixel with a training observation was 52.6% for SIPaKMeD, 71.5% for CRIC, and 34.0% for Cx22 (`results/study2/structural/literal_overlap_p1.csv`). CRIC is exposed on every measure: 29 cells per slide image, essentially every test cell with a training sibling, and slide identity explaining two thirds of the class entropy. That is the structure that predicted, before training, that CRIC's inflation would be the largest.

PART IV

Engineering and Execution

26

The host, the data path, and the training stack

26.1 The host and its constraints

All Study 2 and Amendment 8 computation ran on one shared server (`node1`) with eight NVIDIA RTX A6000 GPUs (49 GB each). It has no job scheduler; work runs directly on the login node. The operating rules for the project were: never more than four GPUs at once, check for vacant GPUs first, keep data that feeds GPUs off the shared network filesystem, use the fast local disk and vacate it afterwards.

Table 26.1: Host constraints discovered or imposed, and the response to each.

Constraint	Consequence and response
Four GPUs at most	Study 2 used GPUs 0, 3, 4, 5 (GPUs 1 and 2 were in use by another user); Amendment 8 used 4–7 after another user started jobs on 0–3 twenty-five minutes before launch. The scheduler refuses more than four.
Four CPU cores per user	Found during Study 2 preparation: the per-user cgroup quota is 400% (four cores) on a 128-core machine. The power simulation ran far slower than its core count suggested, a data-loader benchmark was invalid, and per-image augmentation in loader workers could not feed even one GPU. This is very likely the "host bottleneck" recorded in Study 1 (Chapter 15). Response: decode crops once into caches, upload them to the GPU, augment there (Section 26.3), keep CPU threads low, and run heavy CPU analyses on a workstation.
`/scratch` not writable	All node-local work used `/tmp/rchethan1_cytofair` (the 192 GB root partition shared with the operating system) and `/dev/shm`; a disk guard keeps 40 GB free.
Slow network home	The NFS home can stall under load; a recursive search over it had to be stopped early in Study 2 preparation. Rule adopted: only static artifacts go there, and copies are single bandwidth-limited streams.

Constraint	Consequence and response
Shared GPUs	Other users' jobs can appear at any time. The scheduler checks each GPU's memory before launching (at most 30,000 MiB in use), and an infrastructure failure caused by another process does not count as a retry (Amendment 7).
No scheduler	Background controllers launched with `setsid nohup` and detached from the SSH session; a status file records every running task, completion, and failure.

26.2 The data path

26.2.1 Provenance to splits

Every split is derived from the provenance maps of Study 1 (`provenance/*.parquet`), whose recovery and verification are described in Part II. `cytofair study2-design` writes every Study 2 split, factorial design, transplant plan, and the probe split with SHA-256 hashes into `splits/study2/study2_split_manifest.csv`; `cytofair followup-design` does the same for Amendment 8 (`splits/study2/followup/`). Trainers hash the split file before use and refuse a mismatch.

26.2.2 Packed crop caches

Decoding thousands of small BMP and PNG files on four cores is slow, and reading them from the network home is unreliable. Each image set is therefore packed once into a cache on the node-local disk: one `uint8` array of all pixels (`{name}_pixels.npy`, memory-mapped), an index (`{name}_index.parquet`) with each crop's offset, height, width, and the SHA-256 of its pixels, and a manifest with the SHA-256 of both files. A cache is verified on load; a hash mismatch stops the run.

Table 26.2: Caches used by Study 2 and Amendment 8 (hashes in Appendix P).

Cache	Images	Content
`sipakmed`	4,049	every SIPaKMeD crop
`cric`	11,534	100×100 windows around every CRIC nucleus
`transplant_seed*` (10)	4,096–4,400 each	2B composites of each seed's common test set
`sipakmed_masked`	4,049	background set to the ImageNet mean (F3)
`sipakmed_reinhard50/100`	4,049 each	Reinhard normalisation at strength 0.5 and 1 (F6)
`ablation_seed*` (10)	six per test crop	the F2 carrier ablations

26.3 The GPU-resident input path

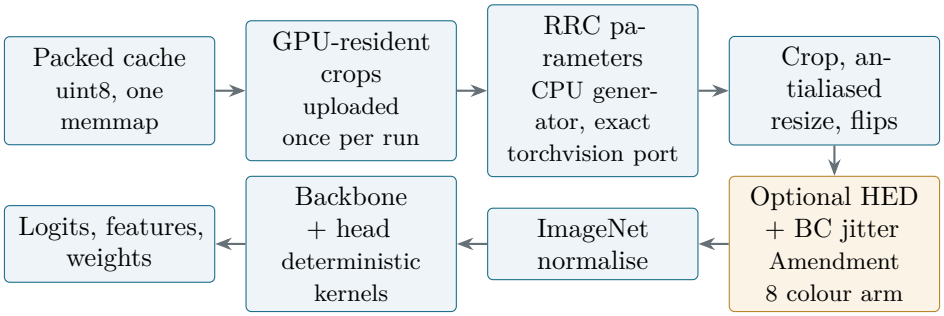


Figure 26.1: The GPU-resident training input path. The host allowed four CPU cores per user, which made per-image PIL augmentation in data-loader workers the bottleneck; moving the registered augmentation onto the GPU removed it while sampling exactly the same crop geometry as torchvision for the same random stream.

The registered augmentation is `RandomResizedCrop(224, scale 0.8-1.0, ratio 3/4-4/3)`, horizontal and vertical flips, and ImageNet normalisation. `src/cytofair/gpu_augment.py` keeps every crop of a split on the GPU as a `uint8` tensor and draws crop parameters with an exact port of torchvision's `RandomResizedCrop.get_params` from an explicit CPU generator, including the float32 exponential torchvision uses for the aspect ratio. A unit test compares the sampled geometry with torchvision's for four image shapes and 25 seeds each and requires identity. Resizing uses antialiased bilinear interpolation, PyTorch's implementation of PIL's filter, which can differ from PIL by uint8 rounding. The Amendment 8 colour arm adds HED optical-density jitter and brightness/contrast jitter, drawn from the same generator after the geometry, so runs without colour augmentation reproduce the registered random stream exactly.

26.4 Determinism

Every run sets cuDNN deterministic mode, disables cuDNN benchmarking, calls `torch.use_deterministic_algorithms(True,warn_only=True)`, and sets `CUBLAS_WORKSPACE_CONFIG=:4096:8`. During smoke tests ViT-B/16 raised a warning that the memory-efficient attention backward pass is non-deterministic; the flash, memory-efficient, and cuDNN attention kernels were disabled in favour of the math kernel, and two identical ViT runs then produced bitwise-identical logits. The follow-up evaluation code avoids `index_add_` (non-deterministic on CUDA) by computing group sums as one-hot matrix products.

Final weights are stored in float16 to halve their size. Re-evaluating the saved Phase 1 checkpoints reproduced the recorded test accuracy exactly for 418 of 480 models; 56 differed by one test crop and six by two or three crops out of 810, all from float16 rounding. Every follow-up comparison is made within one re-evaluation, so this does not bias any registered contrast.

26.5 What every run leaves behind

`src/cytofair/train_study2.py` writes, for every run:

- a JSON record with the dataset, protocol, architecture, seed, accuracy, macro-F1, balanced accuracy, NLL, 15-bin ECE, the confusion matrix, train and test crop and parent counts, the fraction of test crops with a training parent, the recipe and augmentation, adversary settings,

the split file and semantic hashes, cache hashes, the git commit, GPU, start and end times, runtime metadata, a per-epoch history, and the path and SHA-256 of every artifact;

- per-crop test logits (Parquet), float16 final weights, penultimate features of the 1,696 probe crops (NPZ), and logits for any registered extra evaluation set (the 2B composites).

The follow-up evaluation tasks write comparable records (`src/cytofair/followup_eval.py`), verifying each checkpoint's hash before loading it. The exported tables in `results/study2/run_records` keep every field the record plots, including all 45,600 epoch histories.

27

Scheduling, execution, and archiving

27.1 Scheduling, failure handling, and collection

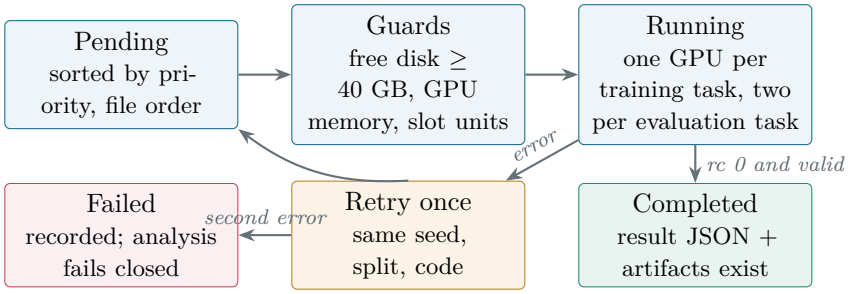


Figure 27.1: The resumable scheduler (`scripts/study2/launch_matrix.py`). A task whose valid result already exists is skipped on restart, so the controller can be relaunched after any interruption without duplicating a run.

A task is a JSON line naming its command, the result file that proves completion, a priority, the GPU share it needs, and an `expect` block of fields its result must carry (dataset, protocol, architecture, seed, split hash, and for the follow-up arms the cache, colour augmentation, and adversary). `scripts/study2/build_tasks.py` wrote the 1,341 Study 2 tasks and `scripts/study2/build_followup_tasks.py` the 1,580 follow-up tasks; both sets are committed and hashed. Training tasks took a whole GPU; evaluation-only tasks took half, so two ran per GPU.

The collectors (`analysis/study2/collect_runs.py`, `analysis/study2/followup/prepare_followup.py`) require exactly one valid result per registered task whose recorded fields match its `expect` block; anything missing, duplicated, or mismatched is listed and the analysis stops. `-allow-incomplete` produces progress reports only. Under Amendment 7, adversarial runs whose every attempt ended in a non-finite loss are recognised from their logs and recorded as diverged rather than missing. Neither matrix had a failure, retry, or divergence.

27.2 Execution

27.2.1 Smoke tests

Before each registration, short runs exercised every code path on splits that cannot reveal a registered contrast: P1 splits only for Study 2; leak cells, single P1 runs, and single evaluation tasks

for Amendment 8 (Table M.6). Their metrics were checked only for finiteness and integrity (for example, that a re-evaluated checkpoint reproduces its recorded accuracy).

27.2.2 The two matrices

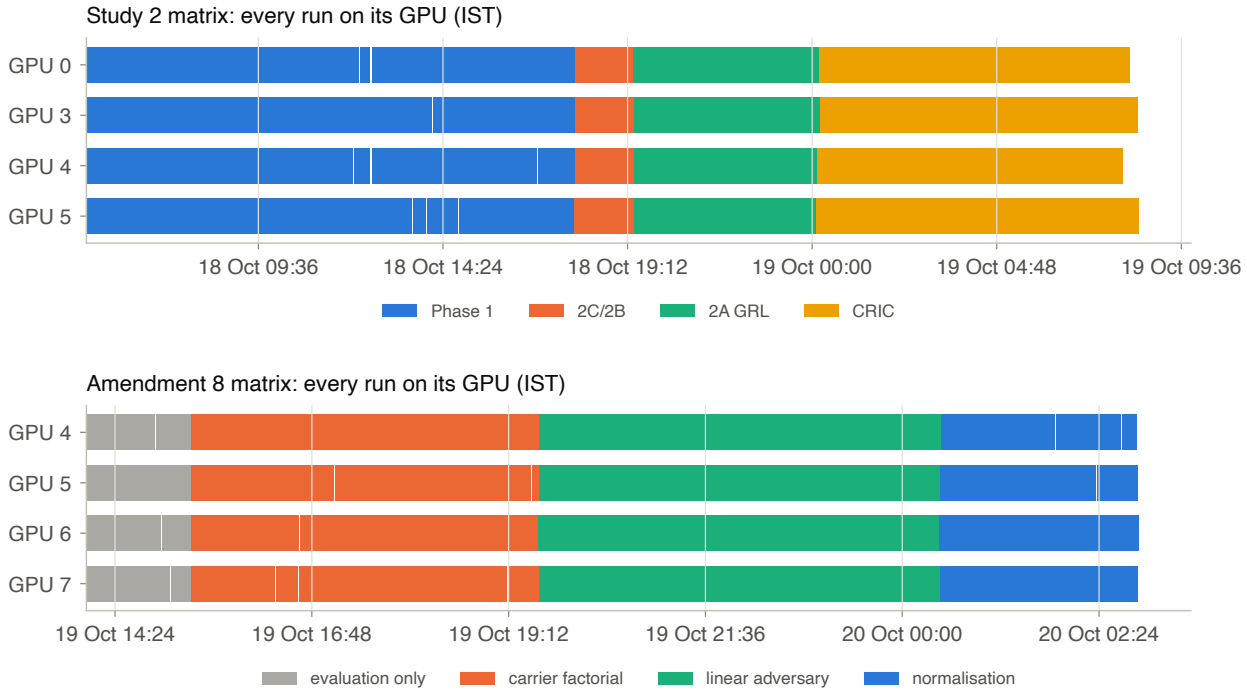


Figure 27.2: Every run of both matrices on its GPU, from the start and end times in the run records (evaluation tasks from the controller log). The arms ran in priority order; white gaps are launch overhead.

Table 27.1: Compute ledger of every Study 2 and Amendment 8 arm (node1, NVIDIA RTX A6000; times in IST). GPU-hours are summed per-run elapsed time; runs shared four GPUs.

Arm	Runs	Mean min	GPU-h	First start	Last finish
Study 2 Phase 1 (H1)	720	3.9	47.1	18 Oct 05:07	18 Oct 17:50
Study 2 2C/2B (H3, H4)	160	2.0	5.4	18 Oct 17:48	18 Oct 19:21
Study 2 2A (H5)	280	3.9	18.2	18 Oct 19:20	19 Oct 00:12
Study 2 CRIC (H2)	180	10.7	32.0	19 Oct 00:06	19 Oct 08:28
A8 carrier factorial (F3, F4)	480	1.9	15.2	19 Oct 15:19	19 Oct 19:34
A8 linear adversary (F7)	300	3.7	18.5	19 Oct 19:33	20 Oct 00:27
A8 stain normalisation (F6)	160	3.4	9.1	20 Oct 00:27	20 Oct 02:52
A8 ablations + Grad-CAM (F2)	160	0.2	0.6	19 Oct 14:03	19 Oct 14:14
A8 LEACE erasure (F5)	480	0.8	6.5	19 Oct 14:13	19 Oct 15:19

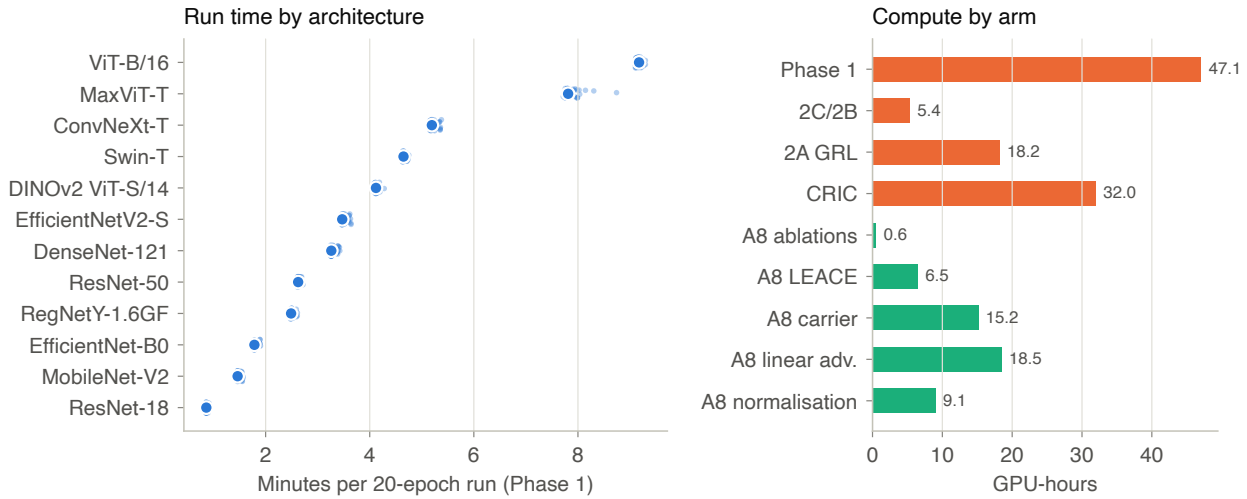


Figure 27.3: Left: wall time of each Phase 1 run by architecture (dots) and its median (ring). Right: GPU-hours by arm.

27.2.3 Software

Table 27.2: Software and hardware recorded in every training run.

Component	Version
python	3.13.9
torch	2.9.0+cu128
torchvision	0.24.0+cu128
timm	1.0.22
cuda	12.8
cudnn	91002
cuda_device	NVIDIA RTX A6000
numpy	2.5.2
pandas	3.0.5
scipy	1.18.1
scikit_learn	1.9.0
opencv	5.0.0
pillow	12.3.0
platform	Linux-4.18.0-372.9.1.el8.x86_64-x86_64-with-glibc2.28
cublas_workspace_config	:4096:8

27.3 Archiving and vacating

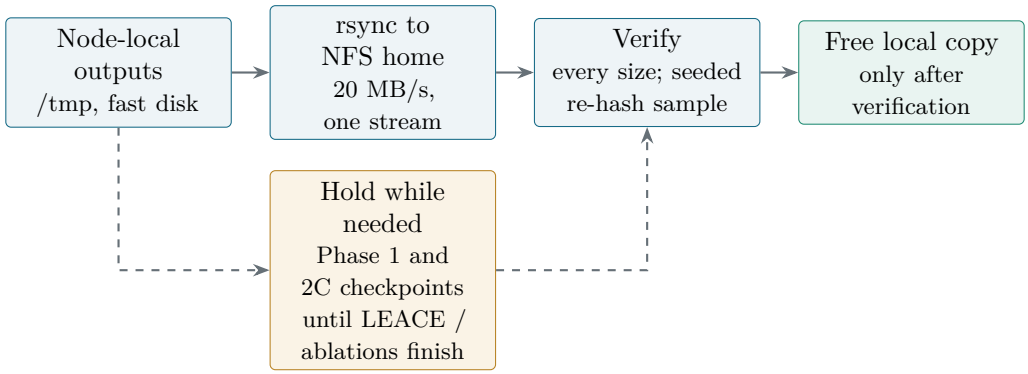


Figure 27.4: Offloading and archiving. The network home is slow and can stall, so copies are single, bandwidth-limited streams; nothing node-local is deleted until its archived copy has been verified.

The follow-up matrix would write about 52 GB of checkpoints into a 71 GB margin, so Study 2 checkpoints were moved to the network home while it ran: the gradient-reversal and CRIC checkpoints first, then the 2C checkpoints once all 160 ablation tasks had finished, then the Phase 1 checkpoints once all 480 LEACE tasks had finished. Each experiment was copied, every checkpoint's size compared, a seeded sample of fifty re-hashed against the hash in its run record, and only then deleted locally. At the end, the follow-up outputs were copied and verified (3,580 artifacts, sizes all matching, 100 re-hashed), the remaining Study 2 outputs re-verified (4,000 artifacts, 200 re-hashed), and the node-local working directory and shared-memory directory deleted. The archive (`~/cytofair_study2_archive`, 121 GB) holds every run record, prediction, feature file, checkpoint, log, and status file.

28

Errors, setbacks, and fixes

Study 1's setbacks (the invalid first classification matrix, the MAnet numerical failure, and the host bottleneck) are recorded in Chapter 15. Everything that went wrong afterwards is listed here, including mistakes in tooling and in procedure; none changed a registered model, split, seed, or endpoint.

Table 28.1: Every error and setback of Study 2 and Amendment 8, and its fix.

#	What went wrong	Fix	Effect on results
Study 2 preparation (before registration)			
1	No PDF text tools on the workstation	Python PDF libraries for reading and rendering	none
2	A recursive search over the slow network home stalled	stopped; rule: never scan the network home	none
3	`/scratch` not writable	node-local `/tmp` and `/dev/shm`	none
4	Four-core CPU quota starved data loaders	GPU-resident augmentation; heavy CPU analysis off-host	none (exact port, unit-tested)
5	Power script mis-named variance components; correlation on its -1 boundary	pattern matching; correlation set to 0 in the simulation	documented in the plan
6	R: S4 slot read with `$` in the protocol model	`@optinfo`	none
7	R: tidy-evaluation operator inside base `subset`	explicit indexing	none
8	2C budget of 240 crops per class infeasible for some seeds	220 crops, low/high ratio 0.70, feasible for 100/100 seeds	chosen before training
9	pandas 3 read-only arrays; Pillow deprecation	copies; argument removed	none
10	Non-deterministic attention backward (ViT)	math attention kernel only; bitwise repeat verified	none
11	Lint findings, a wrong test expectation	fixed	none
12	SSH launches hung on backgrounded command chains	`cd; setsid nohup ... & disown`	none
13	A shell wait loop did not split its input	explicit `read`	none
14	No rule for diverged adversarial runs	Amendment 7, before any 2A run	none (no divergence)
Study 2 analysis			

#	What went wrong	Fix	Effect on results
15	One float16 probe-feature row overflowed (EfficientNetV2-S, one crop)	documented; no follow-up arm reads it	one crop in a descriptive measure
16	A completion watcher never fired	replaced with a file-based check	none
Amendment 8			
17	Inference-mode tensors reused in autograd; non-deterministic `index_add_`	clone; one-hot matrix sums	caught in tests
18	R summary object not serialisable to JSON	converted to a list	caught in dry run
19	Remote `pgrep -f` matched its own command line	file-based completion checks	none
20	Interim preparer crawled on a GPU busy with training	stopped; registered run on a free GPU	none
21	**Preparer labelled standard carrier runs as colour-augmented** (`None` became a truthy NaN)	condition read from each run record; unit test; **Amendment 9**	script halted before output; no estimate seen
22	An archive fetch included smoke checkpoints	refetched records only	none
23	Dose figure plotted out-of-regime levels; labels overlapped	registered regimes only; layout fixed	presentation only
24	Negative numbers typeset with a hyphen	typographic minus in the macro generator	presentation only

The one that mattered

Error 21 is the only one that could have changed a reported number. The synthetic dry run had written condition labels directly and so never exercised the preparer's labelling. The R script stopped at F4 because the contrast term was missing, which exposed the bug before any output existed. The lesson recorded for the future: dry runs must go through every stage of the real pipeline, not start from the tables that stage produces.

PART V

Study 2 Results

29

Overview of the registered outcome

All 1,341 registered runs completed between 18 Oct 05:07 and 19 Oct 08:28 IST. None failed, needed a retry, or diverged; every run was trained from the registration commit; every result matched its registered split hash; and the collector reported no missing, duplicated, or mismatched task. The registered analysis (`analysis/study2/run_all.sh`) ran once, on 19 October at 12:56.

Table 29.1: The five registered Study 2 tests (H5 in pp of gap per pp of recoverability).

	Question	Estimate	Adjusted p	Verdict
H1	SIPaKMeD P1 − P2 accuracy (primary, tested alone)	0.97	<0.001	**supported**
H2	CRIC P1 − P2 accuracy	12.08	<0.001	**supported**
H3	Leakage at matched diversity (2C)	1.73	0.02	**supported**
H4	Own-context route through backgrounds (2B)	0.42	0.23	**not supported**
H5	Gap versus suppressed recoverability (2A)	0.06	0.16	**not supported**

Gate A (H1 supported) means the inflation is reported as certified. Gate B was "murky": H3 was supported but H4 was not and H5 was not supported (one of two architectures failed its manipulation check). That is the result that motivated Amendment 8.

30

H1: crop-level splitting inflates SIPaKMeD accuracy

30.1 The registered test

Mean test accuracy was 97.14% under P1 and 96.18% under P2. The registered residual check flagged heavy tails (residual skewness −0.96, excess kurtosis 3.64, Shapiro–Wilk p <0.001), so, as registered, the two-way cluster bootstrap is the primary inference: 0.97 pp (95% interval 0.59–1.39, bootstrap p <0.001). Every sensitivity analysis agrees (Table 30.1): the mixed model gives the same estimate with Satterthwaite df = 38.8 and p <0.001 under Kenward–Roger, the seed-level sign-flip test gives p <0.001, and the binomial mixed model a log odds ratio of 0.304 (p <0.001).

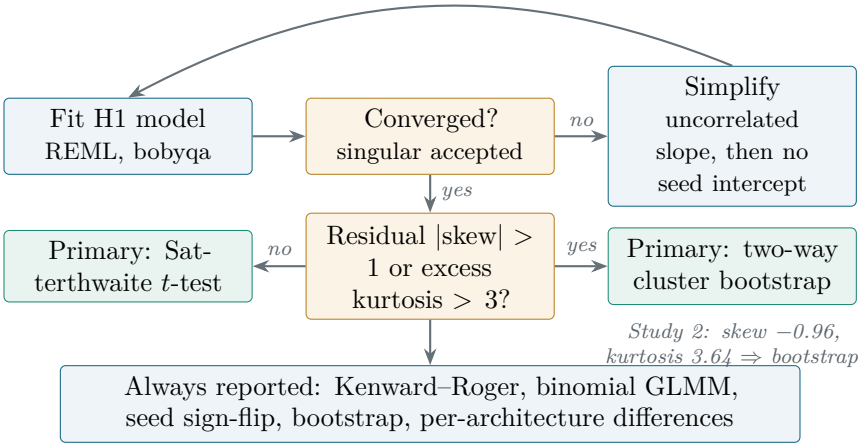


Figure 30.1: The registered H1 decision path. The residual check fired, so the cluster bootstrap became the primary inference exactly as registered; every other method gave the same answer.

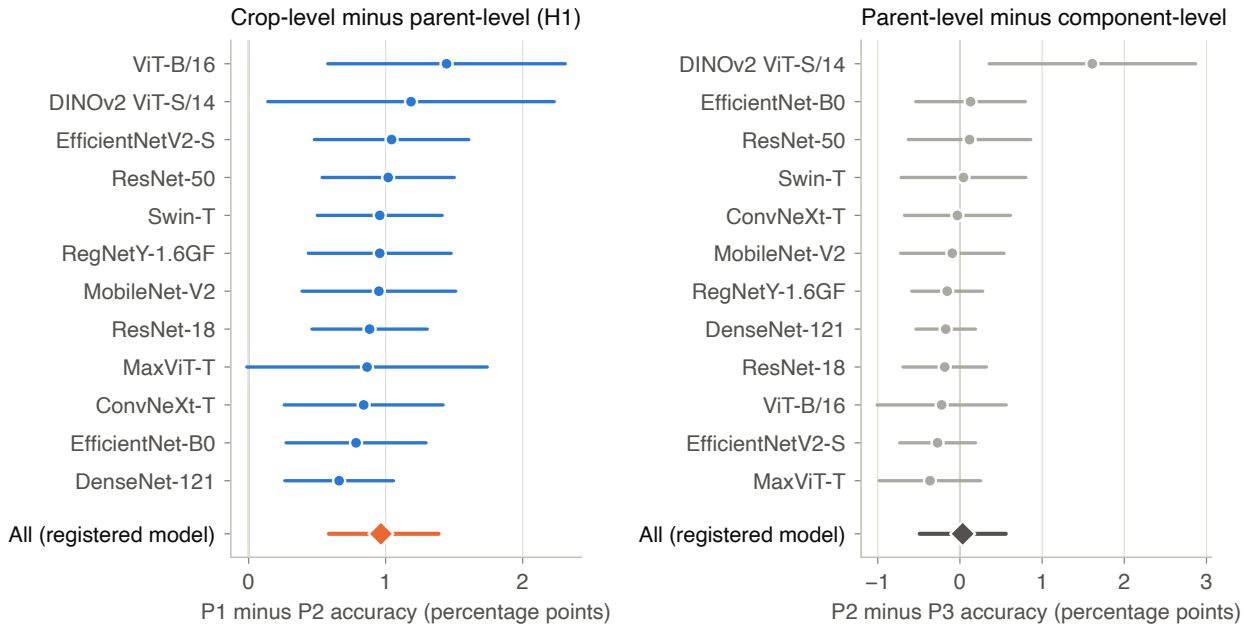


Figure 30.2: P1 − P2 (left) and P2 − P3 (right) per architecture (mean and 95% t interval over 20 seeds) with the registered pooled estimates.

Table 30.1: Every protocol model with its registered sensitivity analyses: Satterthwaite and Kenward–Roger degrees of freedom, two-way cluster bootstrap, and seed sign-flip test. The primary source is the bootstrap when the registered residual check fails.

Contrast	Model	Est.	SE	df_S	p_S	df_{KR}	p_{KR}	Bootstrap CI	Sign-flip p	Primary
H1 accuracy	full	0.97	0.17	38.8	<0.001	18.8	<0.001	0.59–1.39	<0.001	cluster_bootstrap
H1 macro-F1	full	0.96	0.17	38.8	<0.001	18.7	<0.001	0.59–1.39	<0.001	cluster_bootstrap
H1 balanced accuracy	full	0.98	0.17	38.8	<0.001	18.7	<0.001	0.60–1.40	<0.001	cluster_bootstrap
P2 − P3 accuracy	full	0.03	0.25	25.7	0.90	25.2	0.90	−0.45–0.60	0.88	lmm
H2 CRIC accuracy	full	12.08	0.40	8.8	<0.001	8.8	<0.001	11.23–12.92	0.002	lmm
CRIC P1′ − P2	full	12.11	0.43	17.5	<0.001	10.1	<0.001	11.21–13.05	0.002	lmm
CRIC balanced accuracy	full	19.64	0.71	18.0	<0.001	7.2	<0.001	18.00–21.34	0.002	lmm
CRIC macro-F1	full	20.59	0.60	6.3	<0.001	6.3	<0.001	19.19–22.02	0.002	lmm

30.2 Consistency

All 12 architectures scored higher under P1 (0.66–1.44 pp) and 11 of them individually had intervals excluding zero. At the seed level the picture is noisier but one-sided (Figure 30.4). The parent-level error rate is 1.34 times the crop-level one. Macro-F1 (0.96 pp) and balanced accuracy (0.98 pp) moved by the same amount.

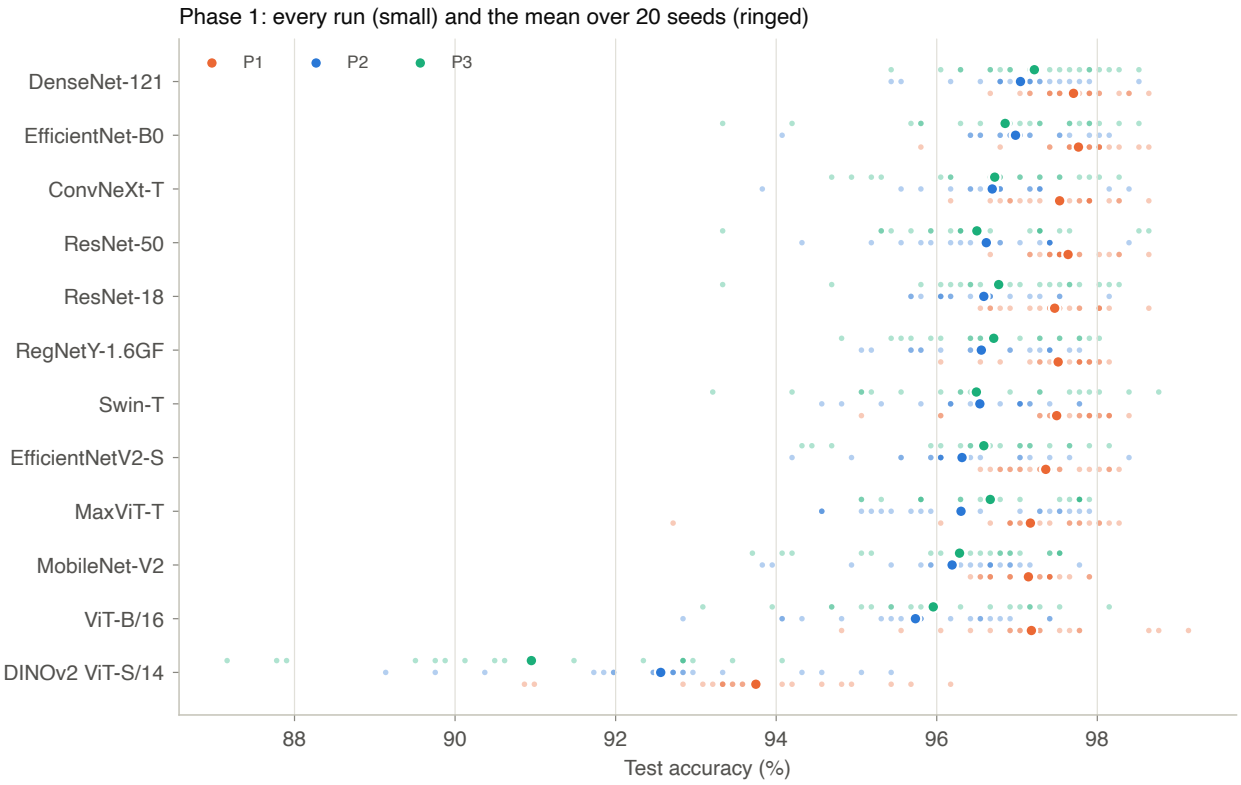


Figure 30.3: Every Phase 1 run and the per-protocol mean over 20 seeds, by architecture.

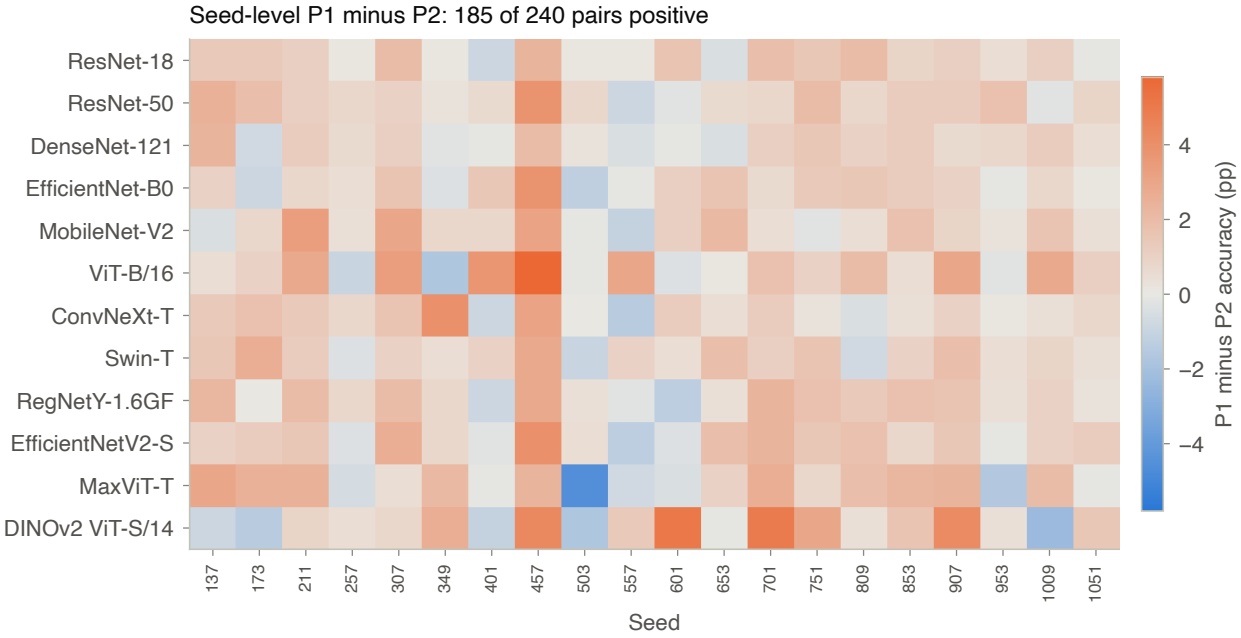


Figure 30.4: P1 − P2 accuracy for every architecture and seed (diverging scale, zero in grey).

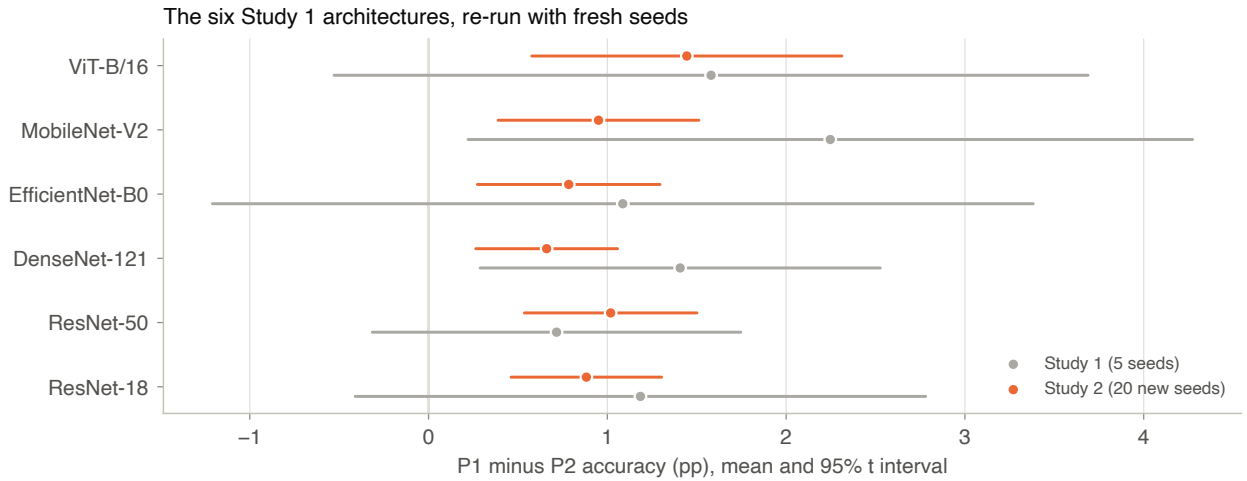


Figure 30.5: The six Study 1 architectures: Study 1's five-seed estimates and Study 2's twenty-seed estimates of P1 − P2. Study 2's intervals are narrower and all exclude zero.

30.3 P2 versus P3, rankings, and the ordering gate

Merging near-duplicate-linked parents on top of parent-level splitting changed nothing (P2 − P3 0.03 pp, −0.49 to 0.56, $p = 0.90$). The registered P3 ≤ P2 ≤ P1 ordering passed for 4 of twelve architectures, as expected when two protocols are equivalent and the order between them is decided by noise. The architecture ranking was largely preserved between P1 and P2 (Kendall $\tau = 0.82$, 0.42–0.86; 6/66 pairs reordered).

30.4 Crop-level evidence

Within crops that were test crops under both protocols in different seeds (3,664 crops), accuracy was 1.29 pp higher under P1 (1.10–1.50; M1). Within P1, a crop sharing source pixels with a training crop was more often correct (log odds 0.45, $p < 0.001$), as was a crop with more training siblings (log odds 0.15 per log sibling, $p = 0.01$), adjusting for class and the crop's difficulty under P2 (M2).

30.5 Training behaviour

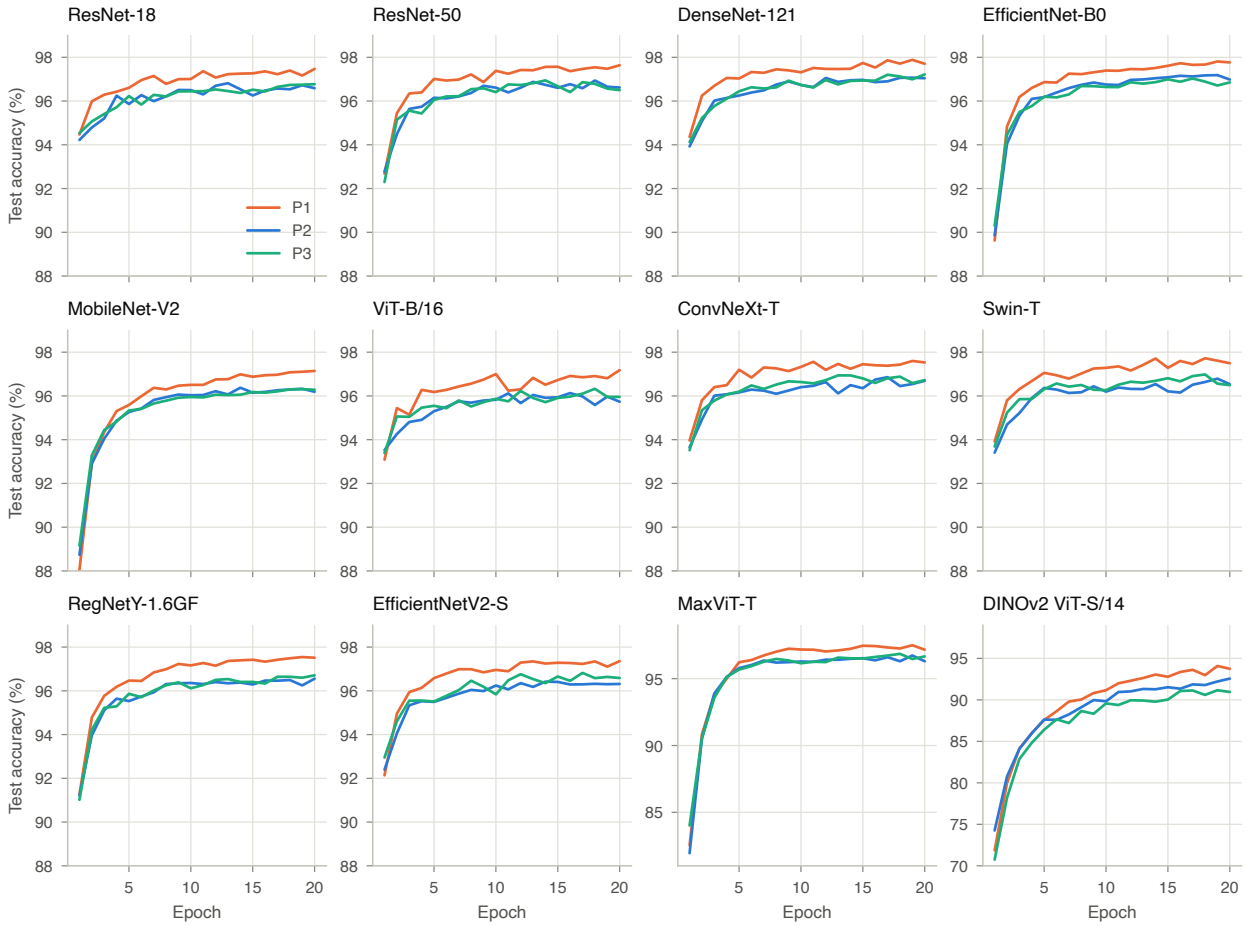


Figure 30.6: Mean test accuracy per epoch under P1, P2, and P3 (20 seeds each). The P1 advantage appears within the first few epochs and persists to the final-epoch endpoint.

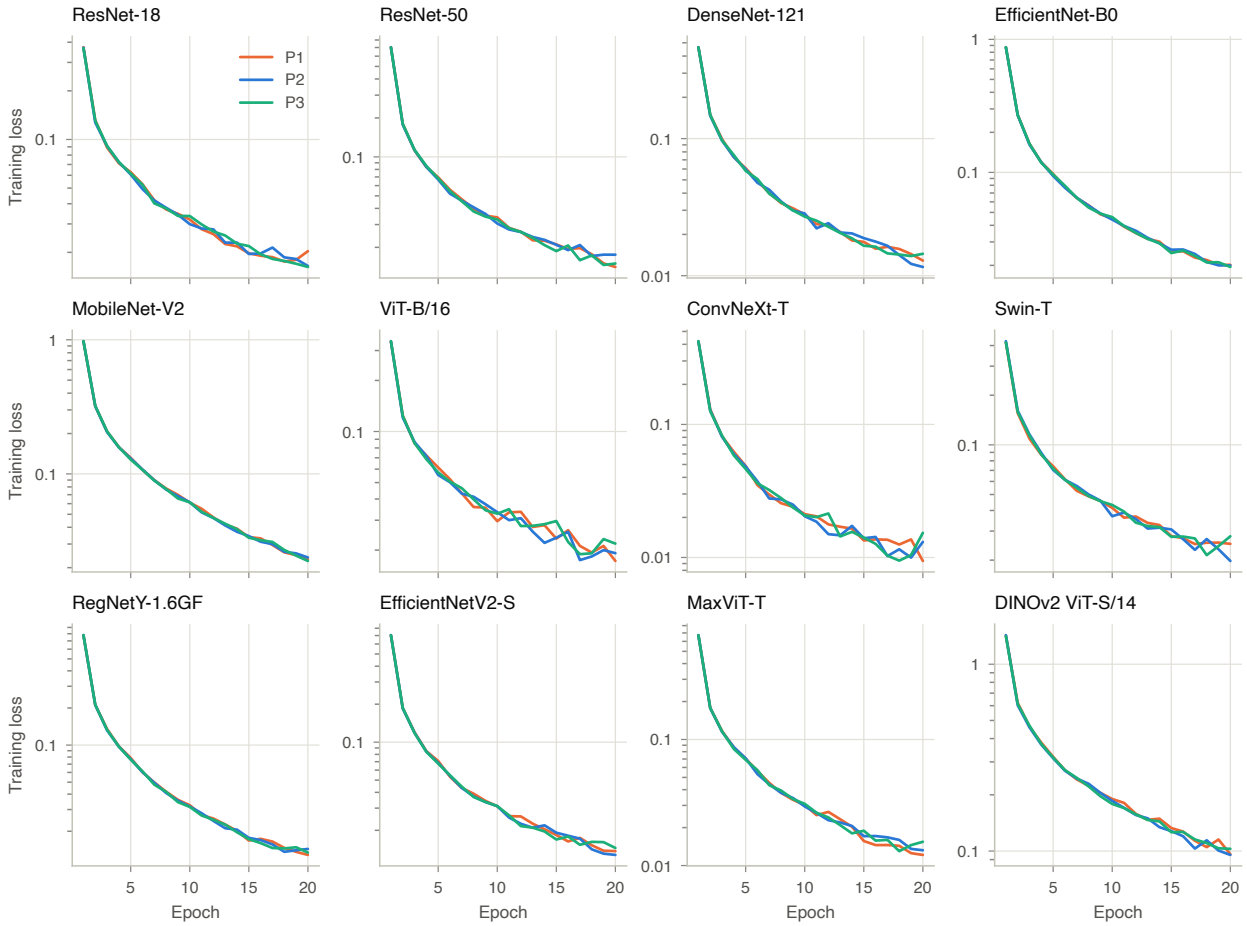


Figure 30.7: Mean training loss per epoch (log scale). Training is indistinguishable across protocols; the difference is in what the test set contains, not in how training proceeds.

30.6 Calibration and errors

Pooled over runs, the 15-bin expected calibration error was 1.78% under P1, 2.43% under P2, and 2.37% under P3, and the negative log-likelihood 0.102, 0.142, and 0.136 nats: crop-level splits flatter calibration as well as accuracy.

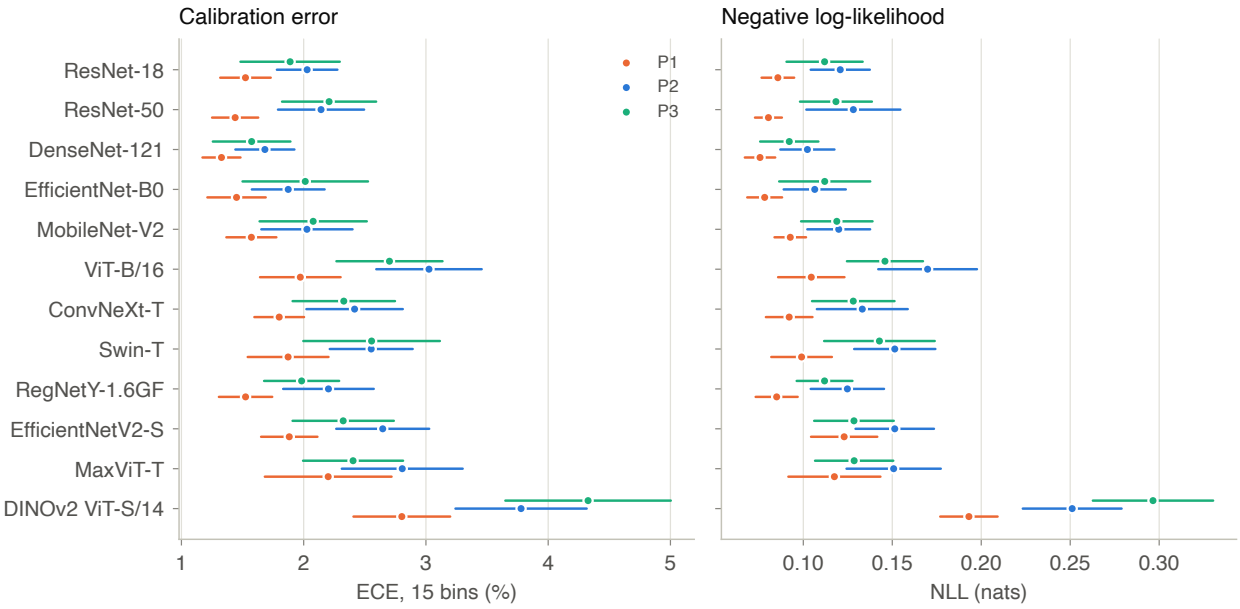


Figure 30.8: Calibration error and negative log-likelihood by architecture and protocol (mean and 95% t interval).

Table 30.2: Phase 1 (SIPaKMeD, 20 seeds per cell): test metrics in percent (mean ± SD over seeds); NLL in nats.

Architecture	Prot.	Accuracy	Macro-F1	Balanced acc.	ECE	NLL
ResNet-18	P1	97.47 ± 0.56	97.48 ± 0.56	97.49 ± 0.56	1.52 ± 0.44	0.086
	P2	96.59 ± 0.68	96.60 ± 0.67	96.59 ± 0.68	2.03 ± 0.53	0.121
	P3	96.77 ± 1.22	96.78 ± 1.21	96.77 ± 1.21	1.89 ± 0.87	0.112
ResNet-50	P1	97.64 ± 0.46	97.65 ± 0.45	97.65 ± 0.46	1.44 ± 0.40	0.080
	P2	96.62 ± 0.94	96.63 ± 0.94	96.62 ± 0.94	2.14 ± 0.75	0.128
	P3	96.50 ± 1.23	96.51 ± 1.23	96.49 ± 1.26	2.21 ± 0.82	0.118
DenseNet-121	P1	97.70 ± 0.51	97.71 ± 0.51	97.72 ± 0.51	1.33 ± 0.33	0.076
	P2	97.04 ± 0.74	97.05 ± 0.74	97.05 ± 0.74	1.68 ± 0.52	0.102
	P3	97.22 ± 0.83	97.23 ± 0.83	97.21 ± 0.83	1.57 ± 0.68	0.092
EfficientNet-B0	P1	97.77 ± 0.61	97.77 ± 0.62	97.78 ± 0.61	1.45 ± 0.51	0.078
	P2	96.98 ± 0.86	96.98 ± 0.87	96.99 ± 0.85	1.87 ± 0.64	0.106
	P3	96.85 ± 1.35	96.86 ± 1.34	96.86 ± 1.34	2.01 ± 1.10	0.112
MobileNet-V2	P1	97.14 ± 0.46	97.15 ± 0.46	97.16 ± 0.46	1.57 ± 0.44	0.093
	P2	96.19 ± 1.01	96.20 ± 1.01	96.19 ± 1.02	2.03 ± 0.80	0.120
	P3	96.28 ± 1.22	96.29 ± 1.22	96.28 ± 1.22	2.08 ± 0.94	0.119
ViT-B/16	P1	97.18 ± 1.03	97.19 ± 1.02	97.19 ± 1.04	1.97 ± 0.70	0.104
	P2	95.73 ± 1.21	95.73 ± 1.23	95.74 ± 1.21	3.02 ± 0.92	0.170
	P3	95.96 ± 1.30	95.96 ± 1.31	95.97 ± 1.30	2.70 ± 0.93	0.146
ConvNeXt-T	P1	97.53 ± 0.64	97.54 ± 0.63	97.54 ± 0.64	1.80 ± 0.43	0.092
	P2	96.69 ± 0.95	96.70 ± 0.94	96.69 ± 0.95	2.42 ± 0.84	0.133
	P3	96.72 ± 1.06	96.72 ± 1.07	96.73 ± 1.05	2.33 ± 0.89	0.128
Swin-T	P1	97.49 ± 0.84	97.50 ± 0.84	97.51 ± 0.84	1.87 ± 0.71	0.099
	P2	96.54 ± 0.89	96.55 ± 0.88	96.55 ± 0.88	2.55 ± 0.72	0.151
	P3	96.49 ± 1.46	96.50 ± 1.47	96.50 ± 1.46	2.56 ± 1.19	0.143
RegNetY-1.6GF	P1	97.51 ± 0.54	97.52 ± 0.54	97.53 ± 0.54	1.52 ± 0.47	0.085
	P2	96.56 ± 0.83	96.57 ± 0.84	96.56 ± 0.83	2.20 ± 0.79	0.125
	P3	96.71 ± 0.93	96.72 ± 0.94	96.71 ± 0.93	1.98 ± 0.66	0.112
EfficientNetV2-S	P1	97.36 ± 0.54	97.37 ± 0.54	97.37 ± 0.53	1.88 ± 0.49	0.123
	P2	96.31 ± 0.95	96.32 ± 0.96	96.32 ± 0.95	2.65 ± 0.81	0.151
	P3	96.59 ± 1.09	96.59 ± 1.09	96.59 ± 1.09	2.32 ± 0.88	0.128
MaxViT-T	P1	97.17 ± 1.19	97.17 ± 1.20	97.19 ± 1.18	2.20 ± 1.11	0.117
	P2	96.30 ± 1.21	96.31 ± 1.20	96.31 ± 1.21	2.81 ± 1.06	0.151
	P3	96.67 ± 0.97	96.67 ± 0.98	96.68 ± 0.97	2.40 ± 0.87	0.129

Architecture	Prot.	Accuracy	Macro-F1	Balanced acc.	ECE	NLL
DINOv2 ViT-S/14	P1	93.75 ± 1.34	93.78 ± 1.33	93.79 ± 1.33	2.80 ± 0.85	0.193
	P2	92.56 ± 1.60	92.61 ± 1.55	92.61 ± 1.60	3.78 ± 1.15	0.251
	P3	90.95 ± 1.95	90.99 ± 1.92	90.99 ± 1.95	4.33 ± 1.45	0.297

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H2: the CRIC inflation is an order of magnitude larger

CRIC accuracy fell from 90.48% under crop-level splitting to 78.40% under slide-level splitting: H2 = 12.08 pp (11.16–12.99; Holm p <0.001). The error rate rose 2.27-fold. Because the classes are imbalanced, balanced accuracy fell further, from 78.53% to 58.89% (19.64 pp), and macro-F1 by 20.59 pp. The size-matched P1-prime gave the same inflation (12.11 pp), so it is not a training-size effect.

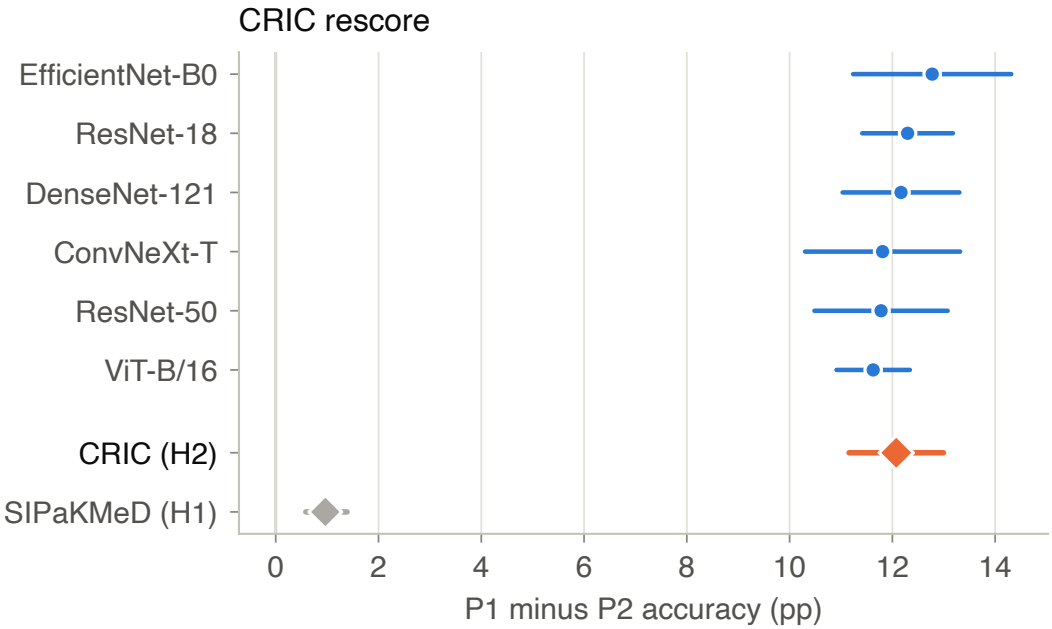


Figure 31.1: CRIC P1 − P2 per architecture, the registered estimate, and the SIPaKMeD estimate for scale.

The loss is concentrated in the abnormal classes (Figure 31.2). Recall for ASC-H, LSIL, and HSIL falls by tens of points when slides are held out, while the majority NILM class barely moves; the crop-level split had been rewarding the recognition of slides whose abnormal cells were already seen.

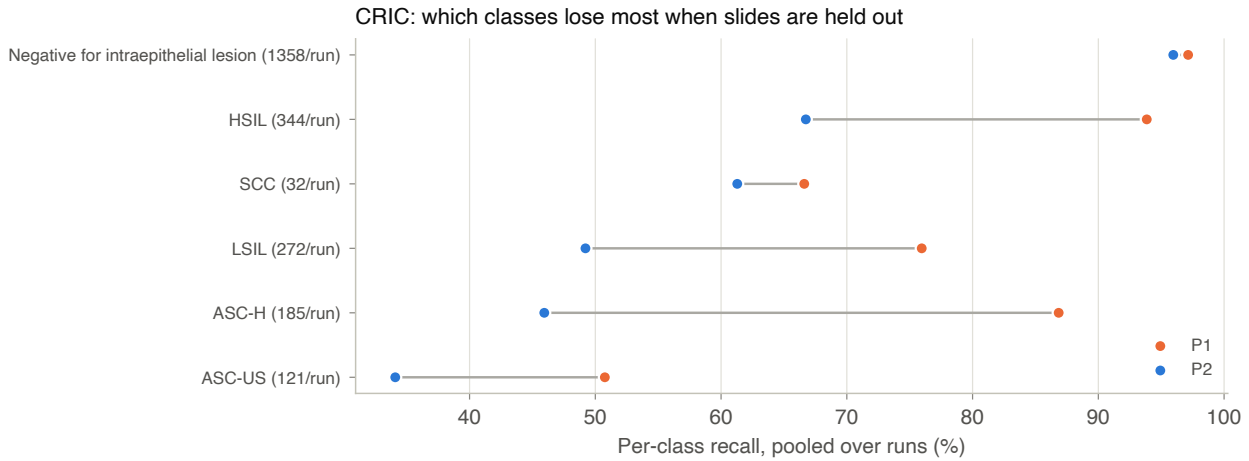


Figure 31.2: CRIC per-class recall under P1 and P2, pooled over all runs, with the mean number of test cells per class.

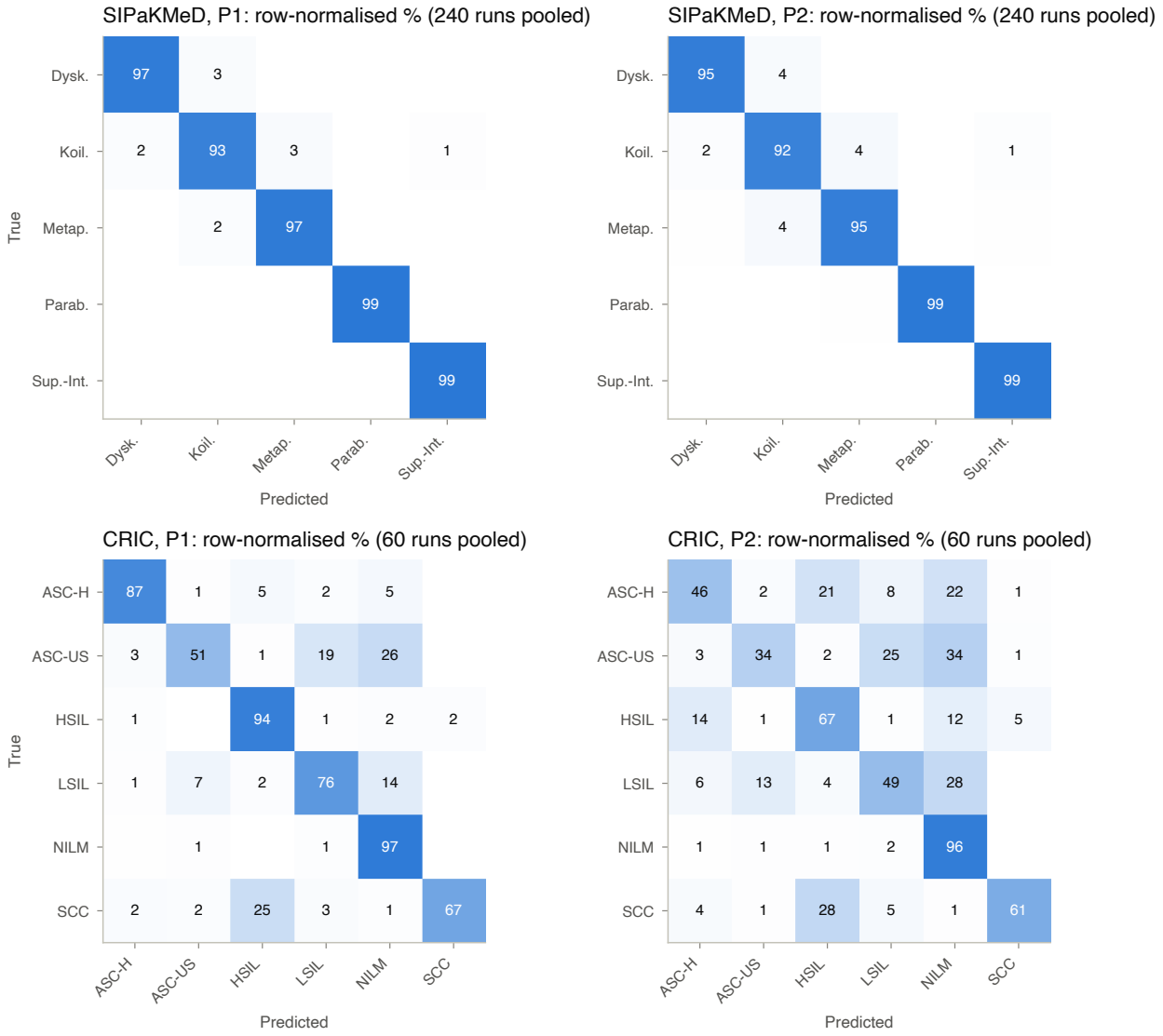


Figure 31.3: Row-normalised confusion matrices pooled over runs. SIPaKMeD errors barely change between protocols; CRIC's abnormal classes collapse toward NILM and toward each other under slide-level splitting.

Table 31.1: CRIC (10 seeds per cell): test metrics in percent (mean ± SD).

Architecture	Prot.	Accuracy	Macro-F1	Balanced acc.	ECE	NLL
ResNet-18	P1	89.91 ± 0.43	79.09 ± 1.31	77.60 ± 2.07	5.50 ± 0.40	0.383
	P1-prime	89.92 ± 0.68	78.94 ± 1.43	77.42 ± 2.25	5.48 ± 0.67	0.387
	P2	77.62 ± 1.14	57.25 ± 1.45	56.79 ± 2.63	14.98 ± 1.11	1.014
ResNet-50	P1	90.54 ± 0.48	80.48 ± 1.63	79.42 ± 2.41	5.85 ± 0.33	0.388
	P1-prime	90.85 ± 0.62	81.01 ± 1.66	79.85 ± 2.02	5.41 ± 0.45	0.371
	P2	78.76 ± 1.80	60.48 ± 3.30	60.34 ± 2.84	14.97 ± 1.82	0.995
DenseNet-121	P1	90.40 ± 0.74	79.66 ± 1.93	77.97 ± 2.71	5.42 ± 0.68	0.374
	P1-prime	90.71 ± 0.48	80.22 ± 1.41	78.32 ± 1.76	5.36 ± 0.46	0.365
	P2	78.23 ± 1.50	59.08 ± 2.36	58.82 ± 2.77	14.95 ± 1.28	1.017
EfficientNet-B0	P1	90.88 ± 0.53	80.74 ± 0.99	79.78 ± 1.46	5.33 ± 0.43	0.370
	P1-prime	90.95 ± 0.60	80.95 ± 1.05	80.83 ± 1.64	5.21 ± 0.54	0.363
	P2	78.10 ± 2.15	58.73 ± 2.78	59.86 ± 2.19	14.90 ± 1.74	1.021
ViT-B/16	P1	89.83 ± 0.63	78.01 ± 2.30	76.37 ± 3.22	6.54 ± 0.95	0.436
	P1-prime	89.54 ± 0.70	78.28 ± 1.81	76.45 ± 2.95	6.55 ± 0.68	0.429
	P2	78.20 ± 1.13	59.54 ± 1.47	57.62 ± 2.28	16.14 ± 1.29	1.054
ConvNeXt-T	P1	91.29 ± 0.63	81.80 ± 1.57	80.03 ± 2.69	5.91 ± 0.58	0.392
	P1-prime	91.10 ± 0.40	80.73 ± 1.97	78.85 ± 2.17	6.02 ± 0.53	0.392
	P2	79.48 ± 1.78	61.15 ± 3.27	59.90 ± 3.99	15.82 ± 1.76	1.082

32

H3: the inflation is leakage, not training diversity

With crop count, field count, and allocation rule held equal, adding the test fields' sibling crops to training raised accuracy on the common test set by 1.73 pp (0.95–2.51; Holm $p = 0.02$). Every architecture showed it at both diversity levels (Figure 32.2). Increasing the number of distinct training fields from 504 to 720 without leakage did nothing (0.00 pp, −0.65 to 0.64), and the leak effect did not depend on diversity (interaction −0.20 pp, $p = 0.60$). The registered approximate decomposition assigns the Phase 1 gap entirely to leakage (diversity component 0.00 pp).

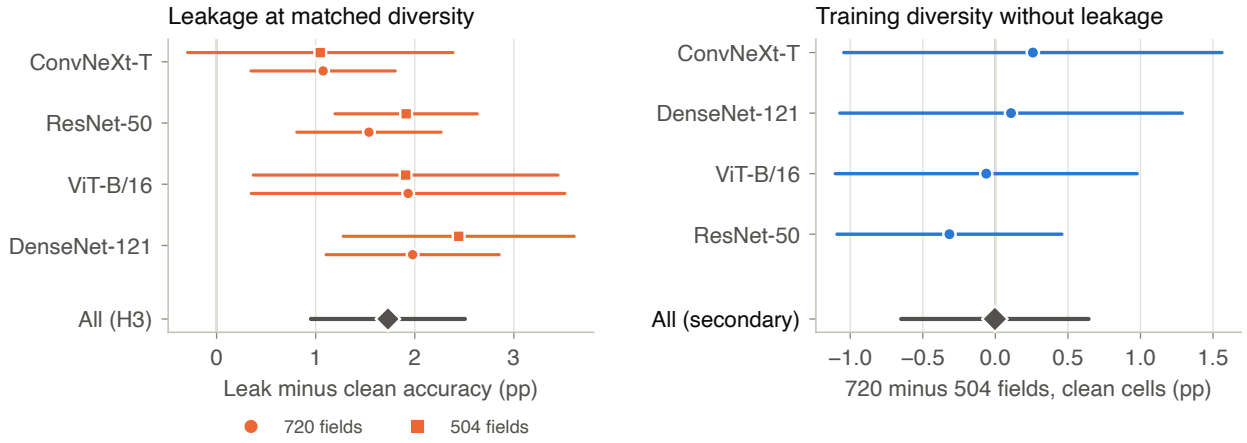


Figure 32.1: Left: leak minus clean accuracy per architecture and diversity level, and the H3 estimate. Right: high minus low diversity within clean cells.

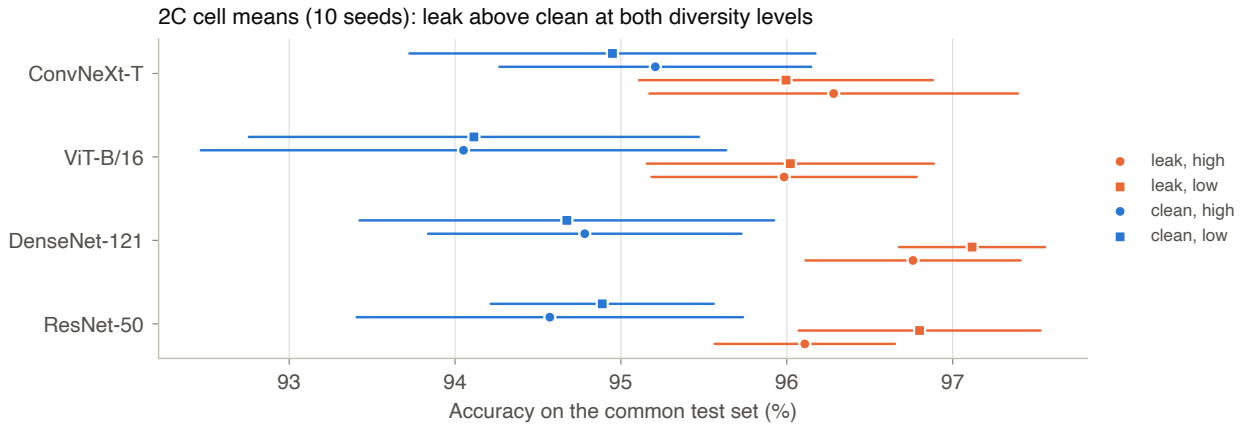


Figure 32.2: All four 2C cell means per architecture (10 seeds each).

The diversity manipulation did change what the training sets contain. In the embedding space of a retrained parent-probe encoder (which identified held-out fields at 24.1% against 0.24% chance), the mean distance between field centroids was 0.499 for 720-field and 0.479 for 504-field clean training sets, and 0.472 (P1, 908 fields) against 0.466 (P2, 773 fields) for the natural splits. More diverse training sets were measurably more diverse; that diversity simply did not improve accuracy.

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H4: background transplants do not isolate the route

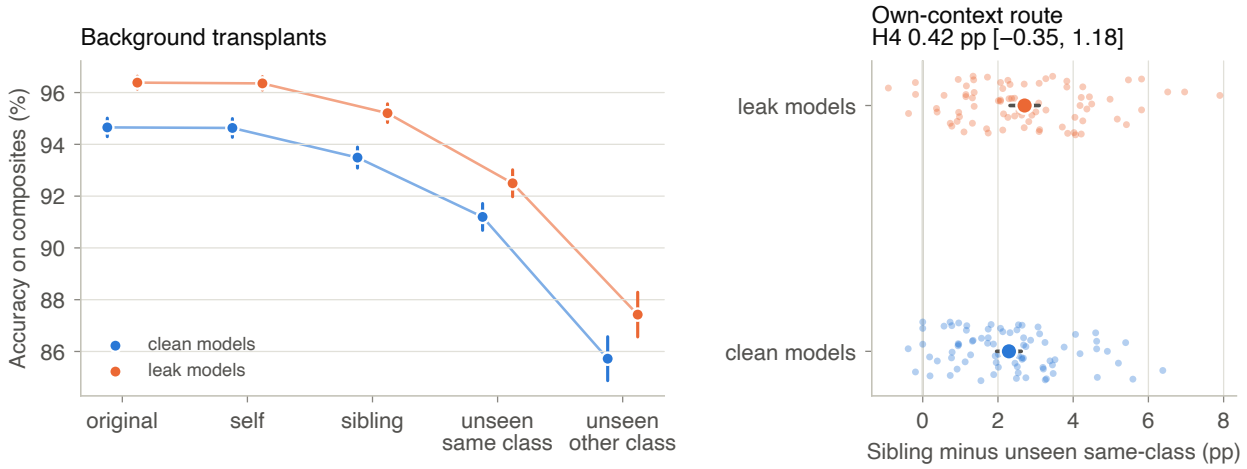


Figure 33.1: Left: accuracy of leak and clean models on composites of the common test set. Right: the per-run own-context route and the registered H4 contrast.

Compositing a cell onto its own inpainted background cost nothing, so seam and inpainting artifacts were negligible. Leak models were 1.73 pp more accurate than clean models on unmodified cells and kept most of that advantage when the background was replaced: 1.72 pp on sibling backgrounds, 1.30 pp on backgrounds from unseen fields of the same class, and 1.70 pp on other-class backgrounds. H4, the leak-minus-clean difference of the own-context route, was 0.42 pp (−0.35 to 1.18; Holm $p = 0.23$): not supported. Its point estimate would assign 24% of the inflation to the background.

Two observations qualify the design. Sibling backgrounds helped even clean models, which had never seen those fields, by 2.29 pp, presumably because a background from the cell's own field matches its stain; that effect is common to both arms but inflates the variance of the contrast. And backgrounds do carry class information: an other-class background moved 3.09 pp of probability toward its class (1.04–5.15, secondary). Restricting H4 to recipients whose background covers at least 30% of the crop gave 0.63 pp (−0.07 to 1.33).

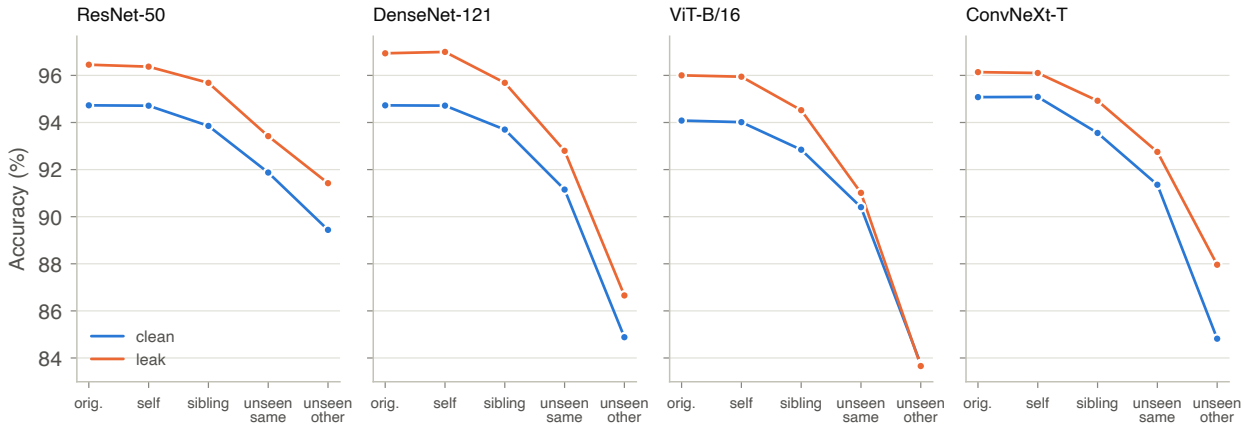


Figure 33.2: Transplant-condition accuracy of leak and clean models per architecture.

Table 33.1: 2C/2B (10 seeds per cell): accuracy on the common test set and on each background-transplant condition (percent), and the donor-class probability shift (pp).

Architecture	Cell	Original	Self	Sibling	Unseen same	Unseen other	Shift
ResNet-50	leak_high	96.11 ± 0.76	95.96 ± 0.77	95.29 ± 0.89	92.61 ± 1.41	90.89 ± 1.24	1.19
	clean_high	94.57 ± 1.63	94.50 ± 1.78	93.43 ± 2.06	91.73 ± 1.94	89.03 ± 2.49	1.35
	leak_low	96.80 ± 1.02	96.78 ± 1.03	96.08 ± 1.03	94.23 ± 1.40	91.95 ± 1.50	1.33
	clean_low	94.89 ± 0.94	94.93 ± 0.83	94.28 ± 0.84	92.02 ± 1.20	89.85 ± 1.16	1.22
DenseNet-121	leak_high	96.76 ± 0.91	96.89 ± 0.77	95.67 ± 0.60	92.80 ± 1.51	86.79 ± 1.96	4.15
	clean_high	94.78 ± 1.32	94.78 ± 1.26	93.72 ± 1.20	90.97 ± 1.99	84.58 ± 2.57	4.50
	leak_low	97.12 ± 0.62	97.10 ± 0.65	95.70 ± 1.28	92.80 ± 2.21	86.52 ± 2.90	4.36
	clean_low	94.67 ± 1.75	94.65 ± 1.83	93.68 ± 1.63	91.33 ± 2.28	85.19 ± 3.46	4.08
ViT-B/16	leak_high	95.98 ± 1.12	95.93 ± 1.22	94.55 ± 2.45	90.44 ± 2.89	84.05 ± 3.05	3.66
	clean_high	94.05 ± 2.21	94.05 ± 2.18	92.82 ± 2.27	90.73 ± 2.43	84.62 ± 4.51	3.36
	leak_low	96.02 ± 1.21	95.96 ± 1.25	94.49 ± 1.36	91.58 ± 2.25	83.26 ± 3.68	4.39
	clean_low	94.11 ± 1.90	93.98 ± 1.97	92.86 ± 2.61	90.07 ± 3.20	82.87 ± 3.39	3.93
ConvNeXt-T	leak_high	96.28 ± 1.55	96.25 ± 1.63	95.01 ± 1.34	92.69 ± 1.81	87.79 ± 3.44	2.71
	clean_high	95.21 ± 1.31	95.19 ± 1.22	93.89 ± 1.32	91.83 ± 2.40	85.63 ± 2.63	3.03
	leak_low	96.00 ± 1.24	95.96 ± 1.17	94.84 ± 2.61	92.81 ± 3.30	88.13 ± 3.33	2.81
	clean_low	94.95 ± 1.71	94.99 ± 1.66	93.22 ± 2.09	90.89 ± 2.82	84.00 ± 4.02	3.40

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H5: gradient reversal was evaded

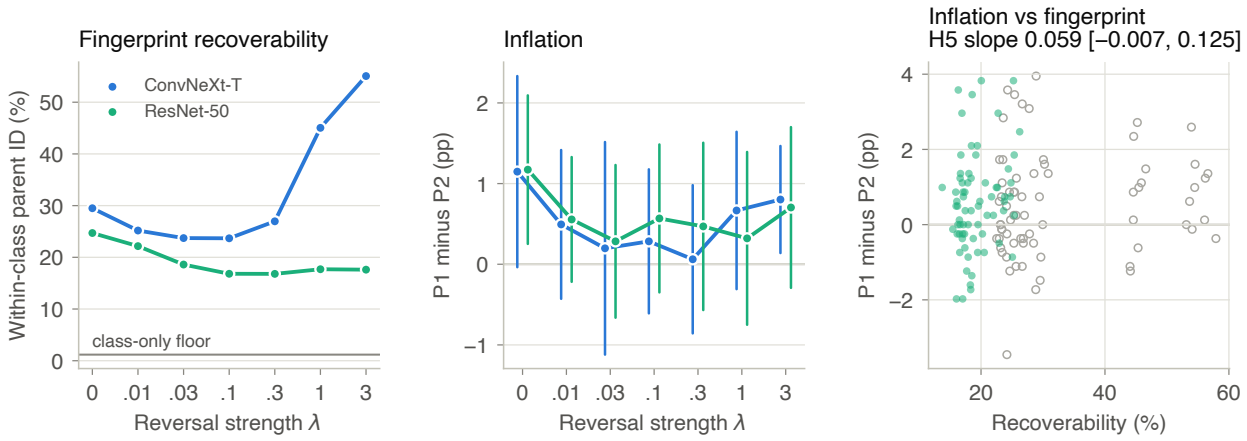


Figure 34.1: The gradient-reversal sweep: probe recoverability, the P1 − P2 gap, and their relation (ConvNeXt-T, which failed its manipulation check, hollow).

For ConvNeXt-T the adversary was evaded: as the reversal strength grew, an independent linear probe identified the field *better* (29.5% at $\lambda = 0$, 55.1% at $\lambda = 3$), while the adversary's own training accuracy fell toward chance (Figure 34.2). The encoder learned features its two-layer adversary could not read but a standardised linear probe could, the failure mode described for adversarial removal in language models [7]. ConvNeXt-T therefore failed the registered manipulation check and was excluded. For ResNet-50, recoverability fell from 24.7% to 17.6% and the mean gap fell from 1.17 pp to between 0.28 and 0.70 pp at every $\lambda > 0$, with P2 accuracy within 1 pp. The registered slope was positive but not significant (0.06 pp per pp, −0.01 to 0.12; Holm $p = 0.16$). Amendment 8's linear adversary later showed the ResNet-50 gap does not fall when recoverability is suppressed on fresh seeds (Part VI), so this drop was most likely noise.

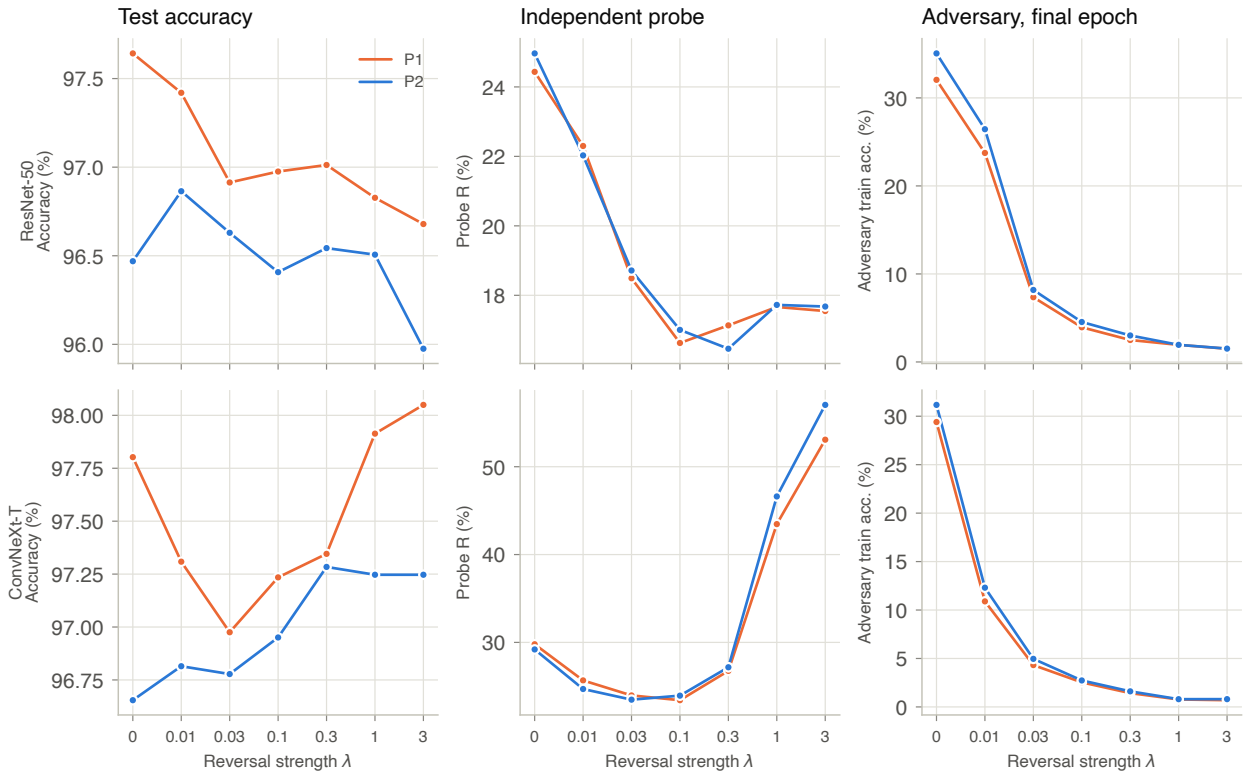


Figure 34.2: The full 2A sweep per architecture: test accuracy, independent-probe recoverability, and the adversary's own within-class training accuracy at the final epoch.

Table 34.1: 2A gradient-reversal sweep (10 seeds): accuracy (percent), mean gap (pp), and within-class parent recoverability R of the frozen features (percent; sklearn probe).

Architecture	λ	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)
ResNet-50	0	97.64 ± 0.46	96.47 ± 1.15	1.17	24.4 ± 1.2	25.0 ± 1.3
	0.01	97.42 ± 0.56	96.86 ± 0.71	0.56	22.3 ± 1.1	22.0 ± 2.0
	0.03	96.91 ± 0.82	96.63 ± 0.84	0.28	18.5 ± 1.7	18.7 ± 1.5
	0.1	96.98 ± 0.38	96.41 ± 1.00	0.57	16.6 ± 1.1	17.0 ± 1.6
	0.3	97.01 ± 0.78	96.54 ± 0.93	0.47	17.1 ± 2.0	16.5 ± 1.0
	1	96.83 ± 0.76	96.51 ± 1.00	0.32	17.7 ± 1.5	17.7 ± 1.7
	3	96.68 ± 0.73	95.98 ± 1.22	0.70	17.5 ± 1.8	17.7 ± 1.5
ConvNeXt-T	0	97.80 ± 0.55	96.65 ± 1.31	1.15	29.8 ± 1.2	29.2 ± 1.5
	0.01	97.31 ± 0.95	96.81 ± 0.73	0.49	25.7 ± 1.3	24.7 ± 1.5
	0.03	96.98 ± 1.00	96.78 ± 1.30	0.20	24.0 ± 1.4	23.5 ± 1.4
	0.1	97.23 ± 0.78	96.95 ± 0.81	0.28	23.4 ± 0.9	23.9 ± 1.1
	0.3	97.35 ± 1.01	97.28 ± 1.05	0.06	26.8 ± 1.5	27.2 ± 1.9
	1	97.91 ± 0.68	97.25 ± 0.82	0.67	43.5 ± 1.0	46.6 ± 1.4
	3	98.05 ± 0.47	97.25 ± 0.65	0.80	53.1 ± 2.0	57.0 ± 2.1

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What Study 2 established

Study 2 in four sentences

Crop-level splitting inflates accuracy, by 0.97 pp on SIPaKMeD across twelve architectures and by 12.08 pp on CRIC. The inflation is leakage: at matched size and diversity, the test fields' siblings alone add 1.73 pp, and diversity alone adds nothing. The size of the inflation follows the dataset's structure, as the closed forms predict. How models use the leak was not settled: the background route was small and uncertain, and the adversary was evaded.

PART VI

Amendment 8: Where the Leaked Signal Lives

36

Why a follow-up, and what was known

Study 2 established *that* crop-level splitting inflates accuracy and that the inflation is leakage, but neither registered mechanism test isolated *how* models use it. Two observations pointed the way. Leak models kept most of their advantage on backgrounds transplanted from unseen fields, which suggested a carrier inside the cell region rather than in the background. And the ConvNeXt-T adversary had been evaded, so the gradient-reversal arm had only one usable architecture.

Amendment 8 (`PREREGISTRATION_STUDY2_AMENDMENT8.md`, commit `80171fe`, 2026-10-19 14:02 IST) registered eight new hypotheses after the Study 2 analysis and before any follow-up model was trained or any follow-up analysis was run on existing outputs. It states what was known and what was not: every registered Study 2 output was known, including transplant condition means and gradient-reversal sweep means; no per-cell CRIC result by slide composition, no ablated image evaluation, no erased-feature evaluation, no Grad-CAM map, and no model trained with masked backgrounds, colour augmentation, stain normalisation, or the linear adversary had been seen. F1–F8 form their own Holm family, separate from H1–H5, whose verdicts the amendment does not touch.

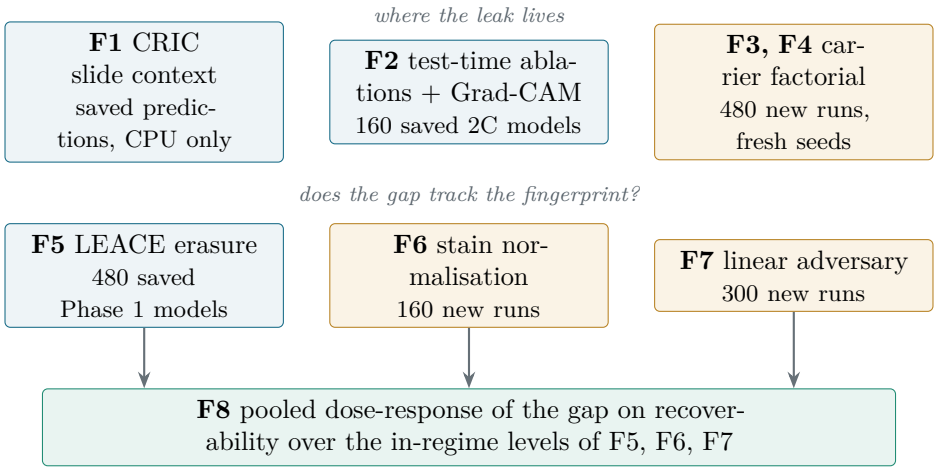


Figure 36.1: The eight Amendment 8 hypotheses. Blue arms reuse saved Study 2 artifacts (no retraining); gold arms train new models on seeds no earlier arm used; F8 pools the three dose-response arms.

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The eight follow-up designs

37.1 F1: slide context in CRIC

CRIC slide images contain cells of several classes (83% of slides have more than one), so a test cell's training siblings can carry a different label. For every cell, the *context class* is the majority class of the other cells of its slide (cells whose slide has no other cell, or a tied majority, are excluded). A *discordant* cell's class differs from its context class. If crop-level models apply a slide-level label prior, discordant cells should receive more probability on the context class when their slide was in training (P1) than when it was not (P2). F1 is that within-cell difference over the 120 saved CRIC P1 and P2 runs, with a percentile cluster bootstrap over slides (10,000 replicates).

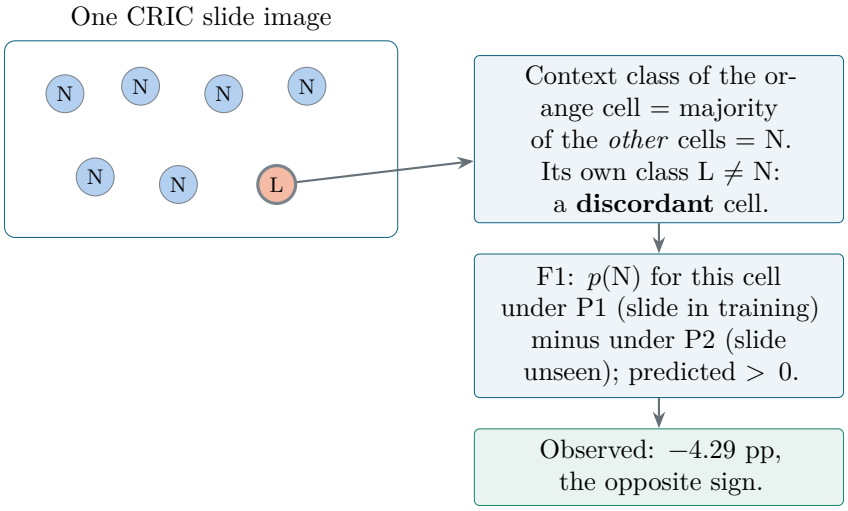


Figure 37.1: How F1 defines a discordant cell. A slide-level label prior would pull discordant cells toward their slide's majority class when the slide was seen in training; the models did the opposite.

37.2 F2: carrier ablations at test time

Each of the 160 saved 2C models is re-evaluated on its seed's common test set under six conditions rendered at native resolution: original; greyscale; Reinhard stain transfer of the crop's LAB mean and standard deviation to those of the 2B unseen same-class donor [26]; low-pass (Gaussian $\sigma = 2$ px at the 224-px input); illumination flattening (subtracting a $\sigma = 16$ px Gaussian); and background replaced by the ImageNet mean. F2 asks whether stain transfer shrinks the

leak-minus-clean advantage. Grad-CAM [31] attribution outside the cell mask is measured on the convolutional models and reported descriptively, with a random-head sanity check [1].

37.3 F3 and F4: carrier removal in training

The 2C factorial is rerun on ten seeds never used for any model (1103, 1151, 1201, 1249, 1301, 1361, 1409, 1451, 1499, 1553) under three conditions: the standard recipe; every training and test crop's background masked to the ImageNet mean; and strong HED optical-density colour augmentation [29, 35] ($\alpha \in [0.8, 1.2]$, $\beta \in [-0.2, 0.2]$ per stain channel, then brightness and contrast in $[0.8, 1.2]$) during training. F3 asks whether leakage survives with no background; F4 whether colour augmentation shrinks it. The standard condition is an independent replication of H3.

37.4 F5: rank-limited LEACE erasure

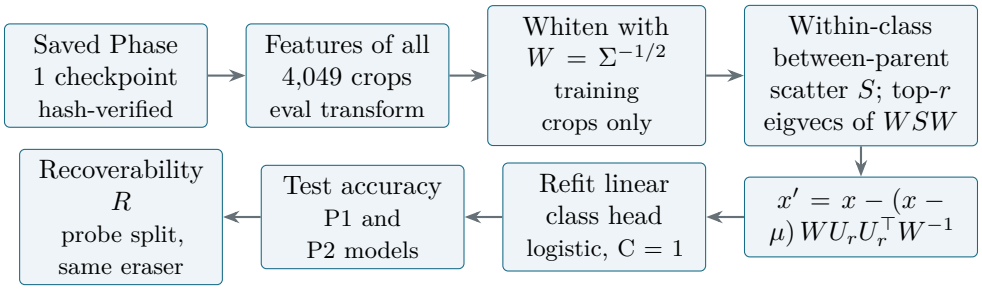


Figure 37.2: Rank-limited LEACE erasure (F5). At full rank the projection is LEACE for the within-class parent indicator: no linear function of the erased features covaries with it on the fitting data. Ranks 0, 4, 16, 64, and 256 give the dose.

Every saved Phase 1 P1 and P2 model (480) is re-evaluated: features of all crops are extracted, the within-class parent directions of the model's own training crops are erased at ranks 0, 4, 16, 64, and 256 [3], the class head is refitted, and test accuracy and recoverability are measured. The regime and manipulation check follow H5. F5 asks whether erasure at the strongest in-regime rank reduces the P1 − P2 gap.

37.5 F6: stain-normalisation dose

Every crop is Reinhard-normalised toward one dataset-wide reference at strength 0.5 and 1.0, in training, test, and probe crops alike. ResNet-50, ConvNeXt-T, DenseNet-121, and EfficientNet-B0 are trained on P1 and P2 for the ten Study 2 seeds no mechanism arm used; strength 0 is the matching Phase 1 run. F6 is the reduction in gap per unit of strength.

37.6 F7: probe-matched linear adversary

The 2A design with a different adversary (Figure 24.4): per-feature standardisation and one linear layer, its own optimiser at 10^{-3}, re-initialised every five epochs. ResNet-50, ConvNeXt-T, and DenseNet-121; $\lambda \in \{0, 0.1, 0.3, 1, 3\}$; P1 and P2; the same ten seeds as F6. F7 is the slope of gap on recoverability, as for H5.

37.7 F8: pooled dose-response

All included architecture-by-level cells of F5, F6, and F7, after their regimes and manipulation checks: gap $\sim$ cell + R_{pair} + (1 | manipulation:seed). F8 is the slope.

37.8 Recoverability, measured the same way everywhere

R for F5–F8 is computed with one GPU implementation of the registered logistic probe (sklearn's objective, C = 1, standardised features, L-BFGS). On the 280 Study 2 gradient-reversal runs it agreed with the registered sklearn probe at $r = 0.999$ (Figure 37.3).

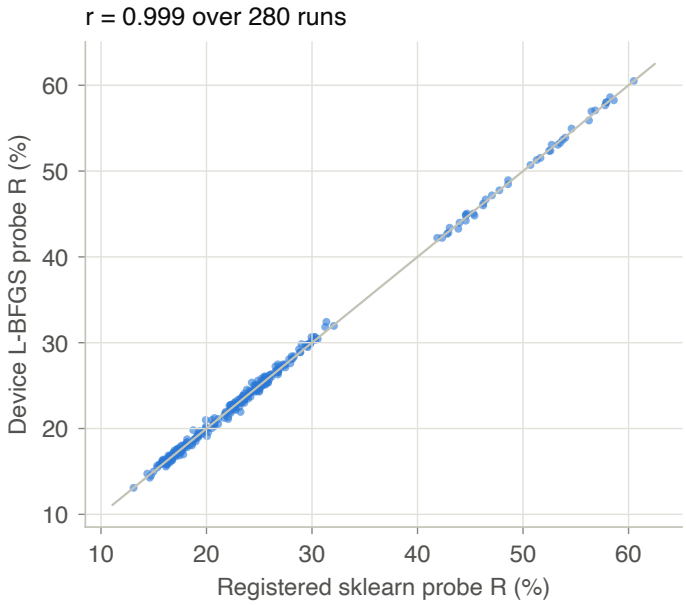


Figure 37.3: Within-class recoverability from the registered sklearn probe and the device probe used for F5–F8, on the 280 Study 2 adversarial runs.

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Execution and the Amendment 9 fix

The new code was unit-tested on the workstation and, for GPU paths, on the host (13 tests, including a check that full-rank erasure leaves no covariance between features and within-class parent identity, and that the device probe matches sklearn to 10^{-3}). A synthetic dry run of the registered models recovered every planted effect. Eleven smoke tasks ran only on leak cells and single P1 runs; they confirmed that re-evaluated checkpoints reproduce their recorded accuracy and that the linear adversary learns. The caches for masking, normalisation, and the ablations were built and hashed, and the registration commit recorded every hash. The matrix of 1,580 tasks then ran from 19 Oct 14:03 to 20 Oct 02:52 without a failure.

Amendment 9: the preparer bug

When the registered models were first run on the completed carrier arm, the R script stopped at F4: the preparer had labelled all 160 standard-recipe runs as colour-augmented, because a missing colour augmentation (`None` in the run record) became a NaN in a DataFrame row, and NaN is truthy. The script halted before writing any output, so no F3 or F4 estimate was seen. The condition is now read from each run record, a unit test covers it, and the fix was registered as Amendment 9 (20 October, 01:25) before the corrected table was analysed. The synthetic dry run had bypassed the preparer, which is why the bug was not caught earlier.

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Follow-up results

Table 39.1: The eight registered follow-up tests (one Holm family). Estimates are oriented so that the predicted direction is positive.

	Question	Estimate	95% CI	Holm p	Verdict
F1	CRIC pull toward the slide's majority class	−4.29	−8.60 to −1.37	0.001	**not supported**
F2	Stain transfer shrinks the leak effect	1.27	0.38 to 2.16	0.03	**supported**
F3	Leak effect with backgrounds masked	1.86	1.42 to 2.30	<0.001	**supported**
F4	Colour augmentation shrinks the leak effect	−0.23	−0.81 to 0.35	1.00	**not supported**
F5	LEACE erasure shrinks the gap	−0.01	−0.10 to 0.07	1.00	**not supported**
F6	Stain normalisation shrinks the gap	0.02	−0.41 to 0.44	1.00	**not supported**
F7	Gap tracks R (linear adversary)	0.000	−0.038 to 0.039	1.00	**not supported**
F8	Gap tracks R (pooled)	0.000	−0.005 to 0.005	1.00	**not supported**

39.1 The leak needs no background (F3)

Trained and tested with every background replaced by a constant, leak models beat clean models by 1.86 pp (1.42–2.30; Holm p <0.001), at least as much as under the standard recipe on the same fresh seeds (1.47 pp), which also replicates H3 (1.47 pp, 0.06–2.89). Masking changed the leak effect by −0.39 pp relative to standard training (−0.96 to 0.19). At test time, removing the background from the original 2C models' inputs left 68% of their advantage.

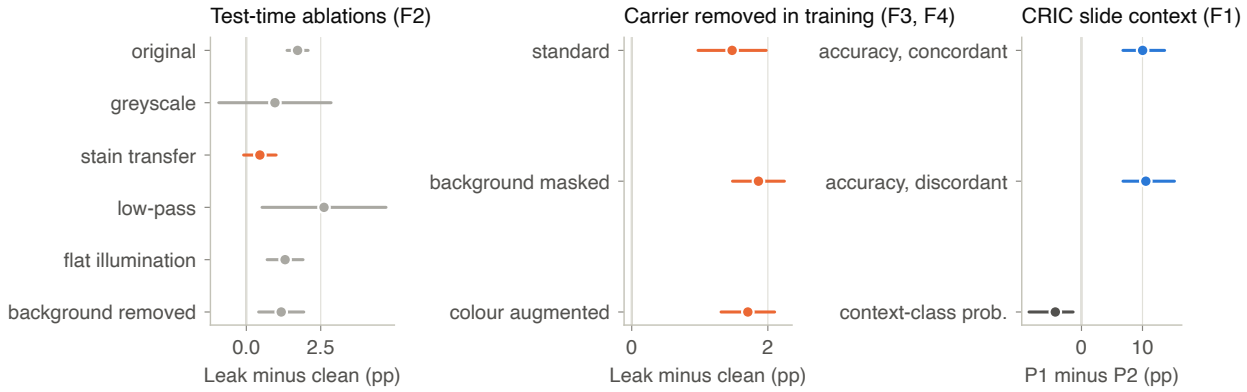


Figure 39.1: Left: the leak advantage under each test-time ablation (F2 highlighted). Middle: the leak effect under each training condition on fresh seeds (F3, F4). Right: CRIC within-cell accuracy gains by concordance, and the context-class probability for discordant cells (F1).

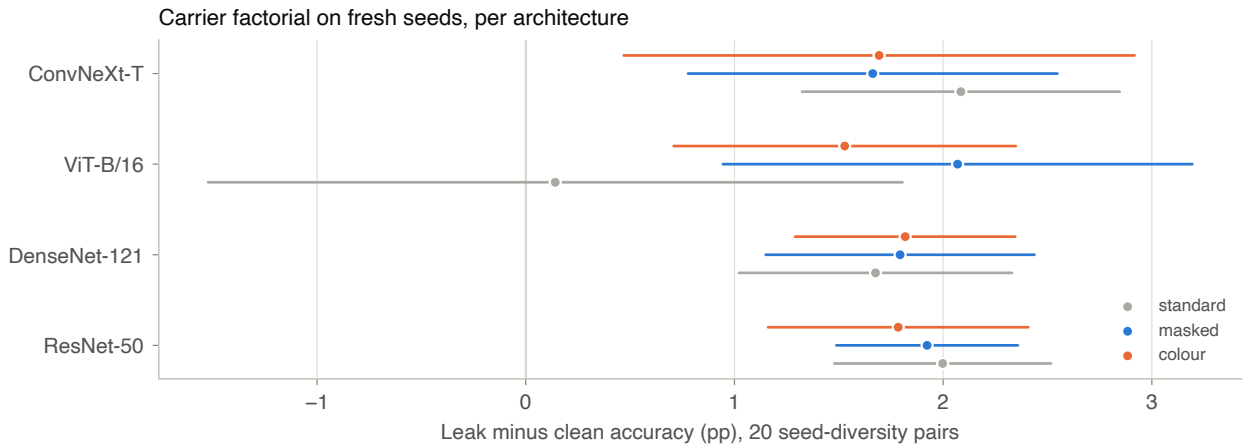


Figure 39.2: The carrier factorial per architecture.

Table 39.2: Amendment 8 carrier factorial (fresh seeds, 10 per cell): accuracy per cell (percent) and the leak effect (pp).

Architecture	Condition	Leak high	Clean high	Leak low	Clean low	Leak effect
ResNet-50	standard	96.06 ± 1.42	93.99 ± 2.08	96.40 ± 0.79	94.48 ± 1.52	2.00
	masked	95.46 ± 1.19	93.42 ± 1.26	95.69 ± 1.36	93.87 ± 1.37	1.92
	colour	95.62 ± 1.29	93.79 ± 1.29	95.84 ± 1.04	94.10 ± 1.45	1.78
DenseNet-121	standard	96.33 ± 1.16	95.14 ± 1.63	97.01 ± 0.66	94.85 ± 1.50	1.68
	masked	95.59 ± 1.32	94.21 ± 1.00	96.19 ± 0.87	93.99 ± 1.84	1.79
	colour	96.20 ± 1.01	94.47 ± 1.48	96.45 ± 0.58	94.54 ± 1.68	1.82
ViT-B/16	standard	93.40 ± 3.70	93.60 ± 1.77	94.97 ± 1.74	94.49 ± 1.32	0.14
	masked	94.92 ± 2.21	92.95 ± 1.97	94.91 ± 1.13	92.74 ± 1.56	2.07
	colour	95.05 ± 1.68	93.53 ± 1.27	95.04 ± 1.67	93.51 ± 1.20	1.53
ConvNeXt-T	standard	96.56 ± 0.95	95.05 ± 1.23	96.66 ± 0.67	94.00 ± 2.32	2.08
	masked	94.80 ± 1.73	93.15 ± 2.00	95.29 ± 1.47	93.61 ± 1.74	1.66
	colour	95.30 ± 1.77	94.24 ± 1.95	95.77 ± 1.51	93.44 ± 2.68	1.69

39.2 The leak is tied to the field's colour rendering at test time (F2), but colour augmentation does not prevent it (F4)

Moving each test cell's colour statistics to those of an unseen field of the same class cut the leak advantage from 1.72 to 0.46 pp, a reduction of 1.27 pp (0.38–2.16; Holm $p = 0.03$), removing 74% of it. Interventions

with comparable accuracy costs did not: with illumination flattened 76% of the advantage remained, and with the background removed 68%. Greyscale and low-pass filtering degraded accuracy too far to be informative (Figure 39.3). Yet models trained with strong colour augmentation leaked as much as standard models (1.71 versus 1.47 pp; F4 −0.23, −0.81 to 0.35).

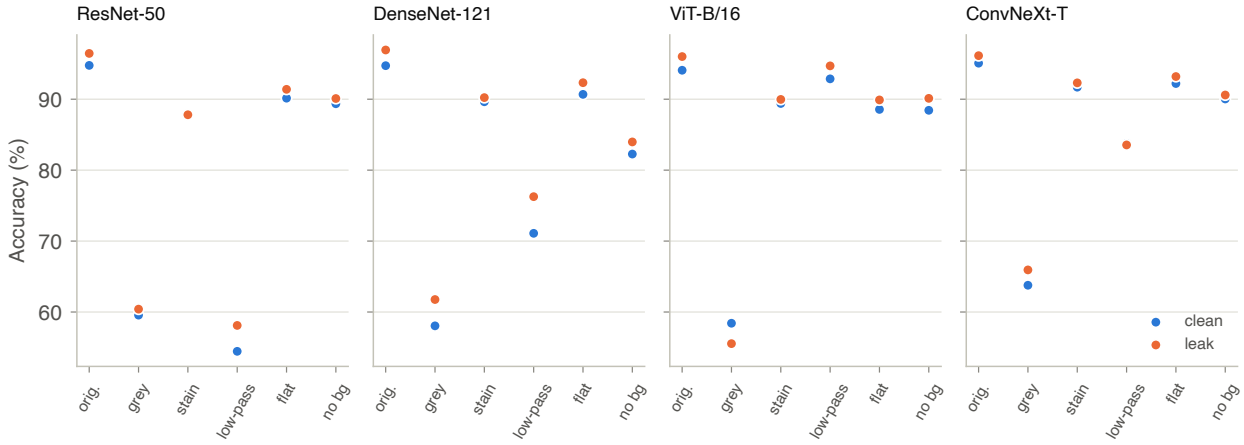


Figure 39.3: Accuracy of leak and clean 2C models under each test-time ablation, per architecture.

Table 39.3: Amendment 8 test-time carrier ablations on the Study 2 2C models: leak and clean accuracy (percent, 20 runs each) and the leak advantage (pp).

Architecture	Ablation	Leak models	Clean models	Leak − clean
ResNet-50	original	96.45 ± 0.94	94.77 ± 1.33	1.69
	grey	60.41 ± 5.61	59.56 ± 7.27	0.84
	stain_transfer	87.81 ± 2.85	87.76 ± 2.77	0.05
	low_pass	58.11 ± 11.63	54.46 ± 11.17	3.65
	illumination_flat	91.39 ± 2.88	90.15 ± 2.18	1.25
	background_grey	90.10 ± 2.57	89.36 ± 1.84	0.74
DenseNet-121	original	96.94 ± 0.78	94.73 ± 1.50	2.21
	grey	61.76 ± 6.30	58.05 ± 8.26	3.71
	stain_transfer	90.23 ± 2.21	89.64 ± 1.92	0.59
	low_pass	76.26 ± 6.79	71.09 ± 5.97	5.17
	illumination_flat	92.33 ± 2.17	90.68 ± 3.18	1.64
	background_grey	83.98 ± 3.72	82.27 ± 5.20	1.71
ViT-B/16	original	96.01 ± 1.15	94.08 ± 2.01	1.93
	grey	55.54 ± 5.12	58.42 ± 7.25	−2.87
	stain_transfer	89.97 ± 2.40	89.40 ± 3.25	0.57
	low_pass	94.70 ± 1.74	92.87 ± 2.95	1.83
	illumination_flat	89.90 ± 2.63	88.56 ± 3.56	1.34
	background_grey	90.12 ± 2.73	88.44 ± 3.54	1.68
ConvNeXt-T	original	96.14 ± 1.38	95.08 ± 1.49	1.06
	grey	65.94 ± 5.38	63.77 ± 5.85	2.16
	stain_transfer	92.31 ± 2.68	91.69 ± 2.52	0.61
	low_pass	83.55 ± 7.47	83.76 ± 6.82	−0.20
	illumination_flat	93.20 ± 1.62	92.21 ± 1.87	0.98
	background_grey	90.59 ± 2.46	90.03 ± 1.98	0.56

39.3 Exploratory: not shared pixels

In leak cells a third or more of the test crops overlap a training sibling in the source image. Splitting the leak advantage by overlap (not registered) shows it is as large for crops that share no pixel with any training crop (1.68 pp) as for crops that do (1.78 pp), in the Study 2 models and in all three carrier conditions; stain transfer removed it in both subsets (0.70 and 0.08 pp). The leak generalises to new cells from a seen field rather than reflecting memorised pixels.

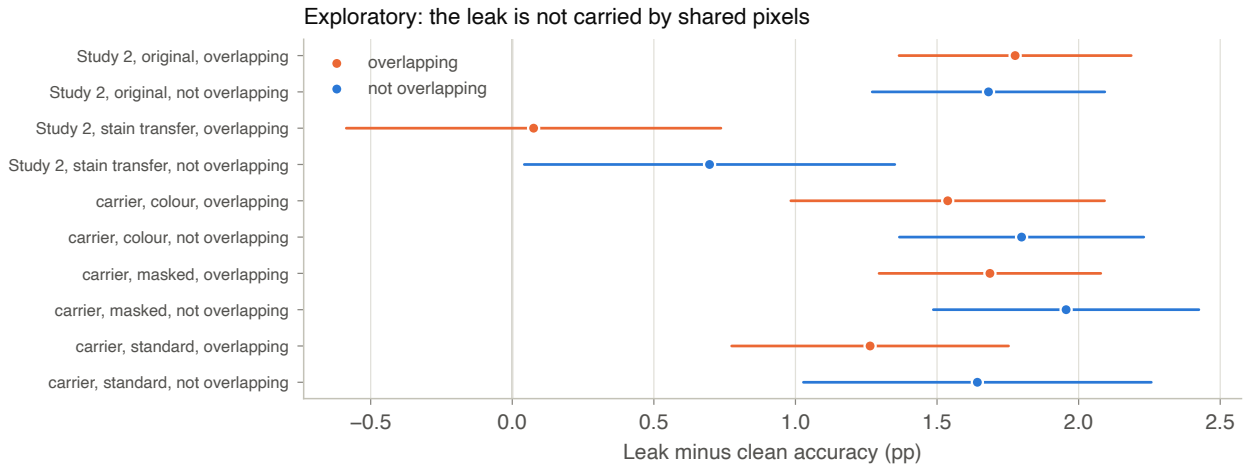


Figure 39.4: Exploratory: the leak advantage on test crops that do or do not share source pixels with a training sibling.

Table 39.4: EXPLORATORY (not registered): leak minus clean accuracy on test crops that do or do not share source pixels with a training sibling (pp; 95% t interval over architecture–seed–diversity pairs).

Condition	Subset	Pairs	Crops	Leak − clean	95% CI
Study 2, original	not overlapping	80	320	1.68	1.27 to 2.09
Study 2, original	overlapping	80	207	1.78	1.37 to 2.19
Study 2, stain transfer	not overlapping	80	320	0.70	0.04 to 1.35
Study 2, stain transfer	overlapping	80	207	0.08	−0.59 to 0.74
carrier, colour	not overlapping	80	329	1.80	1.37 to 2.23
carrier, colour	overlapping	80	206	1.54	0.98 to 2.09
carrier, masked	not overlapping	80	329	1.96	1.49 to 2.42
carrier, masked	overlapping	80	206	1.69	1.30 to 2.08
carrier, standard	not overlapping	80	329	1.64	1.03 to 2.26
carrier, standard	overlapping	80	206	1.26	0.77 to 1.75

39.4 No slide-level label prior (F1)

Over 1,144 discordant CRIC cells from 230 slides, the probability of the slide's majority class was −4.29 pp *lower*, not higher, when the cell's slide was in training (−8.60 to −1.37). All six architectures showed the same sign. Crop-level training raised accuracy as much for discordant cells (10.55 pp) as for concordant ones (10.02 pp; contrast −0.52 pp, −5.75 to 4.66): seen slides help each cell toward its own class, not toward the slide's typical class.

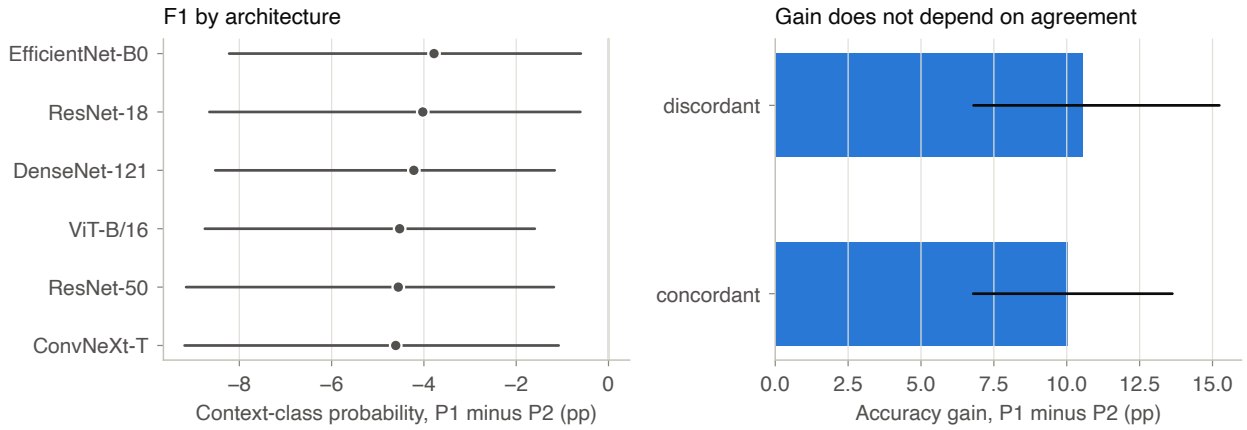


Figure 39.5: F1 per architecture, and the within-cell accuracy gain for concordant and discordant cells.

Table 39.5: F1 per architecture (secondary): within-cell P1 − P2 context-class probability on discordant cells (pp) with slide-bootstrap intervals.

Architecture	Estimate	95% CI	p	Cells
ConvNeXt-T	−4.61	−9.19 to −1.08	0.003	1,144
DenseNet-121	−4.22	−8.52 to −1.17	0.003	1,144
EfficientNet-B0	−3.78	−8.22 to −0.60	0.01	1,144
ResNet-18	−4.02	−8.65 to −0.61	0.01	1,144
ResNet-50	−4.56	−9.15 to −1.18	0.003	1,144
ViT-B/16	−4.53	−8.75 to −1.60	<0.001	1,144

39.5 The gap does not track the fingerprint (F5–F8)

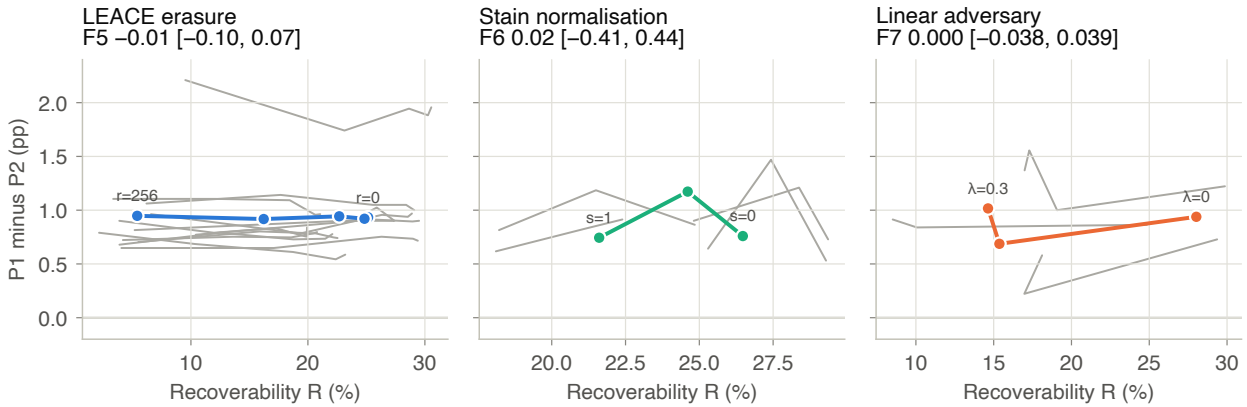


Figure 39.6: The P1 − P2 gap against recoverability along three manipulations (grey: architectures; colour: pooled over architectures that passed the manipulation check, at levels they share; registered in-regime levels only).

Rank-256 LEACE erasure cut recoverability from 25.1% to 5.4% (floor 1.2%) in all twelve architectures while costing P2 0.17 pp, and the gap stayed at 0.93 to 0.95 pp (F5 −0.01, −0.10 to 0.07). The probe-matched adversary passed its check in all three architectures, lowering R from 24.8–29.9% to 8.5–18.1% inside the regime, with a slope of 0.000 (−0.038 to 0.039). Stain normalisation lowered R only from 26.5% to 21.6% while costing P2 1.54 pp, failed its check for every architecture, and left the gap unchanged (F6 0.02, −0.41

to 0.44). Pooled over the 15 included cells, the slope was 0.000 (−0.005 to 0.005): even a 20-pp loss of recoverability would move the gap by at most about 0.1 pp.

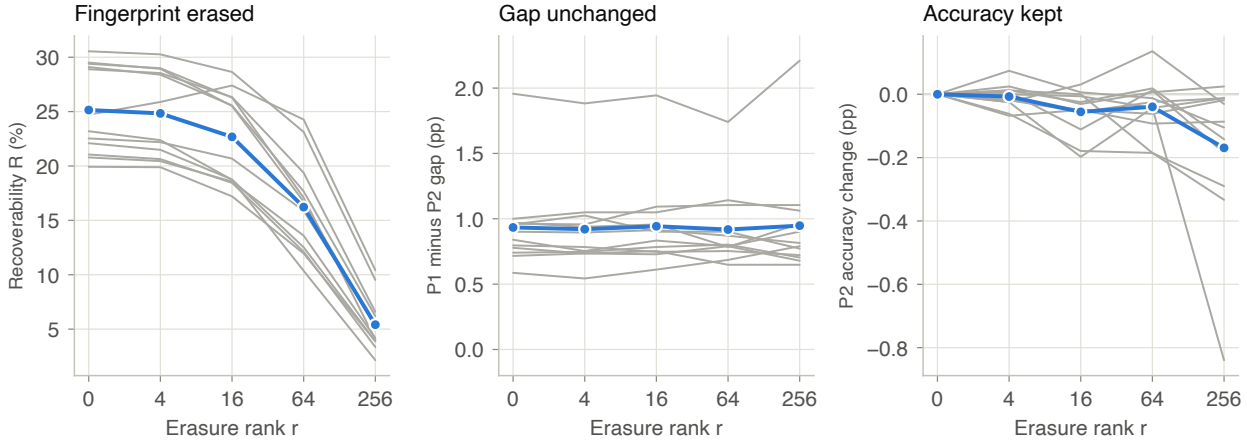


Figure 39.7: LEACE per architecture (grey) and pooled (blue): recoverability, gap, and P2 accuracy change by erasure rank.

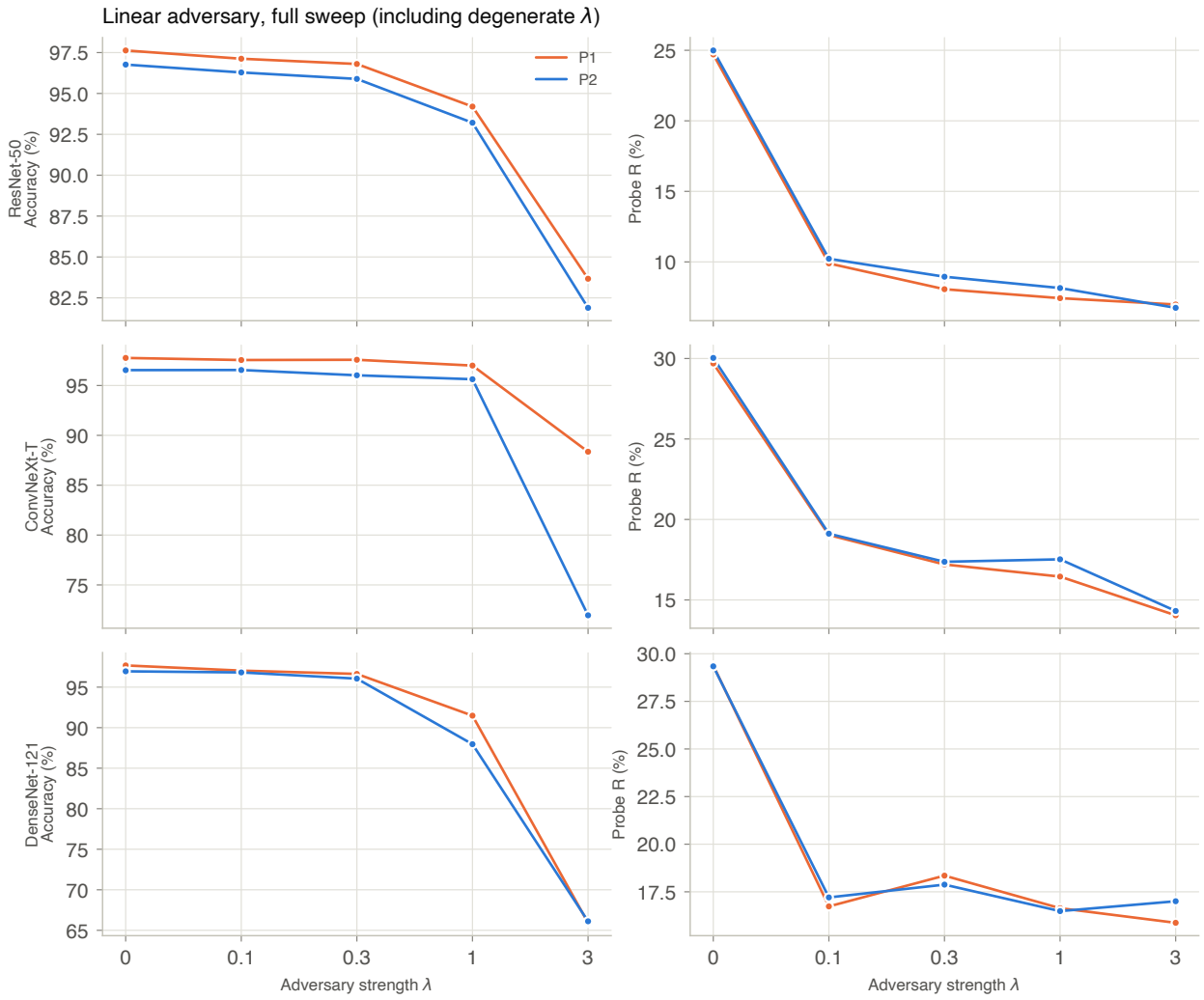


Figure 39.8: The full linear-adversary sweep per architecture, including the degenerate strengths ($\lambda = 1$ for ResNet-50 and DenseNet-121, $\lambda = 3$ for all) where P2 accuracy collapses and which the regime rule excludes.

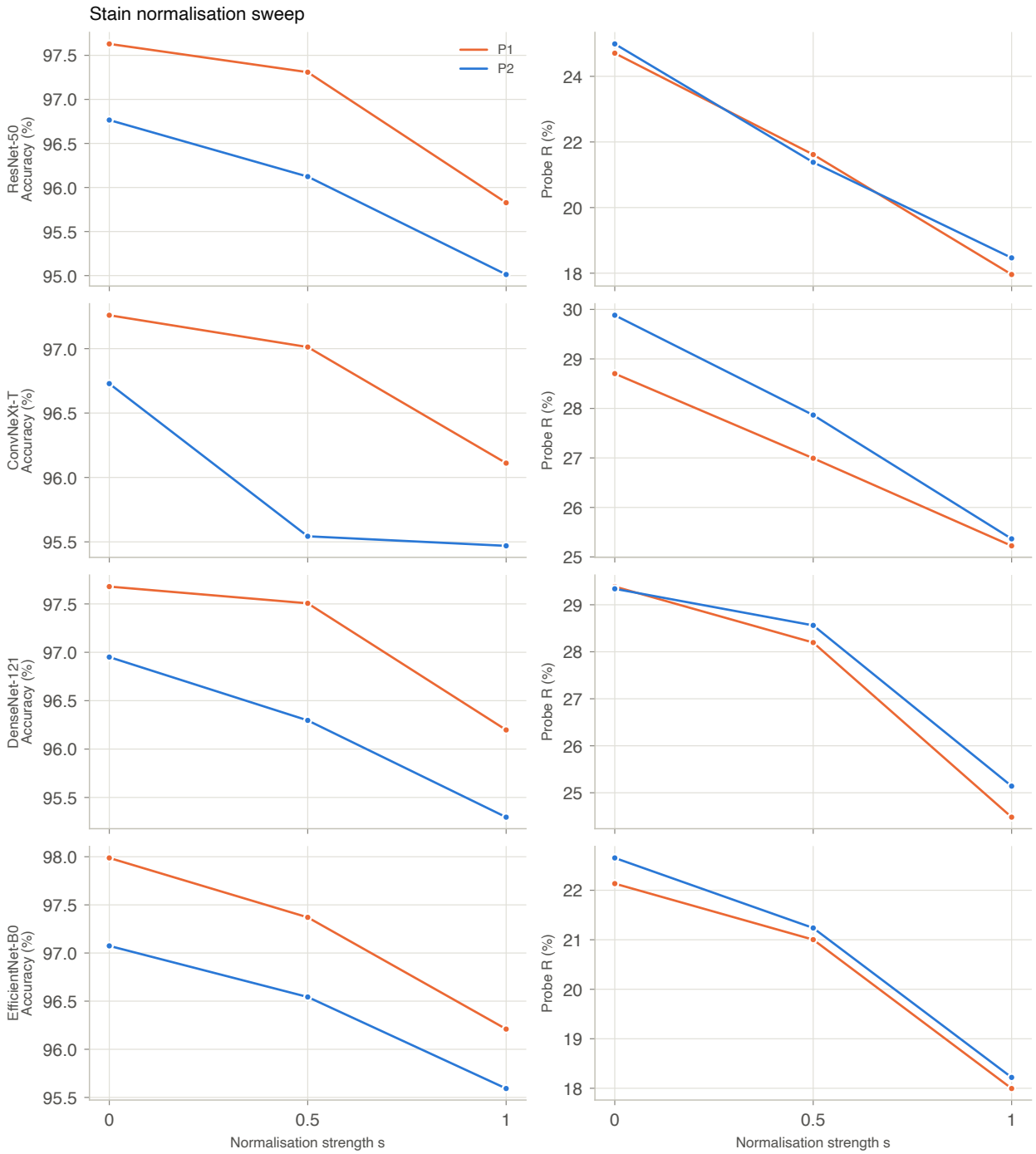


Figure 39.9: The stain-normalisation sweep per architecture: accuracy falls for both protocols, recoverability falls only modestly.

Table 39.6: Regimes and manipulation checks of every dose-response arm: levels whose mean P2 accuracy stays within 1 pp of baseline, the strongest such level, recoverability at baseline and there (percent), the class-only floor, and the check (at least a quarter of the above-floor R removed).

Arm	Architecture	Levels in regime	Strongest	R_0	R^*	Floor	Check
H5 GRL	ResNet-50	0, 0.01, 0.03, 0.1, 0.3, 1, 3	3	24.7	17.6	1.2	**PASS**
	ConvNeXt-T	0, 0.01, 0.03, 0.1, 0.3, 1, 3	3	29.5	55.1	1.2	**FAIL**
F5 LEACE	ResNet-18	0, 4, 16, 64, 256	256	23.2	2.1	1.2	**PASS**
	ResNet-50	0, 4, 16, 64, 256	256	24.7	10.4	1.2	**PASS**

Arm	Architecture	Levels in regime	Strongest	R_0	R^*	Floor	Check
	DenseNet-121	0, 4, 16, 64, 256	256	29.4	4.1	1.2	**PASS**
	EfficientNet-B0	0, 4, 16, 64, 256	256	22.5	5.5	1.2	**PASS**
	MobileNet-V2	0, 4, 16, 64, 256	256	20.8	4.2	1.2	**PASS**
	ViT-B/16	0, 4, 16, 64, 256	256	29.1	6.2	1.2	**PASS**
	ConvNeXt-T	0, 4, 16, 64, 256	256	29.5	5.2	1.2	**PASS**
	Swin-T	0, 4, 16, 64, 256	256	28.9	6.6	1.2	**PASS**
	RegNetY-1.6GF	0, 4, 16, 64, 256	256	21.1	3.4	1.2	**PASS**
	EfficientNetV2-S	0, 4, 16, 64, 256	256	19.9	3.9	1.2	**PASS**
	MaxViT-T	0, 4, 16, 64, 256	256	22.1	3.9	1.2	**PASS**
	DINOv2 ViT-S/14	0, 4, 16, 64, 256	256	30.5	9.5	1.2	**PASS**
F6 normalisation	ResNet-50	0, 0.5	0.5	24.8	21.5	1.2	**FAIL**
	DenseNet-121	0, 0.5	0.5	29.4	28.4	1.2	**FAIL**
	EfficientNet-B0	0, 0.5	0.5	22.4	21.1	1.2	**FAIL**
	ConvNeXt-T	0	0	29.3	29.3	1.2	**FAIL**
F7 linear adv.	ResNet-50	0, 0.1, 0.3	0.3	24.8	8.5	1.2	**PASS**
	DenseNet-121	0, 0.1, 0.3	0.3	29.4	18.1	1.2	**PASS**
	ConvNeXt-T	0, 0.1, 0.3, 1	1	29.9	17.0	1.2	**PASS**

Table 39.7: Amendment 8 rank-limited LEACE erasure on every Phase 1 model (20 seeds): accuracy with a refitted head (percent), gap (pp), logistic recoverability R, and nearest-centroid recoverability (percent).

Architecture	Rank	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)	NCC
ResNet-18	0	97.90 ± 0.37	97.31 ± 0.61	0.59	23.1 ± 1.2	23.2 ± 1.1	21.0 ± 1.0
	4	97.86 ± 0.37	97.32 ± 0.68	0.54	22.2 ± 1.2	22.5 ± 1.1	19.6 ± 1.1
	16	97.92 ± 0.38	97.31 ± 0.70	0.61	18.7 ± 1.1	18.7 ± 1.0	15.2 ± 1.1
	64	97.81 ± 0.37	97.12 ± 0.81	0.69	10.5 ± 0.8	10.2 ± 1.0	8.1 ± 0.8
	256	97.77 ± 0.33	96.98 ± 1.00	0.79	2.0 ± 0.4	2.3 ± 0.5	1.8 ± 0.4
ResNet-50	0	98.12 ± 0.42	97.16 ± 0.67	0.96	24.6 ± 1.1	24.9 ± 1.1	23.0 ± 1.2
	4	98.12 ± 0.45	97.09 ± 0.72	1.02	25.9 ± 1.5	25.9 ± 1.2	23.5 ± 1.2
	16	98.01 ± 0.46	97.11 ± 0.67	0.90	27.6 ± 1.6	27.2 ± 1.2	24.1 ± 1.4
	64	98.00 ± 0.43	97.10 ± 0.76	0.90	24.6 ± 1.8	23.9 ± 1.1	19.9 ± 1.2
	256	97.91 ± 0.43	97.14 ± 0.72	0.77	10.7 ± 0.9	10.1 ± 0.8	8.7 ± 0.8
DenseNet-121	0	98.24 ± 0.32	97.52 ± 0.67	0.72	29.5 ± 1.3	29.2 ± 1.3	23.7 ± 1.3
	4	98.27 ± 0.30	97.53 ± 0.64	0.73	29.1 ± 1.1	28.9 ± 1.2	22.8 ± 1.2
	16	98.17 ± 0.35	97.41 ± 0.73	0.75	26.2 ± 1.3	26.4 ± 0.9	18.8 ± 1.3
	64	98.19 ± 0.25	97.54 ± 0.78	0.65	17.1 ± 0.8	17.0 ± 1.2	10.8 ± 1.1
	256	97.99 ± 0.35	97.34 ± 0.74	0.65	3.7 ± 0.6	4.4 ± 0.6	2.6 ± 0.6
EfficientNet-B0	0	98.04 ± 0.54	97.30 ± 0.88	0.74	22.3 ± 1.1	22.8 ± 1.3	20.1 ± 1.2
	4	98.05 ± 0.53	97.30 ± 0.87	0.75	22.0 ± 1.0	22.4 ± 1.4	19.5 ± 1.2
	16	98.08 ± 0.55	97.30 ± 0.91	0.78	20.6 ± 1.3	20.8 ± 1.4	17.3 ± 1.2
	64	98.06 ± 0.48	97.26 ± 0.94	0.80	16.1 ± 1.2	15.7 ± 1.3	12.0 ± 1.1
	256	97.99 ± 0.49	97.29 ± 0.75	0.70	5.5 ± 0.8	5.5 ± 0.9	4.2 ± 0.6
MobileNet-V2	0	97.41 ± 0.48	96.61 ± 0.88	0.80	20.6 ± 1.2	21.0 ± 0.9	19.3 ± 1.1
	4	97.41 ± 0.52	96.62 ± 0.91	0.78	20.2 ± 1.0	20.7 ± 1.1	18.3 ± 1.0
	16	97.33 ± 0.58	96.59 ± 0.91	0.75	18.5 ± 1.2	18.7 ± 1.0	15.5 ± 1.2
	64	97.38 ± 0.56	96.63 ± 0.85	0.75	13.7 ± 1.0	13.5 ± 0.9	10.8 ± 0.9
	256	97.19 ± 0.49	96.47 ± 0.80	0.72	4.2 ± 0.8	4.2 ± 0.6	3.5 ± 0.6
ViT-B/16	0	97.83 ± 0.75	96.83 ± 0.80	1.00	28.8 ± 1.2	29.4 ± 1.2	25.1 ± 1.3
	4	97.86 ± 0.71	96.81 ± 0.74	1.05	28.1 ± 1.3	28.7 ± 1.1	23.9 ± 1.4
	16	97.83 ± 0.72	96.78 ± 0.82	1.05	25.3 ± 1.4	25.8 ± 1.1	19.9 ± 1.3
	64	97.88 ± 0.68	96.73 ± 0.76	1.14	17.6 ± 1.3	17.7 ± 1.4	12.7 ± 1.2
	256	97.80 ± 0.72	96.74 ± 0.70	1.06	6.1 ± 0.7	6.3 ± 1.0	4.6 ± 0.6
ConvNeXt-T	0	98.20 ± 0.36	97.30 ± 0.72	0.90	29.3 ± 1.3	29.7 ± 1.5	26.3 ± 1.6
	4	98.17 ± 0.40	97.28 ± 0.67	0.90	28.8 ± 1.3	29.0 ± 1.4	25.1 ± 1.6
	16	98.15 ± 0.44	97.24 ± 0.80	0.91	25.5 ± 1.2	25.6 ± 1.5	20.8 ± 1.5
	64	98.14 ± 0.37	97.27 ± 0.73	0.87	16.8 ± 1.1	16.6 ± 1.1	13.0 ± 0.9
	256	98.10 ± 0.38	97.28 ± 0.74	0.81	5.1 ± 0.6	5.3 ± 0.7	4.0 ± 0.6
Swin-T	0	98.17 ± 0.50	97.20 ± 0.81	0.97	29.0 ± 1.4	28.8 ± 1.5	25.5 ± 1.4
	4	98.12 ± 0.49	97.18 ± 0.80	0.94	28.7 ± 1.4	28.4 ± 1.4	24.9 ± 1.3
	16	98.19 ± 0.48	97.23 ± 0.77	0.96	26.5 ± 1.1	26.2 ± 1.5	22.0 ± 1.4
	64	98.12 ± 0.52	97.34 ± 0.79	0.78	19.6 ± 1.3	19.2 ± 1.3	15.0 ± 1.2
	256	98.14 ± 0.41	97.17 ± 0.61	0.96	6.5 ± 0.8	6.7 ± 0.8	5.2 ± 0.8
RegNetY-1.6GF	0	97.87 ± 0.46	96.91 ± 0.83	0.96	21.0 ± 1.4	21.1 ± 1.1	19.4 ± 1.2
	4	97.80 ± 0.42	96.85 ± 0.83	0.96	20.5 ± 1.3	20.8 ± 1.4	18.9 ± 1.1
	16	97.82 ± 0.49	96.73 ± 0.85	1.09	18.5 ± 1.4	18.4 ± 1.3	15.8 ± 1.1
	64	97.83 ± 0.44	96.72 ± 0.92	1.10	12.3 ± 1.1	11.9 ± 1.0	9.9 ± 1.0
	256	97.72 ± 0.42	96.62 ± 0.85	1.10	3.3 ± 0.7	3.4 ± 0.6	2.7 ± 0.7
EfficientNetV2-S	0	97.70 ± 0.50	96.86 ± 1.01	0.84	19.5 ± 1.4	20.4 ± 1.5	17.6 ± 1.1

Architecture	Rank	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)	NCC
	4	97.69 ± 0.48	96.93 ± 0.93	0.75	19.5 ± 1.7	20.2 ± 1.3	15.6 ± 3.1
	16	97.70 ± 0.51	96.86 ± 0.89	0.83	16.7 ± 2.1	17.8 ± 2.0	11.5 ± 4.2
	64	97.64 ± 0.61	96.85 ± 0.97	0.79	11.0 ± 3.3	12.9 ± 2.7	7.6 ± 3.7
	256	97.65 ± 0.51	96.75 ± 0.91	0.90	3.4 ± 1.3	4.4 ± 1.1	2.9 ± 1.1
MaxViT-T	0	97.88 ± 0.46	97.10 ± 0.83	0.78	21.6 ± 1.2	22.6 ± 1.5	19.1 ± 1.3
	4	97.86 ± 0.46	97.12 ± 0.87	0.73	20.9 ± 1.3	22.0 ± 1.5	18.2 ± 1.4
	16	97.80 ± 0.49	97.07 ± 0.86	0.73	18.3 ± 1.2	19.2 ± 1.7	14.8 ± 1.4
	64	97.90 ± 0.49	97.10 ± 0.86	0.79	12.4 ± 1.0	12.6 ± 1.7	9.4 ± 1.0
	256	97.80 ± 0.42	97.12 ± 0.79	0.68	3.8 ± 0.5	4.0 ± 0.9	3.1 ± 0.6
DINOv2 ViT-S/14	0	95.91 ± 0.75	93.96 ± 1.01	1.96	30.8 ± 0.9	30.3 ± 1.4	26.9 ± 1.3
	4	95.81 ± 0.79	93.93 ± 1.02	1.88	30.5 ± 0.9	30.0 ± 1.4	26.1 ± 1.4
	16	95.70 ± 0.76	93.76 ± 0.97	1.94	28.4 ± 1.1	28.9 ± 1.3	23.8 ± 1.4
	64	95.65 ± 0.73	93.91 ± 1.00	1.74	23.2 ± 1.4	23.0 ± 1.4	16.8 ± 1.5
	256	95.33 ± 0.98	93.12 ± 1.17	2.21	9.3 ± 0.8	9.8 ± 1.2	6.3 ± 1.0

Table 39.8: Amendment 8 probe-matched linear adversary (10 seeds): accuracy (percent), gap (pp), and recoverability R (percent; device probe). λ above the regime collapses P2 accuracy.

Architecture	λ	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)
ResNet-50	0	97.63 ± 0.48	96.77 ± 0.71	0.86	24.7 ± 1.2	25.0 ± 0.9
	0.1	97.12 ± 0.29	96.28 ± 0.44	0.84	9.9 ± 1.1	10.2 ± 0.9
	0.3	96.80 ± 0.78	95.89 ± 0.73	0.91	8.1 ± 1.1	9.0 ± 1.1
	1	94.20 ± 1.88	93.21 ± 1.61	0.99	7.4 ± 1.3	8.1 ± 1.4
	3	83.67 ± 8.43	81.89 ± 8.69	1.78	7.0 ± 1.6	6.7 ± 1.3
DenseNet-121	0	97.68 ± 0.55	96.95 ± 0.35	0.73	29.4 ± 1.3	29.3 ± 1.6
	0.1	97.04 ± 0.60	96.81 ± 0.77	0.22	16.7 ± 1.4	17.2 ± 1.5
	0.3	96.63 ± 1.10	96.05 ± 1.30	0.58	18.3 ± 1.5	17.9 ± 1.0
	1	91.48 ± 2.44	87.96 ± 5.59	3.52	16.6 ± 1.0	16.5 ± 1.8
	3	65.93 ± 7.69	66.11 ± 9.63	-0.19	15.9 ± 2.6	17.0 ± 1.1
ConvNeXt-T	0	97.77 ± 0.73	96.54 ± 0.75	1.22	29.7 ± 1.4	30.0 ± 1.1
	0.1	97.56 ± 0.75	96.56 ± 0.78	1.00	19.0 ± 1.4	19.1 ± 1.6
	0.3	97.58 ± 0.65	96.02 ± 1.80	1.56	17.2 ± 1.2	17.4 ± 1.2
	1	97.00 ± 0.69	95.63 ± 1.18	1.37	16.5 ± 1.9	17.5 ± 1.5
	3	88.36 ± 3.04	71.95 ± 28.54	16.41	14.0 ± 1.7	14.3 ± 3.7

Table 39.9: Amendment 8 Reinhard stain normalisation (10 seeds; strength 0 is the matching Phase 1 run): accuracy (percent), gap (pp), and R (percent).

Architecture	s	P1 acc.	P2 acc.	Gap	R (P1)	R (P2)
ResNet-50	0	97.63 ± 0.48	96.77 ± 0.71	0.86	24.7 ± 1.2	25.0 ± 0.9
	0.5	97.31 ± 0.58	96.12 ± 0.68	1.19	21.6 ± 0.7	21.4 ± 0.9
	1	95.83 ± 0.39	95.01 ± 1.40	0.81	18.0 ± 1.0	18.5 ± 1.1
DenseNet-121	0	97.68 ± 0.55	96.95 ± 0.35	0.73	29.4 ± 1.3	29.3 ± 1.6
	0.5	97.51 ± 0.47	96.30 ± 0.61	1.21	28.2 ± 1.1	28.6 ± 1.4
	1	96.20 ± 0.83	95.30 ± 0.92	0.90	24.5 ± 1.0	25.1 ± 0.9
EfficientNet-B0	0	97.99 ± 0.39	97.07 ± 0.52	0.91	22.1 ± 1.2	22.7 ± 1.1
	0.5	97.37 ± 0.56	96.54 ± 0.60	0.83	21.0 ± 1.6	21.2 ± 1.6
	1	96.21 ± 0.56	95.59 ± 1.13	0.62	18.0 ± 1.2	18.2 ± 0.9
ConvNeXt-T	0	97.26 ± 0.63	96.73 ± 0.46	0.53	28.7 ± 1.2	29.9 ± 1.5
	0.5	97.01 ± 0.76	95.54 ± 1.44	1.47	27.0 ± 2.0	27.9 ± 1.0
	1	96.11 ± 1.24	95.47 ± 1.06	0.64	25.2 ± 1.9	25.4 ± 1.1

39.6 Grad-CAM sees no difference

Leak and clean models placed the same share of Grad-CAM attribution outside the cell mask (21.8% on average, against a background pixel share of 34.6%; leak minus clean -0.09 pp, -0.57 to 0.39). The maps

depended on the trained head (median Spearman 0.08 with a random-head map), so the absence of a difference is not a failure of the method's sanity check.

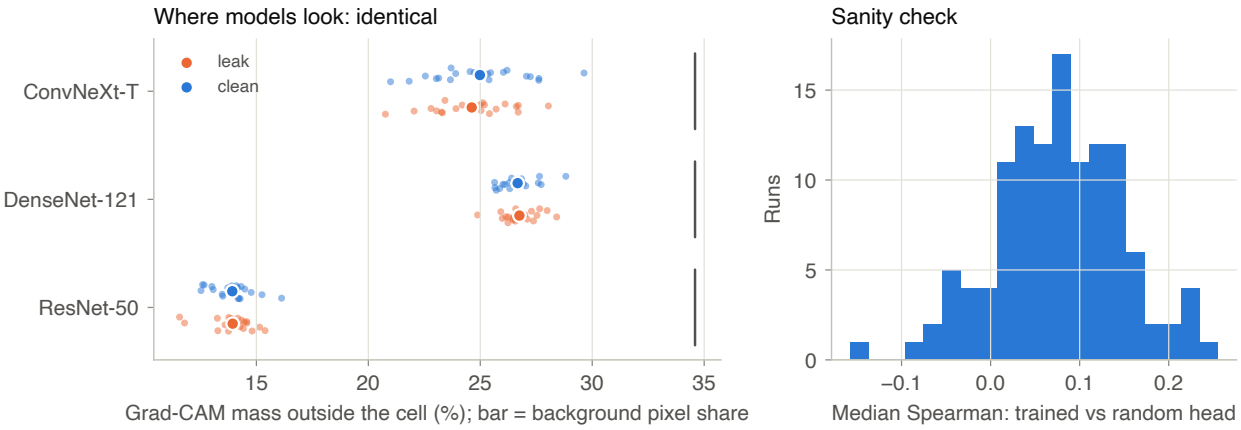


Figure 39.10: Left: per-run Grad-CAM attribution outside the cell for leak and clean models (bar: background pixel share). Right: agreement of each run's maps with random-head maps.

Table 39.10: Grad-CAM (descriptive): share of positive attribution outside the cell mask and background pixel share (percent, 20 runs each), and the median Spearman correlation with a random-head map.

Architecture	Models	Attribution outside	Background pixels	Random-head ρ
ResNet-50	leak	13.9 ± 1.0	34.6 ± 0.5	0.085
	clean	13.9 ± 0.9	34.6 ± 0.5	0.115
DenseNet-121	leak	26.7 ± 0.8	34.6 ± 0.5	0.044
	clean	26.7 ± 0.8	34.6 ± 0.5	0.057
ConvNeXt-T	leak	24.6 ± 1.8	34.6 ± 0.5	0.077
	clean	25.0 ± 2.2	34.6 ± 0.5	0.076

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The mechanism by elimination

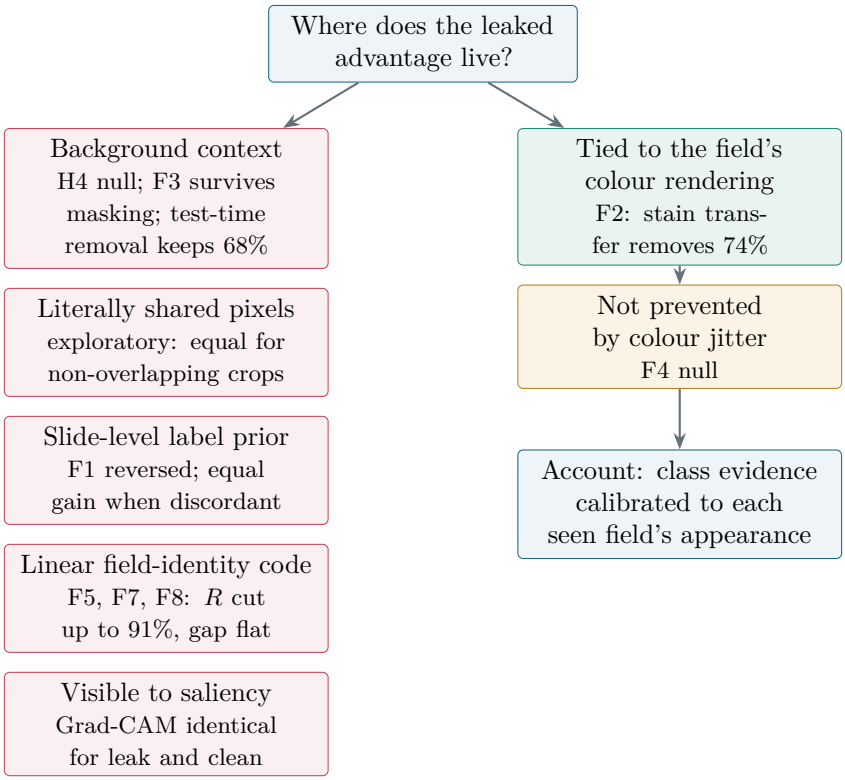


Figure 40.1: The mechanism by elimination. Red: candidate carriers ruled out; green: the one intervention that removed most of the advantage; gold: the qualification that keeps the account modest.

Taken together with Study 2's H4 and H5, the follow-up narrows how models use the leak. The advantage does not need a background (F3, test-time removal, H4). It is not memorisation of shared pixels (exploratory). It is not a slide-level label prior (F1, reversed). It is not carried by a separable, linearly decodable field code: erasing that code in closed form, or suppressing it with an adversary that cannot hide it from a linear probe, cut recoverability by 38–91% without moving the gap (F5, F7, F8). What remains is that the advantage depends on the cell being rendered as its own field renders cells: matching a test cell's colour statistics to another field of the same class removed most of it (F2).

The simplest account consistent with all of this is that a model trained on a field's siblings calibrates its class evidence to that field's appearance (stain, colour balance, and the texture they produce in the cell itself), so cells from a seen field sit further inside their class region. Random colour jitter during training

did not prevent this (F4), presumably because jitter perturbs each image around its own appearance and leaves the field-specific structure of that appearance learnable. This account is a reading of the evidence, not a registered result, and the F2–F4 tension is the part of it most in need of a direct test (for example, stain transfer applied to colour-augmented models).

Decodable is not used, and saliency is not use

Study 1 established *availability*: a probe recognised fields at 109 times chance. The follow-up shows that availability and use come apart: the identity a probe reads can be removed almost entirely without touching the inflation, in line with evidence from language models that probe-decodable information need not be what a model relies on [8, 25]. Grad-CAM showed leak and clean models attending to the same places; a saliency audit would have cleared both.

PART VII

Synthesis

41

What the project established, refuted, and left open

41.1 Claims and their strength

Table 41.1: Every substantive claim of the project and how strongly the evidence supports it.

Claim	Strength	Evidence and boundary
Public cytology benchmarks hide a crop-to-parent hierarchy	established	Study 1: recovered and verified for four datasets; Herlev negative control at zero.
Random crop splits share parents and pixels	established	92.6% (SIPaKMeD), 99.97% (CRIC) of test cells with a training parent; 52.6% / 71.5% / 34.0% with shared pixels.
The parent is recognisable	established	Study 1 probe 109× chance; Study 2 encoder 102× chance.
Crop-level splitting inflates accuracy	established (registered, powered)	H1 +0.97 pp, all twelve architectures; H2 +12.08 pp; five inference methods agree.
The inflation is leakage, not diversity	established (registered, replicated)	H3 +1.73 pp, diversity 0.00 pp; replicated on fresh seeds (+1.47 pp).
Exposure follows structure	supported	CRIC, with 29 cells per slide and high class–slide dependence, inflates ten times more than SIPaKMeD; predicted by the closed forms.
The leak needs no background	established (registered)	F3 +1.86 pp with every background masked; test-time removal leaves 68%.
The leak is tied to the field's colour rendering	supported (registered), qualified	F2 removes 74% at test time; F4 shows colour jitter in training does not prevent it.
A background context route carries the leak	not supported	H4 0.42 pp (−0.35 to 1.18).
Models apply a slide-level label prior (CRIC)	refuted	F1 reversed (−4.29 pp); equal gains for discordant and concordant cells.
The leak is memorised shared pixels	not supported (exploratory)	Equal advantage on non-overlapping crops.
The inflation is carried by linearly decodable field identity	refuted within its scope	F5, F7, F8: recoverability cut 38–91%, gap flat (pooled slope 0.000, CI −0.005 to 0.005).
Saliency maps reveal the leak	not supported (descriptive)	Grad-CAM identical for leak and clean models.
Stronger grouping (P3) is more conservative than P2	not supported	P2 − P3 0.03 pp; the ordering passes for 4 of 12 architectures.

41.2 What we got out of it

Scientific results

A certified, powered estimate of how much crop-level splitting inflates two public cytology benchmarks; a causal separation of leakage from diversity; a mechanism narrowed by elimination; and a methodological warning that the standard audit tools (probes, saliency, augmentation) cannot detect or remove this kind of leakage.

Reusable artifacts

Verified crop-to-parent maps for SIPaKMeD, Cx22, CRIC, and Herlev; registered P1/P1′/P2/P3 splits with hashes for 20 seeds; the 2C factorial designs and transplant plans; a risk calculator (`cytofair risk-report`) that predicts a benchmark's exposure from its parent table; a deterministic GPU-resident training stack with complete per-run artifacts; a fail-closed scheduler and collectors; exported records of every run (`results/study2/run_records`); 121 GB of archived predictions, features, and weights for future analyses without retraining.

Writing

A single-column manuscript (`paper/v2`, 17 pages, every number generated), the earlier ISBI draft, the Study 1 record, and this lifecycle record.

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What worked and what did not

Table 42.1: What worked and what did not, by area.

Area	Worked	Did not work, or cost more than expected
Question design	Separating structure, availability, consequence, and mechanism kept every claim at its true strength.	Study 1 asked the consequence question six times and could not answer it.
Statistics	One crossed mixed model with a split random effect, sized by simulation; registered sensitivity analyses that all agreed; the assumption trigger worked as designed.	Per-architecture testing in Study 1; Satterthwaite *df* for H3 is only 3 with four architectures, so its interval is wide.
Causal designs	The 2C factorial cleanly separated leakage from diversity; evaluation-only follow-ups on saved models were cheap and decisive.	The 2B transplant contrast was noisy (sibling backgrounds help even clean models); the 2A MLP adversary was evaded by ConvNeXt-T.
Mechanism	Elimination with several independent manipulations gave a coherent picture.	No single test identifies the carrier positively; F2 and F4 pull in different directions.
Engineering	GPU-resident augmentation (exact torchvision port); packed hashed caches; bitwise determinism; complete per-run artifacts; zero failed tasks in 2,921.	The four-core CPU quota had to be discovered the hard way; the slow network home forced a strict storage discipline.
Governance	Register, smoke-test on non-contrast splits, dry-run, collect fail-closed, amend with dates; every deviation recorded.	A dry run that bypassed one pipeline stage let a labelling bug through to the first real analysis (Amendment 9). Registrations are local commits, not yet externally timestamped.
Operations	Detached controllers, status files, sequential offloading with verification, clean vacating of node-local storage.	Several shell and remote-wait mistakes (self-matching process searches, hung SSH sessions) cost time but no data.

42.1 Demotivations, and why they mattered

Four results were disappointing at the time. Study 1's six tests failed to survive correction, which looked like a null; Study 2 showed it was underpowered. H4 and H5 failed to isolate a mechanism; Amendment 8 showed why (the leak is neither in the background nor in a linearly readable code). F1 went the opposite way from the prediction; that reversal is itself evidence against a slide-level label prior. And the ConvNeXt-T

adversary was evaded; that failure is a concrete example of why decodability and use must be measured separately. Each setback narrowed the explanation instead of ending it, and none was hidden.

42.2 Lessons

1. Ask a pooled question once, with the dependence structure in the model (the shared test set per seed is a random effect, not twelve independent tests).
2. Keep every per-sample artifact. Half of Amendment 8 cost no training because Study 2 kept logits, features, and weights.
3. Measure manipulations independently of the mechanism they target: the gradient-reversal adversary's own accuracy fell while an independent probe's rose.
4. Dry runs must pass through every stage of the real pipeline.
5. On shared hosts, discover the real resource limits (CPU quota, writable disks, network storage behaviour) before designing the execution.

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Decision log

Every design or execution decision that shaped a result, with the reason it was taken and the alternative it displaced.

Table 43.1: Decisions and their reasons.

Decision	Reason	Alternative not taken
Study 2 statistics		
Add a split random effect to H1	Twelve architectures share each seed's test set; without it the model treats twelve evaluations of one test set as independent.	The roadmap's formula without it.
Twenty seeds	Smallest count meeting the registered power rule (0.90 at pilot variances, 0.75 at 1.5×).	15 seeds (failed the inflated criterion); 25–30 (more compute for little power).
Twelve architectures	Breadth across CNNs, hierarchical transformers, plain ViTs, and a self-supervised model, so the effect is not tied to one family.	The six Study 1 models only.
Bootstrap primary when residuals are heavy-tailed	Registered before data; protects the primary test from non-normal seed-level noise.	Always the Satterthwaite t-test.
H2–H5 in one Holm family; H1 alone	H1 is the question; the mechanism tests are a family of related but separate claims.	One family of five (would have penalised the primary).
Study 2 designs		
220 crops per class in 2C	Largest budget feasible for every registered seed and 100/100 unrelated seeds; 240 failed for seed 307.	240 or 260 crops.
Common test set across all four 2C cells	Every cell is scored on the same crops, so differences are training effects only.	Separate test sets per cell.
Diversity estimated within clean cells only	The low-diversity leak cell gives each test crop more siblings, so the interaction confounds dose with diversity.	A full 2×2 interpretation.
Sibling background as the 2B control	Isolates the cell's own acquisition context from class-typical context.	Only unseen-field backgrounds.
Class-conditional parent adversary	Suppressing parent identity must never require removing class information.	An unconditional 424-way adversary.
CRIC as 100×100 windows	CRIC releases nucleus coordinates on slide images, not crops; fixed windows are the simplest faithful crops.	Variable-size crops from annotations.
Execution		
GPU-resident augmentation	Four-core CPU quota starved data loaders; an exact port keeps the registered recipe.	PIL augmentation in DataLoader workers.

Decision	Reason	Alternative not taken
One training job per GPU	GPUs were saturated by one job; two jobs shared throughput without speed-up.	Two jobs per GPU.
Math attention kernel only	Fused attention kernels have non-deterministic backward passes.	Faster fused kernels.
Float16 checkpoints	Halves storage of about 117 GB of weights; re-evaluation reproduces 418/480 models exactly, the rest within three crops.	Float32 checkpoints.
Amendment 7 before any 2A run	Deterministic divergence would otherwise have failed the whole arm.	Handling divergence after seeing it.
Registered analysis run on the GPU host	Run records point at node-local artifacts; moving gigabytes to a workstation was slower than computing in place.	Pulling all artifacts to the workstation.
Amendment 8		
A separately registered follow-up	Study 2's mechanism tests were inconclusive, and saved artifacts made decisive tests cheap; registering first keeps them confirmatory for new questions.	Unregistered exploration, or re-testing H4/H5 with more seeds.
Reuse saved models where possible	F2 and F5 needed no training; three arms cost 7 GPU-hours in total.	Retraining for every arm.
Fresh seeds for retraining arms	New data for new questions; no reuse of Study 2 mechanism seeds.	Reusing the mechanism seeds.
Probe-matched linear adversary	ConvNeXt-T had evaded the MLP adversary; a standardised linear head cannot hide identity from a linear probe while leaving it readable.	Re-running the MLP adversary.
LEACE ranks up to 256	The smallest feature dimension is 384 (DINOv2 ViT-S); 256 removes most within-class parent structure in every backbone.	Full-rank erasure (exceeds some feature dimensions).
HED-strong colour augmentation	A published, strong, stain-shaped perturbation.	Hue/saturation jitter only.
One device probe for every arm	Recoverability must be measured identically across manipulations; agreement with sklearn verified ($r = 0.999$).	Mixing probe implementations.
Pixel-overlap split labelled exploratory	Added after the registered results to test a specific alternative explanation.	Presenting it as confirmatory.
Close-out		
Offload checkpoints during the follow-up matrix	52 GB of new checkpoints against a 71 GB margin; verified copies before deletion.	Pausing the matrix on the disk guard.
Recommend ISBI framing around audit tools	The registered evidence is strong; the novel take-home for an imaging audience is that probes and saliency cannot detect this leakage.	Framing around "leakage exists".

44

Validity, recommendations, and next steps

44.1 Threats to validity

Parents are not patients. SIPaKMeD fields, Cx22 source images, and CRIC slide images are the deepest released units. Several can come from one patient, so parent-disjoint evaluation is necessary but may not be sufficient for clinical independence; every inflation estimate is a lower bound on what patient-level grouping would remove.

One recipe. Every model used one fixed fine-tuning recipe. Stronger augmentation, longer training, or other optimisers could change the size of the inflation and the models' reliance on acquisition context.

Smaller training sets in the factorial. Matching crop and field counts required 220 crops per class (1,100 in total against 3,239 in Phase 1); the 2C, 2B, and carrier effects are measured on smaller training sets than the headline contrast, and the decomposition that transfers them is approximate.

Imperfect interventions. Foreground masks are threshold rules; inpainting leaves smoothed backgrounds; Reinhard transfer matches only first and second moments; erasure and adversaries act on pooled features and linear structure.

Post-results follow-up. Amendment 8 was motivated by Study 2's results. Its hypotheses and analyses were fixed before any follow-up run, but they are follow-up questions; the pixel-overlap split was added afterwards and is exploratory.

CRIC crops. The CRIC cells are windows cut by this project around published nucleus coordinates, not an official crop release, so absolute CRIC accuracies are not directly comparable with published numbers.

External verification. The registration commits have not yet been pushed or deposited; until they are, their dates rest on the repository and on the commit hashes recorded inside every run.

No clinical claim. The project audits research benchmarks; it does not estimate screening sensitivity, specificity, prevalence-calibrated performance, or effects on readers.

44.2 Recommendations

1. **Release the hierarchy.** Every observation should carry its acquisition parent (field, source image, or slide), a patient or specimen identifier where permitted, and, if audited, a duplicate-component identifier.
2. **Split at the deepest released unit and publish the split**, with its hash.
3. **Report both units**: observation and parent counts per split, and the fraction of test observations with a training sibling.
4. **Check the structure before training** with the closed forms (`cytofair risk-report`).
5. **Treat one question as one test**, with a mixed model that treats architectures and splits as random effects, sized by simulation.
6. **Do not certify a split with a probe, a saliency map, or an augmentation.** Leakage inflated accuracy with field identity erased from the features, with identical Grad-CAM maps, with backgrounds removed, and with strong colour augmentation.
7. **Keep the evidence**: per-sample predictions, final weights, and the exact split for every run.

44.3 Publication options and open decisions

44.3.1 Venues

ISBI 2027. Four pages including figures and tables (references may use a paid fifth page); deadline Monday 26 October 2026, 23:59 US Eastern; single-blind; arXiv preprints allowed; a "Compliance with Ethical Standards" section and a disclosure of AI-generated content in the acknowledgments are required; presentation in person in Lausanne. The evidence is stronger than typical for the venue. The risks are perceived novelty (patient-level splitting is textbook advice) and the page limit. The recommended framing is "how much crop-level splitting inflates public cytology benchmarks, and why the usual audit tools cannot see it": H1–H3 as the result, one figure (per-architecture SIPaKMeD with CRIC), one table of all thirteen registered tests, and a paragraph on F2, F3, and the probe and Grad-CAM findings. Everything else waits for the journal version.

Journal version. The full manuscript (`paper/v2`) with Study 2 and Amendment 8. Check the journal's policy on extended conference papers if ISBI is pursued first; ISBI only forbids simultaneous submission of a substantially similar paper to another conference or workshop.

Preprint. Allowed by ISBI; it would also give the registrations a public timestamp once the repository is pushed.

44.3.2 Decisions only the authors can make

- Push the `study2` branch (and the Cx22 commits) to the public repository, and deposit the preregistrations with a third party (for example OSF) for an external timestamp.
- Choose a software licence; confirm redistribution terms for derived artifacts (SIPaKMeD, CCEDD).
- Co-author approval of every posted or submitted version; fill the e-mail and repository placeholders.
- Whether to submit to ISBI 2027, and who presents in person.
- The AI-use disclosure wording: much of the Study 2 code, analysis, manuscript text, and this record was drafted with an AI assistant under the authors' direction.
- Whether to open Phase 3 (other imaging domains) and the direct test of the F2–F4 tension.

44.3.3 Future work

- Apply test-time stain transfer to the colour-augmented carrier models to resolve the F2–F4 tension directly.
- Non-linear and spatial (pre-pooling) erasure of field identity.
- Patient-level grouping where a dataset's provenance allows it.
- Other recipes (stronger augmentation, longer schedules, stain normalisation as standard practice).
- Other domains where crops share acquisition units (the pre-specified fish-imagery V2).

PART VIII

Appendices

About the appendices

Appendices A–K are the Study 1 appendices, carried over verbatim. The remaining appendices hold the complete ledgers of every Study 2 and Amendment 8 run, the variance components of every registered model, the commit chronology, the registration hashes, the preregistration documents verbatim, and a reproduction cookbook.

A

Complete SIPaKMeD classifier run ledger

The following 120 rows are the entire accepted classifier matrix. Accuracy, macro-F1, and balanced accuracy are final-epoch test values. This appendix makes every seed visible and prevents mean-only reporting from hiding an outlying run.

Table A.1: Complete SIPaKMeD classification run ledger. All values are final-epoch results.

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
densenet121	P1	17	3239	810	98.519	98.521	98.527	4.66	3
densenet121	P1	29	3239	810	98.025	98.023	98.027	4.62	2
densenet121	P1	43	3239	810	97.778	97.776	97.803	4.54	1
densenet121	P1	71	3239	810	98.025	98.031	98.018	4.61	3
densenet121	P1	101	3239	810	98.272	98.284	98.276	4.70	2
densenet121	P1-prime	17	3239	810	98.519	98.521	98.527	4.61	1
densenet121	P1-prime	29	3239	810	98.025	98.023	98.027	4.48	3
densenet121	P1-prime	43	3239	810	97.778	97.776	97.803	4.49	2
densenet121	P1-prime	71	3239	810	98.025	98.031	98.018	4.48	1
densenet121	P1-prime	101	3239	810	98.272	98.284	98.276	4.45	3
densenet121	P2	17	3239	810	95.926	95.972	95.946	4.47	2
densenet121	P2	29	3239	810	97.778	97.766	97.798	4.55	1
densenet121	P2	43	3239	810	96.914	96.927	96.906	4.51	3
densenet121	P2	71	3239	810	96.543	96.561	96.566	4.47	2
densenet121	P2	101	3239	810	96.420	96.446	96.395	4.59	1
densenet121	P3	17	3239	810	97.778	97.781	97.774	4.54	3
densenet121	P3	29	3239	810	97.901	97.895	97.904	4.64	2
densenet121	P3	43	3239	810	97.531	97.538	97.553	4.56	1
densenet121	P3	71	3239	810	96.790	96.824	96.801	4.20	3
densenet121	P3	101	3239	810	96.914	96.945	96.894	4.00	2
efficientnet_b0	P1	17	3239	810	97.654	97.659	97.662	2.84	1
efficientnet_b0	P1	29	3239	810	98.272	98.276	98.273	2.89	3
efficientnet_b0	P1	43	3239	810	97.160	97.183	97.177	2.92	2
efficientnet_b0	P1	71	3239	810	97.531	97.540	97.541	2.98	1
efficientnet_b0	P1	101	3239	810	98.272	98.279	98.288	2.87	3
efficientnet_b0	P1-prime	17	3239	810	97.654	97.659	97.662	2.87	2
efficientnet_b0	P1-prime	29	3239	810	98.272	98.276	98.273	3.00	1
efficientnet_b0	P1-prime	43	3239	810	97.160	97.183	97.177	2.96	3
efficientnet_b0	P1-prime	71	3239	810	97.531	97.540	97.541	2.88	2

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
efficientnet_b0	P1-prime	101	3239	810	98.272	98.279	98.288	2.86	1
efficientnet_b0	P2	17	3239	810	97.778	97.790	97.799	3.01	3
efficientnet_b0	P2	29	3239	810	95.185	95.259	95.163	2.92	2
efficientnet_b0	P2	43	3239	810	98.642	98.641	98.643	2.92	1
efficientnet_b0	P2	71	3239	810	95.432	95.450	95.407	2.97	3
efficientnet_b0	P2	101	3239	810	96.420	96.413	96.422	2.93	2
efficientnet_b0	P3	17	3239	810	98.148	98.144	98.170	2.89	1
efficientnet_b0	P3	29	3239	810	97.901	97.920	97.910	2.98	3
efficientnet_b0	P3	43	3239	810	96.296	96.316	96.325	2.99	2
efficientnet_b0	P3	71	3239	810	97.407	97.428	97.427	2.94	1
efficientnet_b0	P3	101	3239	810	97.284	97.300	97.285	2.80	3
mobilenet_v2	P1	17	3239	810	97.284	97.287	97.278	2.72	2
mobilenet_v2	P1	29	3239	810	97.407	97.406	97.403	2.76	1
mobilenet_v2	P1	43	3239	810	96.914	96.933	96.944	2.74	3
mobilenet_v2	P1	71	3239	810	97.284	97.296	97.289	2.71	2
mobilenet_v2	P1	101	3239	810	98.148	98.149	98.167	2.72	1
mobilenet_v2	P1-prime	17	3239	810	97.284	97.287	97.278	2.69	3
mobilenet_v2	P1-prime	29	3239	810	97.407	97.406	97.403	2.74	2
mobilenet_v2	P1-prime	43	3239	810	96.914	96.933	96.944	2.72	1
mobilenet_v2	P1-prime	71	3239	810	97.284	97.296	97.289	2.71	3
mobilenet_v2	P1-prime	101	3239	810	98.148	98.149	98.167	2.73	2
mobilenet_v2	P2	17	3239	810	96.420	96.432	96.453	2.75	1
mobilenet_v2	P2	29	3239	810	93.704	93.732	93.714	2.73	3
mobilenet_v2	P2	43	3239	810	96.667	96.679	96.637	2.74	2
mobilenet_v2	P2	71	3239	810	93.457	93.486	93.458	2.78	1
mobilenet_v2	P2	101	3239	810	95.556	95.553	95.534	2.77	3
mobilenet_v2	P3	17	3239	810	97.407	97.408	97.433	2.68	2
mobilenet_v2	P3	29	3239	810	96.173	96.163	96.163	2.74	1
mobilenet_v2	P3	43	3239	810	96.667	96.669	96.696	2.74	3
mobilenet_v2	P3	71	3239	810	95.802	95.823	95.813	2.92	2
mobilenet_v2	P3	101	3239	810	94.938	94.930	94.900	2.79	1
resnet18	P1	17	3239	810	97.901	97.900	97.930	2.57	1
resnet18	P1	29	3239	810	97.037	97.028	97.051	2.54	2
resnet18	P1	43	3239	810	96.914	96.922	96.938	2.48	3

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
resnet18	P1	71	3239	810	96.420	96.455	96.410	2.61	1
resnet18	P1	101	3239	810	97.654	97.661	97.674	2.54	2
resnet18	P1-prime	17	3239	810	97.901	97.900	97.930	2.53	3
resnet18	P1-prime	29	3239	810	97.037	97.028	97.051	2.62	1
resnet18	P1-prime	43	3239	810	96.914	96.922	96.938	2.61	2
resnet18	P1-prime	71	3239	810	96.420	96.455	96.410	2.48	3
resnet18	P1-prime	101	3239	810	97.654	97.661	97.674	2.62	1
resnet18	P2	17	3239	810	96.420	96.444	96.447	2.58	2
resnet18	P2	29	3239	810	96.049	96.051	96.058	2.47	3
resnet18	P2	43	3239	810	96.543	96.535	96.502	2.49	1
resnet18	P2	71	3239	810	96.543	96.563	96.528	2.56	2
resnet18	P2	101	3239	810	94.444	94.436	94.384	2.50	3
resnet18	P3	17	3239	810	98.395	98.395	98.400	2.55	1
resnet18	P3	29	3239	810	95.062	95.069	95.014	2.55	3
resnet18	P3	43	3239	810	96.173	96.165	96.215	2.52	2
resnet18	P3	71	3239	810	96.543	96.578	96.561	2.79	1
resnet18	P3	101	3239	810	95.556	95.576	95.526	2.79	3
resnet50	P1	17	3239	810	97.778	97.781	97.792	3.73	2
resnet50	P1	29	3239	810	97.284	97.304	97.275	3.29	1
resnet50	P1	43	3239	810	96.790	96.801	96.805	3.32	3
resnet50	P1	71	3239	810	97.901	97.903	97.904	3.48	2
resnet50	P1	101	3239	810	97.778	97.776	97.777	3.45	1
resnet50	P1-prime	17	3239	810	97.778	97.781	97.792	3.26	3
resnet50	P1-prime	29	3239	810	97.284	97.304	97.275	3.45	2
resnet50	P1-prime	43	3239	810	96.790	96.801	96.805	3.52	1
resnet50	P1-prime	71	3239	810	97.901	97.903	97.904	3.34	3
resnet50	P1-prime	101	3239	810	97.778	97.776	97.777	3.54	2
resnet50	P2	17	3239	810	97.037	97.067	97.053	3.47	1
resnet50	P2	29	3239	810	96.914	96.931	96.916	3.24	3
resnet50	P2	43	3239	810	97.284	97.298	97.290	3.45	2

Architecture	Protocol	Seed	Train	Test	Acc. (percent)	Macro-F1 (percent)	Bal. acc. (percent)	min	GPU
resnet50	P2	71	3239	810	96.420	96.422	96.433	3.47	1
resnet50	P2	101	3239	810	96.296	96.303	96.288	3.29	3
resnet50	P3	17	3239	810	96.543	96.519	96.573	3.47	2
resnet50	P3	29	3239	810	97.037	97.071	97.049	3.46	1
resnet50	P3	43	3239	810	95.679	95.701	95.703	3.25	3
resnet50	P3	71	3239	810	95.926	95.945	95.958	3.73	2
resnet50	P3	101	3239	810	96.173	96.168	96.161	3.89	1
vit_b_16	P1	17	3239	810	97.407	97.414	97.430	9.30	3
vit_b_16	P1	29	3239	810	97.037	97.050	97.061	9.25	2
vit_b_16	P1	43	3239	810	95.556	95.541	95.606	9.35	1
vit_b_16	P1	71	3239	810	98.025	98.036	98.023	9.31	3
vit_b_16	P1	101	3239	810	97.654	97.666	97.684	9.27	2
vit_b_16	P1-prime	17	3239	810	97.407	97.414	97.430	9.29	1
vit_b_16	P1-prime	29	3239	810	97.037	97.050	97.061	9.19	3
vit_b_16	P1-prime	43	3239	810	95.556	95.541	95.606	9.24	2
vit_b_16	P1-prime	71	3239	810	98.025	98.036	98.023	9.30	1
vit_b_16	P1-prime	101	3239	810	97.654	97.666	97.684	9.24	3
vit_b_16	P2	17	3239	810	95.556	95.618	95.526	9.27	2
vit_b_16	P2	29	3239	810	95.802	95.821	95.788	9.36	1
vit_b_16	P2	43	3239	810	96.667	96.693	96.688	9.32	3
vit_b_16	P2	71	3239	810	95.432	95.432	95.392	9.30	2
vit_b_16	P2	101	3239	810	94.321	94.421	94.283	9.28	1
vit_b_16	P3	17	3239	810	96.296	96.258	96.328	9.19	3
vit_b_16	P3	29	3239	810	95.185	95.218	95.214	9.23	2
vit_b_16	P3	43	3239	810	92.840	92.915	92.770	9.26	1
vit_b_16	P3	71	3239	810	97.284	97.292	97.301	9.00	3
vit_b_16	P3	101	3239	810	96.049	96.039	96.066	8.81	2

B

Complete parent-probe run ledger

These 30 rows include five preprocessing arms and the label-shuffled negative control. All use exactly 424 parent classes, 848 training crops, and 848 held-out sibling crops.

Table B.1: Complete parent-identity probe ledger.

Mode	Shuffle	Seed	Parents	Train	Test	Chance (percent)	Accuracy (percent)	min	GPU
background_only	no	17	424	848	848	0.2358	18.514	21.00	3
background_only	no	29	424	848	848	0.2358	20.047	20.55	1
background_only	no	43	424	848	848	0.2358	19.811	20.23	2
background_only	no	71	424	848	848	0.2358	17.925	89.59	3
background_only	no	101	424	848	848	0.2358	17.099	89.40	2
grayscale	no	17	424	848	848	0.2358	15.212	0.93	3
grayscale	no	29	424	848	848	0.2358	13.208	0.88	2
grayscale	no	43	424	848	848	0.2358	14.741	1.02	1
grayscale	no	71	424	848	848	0.2358	13.208	0.97	2
grayscale	no	101	424	848	848	0.2358	12.618	1.02	3
highpass	no	17	424	848	848	0.2358	16.863	15.73	1
highpass	no	29	424	848	848	0.2358	16.156	15.87	1
highpass	no	43	424	848	848	0.2358	16.863	15.93	1
highpass	no	71	424	848	848	0.2358	16.038	15.80	1
highpass	no	101	424	848	848	0.2358	15.448	16.04	1
rgb	no	17	424	848	848	0.2358	25.943	0.89	1
rgb	no	29	424	848	848	0.2358	26.769	0.76	2
rgb	no	43	424	848	848	0.2358	25.236	0.76	3
rgb	no	71	424	848	848	0.2358	25.590	0.77	1
rgb	no	101	424	848	848	0.2358	25.118	0.79	2
rgb	yes	17	424	848	848	0.2358	0.590	0.78	3
rgb	yes	29	424	848	848	0.2358	0.000	0.88	1
rgb	yes	43	424	848	848	0.2358	0.118	0.86	3
rgb	yes	71	424	848	848	0.2358	0.000	0.76	2
rgb	yes	101	424	848	848	0.2358	0.118	0.82	1
stain_normalized	no	17	424	848	848	0.2358	17.335	2.49	1
stain_normalized	no	29	424	848	848	0.2358	17.099	2.41	2
stain_normalized	no	43	424	848	848	0.2358	18.396	2.26	3
stain_normalized	no	71	424	848	848	0.2358	16.745	2.42	1
stain_normalized	no	101	424	848	848	0.2358	16.981	2.20	2

C

Complete Cx22 model run ledger

The following 175 rows are all accepted Cx22 cells: five semantic architectures and two instance architectures, each under five protocols and five seeds. “Primary” is semantic macro-Dice or matched-instance Dice; “Nuc./R70” is nucleus Dice for semantic rows and Recall@0.70 for instance rows.

Table C.1: Complete Cx22 run ledger. Primary is macro Dice for semantic models and matched-instance Dice for instance models; the ninth column is nucleus Dice or recall at Dice 0.70.

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
instance	maskrcnn_resnet50_fpn	OFFICIAL	17	400	100	92.117	87.060	96.991	14.35
instance	maskrcnn_resnet50_fpn	OFFICIAL	29	400	100	92.372	87.354	97.068	14.92
instance	maskrcnn_resnet50_fpn	OFFICIAL	43	400	100	91.844	86.861	96.682	15.89
instance	maskrcnn_resnet50_fpn	OFFICIAL	71	400	100	92.669	87.541	97.608	0.49
instance	maskrcnn_resnet50_fpn	OFFICIAL	101	400	100	92.433	87.559	97.145	1.29
instance	maskrcnn_resnet50_fpn	P1	17	1056	264	92.683	87.370	96.540	33.85
instance	maskrcnn_resnet50_fpn	P1	29	1056	264	91.354	86.327	95.009	33.71
instance	maskrcnn_resnet50_fpn	P1	43	1056	264	92.282	87.191	96.854	35.84
instance	maskrcnn_resnet50_fpn	P1	71	1056	264	92.343	87.320	96.550	37.54
instance	maskrcnn_resnet50_fpn	P1	101	1056	264	91.819	86.735	96.471	2.55
instance	maskrcnn_resnet50_fpn	P1-prime	17	1056	264	92.683	87.370	96.540	33.94
instance	maskrcnn_resnet50_fpn	P1-prime	29	1056	264	91.354	86.327	95.009	33.95
instance	maskrcnn_resnet50_fpn	P1-prime	43	1056	264	92.282	87.191	96.854	35.19
instance	maskrcnn_resnet50_fpn	P1-prime	71	1055	264	91.733	86.958	95.483	38.29
instance	maskrcnn_resnet50_fpn	P1-prime	101	1053	264	92.213	87.257	97.248	2.56
instance	maskrcnn_resnet50_fpn	P2	17	1056	264	91.815	86.503	95.983	33.84
instance	maskrcnn_resnet50_fpn	P2	29	1056	264	91.313	86.228	95.159	34.29
instance	maskrcnn_resnet50_fpn	P2	43	1056	264	92.024	86.559	96.750	35.88
instance	maskrcnn_resnet50_fpn	P2	71	1055	265	90.993	85.998	92.239	0.26
instance	maskrcnn_resnet50_fpn	P2	101	1053	267	91.793	86.639	94.671	2.57
instance	maskrcnn_resnet50_fpn	P3	17	891	429	89.855	84.435	94.937	47.44
instance	maskrcnn_resnet50_fpn	P3	29	825	495	91.606	86.357	95.207	49.79
instance	maskrcnn_resnet50_fpn	P3	43	894	426	90.023	84.807	94.415	47.25
instance	maskrcnn_resnet50_fpn	P3	71	773	547	92.181	86.751	96.205	1.16
instance	maskrcnn_resnet50_fpn	P3	101	791	529	91.910	86.399	96.806	1.98
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	17	400	100	91.970	87.274	95.910	1.31
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	29	400	100	92.548	87.588	96.991	1.33
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	43	400	100	91.041	86.290	94.367	1.32
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	71	400	100	92.442	87.459	97.068	1.69
instance	maskrcnn_resnet50_fpn_v2	OFFICIAL	101	400	100	92.886	87.943	97.531	1.40
instance	maskrcnn_resnet50_fpn_v2	P1	17	1056	264	91.235	86.322	91.816	2.63
instance	maskrcnn_resnet50_fpn_v2	P1	29	1056	264	91.543	86.726	92.276	2.64
instance	maskrcnn_resnet50_fpn_v2	P1	43	1056	264	92.536	87.688	94.870	2.65
instance	maskrcnn_resnet50_fpn_v2	P1	71	1056	264	92.342	87.566	94.856	2.63
instance	maskrcnn_resnet50_fpn_v2	P1	101	1056	264	92.318	87.546	95.131	2.99
instance	maskrcnn_resnet50_fpn_v2	P1-prime	17	1056	264	91.235	86.322	91.816	2.63
instance	maskrcnn_resnet50_fpn_v2	P1-prime	29	1056	264	91.543	86.726	92.276	2.63
instance	maskrcnn_resnet50_fpn_v2	P1-prime	43	1056	264	92.536	87.688	94.870	2.64
instance	maskrcnn_resnet50_fpn_v2	P1-prime	71	1055	264	91.938	87.241	93.915	2.62
instance	maskrcnn_resnet50_fpn_v2	P1-prime	101	1053	264	92.502	87.584	95.272	2.85
instance	maskrcnn_resnet50_fpn_v2	P2	17	1056	264	92.036	87.121	95.303	2.64
instance	maskrcnn_resnet50_fpn_v2	P2	29	1056	264	91.737	86.754	94.870	2.65
instance	maskrcnn_resnet50_fpn_v2	P2	43	1056	264	92.056	86.980	95.297	2.65

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
instance	maskrcnn_resnet50_fpn_v2	P2	71	1055	265	90.232	85.363	87.492	2.64
instance	maskrcnn_resnet50_fpn_v2	P2	101	1053	267	90.762	85.672	92.038	2.70
instance	maskrcnn_resnet50_fpn_v2	P3	17	891	429	91.242	85.867	94.762	2.26
instance	maskrcnn_resnet50_fpn_v2	P3	29	825	495	89.960	84.956	91.273	2.10
instance	maskrcnn_resnet50_fpn_v2	P3	43	894	426	90.317	85.094	91.622	2.29
instance	maskrcnn_resnet50_fpn_v2	P3	71	773	547	90.474	85.440	90.769	1.98
instance	maskrcnn_resnet50_fpn_v2	P3	101	791	529	91.026	85.770	94.358	2.02
semantic	deeplabv3plus_resnet50	OFFICIAL	17	400	100	82.697	72.771	71.084	1.02
semantic	deeplabv3plus_resnet50	OFFICIAL	29	400	100	83.602	73.897	72.700	1.13
semantic	deeplabv3plus_resnet50	OFFICIAL	43	400	100	84.126	74.580	73.775	1.08
semantic	deeplabv3plus_resnet50	OFFICIAL	71	400	100	83.972	74.385	73.418	1.12
semantic	deeplabv3plus_resnet50	OFFICIAL	101	400	100	80.382	70.180	66.995	1.11
semantic	deeplabv3plus_resnet50	P1	17	1056	264	88.570	80.909	81.412	2.09
semantic	deeplabv3plus_resnet50	P1	29	1056	264	88.464	80.728	81.373	1.91
semantic	deeplabv3plus_resnet50	P1	43	1056	264	89.196	81.695	82.572	2.02
semantic	deeplabv3plus_resnet50	P1	71	1056	264	89.542	82.165	83.162	2.14
semantic	deeplabv3plus_resnet50	P1	101	1056	264	88.738	81.195	81.662	2.32
semantic	deeplabv3plus_resnet50	P1-prime	17	1056	264	88.570	80.909	81.412	2.13
semantic	deeplabv3plus_resnet50	P1-prime	29	1056	264	88.464	80.728	81.373	1.93
semantic	deeplabv3plus_resnet50	P1-prime	43	1056	264	89.196	81.695	82.572	2.03
semantic	deeplabv3plus_resnet50	P1-prime	71	1055	264	89.443	82.040	82.958	2.14
semantic	deeplabv3plus_resnet50	P1-prime	101	1053	264	88.514	80.848	81.162	2.28
semantic	deeplabv3plus_resnet50	P2	17	1056	264	88.304	80.549	81.070	1.90
semantic	deeplabv3plus_resnet50	P2	29	1056	264	88.930	81.407	82.016	2.03
semantic	deeplabv3plus_resnet50	P2	43	1056	264	88.731	80.967	81.799	1.99
semantic	deeplabv3plus_resnet50	P2	71	1055	265	88.988	81.513	82.202	2.22
semantic	deeplabv3plus_resnet50	P2	101	1053	267	88.268	80.470	80.952	2.28
semantic	deeplabv3plus_resnet50	P3	17	891	429	88.332	80.528	81.111	2.13
semantic	deeplabv3plus_resnet50	P3	29	825	495	87.707	79.682	80.088	2.57
semantic	deeplabv3plus_resnet50	P3	43	894	426	88.621	80.940	81.705	2.70
semantic	deeplabv3plus_resnet50	P3	71	773	547	88.041	80.050	80.594	3.46
semantic	deeplabv3plus_resnet50	P3	101	791	529	87.790	79.677	80.185	3.43
semantic	fpn_resnet34	OFFICIAL	17	400	100	84.378	74.779	74.746	1.08
semantic	fpn_resnet34	OFFICIAL	29	400	100	84.801	75.377	75.422	1.12
semantic	fpn_resnet34	OFFICIAL	43	400	100	84.389	74.760	74.723	1.10
semantic	fpn_resnet34	OFFICIAL	71	400	100	84.630	75.136	75.152	1.07
semantic	fpn_resnet34	OFFICIAL	101	400	100	83.934	74.231	73.850	1.09
semantic	fpn_resnet34	P1	17	1056	264	88.606	80.936	81.657	2.30
semantic	fpn_resnet34	P1	29	1056	264	88.376	80.651	81.204	2.17
semantic	fpn_resnet34	P1	43	1056	264	89.242	81.731	82.802	2.29
semantic	fpn_resnet34	P1	71	1056	264	89.415	81.991	83.139	2.29
semantic	fpn_resnet34	P1	101	1056	264	88.854	81.286	81.903	2.31
semantic	fpn_resnet34	P1-prime	17	1056	264	88.606	80.936	81.657	2.23
semantic	fpn_resnet34	P1-prime	29	1056	264	88.376	80.651	81.204	2.19
semantic	fpn_resnet34	P1-prime	43	1056	264	89.242	81.731	82.802	2.30
semantic	fpn_resnet34	P1-prime	71	1055	264	89.760	82.473	83.695	2.30
semantic	fpn_resnet34	P1-prime	101	1053	264	88.880	81.350	81.891	2.26

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
semantic	fpn_resnet34	P2	17	1056	264	88.539	80.901	81.452	2.18
semantic	fpn_resnet34	P2	29	1056	264	89.006	81.470	82.385	2.20
semantic	fpn_resnet34	P2	43	1056	264	89.498	82.080	83.321	2.22
semantic	fpn_resnet34	P2	71	1055	265	89.102	81.618	82.649	2.24
semantic	fpn_resnet34	P2	101	1053	267	88.296	80.471	81.211	2.25
semantic	fpn_resnet34	P3	17	891	429	88.029	80.133	80.698	2.84
semantic	fpn_resnet34	P3	29	825	495	88.080	80.127	80.967	3.16
semantic	fpn_resnet34	P3	43	894	426	88.744	81.048	82.174	2.75
semantic	fpn_resnet34	P3	71	773	547	88.510	80.711	81.583	3.46
semantic	fpn_resnet34	P3	101	791	529	88.287	80.367	81.266	3.45
semantic	manet_resnet34	OFFICIAL	17	400	100	83.189	73.517	71.864	1.06
semantic	manet_resnet34	OFFICIAL	29	400	100	86.409	77.571	77.856	1.08
semantic	manet_resnet34	OFFICIAL	43	400	100	86.521	77.608	78.266	1.07
semantic	manet_resnet34	OFFICIAL	71	400	100	86.256	77.306	77.825	1.15
semantic	manet_resnet34	OFFICIAL	101	400	100	85.118	75.878	75.519	1.19
semantic	manet_resnet34	P1	17	1056	264	89.317	81.998	82.837	2.13
semantic	manet_resnet34	P1	29	1056	264	89.510	82.294	83.125	2.19
semantic	manet_resnet34	P1	43	1056	264	90.535	83.620	85.003	2.09
semantic	manet_resnet34	P1	71	1056	264	90.674	83.752	85.088	2.06
semantic	manet_resnet34	P1	101	1056	264	90.128	83.049	84.158	2.14
semantic	manet_resnet34	P1-prime	17	1056	264	89.317	81.998	82.837	2.17
semantic	manet_resnet34	P1-prime	29	1056	264	89.510	82.294	83.125	2.06
semantic	manet_resnet34	P1-prime	43	1056	264	90.535	83.620	85.003	2.05
semantic	manet_resnet34	P1-prime	71	1055	264	90.764	83.957	85.467	2.05
semantic	manet_resnet34	P1-prime	101	1053	264	90.221	83.230	84.212	2.07
semantic	manet_resnet34	P2	17	1056	264	89.337	81.962	83.086	2.14
semantic	manet_resnet34	P2	29	1056	264	90.185	83.160	84.409	2.08
semantic	manet_resnet34	P2	43	1056	264	90.170	83.032	84.460	2.01
semantic	manet_resnet34	P2	71	1055	265	90.458	83.608	84.919	2.09
semantic	manet_resnet34	P2	101	1053	267	89.687	82.410	83.599	2.00
semantic	manet_resnet34	P3	17	891	429	89.762	82.615	83.748	2.75
semantic	manet_resnet34	P3	29	825	495	89.392	82.029	83.305	2.62
semantic	manet_resnet34	P3	43	894	426	89.895	82.761	83.970	2.16
semantic	manet_resnet34	P3	71	773	547	89.469	82.176	83.199	3.03
semantic	manet_resnet34	P3	101	791	529	89.625	82.349	83.609	1.96
semantic	unet_resnet34	OFFICIAL	17	400	100	85.332	76.145	75.734	1.03
semantic	unet_resnet34	OFFICIAL	29	400	100	85.420	76.382	75.839	1.11
semantic	unet_resnet34	OFFICIAL	43	400	100	85.102	75.810	75.351	1.03
semantic	unet_resnet34	OFFICIAL	71	400	100	86.580	77.740	78.197	1.13
semantic	unet_resnet34	OFFICIAL	101	400	100	85.749	76.664	76.744	1.06
semantic	unet_resnet34	P1	17	1056	264	90.196	83.235	84.389	2.09
semantic	unet_resnet34	P1	29	1056	264	89.654	82.414	83.452	2.16
semantic	unet_resnet34	P1	43	1056	264	90.616	83.797	85.081	2.01
semantic	unet_resnet34	P1	71	1056	264	90.880	84.082	85.657	2.09
semantic	unet_resnet34	P1	101	1056	264	90.093	82.967	84.100	2.04
semantic	unet_resnet34	P1-prime	17	1056	264	90.196	83.235	84.389	2.12
semantic	unet_resnet34	P1-prime	29	1056	264	89.654	82.414	83.452	2.20

Task	Architecture	Protocol	Seed	Train	Test	Primary (percent)	IoU (percent)	Nuc./R70 (percent)	hours
semantic	unet_resnet34	P1-prime	43	1056	264	90.616	83.797	85.081	1.98
semantic	unet_resnet34	P1-prime	71	1055	264	91.020	84.317	85.849	2.06
semantic	unet_resnet34	P1-prime	101	1053	264	90.092	83.064	84.113	1.95
semantic	unet_resnet34	P2	17	1056	264	89.429	82.217	83.000	2.10
semantic	unet_resnet34	P2	29	1056	264	90.249	83.272	84.602	2.17
semantic	unet_resnet34	P2	43	1056	264	90.674	83.773	85.275	2.03
semantic	unet_resnet34	P2	71	1055	265	89.519	82.235	83.280	2.16
semantic	unet_resnet34	P2	101	1053	267	90.082	82.951	84.446	2.01
semantic	unet_resnet34	P3	17	891	429	89.911	82.788	83.948	2.23
semantic	unet_resnet34	P3	29	825	495	89.629	82.392	83.516	2.85
semantic	unet_resnet34	P3	43	894	426	89.892	82.748	83.984	2.82
semantic	unet_resnet34	P3	71	773	547	89.840	82.694	83.704	3.12
semantic	unet_resnet34	P3	101	791	529	89.576	82.274	83.557	2.96
semantic	unetplusplus_resnet34	OFFICIAL	17	400	100	85.933	77.036	76.615	1.06
semantic	unetplusplus_resnet34	OFFICIAL	29	400	100	86.864	78.288	78.526	1.15
semantic	unetplusplus_resnet34	OFFICIAL	43	400	100	86.771	78.140	78.396	1.11
semantic	unetplusplus_resnet34	OFFICIAL	71	400	100	88.276	79.991	81.686	1.08
semantic	unetplusplus_resnet34	OFFICIAL	101	400	100	88.645	80.557	82.123	1.08
semantic	unetplusplus_resnet34	P1	17	1056	264	90.270	83.391	84.440	1.97
semantic	unetplusplus_resnet34	P1	29	1056	264	89.890	82.832	83.858	1.88
semantic	unetplusplus_resnet34	P1	43	1056	264	90.909	84.191	85.525	2.05
semantic	unetplusplus_resnet34	P1	71	1056	264	91.190	84.587	86.038	1.89
semantic	unetplusplus_resnet34	P1	101	1056	264	90.564	83.715	84.868	2.26
semantic	unetplusplus_resnet34	P1-prime	17	1056	264	90.270	83.391	84.440	1.97
semantic	unetplusplus_resnet34	P1-prime	29	1056	264	89.890	82.832	83.858	1.85
semantic	unetplusplus_resnet34	P1-prime	43	1056	264	90.909	84.191	85.525	2.03
semantic	unetplusplus_resnet34	P1-prime	71	1055	264	91.266	84.711	86.229	1.85
semantic	unetplusplus_resnet34	P1-prime	101	1053	264	90.576	83.692	84.939	2.27
semantic	unetplusplus_resnet34	P2	17	1056	264	90.077	83.064	84.058	1.87
semantic	unetplusplus_resnet34	P2	29	1056	264	90.324	83.431	84.639	1.91
semantic	unetplusplus_resnet34	P2	43	1056	264	90.205	83.068	84.221	2.03
semantic	unetplusplus_resnet34	P2	71	1055	265	90.977	84.343	85.735	1.81
semantic	unetplusplus_resnet34	P2	101	1053	267	90.300	83.261	84.697	2.02
semantic	unetplusplus_resnet34	P3	17	891	429	90.011	82.987	84.055	2.10
semantic	unetplusplus_resnet34	P3	29	825	495	89.719	82.531	83.578	2.58
semantic	unetplusplus_resnet34	P3	43	894	426	90.126	83.176	84.281	2.19
semantic	unetplusplus_resnet34	P3	71	773	547	90.283	83.365	84.578	2.56
semantic	unetplusplus_resnet34	P3	101	791	529	90.099	83.076	84.302	2.83

D

Owner data files and integrity hashes

This table is generated from `data_sources.json`. A reproduction refuses a file whose byte length or SHA-256 differs, even if its archive name is unchanged.

Table D.1: Declared owner-source files, exact byte sizes, checksums, and use notes.

Dataset	File	Bytes	SHA-256	Terms note
sipakmed	im_Superficial-Intermediate.7z	762,939,737	3b73e406df14d60b1bd9c8ac3b01b594e70bb51ba32fb080141fc6a7f489a599	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Parabasal.7z	548,667,699	13e9c29f016a5b83731e038cfe518671a7815df28836dee706acba6ecd2369a8	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Koilocytotic.7z	1,290,478,065	653578e878e3cc63a8510cd7ec26d918637d9881e913e29a5bb205d0d9cfb70d	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Metaplastic.7z	1,396,480,123	97f1badc5f606f127d2af0d18a008ebd41a2b60f44785cb332a6eaa33004d2af	Public for experimental use; cite Plissiti et al., ICIP 2018
sipakmed	im_Dyskeratotic.7z	1,182,980,549	a8768bc03c9e814063ff946b7ef7548067e16b760168267d2b12e95a66955cce	Public for experimental use; cite Plissiti et al., ICIP 2018
cx22	Cx22-Multi-Test.zip	1,780,461	6d444e6e4bf8734073bc90e6e2bb29b0e36c66f7b783cbad9f969208621624bb	Non-commercial research and education only
cx22	Cx22-Multi-Train.zip	6,361,959	171be9caf6d0709f3fce2b81a07e1974229c5faae355f197efe553ec1d6d1992	Non-commercial research and education only
cx22	Cx22-Pair.zip	3,705,743	130b6a92db9811151a60e27dc70f4180965ecf0f1dbc84146c87d00c41d7ba9f	Non-commercial research and education only
ccedd	CCEDD.zip	1,507,832,267	d85e6a380a15923f0c2b38e13dd31467f75d11b49e8d21740f16ee69082cdcb1	Non-commercial research and education only; redistribution prohibited
cric	classifications.csv	1,224,400	e1f50183998ddb7392d2a0d4c96dce44c00dbe3833df03f266cc36e047445de5	CC BY 4.0; cite Rezende et al. and the Figshare record
herlev	Cervical Cell Image Database.zip	44,688,829	5b5b740abb599710b9d97fe72954540df9ca7255b2b40a52c8b25ac90d0c50da	CC BY 4.0; repackaging of the original Herlev cervical-cell images

E

Released split hashes

Each CSV contains crop ID, parent ID, partition, protocol, and seed. Hashes identify the exact semantic assignment rather than merely naming a random seed.

Table E.1: Released SIPaKMeD split manifests and hashes.

Protocol	Seed	File	SHA-256
P1	17	sipakmed_p1_seed17.csv	28abd1605d72d888f561cecbc4d987b0dafff671653c37d0460bccac8d3c0cac
P1-prime	17	sipakmed_p1_prime_seed17.csv	611fcb7e6ed98e66a86f9cbdbcb27644850e15d93a4143249dc611d00364d784
P2	17	sipakmed_p2_seed17.csv	488077976eb9397b2a772597e4453e2a31e809d67cb1a21595ece5b143b35262
P3	17	sipakmed_p3_seed17.csv	797a4c0d73fd0722aebec95a14cf798ac3cf94575ac7e93812a85b738df0ea14
P1	29	sipakmed_p1_seed29.csv	adef646d2edabf02bf92a00a7a8e68eefee876159ee2d43d1ab4c25fc80e1e75
P1-prime	29	sipakmed_p1_prime_seed29.csv	72f36aa2713e5dbd4a7d16f471ec466f88dacefbf597d5831bf3b409cc9cb256
P2	29	sipakmed_p2_seed29.csv	4088e6c4b03fca9e16ecb7db57b221e1769e09a5bf6f9c2f36b52519332aed21
P3	29	sipakmed_p3_seed29.csv	3722ec8802c1bbfdf4fe7d9aad2ecd99be4ca66da83add1e331f6d816ed936cc
P1	43	sipakmed_p1_seed43.csv	d5087a1636cceebd8943accfbeb2129046772df01276a693b202f02340846beb
P1-prime	43	sipakmed_p1_prime_seed43.csv	a7417cb3eeccbc4629bd212afffc2c3f42045e9d864f10b232b2c1f15e3a4b1b
P2	43	sipakmed_p2_seed43.csv	bd1a079eadacd51adbeb0dc169d0bcca5cf3b011d3d6b71bb98e4fa71988a62f
P3	43	sipakmed_p3_seed43.csv	877f62465214f160aaa2776bd4ba92c2054c78d744261ae9ada9a4ce66343de8
P1	71	sipakmed_p1_seed71.csv	5148ddbc11d90a927f6cf7d9bc5213d0b1431b9d80cb58fa0d870de76c889198
P1-prime	71	sipakmed_p1_prime_seed71.csv	21ce3cab719badee5dad0bb192c2e55e695fe44d35d58c0a9c6921e4aa7c5b3a
P2	71	sipakmed_p2_seed71.csv	52095a0a39a27f439d64646ed50b07f15bfc729dee195a42f0dcb4c4739d0e25
P3	71	sipakmed_p3_seed71.csv	bbf975f39f790159dcc903bb3d53eb87968e6d612066bfe0cd5c1e8fb6cc860e
P1	101	sipakmed_p1_seed101.csv	1dadec3ecdbce04080b796d735ca7ecfe84cc4b74342d8480326b7c95dd9ff44
P1-prime	101	sipakmed_p1_prime_seed101.csv	d9e096b00642b40eb979f3d75e827f6929e15d0c62a3a9a3738eb4490526185f
P2	101	sipakmed_p2_seed101.csv	178418b059a5c924b305ca5a05472b2076f3423626538e27416ad39f6e1f891f
P3	101	sipakmed_p3_seed101.csv	269ac8e1badbad7c11a5a45cd72b2c2d17ac0fa2c09cb0fa007b45ce51bbef90

F

Git decision chronology

The chronology below is descriptive provenance for major decisions and corrections. The authoritative ordering remains the Git object database. In particular, the first entry is the original preregistration commit.

Table F.1: Complete repository decision chronology through the frozen result set.

Commit	Date	Recorded change
67e8ded	2026-08-22	preregister confirmatory analysis
8ebc6d5	2026-08-22	build leakage verification pipeline
12c9de7	2026-08-22	add parent identity probe and complete source checksums
4ba6a96	2026-08-22	make image provenance recovery independent
1dbd6ac	2026-08-22	separate provenance recovery stages
c551f2d	2026-08-22	add matched classification harness
14a1569	2026-08-22	add near duplicate audit and P3 grouping
79b798a	2026-08-22	amend SIPaKMeD class order before analysis
70b5ffa	2026-08-22	add bounded GPU probe launcher
4aa3e60	2026-08-22	fail closed on image provenance confidence
c18d393	2026-08-22	validate Cx22 against declared source images
f7ca331	2026-08-22	add bounded matched evaluation launcher
318120f	2026-08-22	summarize matched runs in one correction family
8f6f76e	2026-08-22	complete preregistered probe ablations
680fa53	2026-08-22	record reproducible model run metadata
0c5166e	2026-08-22	checkpoint exhaustive provenance matching
ea053e9	2026-08-22	verify structural protocol ordering
65b0230	2026-08-22	export fixed protocol split assets
3261dc0	2026-08-22	render fail-closed verification contract
79e5b31	2026-08-22	require named provenance evidence in report
cded044	2026-08-22	measure literal cross-split pixel overlap
55708d4	2026-08-22	keep training commands free of analysis-only imports
6369698	2026-08-22	generate self-contained thesis figure
da548a4	2026-08-22	bound clean reproduction matcher to three workers
83d56af	2026-08-22	verify published class crop counts
03e7442	2026-08-22	document fail-closed release workflow
be46a53	2026-08-22	separate canonical and audit provenance maps
c617374	2026-08-22	bound provenance image cache to class working set
4c6282e	2026-08-22	use multiscale provenance matching only as fallback
67892b7	2026-08-22	align provenance validation with registered criteria
268e56a	2026-08-22	build explicit section 11 acceptance ledger
438fa3f	2026-08-22	measure duplicate detector precision and recall
62a51ef	2026-08-22	record virgin-host reproduction evidence
dd0a56e	2026-08-22	generate five-seed parent probe figure
b90805a	2026-08-22	reproduce P3 splits and pixel overlap end to end
82e541d	2026-08-22	add matched protocol results figure
adf5539	2026-08-22	document full matched classifier matrix
bf8a4ec	2026-08-22	enforce paper table reporting contract
ab40681	2026-08-22	separate Figure 2 control labels
a72fb5f	2026-08-22	record split and image library versions
4da9f62	2026-08-22	pin CUDA 12.8 training environment
f936802	2026-08-22	verify classification matrix integrity
157b882	2026-08-22	report secondary ordering violations

Commit	Date	Recorded change
10fb2a3	2026-08-22	pin reproducible split runtime
970c74e	2026-08-22	gate acceptance on matrix integrity
8ebb2b9	2026-08-22	simulate exact P1 allocation rule
525df1d	2026-08-22	report architecture ranking stability
1c2da7f	2026-08-22	record preregistered provenance and five-seed results
2c5d5c3	2026-08-22	record post-result acceptance evidence
d347b07	2026-08-24	preregister Cx22 segmentation rescore
a58f72f	2026-08-24	add resumable Cx22 segmentation extension
635e660	2026-08-24	decode Cx22 instance labels with bounded memory
2fdb136	2026-08-24	refresh Cx22 controller heartbeat during epochs
d66dba4	2026-08-24	add memory-mapped Cx22 training cache
a18ddef	2026-08-24	preregister mixed-precision Cx22 execution
f10c9c5	2026-08-24	enable preregistered mixed precision for Cx22
48c9b65	2026-08-28	stabilize Cx22 MAnet training in FP32
373276a	2026-08-28	enforce deterministic CUDA workspace for Cx22
a9594b3	2026-09-06	remove redundant Cx22 instance runtime overhead
01c4c31	2026-09-06	stage Cx22 checkpoints off network storage

G

Exact command cookbook

G.1 Local checks

```
uv sync --extra dev
uv run pytest
uv run ruff check .
uv run cytofair --help
```

G.2 Recover and verify structural evidence

```
uv run cytofair recover-sipakmed data/extracted/sipakmed --workers 3
uv run cytofair recover-cx22 data/extracted/cx22 \
  --image-root data/extracted/ccedd/png
uv run cytofair recover-cric data/raw/cric/classifications.csv
uv run cytofair recover-herlev data/extracted/herlev

uv run cytofair audit-duplicates \
  provenance/sipakmed_parent_map.parquet \
  --hashes results/duplicates/sipakmed_hashes.parquet \
  --pairs results/duplicates/sipakmed_near_duplicate_pairs.csv \
  --output provenance/sipakmed_parent_map_p3.parquet

uv run cytofair make-splits \
  provenance/sipakmed_parent_map_p3.parquet \
  --dataset SIPaKMeD --output splits/generated

uv run cytofair pixel-overlap \
  provenance/sipakmed_branch_b.parquet --dataset SIPaKMeD
```

G.3 Launch the probe and classifier matrices

```
python scripts/launch_probe_matrix.py \
  provenance/sipakmed_parent_map_p3.parquet results/probe \
  --max-gpus 3 \
  --modes rgb grayscale stain_normalized background_only highpass

```

```
python scripts/launch_classification_matrix.py \
  --dataset-map SIPaKMeD=provenance/sipakmed_parent_map_p3.parquet \
  --output results/classification --max-gpus 3 \
  --architectures resnet18 resnet50 densenet121 \
    efficientnet_b0 mobilenet_v2 vit_b_16 \
  --protocols P1 P1-prime P2 P3
```

G.4 Materialize and launch Cx22

```
cytofair materialize-cx22 provenance/cx22_parent_map.parquet \
  --labels /path/to/extracted/cx22 \
  --assets /path/to/cx22_assets \
  --output provenance/cx22_model_map.parquet

cytofair audit-duplicates provenance/cx22_model_map.parquet \
  --hashes results/cx22_duplicates/hashes.parquet \
  --pairs results/cx22_duplicates/pairs.csv \
  --output provenance/cx22_model_map_p3.parquet

cytofair cache-cx22 provenance/cx22_model_map_p3.parquet \
  --cache /path/to/cx22_assets/cache256 \
  --output provenance/cx22_cached_model_map_p3.parquet

nohup python scripts/launch_cx22_segmentation_matrix.py \
  --map provenance/cx22_cached_model_map_p3.parquet \
  --output results/cx22_segmentation --max-gpus 3 \
  > results/cx22_segmentation/controller.log 2>&1 < /dev/null &

python scripts/check_cx22_segmentation.py results/cx22_segmentation
```

H

Artifact index and evidence status

Table H.1: Principal repository artifacts.

Artifact	Status	Purpose
`PREREGISTRATION.md`	Frozen plan	Endpoints, seeds, protocols, exclusions, corrections, amendments.
`data_sources.json`	Versioned	Owner URLs, filenames, sizes, hashes, and terms notes.
`provenance/*.parquet`	Complete derived maps	Observation-to-parent identities, match diagnostics, boxes, hashes, and P3 components.
`splits/*.csv`	Versioned interface	Exact per-seed assignments joined by crop ID.
`results/classification/*.json`	120 accepted	Classifier metrics, environment, split hash, and history.
`results/probe/*.json`	30 accepted	Mechanism probe metrics, chance, environment, and history.
`results/cx22_segmentation/*.json`	175 accepted	Semantic/instance metrics, manifests, runtime, and history.
`results/attempts/`	Excluded, retained	Twelve incompatible split-library runs and three smoke tests.
`reports/generated/acceptance_contract.*`	Current	Machine-readable PASS/FAIL/PENDING gates.
`scripts/reproduce_table1.sh`	Implemented	GPU-free raw-download-to-structural-table workflow.
`scripts/verify_clean_clone.sh`	Implemented, unattested	Non-local clone wrapper and optional virgin-host attestation.
`reports/lab_paper_self_audit.md`	Pending	Nine-record title/protocol crosswalk requiring source-paper checks.
`reports/figure1_reader_test.md`	Pending	Independent reader protocol and blank response record.

I

Complete derived-result ledgers

These tables expose every structural seed, literal-overlap seed/protocol, registered classifier contrast, registered Cx22 contrast, and rank-stability result. The model run ledgers in Appendices A–C are the underlying run-level evidence; this appendix is the complete derived statistical layer.

I.1 Structural and literal-overlap seeds

Table I.1: Complete structural seed ledger.

Dataset	Seed	P1 parent straddle (percent)	P1 test parent in train (percent)	P2 straddle (percent)	P3 straddle (percent)
SIPaKMeD	17	45.549	92.222	0.000	0.000
SIPaKMeD	29	45.445	92.840	0.000	0.000
SIPaKMeD	43	44.203	92.469	0.000	0.000
SIPaKMeD	71	47.101	93.210	0.000	0.000
SIPaKMeD	101	45.859	92.346	0.000	0.000
Cx22	17	34.241	85.985	0.000	0.000
Cx22	29	32.685	80.682	0.000	0.000
Cx22	43	35.214	89.773	0.000	0.000
Cx22	71	34.825	86.742	0.000	0.000
Cx22	101	33.074	82.955	0.000	0.000
CRIC	17	94.750	99.827	0.000	0.000
CRIC	29	95.000	100.000	0.000	0.000
CRIC	43	94.250	100.000	0.000	0.000
CRIC	71	92.500	100.000	0.000	0.000
CRIC	101	95.250	100.000	0.000	0.000
Herlev	17	0.000	0.000	0.000	0.000
Herlev	29	0.000	0.000	0.000	0.000
Herlev	43	0.000	0.000	0.000	0.000
Herlev	71	0.000	0.000	0.000	0.000
Herlev	101	0.000	0.000	0.000	0.000

Table I.2: Complete SIPaKMeD literal source-pixel overlap ledger.

Protocol	Seed	Cross-split pairs	Overlapping pairs	Pair overlap (percent)	Test crops overlapping train	Test-crop overlap (percent)
P1	17	4294	673	15.673	411	50.741

Protocol	Seed	Cross-split pairs	Overlapping pairs	Pair overlap (percent)	Test crops overlapping train	Test-crop overlap (percent)
P2	17	0	0	0.000	0	0.000
P3	17	0	0	0.000	0	0.000
P1	29	4153	695	16.735	415	51.235
P2	29	0	0	0.000	0	0.000
P3	29	0	0	0.000	0	0.000
P1	43	4222	749	17.740	443	54.691
P2	43	0	0	0.000	0	0.000
P3	43	0	0	0.000	0	0.000
P1	71	4240	705	16.627	418	51.605
P2	71	0	0	0.000	0	0.000
P3	71	0	0	0.000	0	0.000
P1	101	4199	724	17.242	444	54.815
P2	101	0	0	0.000	0	0.000
P3	101	0	0	0.000	0	0.000

I.2 Classifier contrasts

Table I.3: Complete SIPaKMeD classifier contrast ledger. Secondary tests have no confirmatory BH q-value.

Family	Architecture	Left	Right	n	Mean diff. (pp)	SD (pp)	Test	p	BH q	Effect	Value	Effect 95 percent CI	Shapiro p
primary	densenet121	P1	P2	5	1.407	0.901	paired_t	0.0251	0.1109	cohens_dz	1.561	[0.985, 4.747]	0.9922
primary	efficientnet_b0	P1	P2	5	1.086	1.849	paired_t	0.2591	0.2591	cohens_dz	0.588	[-0.316, 4.668]	0.6215
primary	mobilenet_v2	P1	P2	5	2.247	1.632	paired_t	0.0370	0.1109	cohens_dz	1.377	[0.809, 6.686]	0.3183
primary	resnet18	P1	P2	5	1.185	1.285	paired_t	0.1081	0.1511	cohens_dz	0.922	[0.442, 2.365]	0.6256
primary	resnet50	P1	P2	5	0.716	0.830	paired_t	0.1260	0.1511	cohens_dz	0.863	[0.085, 4.025]	0.4504
primary	vit_b_16	P1	P2	5	1.580	1.698	paired_t	0.1059	0.1511	cohens_dz	0.931	[0.057, 4.950]	0.6216
primary	densenet121	P1-prime	P2	5	1.407	0.901	paired_t	0.0251	0.1109	cohens_dz	1.561	[0.932, 4.747]	0.9922
primary	efficientnet_b0	P1-prime	P2	5	1.086	1.849	paired_t	0.2591	0.2591	cohens_dz	0.588	[-0.316, 4.610]	0.6215
primary	mobilenet_v2	P1-prime	P2	5	2.247	1.632	paired_t	0.0370	0.1109	cohens_dz	1.377	[0.809, 6.686]	0.3183
primary	resnet18	P1-prime	P2	5	1.185	1.285	paired_t	0.1081	0.1511	cohens_dz	0.922	[0.426, 2.442]	0.6256
primary	resnet50	P1-prime	P2	5	0.716	0.830	paired_t	0.1260	0.1511	cohens_dz	0.863	[0.085, 4.025]	0.4504
primary	vit_b_16	P1-prime	P2	5	1.580	1.698	paired_t	0.1059	0.1511	cohens_dz	0.931	[0.120, 4.950]	0.6216
secondary	densenet121	P2	P3	5	-0.667	0.691	paired_t	0.0971	–	cohens_dz	-0.965	[-3.266, -0.725]	0.0863
secondary	efficientnet_b0	P2	P3	5	-0.716	1.943	paired_t	0.4562	–	cohens_dz	-0.369	[-2.708, 0.467]	0.6101
secondary	mobilenet_v2	P2	P3	5	-1.037	1.376	paired_t	0.1673	–	cohens_dz	-0.753	[-3.329, 0.075]	0.4329
secondary	resnet18	P2	P3	5	-0.346	1.189	paired_t	0.5510	–	cohens_dz	-0.291	[-1.516, 1.255]	0.7941
secondary	resnet50	P2	P3	5	0.519	0.661	paired_t	0.1545	–	cohens_dz	0.784	[0.361, 2.534]	0.2922
secondary	vit_b_16	P2	P3	5	0.025	2.345	paired_t	0.9823	–	cohens_dz	0.011	[-3.338, 0.817]	0.2169

I.3 Cx22 contrasts

Table I.4: Complete Cx22 contrast ledger. Semantic and instance primary families were corrected separately by task.

Family	Architecture	Left	Right	n	Mean diff. (pp)	SD (pp)	Test	p	BH q	Effect	Value	Effect 95 percent CI	Shapiro p
primary	deeplabv3plus_resnet50	P1	P2	5	0.258	0.418	wilcoxon	0.3125	0.8458	rank_biserial	0.600	[-0.200, 1.000]	0.0338
primary	fpn_resnet34	P1	P2	5	0.010	0.468	paired_t	0.9625	0.9625	cohens_dz	0.022	[-1.157, 1.799]	0.9413
primary	manet_resnet34	P1	P2	5	0.065	0.450	paired_t	0.7612	0.8458	cohens_dz	0.146	[-0.653, 3.986]	0.2308
primary	unet_resnet34	P1	P2	5	0.297	0.767	paired_t	0.4353	0.8458	cohens_dz	0.387	[-0.840, 1.537]	0.7640
primary	unetplusplus_resnet34	P1	P2	5	0.188	0.406	paired_t	0.3582	0.8458	cohens_dz	0.464	[-0.447, 7.482]	0.4491
primary	deeplabv3plus_resnet50	P1-prime	P2	5	0.193	0.382	wilcoxon	0.6250	0.8458	rank_biserial	0.333	[-0.600, 1.000]	0.0411
primary	fpn_resnet34	P1-prime	P2	5	0.085	0.549	paired_t	0.7474	0.8458	cohens_dz	0.154	[-1.157, 2.035]	0.5968
primary	manet_resnet34	P1-prime	P2	5	0.102	0.479	paired_t	0.6583	0.8458	cohens_dz	0.213	[-0.694, 4.502]	0.2960
primary	unet_resnet34	P1-prime	P2	5	0.325	0.817	paired_t	0.4240	0.8458	cohens_dz	0.398	[-0.842, 1.506]	0.7316
primary	unetplusplus_resnet34	P1-prime	P2	5	0.206	0.409	paired_t	0.3233	0.8458	cohens_dz	0.503	[-0.375, 6.594]	0.4512
primary	maskrcnn_resnet50_fpn	P1	P2	5	0.509	0.581	paired_t	0.1220	0.2440	cohens_dz	0.875	[0.516, 2.175]	0.2370
primary	maskrcnn_resnet50_fpn_v2	P1	P2	5	0.630	1.204	paired_t	0.3068	0.3116	cohens_dz	0.523	[-0.568, 2.360]	0.7974
primary	maskrcnn_resnet50_fpn	P1-prime	P2	5	0.466	0.340	paired_t	0.0377	0.1508	cohens_dz	1.368	[0.769, 4.031]	0.7993
primary	maskrcnn_resnet50_fpn_v2	P1-prime	P2	5	0.586	1.133	paired_t	0.3116	0.3116	cohens_dz	0.518	[-0.568, 2.651]	0.4050
secondary	deeplabv3plus_resnet50	P2	P3	5	0.546	0.534	paired_t	0.0843	–	cohens_dz	1.022	[0.461, 3.165]	0.6257
secondary	fpn_resnet34	P2	P3	5	0.558	0.346	paired_t	0.0226	–	cohens_dz	1.613	[0.738, 7.765]	0.6155
secondary	manet_resnet34	P2	P3	5	0.339	0.569	paired_t	0.2537	–	cohens_dz	0.596	[-0.284, 2.625]	0.8303
secondary	unet_resnet34	P2	P3	5	0.221	0.579	paired_t	0.4421	–	cohens_dz	0.381	[-0.533, 6.386]	0.1888
secondary	unetplusplus_resnet34	P2	P3	5	0.329	0.299	paired_t	0.0696	–	cohens_dz	1.101	[0.722, 2.785]	0.1372
secondary	maskrcnn_resnet50_fpn	P2	P3	5	0.473	1.435	paired_t	0.5021	–	cohens_dz	0.330	[-1.325, 1.664]	0.2311
secondary	maskrcnn_resnet50_fpn_v2	P2	P3	5	0.761	1.006	paired_t	0.1661	–	cohens_dz	0.756	[-0.093, 3.627]	0.1440

I.4 Rank stability

Table I.5: Complete secondary P1-versus-P2 rank-stability ledger.

Task	Arch.	Seeds	Kendall tau	95 percent CI	p	Reordered pairs	Tie changes
classification	6	5	0.467	[0.071, 0.733]	0.2722	4/15	0
semantic segmentation	5	5	0.800	[0.600, 1.000]	0.0833	1/10	0
instance segmentation	2	5	1.000	[-1.000, 1.000]	1.0000	0/1	0

J

Excluded attempt ledger

No row below was excluded for its score or direction. The twelve twenty-epoch rows belong to the incompatible scikit-learn 1.7.2 split matrix; all were excluded when the entire matrix restarted. The three one-epoch rows are execution smoke tests that were never eligible for confirmatory summaries.

Table J.1: Complete excluded-attempt ledger. Metrics are shown only to demonstrate that exclusion was rule-based, not score-based.

Reason	Architecture/mode	Protocol	Seed	Epochs	Train	Test	Recorded metric (percent)	Eligible
split-library failure	resnet18	P1-prime	101	20	3223	810	98.395	No
split-library failure	resnet18	P1-prime	17	20	3214	810	96.914	No
split-library failure	resnet18	P1-prime	29	20	3237	810	97.901	No
split-library failure	resnet18	P1-prime	43	20	3215	810	97.284	No
split-library failure	resnet18	P1-prime	71	20	3236	810	98.148	No
split-library failure	resnet18	P1	101	20	3239	810	97.654	No
split-library failure	resnet18	P1	17	20	3239	810	97.901	No
split-library failure	resnet18	P1	29	20	3239	810	97.037	No
split-library failure	resnet18	P1	43	20	3239	810	96.914	No
split-library failure	resnet18	P1	71	20	3239	810	96.420	No
split-library failure	resnet18	P2	17	20	3214	835	96.647	No
split-library failure	resnet18	P2	29	20	3237	812	95.936	No
smoke test	resnet18/rgb	–	17	1	848	848	0.590	No
smoke test	resnet18	P1	17	1	3239	810	95.556	No
smoke test	vit_b_16	P1	17	1	3239	810	95.679	No

K

Reading guide for future maintainers

Start with the acceptance contract, not the paper prose. If a gate fails, follow its evidence path into the structured output. Regenerate a summary rather than editing a number. If a library upgrade changes a split, do not overwrite the published CSV: create a new version and preserve the previous hash. If new data supply patient IDs, introduce a stronger protocol rather than relabeling source-disjoint as patient-disjoint. If a model is rerun, retain its per-sample predictions and checkpoint so confusion matrices and qualitative panels become reproducible first-class artifacts.

The simplest invariant is also the most important: an observation may be the tensor consumed by a network, but it is not automatically the independent unit represented by the test set.

L

Study 2 run ledgers

Every registered Study 2 run, with its metrics in percent, wall time in minutes, and the GPU it ran on.

Table L.1: Complete Phase 1 run ledger (720 runs; metrics in percent; minutes of wall time; GPU index).

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-18	137	P1	97.41	97.42	97.43	1.36	0.9	3
ResNet-18	137	P2	96.05	96.08	96.05	2.29	0.9	4
ResNet-18	137	P3	95.80	95.84	95.80	2.53	0.9	5
ResNet-18	173	P1	97.41	97.39	97.45	1.66	0.9	3
ResNet-18	173	P2	96.05	96.08	96.07	2.06	0.9	0
ResNet-18	173	P3	96.42	96.41	96.45	1.86	0.9	3
ResNet-18	211	P1	96.91	96.91	96.92	1.61	0.9	4
ResNet-18	211	P2	95.80	95.83	95.79	2.33	0.9	0
ResNet-18	211	P3	94.69	94.73	94.69	3.21	0.9	4
ResNet-18	257	P1	96.67	96.67	96.70	2.53	0.9	3
ResNet-18	257	P2	96.54	96.56	96.54	2.11	0.9	5
ResNet-18	257	P3	97.04	97.05	97.05	1.65	0.9	3
ResNet-18	307	P1	98.64	98.64	98.65	0.65	0.9	0
ResNet-18	307	P2	96.67	96.67	96.64	1.50	0.9	5
ResNet-18	307	P3	97.78	97.80	97.78	1.28	0.9	0
ResNet-18	349	P1	96.54	96.55	96.55	1.95	0.9	5
ResNet-18	349	P2	96.42	96.46	96.42	1.74	0.9	0
ResNet-18	349	P3	98.27	98.29	98.29	0.97	0.9	5
ResNet-18	401	P1	97.28	97.28	97.32	1.47	0.9	4
ResNet-18	401	P2	98.15	98.17	98.16	1.12	0.9	3
ResNet-18	401	P3	97.53	97.55	97.52	1.08	0.9	4
ResNet-18	457	P1	98.02	98.03	98.05	0.98	0.9	3
ResNet-18	457	P2	95.68	95.65	95.69	2.81	0.9	0
ResNet-18	457	P3	97.65	97.66	97.67	1.36	0.9	3
ResNet-18	503	P1	96.67	96.70	96.69	2.06	0.9	4
ResNet-18	503	P2	96.54	96.54	96.58	2.56	0.9	0
ResNet-18	503	P3	96.54	96.54	96.55	2.35	0.9	4
ResNet-18	557	P1	97.65	97.67	97.67	1.29	0.9	0
ResNet-18	557	P2	97.53	97.53	97.52	1.55	0.9	5
ResNet-18	557	P3	97.41	97.41	97.42	1.51	0.9	0
ResNet-18	601	P1	97.78	97.78	97.80	1.13	0.9	3
ResNet-18	601	P2	96.17	96.19	96.19	1.83	0.9	0
ResNet-18	601	P3	96.91	96.95	96.92	1.58	0.9	3
ResNet-18	653	P1	96.79	96.81	96.81	1.68	0.9	0
ResNet-18	653	P2	97.28	97.30	97.31	1.86	0.9	3
ResNet-18	653	P3	96.42	96.36	96.38	1.96	0.9	0
ResNet-18	701	P1	97.53	97.53	97.54	1.99	0.9	3
ResNet-18	701	P2	95.68	95.71	95.70	2.77	0.9	5
ResNet-18	701	P3	93.33	93.38	93.35	4.43	0.9	3
ResNet-18	751	P1	97.53	97.53	97.56	1.92	0.9	5
ResNet-18	751	P2	96.05	96.05	96.01	2.75	0.9	3
ResNet-18	751	P3	96.05	96.06	96.04	1.94	0.9	4
ResNet-18	809	P1	98.15	98.14	98.18	1.87	0.9	4
ResNet-18	809	P2	96.17	96.17	96.17	2.79	0.9	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-18	809	P3	96.30	96.28	96.34	2.72	0.9	3
ResNet-18	853	P1	98.02	98.02	98.04	1.10	0.9	5
ResNet-18	853	P2	97.16	97.16	97.17	1.73	0.9	4
ResNet-18	853	P3	96.17	96.16	96.17	2.50	0.9	5
ResNet-18	907	P1	97.78	97.79	97.79	1.23	0.9	0
ResNet-18	907	P2	96.67	96.65	96.66	1.95	0.9	3
ResNet-18	907	P3	98.02	98.04	98.01	1.14	0.9	0
ResNet-18	953	P1	97.16	97.17	97.17	1.25	0.9	3
ResNet-18	953	P2	96.67	96.70	96.65	1.72	0.9	0
ResNet-18	953	P3	98.15	98.16	98.14	1.29	0.9	3
ResNet-18	1009	P1	98.02	98.04	98.03	1.36	0.9	4
ResNet-18	1009	P2	96.91	96.91	96.92	2.03	0.9	0
ResNet-18	1009	P3	97.65	97.65	97.62	1.00	0.9	4
ResNet-18	1051	P1	97.41	97.42	97.41	1.38	0.9	5
ResNet-18	1051	P2	97.53	97.53	97.54	1.06	0.9	0
ResNet-18	1051	P3	97.28	97.28	97.30	1.42	0.9	5
ResNet-50	137	P1	97.65	97.66	97.68	1.05	2.6	0
ResNet-50	137	P2	95.19	95.24	95.18	2.87	2.6	3
ResNet-50	137	P3	97.28	97.30	97.31	1.62	2.6	4
ResNet-50	173	P1	97.78	97.80	97.79	1.03	2.6	5
ResNet-50	173	P2	95.93	95.97	95.96	2.05	2.6	4
ResNet-50	173	P3	97.28	97.32	97.29	1.46	2.6	0
ResNet-50	211	P1	97.41	97.42	97.42	1.47	2.6	0
ResNet-50	211	P2	96.30	96.32	96.27	2.20	2.6	3
ResNet-50	211	P3	95.31	95.32	95.29	3.36	2.6	5
ResNet-50	257	P1	97.41	97.42	97.44	1.59	2.6	4
ResNet-50	257	P2	96.67	96.68	96.66	2.07	2.6	0
ResNet-50	257	P3	97.65	97.65	97.67	1.18	2.6	5
ResNet-50	307	P1	98.27	98.28	98.28	1.29	2.6	3
ResNet-50	307	P2	97.28	97.28	97.29	1.75	2.6	5
ResNet-50	307	P3	98.64	98.66	98.64	1.10	2.6	4
ResNet-50	349	P1	96.67	96.69	96.67	2.11	2.6	4
ResNet-50	349	P2	96.42	96.45	96.42	2.23	2.6	0
ResNet-50	349	P3	95.43	95.47	95.43	2.67	2.6	3
ResNet-50	401	P1	98.02	98.03	98.03	1.38	2.6	0
ResNet-50	401	P2	97.41	97.42	97.43	1.32	2.6	3
ResNet-50	401	P3	93.33	93.36	93.20	3.95	2.6	5
ResNet-50	457	P1	98.15	98.14	98.18	1.18	2.6	5
ResNet-50	457	P2	94.32	94.31	94.35	3.88	2.6	4
ResNet-50	457	P3	96.30	96.30	96.33	2.63	2.6	0
ResNet-50	503	P1	97.53	97.54	97.56	1.35	2.6	3
ResNet-50	503	P2	96.79	96.79	96.78	2.09	2.6	0
ResNet-50	503	P3	96.17	96.16	96.17	2.54	2.6	5
ResNet-50	557	P1	97.53	97.53	97.55	1.29	2.6	4
ResNet-50	557	P2	98.40	98.40	98.40	1.16	2.6	3
ResNet-50	557	P3	96.30	96.35	96.28	2.26	2.6	5
ResNet-50	601	P1	97.16	97.18	97.19	1.88	2.6	5
ResNet-50	601	P2	97.41	97.42	97.42	1.68	2.6	0
ResNet-50	601	P3	97.28	97.29	97.31	1.50	2.6	4
ResNet-50	653	P1	97.28	97.31	97.29	2.18	2.6	4
ResNet-50	653	P2	96.67	96.69	96.68	2.23	2.6	5
ResNet-50	653	P3	95.68	95.66	95.61	3.04	2.6	3
ResNet-50	701	P1	97.53	97.54	97.53	1.47	2.6	5
ResNet-50	701	P2	96.79	96.81	96.81	2.25	2.7	3
ResNet-50	701	P3	95.31	95.32	95.31	3.15	2.7	0
ResNet-50	751	P1	97.53	97.56	97.55	1.31	2.6	0
ResNet-50	751	P2	95.56	95.55	95.55	3.75	2.7	5
ResNet-50	751	P3	95.93	95.95	95.92	2.45	2.7	3
ResNet-50	809	P1	97.78	97.77	97.80	1.37	2.7	4
ResNet-50	809	P2	97.04	97.05	97.02	1.90	2.7	5
ResNet-50	809	P3	96.67	96.64	96.70	2.43	2.7	0
ResNet-50	853	P1	97.41	97.41	97.44	2.18	2.7	3
ResNet-50	853	P2	96.17	96.17	96.18	2.40	2.7	0
ResNet-50	853	P3	95.93	95.96	95.91	2.35	2.7	4
ResNet-50	907	P1	98.64	98.65	98.65	0.84	2.6	5
ResNet-50	907	P2	97.41	97.41	97.43	1.60	2.7	3
ResNet-50	907	P3	97.53	97.56	97.53	1.40	2.7	4
ResNet-50	953	P1	97.53	97.55	97.53	1.28	2.6	0
ResNet-50	953	P2	95.80	95.87	95.79	2.85	2.6	4
ResNet-50	953	P3	98.52	98.52	98.51	0.83	2.6	5
ResNet-50	1009	P1	97.16	97.18	97.16	1.72	2.6	3
ResNet-50	1009	P2	97.41	97.42	97.41	1.58	2.6	5
ResNet-50	1009	P3	97.16	97.16	97.13	2.12	2.6	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	1051	P1	98.27	98.29	98.28	0.81	2.7	0
ResNet-50	1051	P2	97.41	97.40	97.43	0.99	2.6	3
ResNet-50	1051	P3	96.30	96.28	96.28	2.12	2.6	4
DenseNet-121	137	P1	97.90	97.91	97.92	1.44	3.3	5
DenseNet-121	137	P2	95.56	95.54	95.57	2.34	3.3	0
DenseNet-121	137	P3	96.67	96.69	96.66	2.19	3.3	3
DenseNet-121	173	P1	97.78	97.79	97.79	1.43	3.3	3
DenseNet-121	173	P2	98.52	98.52	98.51	1.03	3.3	5
DenseNet-121	173	P3	98.52	98.53	98.52	0.83	3.3	4
DenseNet-121	211	P1	98.40	98.40	98.41	1.04	3.3	4
DenseNet-121	211	P2	97.16	97.18	97.15	1.83	3.3	0
DenseNet-121	211	P3	96.67	96.69	96.66	2.43	3.3	3
DenseNet-121	257	P1	97.16	97.17	97.19	1.74	3.3	3
DenseNet-121	257	P2	96.54	96.54	96.52	1.39	3.3	4
DenseNet-121	257	P3	97.78	97.78	97.78	0.90	3.3	0
DenseNet-121	307	P1	98.64	98.65	98.65	0.70	3.3	0
DenseNet-121	307	P2	97.53	97.53	97.51	1.61	3.3	3
DenseNet-121	307	P3	98.27	98.29	98.28	0.88	3.3	5
DenseNet-121	349	P1	97.53	97.53	97.53	1.17	3.3	5
DenseNet-121	349	P2	97.78	97.79	97.79	1.24	3.3	4
DenseNet-121	349	P3	98.15	98.15	98.14	1.18	3.3	0
DenseNet-121	401	P1	97.16	97.18	97.19	1.83	3.3	4
DenseNet-121	401	P2	97.28	97.29	97.30	1.36	3.3	0
DenseNet-121	401	P3	96.91	96.90	96.86	1.59	3.3	3
DenseNet-121	457	P1	97.41	97.39	97.45	1.48	3.3	3
DenseNet-121	457	P2	95.43	95.43	95.43	3.19	3.3	5
DenseNet-121	457	P3	96.30	96.30	96.32	1.75	3.3	4
DenseNet-121	503	P1	97.90	97.91	97.93	1.22	3.3	4
DenseNet-121	503	P2	97.65	97.65	97.67	1.54	3.3	3
DenseNet-121	503	P3	96.05	96.05	96.05	2.20	3.3	0
DenseNet-121	557	P1	97.41	97.43	97.42	0.92	3.3	0
DenseNet-121	557	P2	97.90	97.90	97.92	1.18	3.3	4
DenseNet-121	557	P3	97.90	97.91	97.89	0.88	3.3	3
DenseNet-121	601	P1	97.04	97.05	97.06	1.45	3.3	3
DenseNet-121	601	P2	97.16	97.17	97.18	1.39	3.3	5
DenseNet-121	601	P3	96.91	96.94	96.92	1.53	3.3	0
DenseNet-121	653	P1	96.67	96.70	96.67	1.99	3.3	0
DenseNet-121	653	P2	97.16	97.18	97.15	1.27	3.3	4
DenseNet-121	653	P3	97.41	97.42	97.39	1.92	3.3	5
DenseNet-121	701	P1	97.90	97.90	97.91	1.39	3.4	4
DenseNet-121	701	P2	96.79	96.80	96.83	2.02	3.4	5
DenseNet-121	701	P3	95.43	95.43	95.44	2.90	3.4	3
DenseNet-121	751	P1	98.27	98.29	98.30	0.91	3.4	4
DenseNet-121	751	P2	96.79	96.78	96.81	1.87	3.4	0
DenseNet-121	751	P3	96.79	96.80	96.78	1.56	3.4	5
DenseNet-121	809	P1	98.40	98.40	98.41	0.96	3.4	3
DenseNet-121	809	P2	97.41	97.39	97.39	1.62	3.4	4
DenseNet-121	809	P3	97.16	97.15	97.20	2.00	3.4	5
DenseNet-121	853	P1	97.41	97.41	97.43	1.44	3.4	5
DenseNet-121	853	P2	96.17	96.19	96.17	2.47	3.3	3
DenseNet-121	853	P3	96.30	96.34	96.30	2.43	3.4	0
DenseNet-121	907	P1	97.53	97.54	97.55	1.59	3.3	0
DenseNet-121	907	P2	96.91	96.91	96.91	1.25	3.3	5
DenseNet-121	907	P3	97.90	97.93	97.90	1.97	3.3	3
DenseNet-121	953	P1	98.02	98.03	98.04	1.33	3.3	3
DenseNet-121	953	P2	97.28	97.31	97.26	1.54	3.3	0
DenseNet-121	953	P3	97.53	97.54	97.51	1.13	3.3	4
DenseNet-121	1009	P1	98.02	98.03	98.03	1.47	3.3	4
DenseNet-121	1009	P2	96.79	96.80	96.80	1.75	3.3	3
DenseNet-121	1009	P3	97.65	97.64	97.63	0.63	3.3	5
DenseNet-121	1051	P1	97.53	97.55	97.55	1.06	3.3	5
DenseNet-121	1051	P2	97.04	97.03	97.06	1.77	3.3	0
DenseNet-121	1051	P3	98.02	98.03	98.02	0.57	3.3	3
EfficientNet-B0	137	P1	98.15	98.15	98.16	0.95	1.8	4
EfficientNet-B0	137	P2	97.16	97.16	97.16	1.94	1.8	5
EfficientNet-B0	137	P3	97.28	97.30	97.30	1.47	1.8	4
EfficientNet-B0	173	P1	96.79	96.79	96.82	1.83	1.8	0
EfficientNet-B0	173	P2	97.65	97.66	97.66	1.15	1.8	3
EfficientNet-B0	173	P3	97.04	97.06	97.05	1.67	1.8	0
EfficientNet-B0	211	P1	97.90	97.92	97.91	1.12	1.8	5
EfficientNet-B0	211	P2	97.16	97.19	97.15	1.87	1.8	4
EfficientNet-B0	211	P3	93.33	93.42	93.32	4.89	1.8	5
EfficientNet-B0	257	P1	97.41	97.41	97.42	1.76	1.8	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
EfficientNet-B0	257	P2	96.91	96.93	96.91	2.38	1.8	3
EfficientNet-B0	257	P3	97.28	97.28	97.27	2.10	1.8	5
EfficientNet-B0	307	P1	98.02	98.02	98.05	1.43	1.8	4
EfficientNet-B0	307	P2	96.42	96.41	96.41	1.84	1.8	0
EfficientNet-B0	307	P3	98.52	98.54	98.52	0.94	1.8	4
EfficientNet-B0	349	P1	97.65	97.67	97.67	1.63	1.8	3
EfficientNet-B0	349	P2	98.02	98.03	98.03	0.87	1.8	5
EfficientNet-B0	349	P3	97.65	97.66	97.65	0.98	1.8	3
EfficientNet-B0	401	P1	98.02	98.04	98.04	1.22	1.8	5
EfficientNet-B0	401	P2	96.54	96.57	96.52	2.20	1.8	4
EfficientNet-B0	401	P3	97.78	97.77	97.75	1.55	1.8	5
EfficientNet-B0	457	P1	97.90	97.89	97.94	1.28	1.8	0
EfficientNet-B0	457	P2	94.07	94.00	94.12	3.74	1.8	3
EfficientNet-B0	457	P3	96.30	96.28	96.34	2.66	1.8	0
EfficientNet-B0	503	P1	95.80	95.79	95.84	3.09	1.8	5
EfficientNet-B0	503	P2	97.04	97.05	97.06	1.72	1.8	4
EfficientNet-B0	503	P3	95.68	95.68	95.69	2.87	1.8	5
EfficientNet-B0	557	P1	97.78	97.78	97.78	1.24	1.8	5
EfficientNet-B0	557	P2	97.90	97.92	97.91	1.26	1.8	0
EfficientNet-B0	557	P3	97.65	97.66	97.67	1.03	1.8	5
EfficientNet-B0	601	P1	97.90	97.91	97.92	1.42	1.8	4
EfficientNet-B0	601	P2	96.79	96.81	96.80	1.58	1.8	3
EfficientNet-B0	601	P3	97.90	97.91	97.93	1.31	1.8	4
EfficientNet-B0	653	P1	98.64	98.65	98.65	0.91	1.8	3
EfficientNet-B0	653	P2	97.04	97.04	97.04	1.93	1.8	0
EfficientNet-B0	653	P3	95.80	95.81	95.79	2.28	1.8	3
EfficientNet-B0	701	P1	97.41	97.41	97.41	1.97	1.9	0
EfficientNet-B0	701	P2	96.79	96.81	96.81	1.93	1.9	4
EfficientNet-B0	701	P3	94.20	94.14	94.23	4.03	1.9	5
EfficientNet-B0	751	P1	98.27	98.28	98.29	1.16	1.9	3
EfficientNet-B0	751	P2	96.91	96.90	96.91	2.37	1.9	4
EfficientNet-B0	751	P3	97.16	97.18	97.17	1.94	1.9	3
EfficientNet-B0	809	P1	98.02	98.03	98.04	1.45	1.9	0
EfficientNet-B0	809	P2	96.54	96.55	96.54	2.17	1.8	3
EfficientNet-B0	809	P3	95.80	95.77	95.84	3.39	1.9	0
EfficientNet-B0	853	P1	97.65	97.65	97.67	1.88	1.8	4
EfficientNet-B0	853	P2	96.42	96.41	96.39	2.46	1.8	5
EfficientNet-B0	853	P3	96.54	96.55	96.53	2.31	1.8	4
EfficientNet-B0	907	P1	98.52	98.52	98.51	0.69	1.8	4
EfficientNet-B0	907	P2	97.53	97.53	97.54	1.65	1.8	0
EfficientNet-B0	907	P3	97.90	97.90	97.89	1.40	1.8	4
EfficientNet-B0	953	P1	98.02	98.04	98.03	1.15	1.8	5
EfficientNet-B0	953	P2	98.15	98.15	98.15	1.17	1.8	3
EfficientNet-B0	953	P3	98.02	98.02	98.02	1.05	1.8	5
EfficientNet-B0	1009	P1	97.78	97.79	97.79	1.38	1.8	0
EfficientNet-B0	1009	P2	97.04	97.04	97.06	2.13	1.8	4
EfficientNet-B0	1009	P3	96.91	96.92	96.89	1.58	1.8	0
EfficientNet-B0	1051	P1	97.65	97.66	97.67	1.47	1.8	4
EfficientNet-B0	1051	P2	97.53	97.53	97.56	1.13	1.8	5
EfficientNet-B0	1051	P3	98.27	98.28	98.27	0.78	1.8	4
MobileNet-V2	137	P1	96.42	96.41	96.45	2.48	1.5	0
MobileNet-V2	137	P2	96.91	96.91	96.92	1.66	1.5	5
MobileNet-V2	137	P3	97.53	97.54	97.54	1.72	1.5	3
MobileNet-V2	173	P1	97.41	97.42	97.41	1.21	1.5	3
MobileNet-V2	173	P2	96.67	96.69	96.67	1.23	1.5	5
MobileNet-V2	173	P3	97.04	97.06	97.05	1.75	1.5	4
MobileNet-V2	211	P1	97.28	97.28	97.31	1.28	1.5	4
MobileNet-V2	211	P2	93.95	94.03	93.92	3.68	1.5	0
MobileNet-V2	211	P3	94.07	94.10	94.05	3.69	1.5	3
MobileNet-V2	257	P1	96.91	96.91	96.94	1.77	1.5	3
MobileNet-V2	257	P2	96.54	96.55	96.53	1.96	1.5	4
MobileNet-V2	257	P3	96.79	96.81	96.78	1.51	1.5	0
MobileNet-V2	307	P1	97.90	97.89	97.92	0.63	1.5	0
MobileNet-V2	307	P2	94.94	94.93	94.92	2.44	1.5	3
MobileNet-V2	307	P3	97.41	97.43	97.41	1.30	1.5	5
MobileNet-V2	349	P1	96.67	96.68	96.67	1.55	1.5	5
MobileNet-V2	349	P2	95.93	95.95	95.95	2.51	1.5	0
MobileNet-V2	349	P3	97.53	97.54	97.53	1.11	1.5	4
MobileNet-V2	401	P1	97.41	97.42	97.41	1.52	1.5	0
MobileNet-V2	401	P2	96.67	96.68	96.67	1.62	1.5	4
MobileNet-V2	401	P3	96.79	96.82	96.77	2.22	1.5	3
MobileNet-V2	457	P1	96.91	96.90	96.93	1.63	1.5	3
MobileNet-V2	457	P2	93.83	93.82	93.85	4.12	1.5	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
MobileNet-V2	457	P3	95.93	95.92	95.97	2.52	1.5	4
MobileNet-V2	503	P1	96.54	96.54	96.57	2.28	1.5	4
MobileNet-V2	503	P2	96.67	96.67	96.68	2.39	1.5	3
MobileNet-V2	503	P3	94.20	94.22	94.19	3.67	1.5	0
MobileNet-V2	557	P1	96.67	96.68	96.69	1.63	1.5	0
MobileNet-V2	557	P2	97.78	97.79	97.78	1.23	1.5	4
MobileNet-V2	557	P3	96.42	96.45	96.40	1.47	1.5	3
MobileNet-V2	601	P1	97.41	97.42	97.42	1.07	1.5	3
MobileNet-V2	601	P2	96.30	96.31	96.31	1.96	1.5	5
MobileNet-V2	601	P3	96.91	96.92	96.93	1.61	1.5	0
MobileNet-V2	653	P1	97.53	97.54	97.53	1.33	1.5	0
MobileNet-V2	653	P2	95.43	95.47	95.42	2.16	1.5	4
MobileNet-V2	653	P3	96.54	96.53	96.53	2.42	1.5	5
MobileNet-V2	701	P1	96.67	96.66	96.69	2.22	1.6	3
MobileNet-V2	701	P2	96.17	96.19	96.19	2.08	1.5	0
MobileNet-V2	701	P3	93.70	93.68	93.69	3.99	1.5	4
MobileNet-V2	751	P1	96.54	96.56	96.57	1.92	1.6	0
MobileNet-V2	751	P2	96.79	96.79	96.81	2.51	1.5	5
MobileNet-V2	751	P3	95.19	95.21	95.15	2.44	1.5	4
MobileNet-V2	809	P1	96.91	96.92	96.92	1.46	1.5	4
MobileNet-V2	809	P2	96.42	96.42	96.43	1.45	1.6	3
MobileNet-V2	809	P3	96.05	96.02	96.10	2.36	1.5	5
MobileNet-V2	853	P1	97.65	97.65	97.66	1.37	1.5	3
MobileNet-V2	853	P2	95.93	95.91	95.92	2.05	1.5	5
MobileNet-V2	853	P3	95.06	95.08	95.03	2.90	1.5	0
MobileNet-V2	907	P1	97.90	97.91	97.90	1.14	1.5	0
MobileNet-V2	907	P2	97.04	97.05	97.04	1.57	1.5	5
MobileNet-V2	907	P3	96.67	96.70	96.64	1.76	1.5	3
MobileNet-V2	953	P1	97.41	97.43	97.42	1.51	1.5	3
MobileNet-V2	953	P2	97.16	97.19	97.16	1.05	1.5	0
MobileNet-V2	953	P3	97.41	97.42	97.37	1.04	1.5	4
MobileNet-V2	1009	P1	97.41	97.42	97.41	1.77	1.5	4
MobileNet-V2	1009	P2	95.80	95.80	95.82	1.89	1.5	3
MobileNet-V2	1009	P3	96.91	96.91	96.87	1.59	1.5	5
MobileNet-V2	1051	P1	97.28	97.31	97.30	1.68	1.5	0
MobileNet-V2	1051	P2	96.91	96.92	96.91	0.98	1.5	5
MobileNet-V2	1051	P3	97.53	97.53	97.53	0.48	1.5	3
ViT-B/16	137	P1	97.16	97.17	97.18	1.82	9.2	4
ViT-B/16	137	P2	96.67	96.69	96.66	2.06	9.1	0
ViT-B/16	137	P3	97.53	97.56	97.53	2.08	9.2	5
ViT-B/16	173	P1	97.78	97.77	97.80	1.30	9.1	0
ViT-B/16	173	P2	96.79	96.81	96.80	2.95	9.2	3
ViT-B/16	173	P3	94.69	94.80	94.70	2.84	9.2	5
ViT-B/16	211	P1	97.16	97.18	97.17	1.87	9.2	5
ViT-B/16	211	P2	94.32	94.40	94.27	4.43	9.2	4
ViT-B/16	211	P3	95.06	95.06	95.09	3.43	9.1	0
ViT-B/16	257	P1	95.56	95.54	95.54	3.36	9.2	5
ViT-B/16	257	P2	96.54	96.53	96.53	2.09	9.2	3
ViT-B/16	257	P3	97.16	97.16	97.19	2.39	9.2	4
ViT-B/16	307	P1	99.14	99.14	99.14	0.59	9.2	4
ViT-B/16	307	P2	95.80	95.80	95.78	3.32	9.1	0
ViT-B/16	307	P3	97.16	97.17	97.19	1.57	9.2	3
ViT-B/16	349	P1	94.81	94.87	94.74	2.85	9.1	3
ViT-B/16	349	P2	96.54	96.56	96.58	2.44	9.2	5
ViT-B/16	349	P3	97.28	97.29	97.30	1.60	9.1	0
ViT-B/16	401	P1	97.78	97.79	97.79	1.67	9.2	5
ViT-B/16	401	P2	94.07	93.89	94.18	4.11	9.1	0
ViT-B/16	401	P3	96.91	96.93	96.87	1.84	9.2	4
ViT-B/16	457	P1	98.64	98.65	98.65	0.97	9.1	0
ViT-B/16	457	P2	92.84	92.82	92.78	5.49	9.2	3
ViT-B/16	457	P3	96.30	96.31	96.30	2.08	9.2	5
ViT-B/16	503	P1	96.05	96.08	96.10	2.46	9.2	5
ViT-B/16	503	P2	96.17	96.17	96.21	2.67	9.2	4
ViT-B/16	503	P3	94.69	94.68	94.71	3.92	9.2	3
ViT-B/16	557	P1	97.04	97.05	97.06	2.17	9.2	5
ViT-B/16	557	P2	94.07	94.07	94.13	3.26	9.1	0
ViT-B/16	557	P3	95.68	95.69	95.72	3.13	9.2	4
ViT-B/16	601	P1	97.04	97.06	97.06	2.04	9.2	4
ViT-B/16	601	P2	97.41	97.40	97.43	2.25	9.2	3
ViT-B/16	601	P3	98.15	98.16	98.16	1.31	9.2	5
ViT-B/16	653	P1	97.53	97.54	97.54	1.86	9.2	3
ViT-B/16	653	P2	97.41	97.39	97.38	1.77	9.1	0
ViT-B/16	653	P3	95.43	95.45	95.43	2.88	9.2	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ViT-B/16	701	P1	97.28	97.27	97.29	1.93	9.2	5
ViT-B/16	701	P2	95.56	95.59	95.58	3.31	9.2	3
ViT-B/16	701	P3	93.09	92.99	93.15	4.48	9.1	0
ViT-B/16	751	P1	96.42	96.48	96.42	2.65	9.2	3
ViT-B/16	751	P2	95.43	95.47	95.44	2.90	9.2	0
ViT-B/16	751	P3	95.43	95.42	95.44	2.79	9.2	5
ViT-B/16	809	P1	97.28	97.30	97.33	2.11	9.2	0
ViT-B/16	809	P2	95.31	95.35	95.31	3.32	9.3	4
ViT-B/16	809	P3	96.67	96.66	96.71	2.39	9.2	3
ViT-B/16	853	P1	96.91	96.91	96.91	2.27	9.2	4
ViT-B/16	853	P2	96.42	96.39	96.42	2.64	9.3	3
ViT-B/16	853	P3	93.95	94.03	93.92	4.64	9.3	5
ViT-B/16	907	P1	98.77	98.78	98.78	0.87	9.2	4
ViT-B/16	907	P2	95.80	95.78	95.82	3.27	9.1	0
ViT-B/16	907	P3	97.04	97.03	97.07	1.89	9.2	5
ViT-B/16	953	P1	96.67	96.68	96.70	2.87	9.2	5
ViT-B/16	953	P2	96.91	96.95	96.92	2.34	9.2	3
ViT-B/16	953	P3	95.80	95.83	95.78	2.81	9.1	0
ViT-B/16	1009	P1	97.65	97.67	97.67	1.47	9.1	0
ViT-B/16	1009	P2	94.81	94.79	94.89	3.77	9.2	4
ViT-B/16	1009	P3	95.19	95.16	95.19	3.31	9.2	3
ViT-B/16	1051	P1	96.91	96.93	96.94	2.33	9.2	4
ViT-B/16	1051	P2	95.80	95.81	95.78	2.10	9.1	0
ViT-B/16	1051	P3	95.93	95.89	95.90	2.65	9.2	3
ConvNeXt-T	137	P1	98.15	98.16	98.16	1.55	5.2	3
ConvNeXt-T	137	P2	96.79	96.79	96.81	2.44	5.2	3
ConvNeXt-T	137	P3	96.67	96.67	96.68	2.41	5.2	4
ConvNeXt-T	173	P1	97.90	97.90	97.89	1.43	5.2	4
ConvNeXt-T	173	P2	96.17	96.19	96.17	2.90	5.2	4
ConvNeXt-T	173	P3	96.79	96.81	96.81	2.43	5.2	0
ConvNeXt-T	211	P1	97.90	97.90	97.91	1.64	5.2	3
ConvNeXt-T	211	P2	96.54	96.56	96.54	2.68	5.2	3
ConvNeXt-T	211	P3	95.31	95.31	95.32	4.00	5.2	5
ConvNeXt-T	257	P1	97.90	97.90	97.91	1.83	5.2	0
ConvNeXt-T	257	P2	97.16	97.16	97.16	2.07	5.2	0
ConvNeXt-T	257	P3	97.78	97.78	97.78	1.10	5.2	5
ConvNeXt-T	307	P1	98.27	98.28	98.28	1.22	5.2	5
ConvNeXt-T	307	P2	96.67	96.68	96.66	2.58	5.2	5
ConvNeXt-T	307	P3	98.27	98.27	98.29	1.32	5.2	4
ConvNeXt-T	349	P1	97.78	97.77	97.80	1.83	5.2	4
ConvNeXt-T	349	P2	93.83	93.89	93.83	5.03	5.2	4
ConvNeXt-T	349	P3	97.28	97.31	97.26	2.19	5.2	3
ConvNeXt-T	401	P1	97.53	97.55	97.56	1.51	5.2	3
ConvNeXt-T	401	P2	98.40	98.40	98.41	1.20	5.2	3
ConvNeXt-T	401	P3	97.53	97.55	97.51	1.88	5.2	5
ConvNeXt-T	457	P1	98.64	98.65	98.66	1.16	5.2	4
ConvNeXt-T	457	P2	95.56	95.55	95.57	3.12	5.2	4
ConvNeXt-T	457	P3	96.79	96.79	96.82	2.00	5.1	0
ConvNeXt-T	503	P1	97.28	97.30	97.31	1.90	5.2	0
ConvNeXt-T	503	P2	97.28	97.28	97.29	1.58	5.1	0
ConvNeXt-T	503	P3	94.69	94.70	94.74	4.10	5.2	5
ConvNeXt-T	557	P1	96.67	96.70	96.67	2.26	5.2	3
ConvNeXt-T	557	P2	98.15	98.16	98.14	1.33	5.2	3
ConvNeXt-T	557	P3	97.90	97.92	97.90	1.52	5.2	5
ConvNeXt-T	601	P1	97.65	97.67	97.68	2.02	5.2	0
ConvNeXt-T	601	P2	96.42	96.43	96.44	2.15	5.2	0
ConvNeXt-T	601	P3	98.02	98.03	98.04	1.23	5.2	4
ConvNeXt-T	653	P1	97.78	97.78	97.80	1.74	5.2	5
ConvNeXt-T	653	P2	97.28	97.26	97.31	1.68	5.2	5
ConvNeXt-T	653	P3	94.94	94.88	94.96	3.55	5.2	3
ConvNeXt-T	701	P1	97.04	97.03	97.03	2.29	5.4	4
ConvNeXt-T	701	P2	95.80	95.80	95.85	3.44	5.4	4
ConvNeXt-T	701	P3	95.19	95.14	95.25	3.72	5.3	5
ConvNeXt-T	751	P1	96.91	96.96	96.93	1.91	5.3	4
ConvNeXt-T	751	P2	96.67	96.67	96.64	2.67	5.4	4
ConvNeXt-T	751	P3	96.05	96.06	96.05	2.25	5.3	3
ConvNeXt-T	809	P1	96.17	96.25	96.16	2.72	5.4	5
ConvNeXt-T	809	P2	96.67	96.65	96.64	2.52	5.4	5
ConvNeXt-T	809	P3	97.04	97.03	97.03	2.25	5.3	0
ConvNeXt-T	853	P1	97.16	97.17	97.15	1.82	5.2	0
ConvNeXt-T	853	P2	96.79	96.79	96.75	2.51	5.2	0
ConvNeXt-T	853	P3	96.17	96.14	96.16	2.93	5.2	4
ConvNeXt-T	907	P1	98.27	98.28	98.29	1.17	5.2	3

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	907	P2	97.28	97.28	97.30	2.01	5.2	3
ConvNeXt-T	907	P3	97.28	97.29	97.29	2.04	5.2	4
ConvNeXt-T	953	P1	96.91	96.90	96.94	2.07	5.3	4
ConvNeXt-T	953	P2	96.79	96.83	96.80	2.37	5.2	4
ConvNeXt-T	953	P3	96.17	96.21	96.14	1.89	5.2	5
ConvNeXt-T	1009	P1	96.79	96.81	96.82	2.52	5.2	5
ConvNeXt-T	1009	P2	96.42	96.43	96.42	2.26	5.2	5
ConvNeXt-T	1009	P3	97.04	97.06	96.99	2.04	5.2	0
ConvNeXt-T	1051	P1	97.90	97.91	97.92	1.40	5.2	5
ConvNeXt-T	1051	P2	97.16	97.19	97.15	1.79	5.2	5
ConvNeXt-T	1051	P3	97.53	97.53	97.52	1.71	5.2	4
Swin-T	137	P1	96.05	96.01	96.11	3.18	4.6	0
Swin-T	137	P2	94.57	94.59	94.61	4.07	4.6	5
Swin-T	137	P3	97.28	97.31	97.28	2.13	4.7	3
Swin-T	173	P1	97.90	97.91	97.92	1.16	4.6	3
Swin-T	173	P2	95.31	95.37	95.32	3.21	4.7	5
Swin-T	173	P3	96.30	96.34	96.32	2.60	4.7	4
Swin-T	211	P1	97.78	97.77	97.79	1.72	4.7	4
Swin-T	211	P2	96.54	96.58	96.53	2.72	4.6	0
Swin-T	211	P3	94.20	94.24	94.14	4.49	4.6	3
Swin-T	257	P1	97.41	97.41	97.44	2.05	4.6	3
Swin-T	257	P2	97.78	97.78	97.79	1.59	4.7	4
Swin-T	257	P3	96.91	96.92	96.88	1.78	4.6	0
Swin-T	307	P1	97.41	97.40	97.43	1.77	4.6	0
Swin-T	307	P2	96.42	96.43	96.42	2.47	4.6	3
Swin-T	307	P3	97.90	97.90	97.92	1.05	4.7	5
Swin-T	349	P1	97.53	97.53	97.56	1.93	4.6	0
Swin-T	349	P2	97.04	97.05	97.05	2.27	4.6	5
Swin-T	349	P3	97.65	97.69	97.69	1.80	4.6	4
Swin-T	401	P1	98.15	98.16	98.17	1.65	4.6	0
Swin-T	401	P2	97.16	97.17	97.17	2.06	4.6	4
Swin-T	401	P3	97.78	97.77	97.76	1.71	4.7	3
Swin-T	457	P1	97.65	97.64	97.70	2.13	4.6	3
Swin-T	457	P2	94.81	94.77	94.85	4.23	4.6	5
Swin-T	457	P3	96.54	96.55	96.55	2.43	4.7	4
Swin-T	503	P1	96.05	96.03	96.10	2.84	4.7	4
Swin-T	503	P2	97.04	97.06	97.05	2.19	4.6	3
Swin-T	503	P3	95.06	95.08	95.07	3.60	4.6	0
Swin-T	557	P1	98.15	98.16	98.16	1.69	4.6	0
Swin-T	557	P2	97.16	97.16	97.20	1.70	4.6	4
Swin-T	557	P3	98.02	98.03	98.04	1.76	4.6	3
Swin-T	601	P1	97.90	97.90	97.92	1.24	4.7	3
Swin-T	601	P2	97.41	97.42	97.40	1.79	4.6	5
Swin-T	601	P3	98.77	98.78	98.78	0.81	4.6	0
Swin-T	653	P1	98.02	98.03	98.04	1.35	4.6	0
Swin-T	653	P2	96.17	96.23	96.18	2.37	4.7	4
Swin-T	653	P3	96.05	96.06	96.05	2.06	4.6	5
Swin-T	701	P1	97.28	97.29	97.29	2.32	4.7	3
Swin-T	701	P2	96.17	96.19	96.20	2.84	4.7	0
Swin-T	701	P3	93.21	93.08	93.22	5.46	4.7	4
Swin-T	751	P1	97.78	97.79	97.80	1.76	4.7	0
Swin-T	751	P2	96.17	96.20	96.14	2.84	4.7	5
Swin-T	751	P3	95.06	95.13	95.03	4.27	4.8	4
Swin-T	809	P1	95.06	95.13	95.04	3.73	4.7	4
Swin-T	809	P2	95.80	95.77	95.82	3.36	4.7	3
Swin-T	809	P3	96.42	96.41	96.46	2.67	4.7	5
Swin-T	853	P1	98.02	98.03	98.03	1.29	4.7	3
Swin-T	853	P2	97.04	97.01	97.01	2.29	4.7	5
Swin-T	853	P3	95.56	95.55	95.60	3.12	4.7	0
Swin-T	907	P1	98.40	98.40	98.42	1.04	4.6	0
Swin-T	907	P2	96.54	96.60	96.55	2.52	4.6	5
Swin-T	907	P3	98.40	98.40	98.39	1.49	4.7	3
Swin-T	953	P1	97.28	97.29	97.31	1.93	4.6	3
Swin-T	953	P2	96.79	96.80	96.81	2.44	4.6	0
Swin-T	953	P3	96.30	96.33	96.29	2.96	4.7	4
Swin-T	1009	P1	97.90	97.92	97.89	1.06	4.7	4
Swin-T	1009	P2	97.04	97.02	97.07	2.41	4.6	3
Swin-T	1009	P3	97.28	97.26	97.25	1.78	4.7	5
Swin-T	1051	P1	98.15	98.16	98.16	1.62	4.6	0
Swin-T	1051	P2	97.78	97.77	97.80	1.69	4.7	3
Swin-T	1051	P3	95.19	95.24	95.20	3.15	4.7	5
RegNetY-1.6GF	137	P1	97.90	97.91	97.91	1.50	2.5	0
RegNetY-1.6GF	137	P2	95.68	95.71	95.68	3.53	2.5	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
RegNetY-1.6GF	137	P3	96.05	96.06	96.06	2.35	2.5	4
RegNetY-1.6GF	173	P1	97.78	97.79	97.80	1.18	2.5	0
RegNetY-1.6GF	173	P2	97.78	97.80	97.80	1.49	2.5	3
RegNetY-1.6GF	173	P3	97.53	97.55	97.53	1.15	2.5	5
RegNetY-1.6GF	211	P1	97.65	97.67	97.67	1.33	2.5	5
RegNetY-1.6GF	211	P2	95.68	95.75	95.66	3.02	2.5	4
RegNetY-1.6GF	211	P3	95.68	95.69	95.68	2.82	2.5	0
RegNetY-1.6GF	257	P1	96.54	96.54	96.57	2.24	2.5	5
RegNetY-1.6GF	257	P2	95.80	95.82	95.78	2.47	2.5	3
RegNetY-1.6GF	257	P3	96.54	96.55	96.54	2.06	2.5	4
RegNetY-1.6GF	307	P1	97.78	97.77	97.81	1.61	2.5	0
RegNetY-1.6GF	307	P2	95.80	95.78	95.78	2.70	2.5	4
RegNetY-1.6GF	307	P3	97.78	97.82	97.78	1.34	2.5	3
RegNetY-1.6GF	349	P1	98.02	98.03	98.03	1.35	2.5	3
RegNetY-1.6GF	349	P2	97.28	97.30	97.28	1.07	2.5	0
RegNetY-1.6GF	349	P3	98.02	98.04	98.03	0.93	2.5	5
RegNetY-1.6GF	401	P1	96.79	96.81	96.80	1.75	2.5	0
RegNetY-1.6GF	401	P2	97.65	97.67	97.66	1.83	2.5	5
RegNetY-1.6GF	401	P3	97.04	97.05	96.99	1.88	2.5	4
RegNetY-1.6GF	457	P1	97.90	97.89	97.94	0.93	2.5	0
RegNetY-1.6GF	457	P2	95.06	95.01	95.10	3.76	2.5	3
RegNetY-1.6GF	457	P3	96.67	96.67	96.67	1.90	2.5	5
RegNetY-1.6GF	503	P1	97.28	97.28	97.31	1.85	2.5	5
RegNetY-1.6GF	503	P2	96.91	96.93	96.91	1.53	2.5	4
RegNetY-1.6GF	503	P3	95.43	95.44	95.42	3.12	2.5	3
RegNetY-1.6GF	557	P1	97.16	97.17	97.17	1.77	2.5	0
RegNetY-1.6GF	557	P2	97.41	97.40	97.41	1.22	2.5	5
RegNetY-1.6GF	557	P3	96.54	96.57	96.54	1.78	2.5	4
RegNetY-1.6GF	601	P1	96.05	96.08	96.05	2.62	2.5	3
RegNetY-1.6GF	601	P2	97.41	97.43	97.40	1.33	2.5	4
RegNetY-1.6GF	601	P3	97.28	97.29	97.31	1.79	2.5	5
RegNetY-1.6GF	653	P1	97.28	97.30	97.27	2.13	2.5	3
RegNetY-1.6GF	653	P2	96.91	96.94	96.92	2.07	2.5	0
RegNetY-1.6GF	653	P3	95.56	95.56	95.52	2.45	2.5	4
RegNetY-1.6GF	701	P1	97.53	97.54	97.54	1.33	2.5	5
RegNetY-1.6GF	701	P2	95.19	95.20	95.23	3.04	2.6	3
RegNetY-1.6GF	701	P3	94.81	94.80	94.85	3.17	2.6	0
RegNetY-1.6GF	751	P1	97.78	97.80	97.80	1.38	2.5	0
RegNetY-1.6GF	751	P2	96.05	96.04	96.04	2.72	2.6	3
RegNetY-1.6GF	751	P3	95.93	95.92	95.94	2.36	2.5	5
RegNetY-1.6GF	809	P1	97.90	97.92	97.91	1.40	2.6	4
RegNetY-1.6GF	809	P2	96.54	96.54	96.55	2.13	2.6	0
RegNetY-1.6GF	809	P3	96.42	96.42	96.46	2.81	2.6	3
RegNetY-1.6GF	853	P1	98.15	98.15	98.16	0.57	2.5	3
RegNetY-1.6GF	853	P2	96.42	96.41	96.39	2.59	2.5	5
RegNetY-1.6GF	853	P3	96.42	96.42	96.40	1.91	2.5	4
RegNetY-1.6GF	907	P1	98.02	98.03	98.04	1.18	2.5	0
RegNetY-1.6GF	907	P2	96.42	96.45	96.43	2.55	2.5	4
RegNetY-1.6GF	907	P3	97.90	97.91	97.88	1.54	2.5	5
RegNetY-1.6GF	953	P1	97.78	97.78	97.79	1.34	2.6	3
RegNetY-1.6GF	953	P2	97.41	97.44	97.39	1.37	2.6	5
RegNetY-1.6GF	953	P3	97.28	97.29	97.24	1.72	2.5	0
RegNetY-1.6GF	1009	P1	97.53	97.55	97.53	1.24	2.5	0
RegNetY-1.6GF	1009	P2	96.54	96.56	96.54	2.24	2.5	4
RegNetY-1.6GF	1009	P3	97.53	97.54	97.51	1.15	2.5	3
RegNetY-1.6GF	1051	P1	97.41	97.42	97.42	1.80	2.5	4
RegNetY-1.6GF	1051	P2	97.16	97.18	97.15	1.39	2.5	0
RegNetY-1.6GF	1051	P3	97.78	97.77	97.77	1.44	2.5	3
EfficientNetV2-S	137	P1	96.91	96.92	96.94	2.59	3.5	3
EfficientNetV2-S	137	P2	95.93	95.91	95.94	2.54	3.5	0
EfficientNetV2-S	137	P3	95.93	95.95	95.92	2.71	3.5	5
EfficientNetV2-S	173	P1	97.28	97.29	97.30	1.74	3.5	4
EfficientNetV2-S	173	P2	96.05	96.04	96.09	2.34	3.5	0
EfficientNetV2-S	173	P3	96.42	96.43	96.45	2.60	3.5	3
EfficientNetV2-S	211	P1	97.90	97.91	97.90	1.00	3.5	3
EfficientNetV2-S	211	P2	96.42	96.47	96.42	2.50	3.5	5
EfficientNetV2-S	211	P3	94.32	94.34	94.28	4.22	3.5	4
EfficientNetV2-S	257	P1	96.67	96.66	96.70	2.13	3.5	0
EfficientNetV2-S	257	P2	97.04	97.03	97.03	1.88	3.5	5
EfficientNetV2-S	257	P3	97.04	97.04	97.05	1.85	3.5	3
EfficientNetV2-S	307	P1	98.15	98.16	98.17	1.38	3.5	5
EfficientNetV2-S	307	P2	95.56	95.57	95.54	2.95	3.5	0
EfficientNetV2-S	307	P3	97.65	97.67	97.66	1.91	3.5	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
EfficientNetV2-S	349	P1	97.04	97.07	97.04	2.37	3.5	4
EfficientNetV2-S	349	P2	96.30	96.32	96.27	2.74	3.5	3
EfficientNetV2-S	349	P3	97.65	97.68	97.65	1.51	3.5	0
EfficientNetV2-S	401	P1	97.16	97.17	97.17	2.13	3.5	3
EfficientNetV2-S	401	P2	97.41	97.44	97.41	1.66	3.5	0
EfficientNetV2-S	401	P3	97.16	97.16	97.14	1.52	3.5	5
EfficientNetV2-S	457	P1	98.15	98.15	98.16	1.58	3.5	4
EfficientNetV2-S	457	P2	94.20	94.19	94.21	3.90	3.5	0
EfficientNetV2-S	457	P3	96.05	96.05	96.08	2.50	3.5	3
EfficientNetV2-S	503	P1	96.54	96.55	96.58	2.61	3.5	0
EfficientNetV2-S	503	P2	96.05	96.08	96.04	3.36	3.5	5
EfficientNetV2-S	503	P3	96.79	96.78	96.78	2.22	3.5	4
EfficientNetV2-S	557	P1	97.04	97.07	97.06	2.13	3.5	3
EfficientNetV2-S	557	P2	98.40	98.41	98.39	0.99	3.5	0
EfficientNetV2-S	557	P3	97.90	97.91	97.91	1.61	3.5	5
EfficientNetV2-S	601	P1	96.79	96.82	96.82	2.44	3.5	0
EfficientNetV2-S	601	P2	97.16	97.16	97.19	2.40	3.5	3
EfficientNetV2-S	601	P3	97.16	97.17	97.16	1.86	3.5	4
EfficientNetV2-S	653	P1	97.41	97.42	97.41	1.92	3.5	5
EfficientNetV2-S	653	P2	95.56	95.57	95.54	3.36	3.5	3
EfficientNetV2-S	653	P3	96.67	96.65	96.65	2.04	3.5	0
EfficientNetV2-S	701	P1	97.28	97.30	97.29	1.83	3.6	5
EfficientNetV2-S	701	P2	94.94	94.98	94.96	4.05	3.6	3
EfficientNetV2-S	701	P3	94.44	94.43	94.44	4.17	3.6	0
EfficientNetV2-S	751	P1	97.53	97.56	97.54	1.38	3.5	4
EfficientNetV2-S	751	P2	96.05	96.06	96.05	3.48	3.6	0
EfficientNetV2-S	751	P3	97.04	97.07	97.03	2.06	3.6	5
EfficientNetV2-S	809	P1	97.78	97.77	97.82	1.45	3.6	5
EfficientNetV2-S	809	P2	96.05	96.05	96.07	2.84	3.6	4
EfficientNetV2-S	809	P3	94.69	94.62	94.75	4.03	3.6	0
EfficientNetV2-S	853	P1	96.79	96.77	96.82	2.29	3.5	0
EfficientNetV2-S	853	P2	96.05	96.03	96.03	3.09	3.5	3
EfficientNetV2-S	853	P3	96.42	96.43	96.43	2.35	3.5	5
EfficientNetV2-S	907	P1	98.02	98.04	98.02	1.21	3.5	3
EfficientNetV2-S	907	P2	96.54	96.55	96.55	2.80	3.5	0
EfficientNetV2-S	907	P3	96.54	96.59	96.55	2.28	3.5	4
EfficientNetV2-S	953	P1	97.53	97.53	97.56	2.13	3.5	4
EfficientNetV2-S	953	P2	97.65	97.67	97.63	1.48	3.6	0
EfficientNetV2-S	953	P3	98.15	98.16	98.15	1.11	3.5	3
EfficientNetV2-S	1009	P1	96.91	96.95	96.91	2.16	3.5	5
EfficientNetV2-S	1009	P2	95.93	95.93	95.94	2.92	3.5	0
EfficientNetV2-S	1009	P3	96.30	96.30	96.27	2.42	3.5	3
EfficientNetV2-S	1051	P1	98.27	98.28	98.27	1.18	3.5	5
EfficientNetV2-S	1051	P2	97.04	97.04	97.00	1.66	3.5	4
EfficientNetV2-S	1051	P3	97.41	97.41	97.42	1.51	3.5	0
MaxViT-T	137	P1	98.02	98.02	98.04	1.40	7.8	4
MaxViT-T	137	P2	95.06	95.04	95.09	3.02	7.8	3
MaxViT-T	137	P3	95.31	95.30	95.35	3.22	7.8	0
MaxViT-T	173	P1	97.04	97.03	97.08	2.22	7.8	5
MaxViT-T	173	P2	94.57	94.61	94.62	4.09	7.8	4
MaxViT-T	173	P3	97.04	97.05	97.04	1.83	7.8	0
MaxViT-T	211	P1	98.15	98.16	98.16	1.52	7.8	0
MaxViT-T	211	P2	95.68	95.68	95.69	3.31	7.8	3
MaxViT-T	211	P3	95.80	95.84	95.79	3.59	7.8	5
MaxViT-T	257	P1	96.91	96.91	96.93	2.33	7.8	4
MaxViT-T	257	P2	97.53	97.53	97.54	2.32	7.7	0
MaxViT-T	257	P3	97.78	97.78	97.81	1.12	7.8	5
MaxViT-T	307	P1	97.78	97.76	97.81	1.65	7.8	3
MaxViT-T	307	P2	97.28	97.29	97.30	2.25	7.8	5
MaxViT-T	307	P3	97.78	97.78	97.80	1.74	7.8	0
MaxViT-T	349	P1	96.67	96.68	96.69	2.59	7.8	5
MaxViT-T	349	P2	94.57	94.61	94.59	4.08	7.8	4
MaxViT-T	349	P3	97.90	97.94	97.89	1.41	7.8	3
MaxViT-T	401	P1	97.65	97.67	97.68	1.87	7.8	4
MaxViT-T	401	P2	97.78	97.79	97.79	1.44	7.8	3
MaxViT-T	401	P3	95.80	95.76	95.84	2.90	7.8	0
MaxViT-T	457	P1	96.91	96.87	96.96	2.26	7.8	5
MaxViT-T	457	P2	94.57	94.56	94.58	4.07	7.8	4
MaxViT-T	457	P3	96.67	96.66	96.70	2.67	7.8	0
MaxViT-T	503	P1	92.72	92.69	92.80	6.48	7.8	3
MaxViT-T	503	P2	97.28	97.28	97.30	1.96	7.7	0
MaxViT-T	503	P3	95.80	95.79	95.84	3.11	7.8	5
MaxViT-T	557	P1	97.16	97.17	97.19	1.92	7.8	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
MaxViT-T	557	P2	97.90	97.90	97.93	1.38	7.7	3
MaxViT-T	557	P3	97.78	97.79	97.78	1.10	7.8	0
MaxViT-T	601	P1	96.91	96.94	96.93	2.42	7.8	5
MaxViT-T	601	P2	97.41	97.40	97.42	1.64	7.8	0
MaxViT-T	601	P3	96.67	96.67	96.70	2.22	7.8	3
MaxViT-T	653	P1	98.02	98.03	98.05	1.72	8.3	4
MaxViT-T	653	P2	97.04	97.04	97.06	1.92	8.7	5
MaxViT-T	653	P3	97.53	97.52	97.53	1.58	8.1	3
MaxViT-T	701	P1	97.90	97.90	97.90	1.28	8.0	4
MaxViT-T	701	P2	95.31	95.32	95.34	4.04	7.9	5
MaxViT-T	701	P3	95.06	95.02	95.10	3.72	8.0	3
MaxViT-T	751	P1	97.28	97.30	97.31	1.83	7.9	3
MaxViT-T	751	P2	96.54	96.55	96.51	2.90	8.0	4
MaxViT-T	751	P3	96.54	96.52	96.57	2.35	7.9	0
MaxViT-T	809	P1	97.04	97.02	97.06	2.40	8.0	3
MaxViT-T	809	P2	95.19	95.23	95.16	4.25	8.0	5
MaxViT-T	809	P3	96.30	96.29	96.34	3.00	8.0	4
MaxViT-T	853	P1	97.65	97.66	97.67	1.55	7.9	4
MaxViT-T	853	P2	95.43	95.42	95.36	3.46	7.8	0
MaxViT-T	853	P3	95.06	95.08	95.00	3.88	7.8	3
MaxViT-T	907	P1	98.27	98.27	98.29	1.62	7.8	5
MaxViT-T	907	P2	95.93	95.91	95.91	3.36	7.8	3
MaxViT-T	907	P3	97.65	97.66	97.66	1.52	7.8	0
MaxViT-T	953	P1	96.05	96.08	96.07	3.18	7.9	5
MaxViT-T	953	P2	97.65	97.67	97.66	1.80	7.8	4
MaxViT-T	953	P3	97.78	97.79	97.77	1.64	7.8	0
MaxViT-T	1009	P1	97.78	97.79	97.80	1.96	7.8	4
MaxViT-T	1009	P2	95.80	95.76	95.86	3.65	7.9	5
MaxViT-T	1009	P3	96.79	96.78	96.75	2.84	7.8	0
MaxViT-T	1051	P1	97.41	97.41	97.42	1.81	7.9	3
MaxViT-T	1051	P2	97.53	97.53	97.55	1.19	7.8	5
MaxViT-T	1051	P3	96.30	96.30	96.32	2.63	7.8	4
DINOv2 ViT-S/14	137	P1	90.86	90.96	90.94	3.71	4.1	5
DINOv2 ViT-S/14	137	P2	91.73	91.82	91.73	5.24	4.1	5
DINOv2 ViT-S/14	137	P3	92.84	92.86	92.88	2.65	4.1	4
DINOv2 ViT-S/14	173	P1	90.99	91.07	91.05	3.02	4.1	3
DINOv2 ViT-S/14	173	P2	92.47	92.53	92.53	3.25	4.1	3
DINOv2 ViT-S/14	173	P3	92.84	92.82	92.94	3.57	4.1	5
DINOv2 ViT-S/14	211	P1	93.58	93.60	93.63	3.08	4.1	4
DINOv2 ViT-S/14	211	P2	92.72	92.76	92.81	4.67	4.1	4
DINOv2 ViT-S/14	211	P3	90.49	90.52	90.48	4.98	4.1	0
DINOv2 ViT-S/14	257	P1	93.33	93.24	93.40	3.89	4.1	3
DINOv2 ViT-S/14	257	P2	92.84	92.82	92.95	4.11	4.1	3
DINOv2 ViT-S/14	257	P3	93.46	93.45	93.48	2.71	4.1	4
DINOv2 ViT-S/14	307	P1	93.46	93.42	93.54	3.44	4.1	4
DINOv2 ViT-S/14	307	P2	92.72	92.73	92.67	1.98	4.1	4
DINOv2 ViT-S/14	307	P3	91.48	91.48	91.48	3.68	4.1	3
DINOv2 ViT-S/14	349	P1	95.43	95.48	95.46	1.27	4.1	0
DINOv2 ViT-S/14	349	P2	92.84	92.91	92.85	2.93	4.1	0
DINOv2 ViT-S/14	349	P3	92.96	93.02	93.03	3.91	4.1	5
DINOv2 ViT-S/14	401	P1	93.46	93.50	93.55	3.28	4.1	5
DINOv2 ViT-S/14	401	P2	94.57	94.54	94.63	3.61	4.1	5
DINOv2 ViT-S/14	401	P3	90.62	90.83	90.65	4.24	4.1	4
DINOv2 ViT-S/14	457	P1	96.17	96.18	96.21	1.79	4.1	3
DINOv2 ViT-S/14	457	P2	91.85	91.92	91.86	4.79	4.1	3
DINOv2 ViT-S/14	457	P3	92.84	92.84	92.87	4.81	4.1	5
DINOv2 ViT-S/14	503	P1	93.33	93.34	93.38	3.72	4.1	4
DINOv2 ViT-S/14	503	P2	95.06	95.07	95.04	2.49	4.1	4
DINOv2 ViT-S/14	503	P3	87.78	87.89	87.88	6.13	4.1	3
DINOv2 ViT-S/14	557	P1	95.68	95.70	95.71	1.78	4.1	5
DINOv2 ViT-S/14	557	P2	94.32	94.35	94.36	2.11	4.1	5
DINOv2 ViT-S/14	557	P3	90.12	89.89	90.12	4.24	4.1	4
DINOv2 ViT-S/14	601	P1	94.81	94.85	94.84	2.80	4.1	4
DINOv2 ViT-S/14	601	P2	89.75	90.10	89.76	4.11	4.1	4
DINOv2 ViT-S/14	601	P3	94.07	94.10	94.11	2.52	4.1	5
DINOv2 ViT-S/14	653	P1	93.21	93.28	93.22	3.19	4.3	0
DINOv2 ViT-S/14	653	P2	93.33	93.39	93.35	2.63	4.1	0
DINOv2 ViT-S/14	653	P3	90.99	90.96	90.98	3.67	4.1	4
DINOv2 ViT-S/14	701	P1	94.07	94.09	94.14	2.47	4.2	0
DINOv2 ViT-S/14	701	P2	89.14	89.33	89.22	5.85	4.2	4
DINOv2 ViT-S/14	701	P3	87.90	87.92	87.92	7.21	4.2	0
DINOv2 ViT-S/14	751	P1	94.94	94.94	94.98	2.01	4.2	5
DINOv2 ViT-S/14	751	P2	91.98	92.02	92.06	3.51	4.2	3

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
DINOv2 ViT-S/14	751	P3	87.16	87.43	87.21	7.66	4.2	5
DINOv2 ViT-S/14	809	P1	93.33	93.46	93.35	1.01	4.2	0
DINOv2 ViT-S/14	809	P2	92.96	93.02	93.01	3.23	4.2	3
DINOv2 ViT-S/14	809	P3	89.88	89.96	89.96	2.90	4.2	0
DINOv2 ViT-S/14	853	P1	93.58	93.59	93.64	3.65	4.1	5
DINOv2 ViT-S/14	853	P2	91.98	91.67	92.07	4.62	4.1	5
DINOv2 ViT-S/14	853	P3	89.51	89.47	89.65	5.47	4.1	4
DINOv2 ViT-S/14	907	P1	94.57	94.64	94.56	2.42	4.1	4
DINOv2 ViT-S/14	907	P2	90.37	90.54	90.46	5.40	4.1	4
DINOv2 ViT-S/14	907	P3	90.99	91.08	91.01	3.76	4.1	5
DINOv2 ViT-S/14	953	P1	92.84	92.87	92.91	3.51	4.1	3
DINOv2 ViT-S/14	953	P2	92.47	92.45	92.51	4.31	4.1	3
DINOv2 ViT-S/14	953	P3	92.35	92.36	92.40	3.91	4.1	5
DINOv2 ViT-S/14	1009	P1	93.09	93.12	93.17	2.99	4.1	3
DINOv2 ViT-S/14	1009	P2	95.43	95.44	95.49	2.32	4.1	3
DINOv2 ViT-S/14	1009	P3	90.99	91.00	91.08	5.31	4.1	4
DINOv2 ViT-S/14	1051	P1	94.20	94.24	94.21	3.02	4.1	0
DINOv2 ViT-S/14	1051	P2	92.72	92.74	92.78	4.40	4.1	0
DINOv2 ViT-S/14	1051	P3	89.75	89.90	89.75	3.21	4.1	3

Table L.2: Complete CRIC run ledger (180 runs).

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-18	137	P1	90.29	79.91	80.99	5.17	2.4	5
ResNet-18	137	P1-prime	90.42	79.96	78.81	4.90	2.4	4
ResNet-18	137	P2	77.21	56.93	56.06	15.34	2.4	5
ResNet-18	173	P1	90.33	79.33	77.66	5.38	2.4	0
ResNet-18	173	P1-prime	89.42	77.91	78.92	6.32	2.4	0
ResNet-18	173	P2	79.19	54.72	52.76	13.90	2.4	0
ResNet-18	211	P1	89.47	79.21	77.35	5.56	2.4	5
ResNet-18	211	P1-prime	89.03	78.04	76.41	6.41	2.4	5
ResNet-18	211	P2	79.30	57.93	56.26	13.97	2.4	5
ResNet-18	257	P1	90.68	81.52	80.34	4.79	2.4	4
ResNet-18	257	P1-prime	90.51	80.46	80.79	5.23	2.4	4
ResNet-18	257	P2	76.61	56.24	54.11	16.74	2.4	4
ResNet-18	307	P1	90.16	79.43	76.56	5.47	2.4	5
ResNet-18	307	P1-prime	90.81	81.98	80.34	4.49	2.4	5
ResNet-18	307	P2	77.57	58.26	59.42	14.68	2.4	5
ResNet-18	349	P1	89.47	77.77	76.57	5.20	2.4	3
ResNet-18	349	P1-prime	90.38	78.70	77.45	4.87	2.4	3
ResNet-18	349	P2	76.45	56.43	58.27	16.90	2.4	3
ResNet-18	401	P1	89.47	78.15	74.99	6.14	2.4	0
ResNet-18	401	P1-prime	89.21	78.43	75.35	5.71	2.4	0
ResNet-18	401	P2	77.31	56.70	53.65	14.96	2.4	0
ResNet-18	457	P1	89.68	76.69	74.80	5.72	2.4	4
ResNet-18	457	P1-prime	90.51	78.75	75.63	5.12	2.4	4
ResNet-18	457	P2	76.80	58.06	60.26	15.04	2.4	4
ResNet-18	503	P1	89.81	79.25	79.18	5.98	2.4	5
ResNet-18	503	P1-prime	89.08	77.64	76.56	6.20	2.4	5
ResNet-18	503	P2	79.08	60.12	58.06	13.66	2.4	5
ResNet-18	557	P1	89.77	79.63	77.54	5.56	2.4	3
ResNet-18	557	P1-prime	89.81	77.59	73.91	5.51	2.4	3
ResNet-18	557	P2	76.66	57.06	59.07	14.67	2.4	3
ResNet-50	137	P1	89.86	80.65	81.45	6.36	7.3	4
ResNet-50	137	P1-prime	90.33	80.73	81.13	5.40	7.3	0
ResNet-50	137	P2	77.94	58.33	56.70	16.41	7.3	5
ResNet-50	173	P1	91.33	83.04	83.86	5.13	7.3	0
ResNet-50	173	P1-prime	92.02	83.50	82.32	4.72	7.3	5
ResNet-50	173	P2	81.05	64.25	63.43	13.25	7.3	0
ResNet-50	211	P1	90.03	81.34	79.72	5.84	7.3	5
ResNet-50	211	P1-prime	90.29	79.45	79.32	5.44	7.3	4
ResNet-50	211	P2	80.38	62.61	61.35	13.78	7.3	5
ResNet-50	257	P1	90.46	80.21	78.47	6.06	7.4	4
ResNet-50	257	P1-prime	91.33	80.74	80.77	5.01	7.3	3
ResNet-50	257	P2	74.84	52.48	55.44	19.13	7.3	4
ResNet-50	307	P1	91.16	82.30	79.82	6.02	7.3	5
ResNet-50	307	P1-prime	91.68	83.75	81.52	4.96	7.3	3
ResNet-50	307	P2	78.87	61.31	62.73	14.49	7.3	5
ResNet-50	349	P1	90.42	78.92	78.72	5.90	7.3	3

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	349	P1-prime	90.29	79.95	79.11	6.25	7.3	0
ResNet-50	349	P2	78.30	60.27	60.23	15.55	7.3	3
ResNet-50	401	P1	90.29	79.95	79.56	5.80	7.3	0
ResNet-50	401	P1-prime	90.59	82.00	80.65	5.37	7.3	4
ResNet-50	401	P2	79.52	61.94	58.62	13.71	7.3	0
ResNet-50	457	P1	90.59	78.99	76.56	5.66	7.3	4
ResNet-50	457	P1-prime	90.90	78.63	75.56	5.78	7.3	5
ResNet-50	457	P2	78.35	60.92	60.86	14.85	7.3	4
ResNet-50	503	P1	90.98	81.56	80.72	5.65	7.3	5
ResNet-50	503	P1-prime	90.46	81.15	80.52	5.37	7.3	3
ResNet-50	503	P2	77.78	59.65	59.78	15.47	7.3	5
ResNet-50	557	P1	90.25	77.83	75.33	6.06	7.3	3
ResNet-50	557	P1-prime	90.59	80.18	77.63	5.76	7.3	4
ResNet-50	557	P2	80.55	63.07	64.28	13.03	7.3	3
DenseNet-121	137	P1	90.77	81.40	79.36	5.65	9.2	3
DenseNet-121	137	P1-prime	90.12	78.92	78.57	5.74	9.2	4
DenseNet-121	137	P2	77.29	56.35	53.39	15.86	9.2	0
DenseNet-121	173	P1	91.55	82.97	83.22	4.36	9.2	5
DenseNet-121	173	P1-prime	90.94	81.93	81.19	5.28	9.2	0
DenseNet-121	173	P2	80.18	61.63	59.67	13.70	9.2	3
DenseNet-121	211	P1	90.12	81.50	80.12	5.08	9.2	4
DenseNet-121	211	P1-prime	90.59	79.02	75.99	5.53	9.2	0
DenseNet-121	211	P2	81.07	63.31	60.96	12.27	9.2	5
DenseNet-121	257	P1	91.07	79.90	78.63	5.07	9.2	3
DenseNet-121	257	P1-prime	90.20	80.04	77.99	5.83	9.2	4
DenseNet-121	257	P2	78.13	59.00	58.60	14.92	9.2	0
DenseNet-121	307	P1	90.20	77.44	74.42	5.57	9.2	3
DenseNet-121	307	P1-prime	91.76	81.93	79.65	4.44	9.2	5
DenseNet-121	307	P2	76.01	55.79	60.52	16.43	9.2	4
DenseNet-121	349	P1	90.20	78.60	78.54	5.42	9.1	0
DenseNet-121	349	P1-prime	91.07	80.37	78.54	4.82	9.2	3
DenseNet-121	349	P2	78.77	60.09	63.41	14.41	9.1	5
DenseNet-121	401	P1	89.12	78.75	76.82	6.43	9.2	4
DenseNet-121	401	P1-prime	90.51	79.45	76.96	5.44	9.1	0
DenseNet-121	401	P2	78.22	60.38	59.59	14.62	9.2	3
DenseNet-121	457	P1	91.20	79.85	78.11	4.71	9.1	5
DenseNet-121	457	P1-prime	90.42	78.11	75.80	5.87	9.1	0
DenseNet-121	457	P2	77.14	59.19	58.14	16.27	9.2	4
DenseNet-121	503	P1	90.16	79.54	76.27	5.40	9.2	3
DenseNet-121	503	P1-prime	90.72	82.20	80.36	5.50	9.1	5
DenseNet-121	503	P2	78.30	57.31	55.94	15.11	9.1	0
DenseNet-121	557	P1	89.64	76.65	74.17	6.54	9.1	4
DenseNet-121	557	P1-prime	90.72	80.22	78.14	5.12	9.1	3
DenseNet-121	557	P2	77.23	57.79	58.03	15.87	9.2	5
EfficientNet-B0	137	P1	90.29	80.62	79.53	5.65	5.0	5
EfficientNet-B0	137	P1-prime	89.99	79.20	78.79	6.21	5.0	3
EfficientNet-B0	137	P2	77.16	57.77	57.65	15.12	5.0	5
EfficientNet-B0	173	P1	91.89	82.77	81.66	4.80	5.0	4
EfficientNet-B0	173	P1-prime	91.72	82.68	82.66	4.72	5.0	5
EfficientNet-B0	173	P2	81.48	63.09	61.81	12.47	5.0	4
EfficientNet-B0	211	P1	90.68	79.19	77.49	5.24	5.0	3
EfficientNet-B0	211	P1-prime	91.29	81.66	81.54	4.73	5.0	4
EfficientNet-B0	211	P2	81.77	62.82	63.40	11.79	5.0	3
EfficientNet-B0	257	P1	91.55	81.41	81.44	4.73	5.0	5
EfficientNet-B0	257	P1-prime	91.72	82.07	82.64	4.60	5.0	5
EfficientNet-B0	257	P2	76.05	57.29	59.00	16.76	5.0	3
EfficientNet-B0	307	P1	90.55	81.10	79.85	5.19	5.0	0
EfficientNet-B0	307	P1-prime	91.20	81.63	82.06	5.40	5.0	3
EfficientNet-B0	307	P2	77.83	57.52	60.43	15.45	5.0	0
EfficientNet-B0	349	P1	90.46	80.08	78.06	6.16	5.0	4
EfficientNet-B0	349	P1-prime	90.29	80.27	79.07	5.81	5.0	0
EfficientNet-B0	349	P2	78.65	59.36	61.94	14.72	5.0	4
EfficientNet-B0	401	P1	90.85	80.96	80.29	5.21	5.0	5
EfficientNet-B0	401	P1-prime	90.55	80.38	81.88	5.52	5.0	4
EfficientNet-B0	401	P2	78.61	60.63	59.86	14.30	5.0	5
EfficientNet-B0	457	P1	90.51	79.73	79.60	5.56	5.0	3
EfficientNet-B0	457	P1-prime	91.24	80.90	80.88	4.81	5.0	5
EfficientNet-B0	457	P2	76.06	57.36	60.23	15.85	5.0	3
EfficientNet-B0	503	P1	90.64	81.14	81.41	5.64	5.0	4
EfficientNet-B0	503	P1-prime	90.46	80.65	80.62	5.39	5.0	3
EfficientNet-B0	503	P2	77.96	57.44	57.99	15.16	5.0	4
EfficientNet-B0	557	P1	91.33	80.42	78.46	5.08	5.0	0
EfficientNet-B0	557	P1-prime	91.03	80.06	78.17	4.89	5.0	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
EfficientNet-B0	557	P2	75.45	54.07	56.27	17.44	5.0	0
ViT-B/16	137	P1	90.55	79.71	78.55	5.74	25.9	5
ViT-B/16	137	P1-prime	90.20	80.75	80.50	6.32	25.8	3
ViT-B/16	137	P2	78.29	58.37	54.71	16.29	25.9	4
ViT-B/16	173	P1	89.99	78.02	79.06	6.45	26.0	4
ViT-B/16	173	P1-prime	89.60	78.42	75.28	6.52	25.7	0
ViT-B/16	173	P2	79.27	59.92	60.53	15.11	25.9	3
ViT-B/16	211	P1	90.51	80.80	79.57	6.04	25.9	3
ViT-B/16	211	P1-prime	89.94	77.45	75.29	6.05	25.7	0
ViT-B/16	211	P2	80.21	60.21	58.31	14.40	25.9	5
ViT-B/16	257	P1	89.03	77.25	75.47	7.99	25.9	3
ViT-B/16	257	P1-prime	90.12	77.30	75.69	6.20	26.0	4
ViT-B/16	257	P2	76.18	60.00	60.82	16.76	25.7	0
ViT-B/16	307	P1	90.20	80.76	79.34	5.77	25.7	0
ViT-B/16	307	P1-prime	88.99	77.43	76.11	6.22	25.9	5
ViT-B/16	307	P2	77.22	59.71	57.55	17.49	25.9	4
ViT-B/16	349	P1	89.55	73.29	70.14	7.18	26.0	4
ViT-B/16	349	P1-prime	88.77	74.45	70.14	8.36	25.8	3
ViT-B/16	349	P2	77.83	57.24	55.76	17.61	25.8	5
ViT-B/16	401	P1	88.77	77.35	71.95	8.24	25.9	5
ViT-B/16	401	P1-prime	90.03	80.54	79.28	6.51	25.7	0
ViT-B/16	401	P2	78.65	59.91	56.37	16.21	25.8	3
ViT-B/16	457	P1	90.51	79.10	77.73	5.66	25.8	3
ViT-B/16	457	P1-prime	89.81	78.89	75.15	6.03	25.7	0
ViT-B/16	457	P2	78.70	61.55	58.62	15.28	26.0	4
ViT-B/16	503	P1	89.47	75.79	75.20	6.49	26.0	4
ViT-B/16	503	P1-prime	88.08	79.19	78.85	6.87	25.8	5
ViT-B/16	503	P2	77.35	57.28	54.37	17.84	25.7	0
ViT-B/16	557	P1	89.73	78.02	76.66	5.90	25.7	0
ViT-B/16	557	P1-prime	89.86	78.34	78.26	6.38	25.8	3
ViT-B/16	557	P2	78.35	61.15	59.20	14.42	25.8	5
ConvNeXt-T	137	P1	91.55	83.50	79.91	5.85	14.3	3
ConvNeXt-T	137	P1-prime	90.77	80.41	77.64	6.69	14.4	4
ConvNeXt-T	137	P2	78.63	60.52	56.01	17.18	14.2	0
ConvNeXt-T	173	P1	91.94	83.44	83.16	5.38	14.3	0
ConvNeXt-T	173	P1-prime	90.98	82.30	79.95	5.54	14.3	3
ConvNeXt-T	173	P2	81.52	61.62	60.22	14.36	14.3	5
ConvNeXt-T	211	P1	90.38	81.37	80.77	5.95	14.3	0
ConvNeXt-T	211	P1-prime	91.29	83.19	81.53	5.62	14.3	5
ConvNeXt-T	211	P2	81.81	62.91	59.70	13.85	14.4	4
ConvNeXt-T	257	P1	91.55	82.04	81.67	5.60	14.4	4
ConvNeXt-T	257	P1-prime	91.63	81.60	80.39	5.65	14.2	0
ConvNeXt-T	257	P2	80.77	64.19	65.05	14.61	14.3	5
ConvNeXt-T	307	P1	91.68	83.12	82.39	5.21	14.3	5
ConvNeXt-T	307	P1-prime	91.68	81.84	79.38	5.67	14.3	4
ConvNeXt-T	307	P2	77.78	61.31	61.71	16.69	14.3	3
ConvNeXt-T	349	P1	91.33	80.34	76.30	6.26	14.3	3
ConvNeXt-T	349	P1-prime	90.98	77.69	75.88	5.88	14.2	5
ConvNeXt-T	349	P2	79.08	60.47	60.42	16.43	14.2	0
ConvNeXt-T	401	P1	90.59	80.19	75.89	6.90	14.2	0
ConvNeXt-T	401	P1-prime	90.94	81.39	81.60	5.84	14.3	3
ConvNeXt-T	401	P2	77.79	58.68	56.94	16.55	14.4	4
ConvNeXt-T	457	P1	90.64	78.90	76.97	6.27	14.3	0
ConvNeXt-T	457	P1-prime	90.55	77.45	75.59	6.99	14.4	4
ConvNeXt-T	457	P2	80.39	64.21	65.34	14.58	14.3	5
ConvNeXt-T	503	P1	90.98	82.89	82.26	6.52	14.3	5
ConvNeXt-T	503	P1-prime	90.68	82.10	79.29	6.61	14.2	0
ConvNeXt-T	503	P2	80.51	64.10	61.27	14.46	14.3	3
ConvNeXt-T	557	P1	92.28	82.23	81.03	5.16	14.3	3
ConvNeXt-T	557	P1-prime	91.46	79.34	77.21	5.69	14.3	5
ConvNeXt-T	557	P2	76.53	53.49	52.30	19.50	14.3	4

Table L.3: Complete 2C/2B run ledger (160 runs; protocol is the factorial cell).

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	137	clean_high	97.13	97.00	97.04	2.09	1.1	4
ResNet-50	137	clean_low	96.36	96.13	96.18	2.09	1.1	0
ResNet-50	137	leak_high	97.13	96.93	96.91	1.40	1.1	5
ResNet-50	137	leak_low	97.89	97.77	97.77	1.18	1.1	5

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	173	clean_high	95.55	95.56	95.51	1.42	1.1	4
ResNet-50	173	clean_low	94.97	94.95	95.03	2.09	1.1	0
ResNet-50	173	leak_high	96.71	96.64	96.74	1.29	1.1	3
ResNet-50	173	leak_low	96.71	96.73	96.70	1.46	1.1	5
ResNet-50	211	clean_high	94.41	94.31	94.35	3.02	1.1	4
ResNet-50	211	clean_low	94.41	94.25	94.31	2.29	1.1	0
ResNet-50	211	leak_high	96.15	96.06	96.05	2.16	1.1	3
ResNet-50	211	leak_low	97.50	97.44	97.44	2.04	1.1	5
ResNet-50	257	clean_high	94.07	93.82	93.79	3.46	1.1	4
ResNet-50	257	clean_low	95.03	94.84	94.89	2.01	1.1	3
ResNet-50	257	leak_high	96.56	96.35	96.21	1.61	1.1	3
ResNet-50	257	leak_low	96.75	96.62	96.76	2.22	1.1	5
ResNet-50	307	clean_high	94.92	94.78	94.64	1.98	1.1	4
ResNet-50	307	clean_low	93.95	93.71	93.66	3.47	1.1	3
ResNet-50	307	leak_high	95.51	95.31	95.24	1.80	1.1	0
ResNet-50	307	leak_low	97.85	97.72	97.70	1.27	1.1	5
ResNet-50	349	clean_high	92.29	92.11	92.19	3.53	1.1	4
ResNet-50	349	clean_low	94.74	94.59	94.64	2.75	1.1	3
ResNet-50	349	leak_high	94.92	94.81	94.88	1.91	1.1	0
ResNet-50	349	leak_low	97.18	97.05	97.09	1.64	1.1	5
ResNet-50	401	clean_high	92.07	91.62	91.82	4.05	1.1	4
ResNet-50	401	clean_low	93.23	93.02	93.01	4.14	1.1	0
ResNet-50	401	leak_high	95.16	94.97	94.93	2.54	1.1	0
ResNet-50	401	leak_low	94.78	94.56	94.54	3.25	1.1	5
ResNet-50	457	clean_high	96.34	96.18	96.14	2.44	1.1	4
ResNet-50	457	clean_low	95.80	95.61	95.50	3.17	1.1	0
ResNet-50	457	leak_high	96.89	96.73	96.75	1.29	1.1	3
ResNet-50	457	leak_low	96.16	96.03	95.93	1.85	1.1	5
ResNet-50	503	clean_high	95.28	95.18	95.09	2.20	1.1	4
ResNet-50	503	clean_low	95.85	95.80	95.82	1.97	1.1	3
ResNet-50	503	leak_high	96.42	96.36	96.35	2.08	1.1	3
ResNet-50	503	leak_low	97.55	97.48	97.54	1.91	1.1	5
ResNet-50	557	clean_high	93.64	93.51	93.40	3.58	1.1	4
ResNet-50	557	clean_low	94.55	94.45	94.41	2.81	1.1	3
ResNet-50	557	leak_high	95.64	95.50	95.55	2.45	1.1	0
ResNet-50	557	leak_low	95.64	95.56	95.46	2.43	1.1	5
DenseNet-121	137	clean_high	96.93	96.76	96.84	1.27	1.3	4
DenseNet-121	137	clean_low	94.25	93.97	93.92	2.00	1.3	0
DenseNet-121	137	leak_high	97.51	97.29	97.19	0.98	1.3	3
DenseNet-121	137	leak_low	96.36	96.06	96.03	1.35	1.3	5
DenseNet-121	173	clean_high	94.58	94.54	94.62	2.99	1.3	4
DenseNet-121	173	clean_low	96.32	96.26	96.36	2.13	1.3	0
DenseNet-121	173	leak_high	96.52	96.44	96.36	1.65	1.3	3
DenseNet-121	173	leak_low	97.68	97.69	97.69	1.08	1.3	5
DenseNet-121	211	clean_high	95.95	95.89	95.92	1.19	1.3	4
DenseNet-121	211	clean_low	95.18	95.03	95.06	2.73	1.3	0
DenseNet-121	211	leak_high	97.69	97.62	97.61	1.65	1.4	3
DenseNet-121	211	leak_low	97.69	97.62	97.63	1.82	1.3	5
DenseNet-121	257	clean_high	94.84	94.62	94.75	1.74	1.3	4
DenseNet-121	257	clean_low	96.75	96.63	96.80	1.67	1.4	3
DenseNet-121	257	leak_high	95.98	95.70	95.76	1.42	1.3	0
DenseNet-121	257	leak_low	97.51	97.38	97.38	0.95	1.3	5
DenseNet-121	307	clean_high	94.53	94.19	94.12	3.35	1.3	4
DenseNet-121	307	clean_low	94.14	93.89	93.88	2.61	1.3	3
DenseNet-121	307	leak_high	98.05	97.99	98.03	1.11	1.3	0
DenseNet-121	307	leak_low	97.85	97.76	97.73	1.36	1.3	5
DenseNet-121	349	clean_high	92.67	92.40	92.46	3.21	1.3	4
DenseNet-121	349	clean_low	90.41	90.56	90.38	4.62	1.3	3
DenseNet-121	349	leak_high	96.80	96.66	96.69	2.06	1.3	0
DenseNet-121	349	leak_low	96.80	96.69	96.71	2.57	1.3	5
DenseNet-121	401	clean_high	93.04	92.69	92.79	3.22	1.3	4
DenseNet-121	401	clean_low	94.58	94.40	94.42	3.65	1.3	0
DenseNet-121	401	leak_high	95.94	95.77	95.79	2.01	1.3	3
DenseNet-121	401	leak_low	95.94	95.77	95.74	2.37	1.3	5
DenseNet-121	457	clean_high	95.61	95.36	95.26	2.48	1.3	4
DenseNet-121	457	clean_low	94.33	94.15	94.03	3.07	1.3	0
DenseNet-121	457	leak_high	97.62	97.49	97.51	1.29	1.3	3
DenseNet-121	457	leak_low	97.07	96.91	96.80	1.15	1.3	5
DenseNet-121	503	clean_high	95.66	95.55	95.45	2.13	1.3	4
DenseNet-121	503	clean_low	95.85	95.78	95.85	2.13	1.3	3
DenseNet-121	503	leak_high	96.04	95.94	95.98	1.72	1.3	0
DenseNet-121	503	leak_low	97.17	97.14	97.18	1.40	1.3	5
DenseNet-121	557	clean_high	94.00	93.84	93.68	2.54	1.3	4

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	557	clean_low	94.91	94.81	94.73	3.56	1.4	3
DenseNet-121	557	leak_high	95.45	95.34	95.28	2.35	1.3	0
DenseNet-121	557	leak_low	97.09	97.00	96.97	2.05	1.3	5
ViT-B/16	137	clean_high	95.79	95.58	95.56	2.85	3.6	4
ViT-B/16	137	clean_low	96.36	96.15	96.23	2.00	3.6	0
ViT-B/16	137	leak_high	98.08	97.96	97.96	1.33	3.6	3
ViT-B/16	137	leak_low	94.83	94.52	94.35	3.53	3.6	5
ViT-B/16	173	clean_high	88.97	88.31	89.87	7.64	3.6	4
ViT-B/16	173	clean_low	90.52	90.26	90.81	6.02	3.5	0
ViT-B/16	173	leak_high	95.94	95.97	96.03	2.33	3.6	3
ViT-B/16	173	leak_low	95.16	95.20	95.13	2.87	3.6	5
ViT-B/16	211	clean_high	96.15	96.09	96.13	2.76	3.6	4
ViT-B/16	211	clean_low	95.38	95.35	95.37	2.24	3.5	0
ViT-B/16	211	leak_high	97.30	97.19	97.19	1.57	3.6	3
ViT-B/16	211	leak_low	98.46	98.41	98.39	1.27	3.6	5
ViT-B/16	257	clean_high	93.88	93.54	93.82	4.53	3.6	4
ViT-B/16	257	clean_low	93.69	93.40	94.04	4.79	3.6	3
ViT-B/16	257	leak_high	94.84	94.62	94.61	3.77	3.6	0
ViT-B/16	257	leak_low	97.32	97.21	97.29	1.43	3.6	5
ViT-B/16	307	clean_high	95.90	95.73	95.76	3.03	3.6	4
ViT-B/16	307	clean_low	93.36	93.28	93.59	4.21	3.6	3
ViT-B/16	307	leak_high	95.90	95.83	95.85	2.28	3.6	0
ViT-B/16	307	leak_low	95.51	95.30	95.33	3.71	3.5	5
ViT-B/16	349	clean_high	92.29	92.04	92.11	5.83	3.6	4
ViT-B/16	349	clean_low	94.92	94.73	94.76	3.40	3.6	3
ViT-B/16	349	leak_high	95.11	94.96	95.08	3.71	3.6	0
ViT-B/16	349	leak_low	96.80	96.68	96.69	2.30	3.6	5
ViT-B/16	401	clean_high	93.62	93.37	93.43	4.64	3.6	4
ViT-B/16	401	clean_low	92.84	92.63	92.71	4.85	3.5	0
ViT-B/16	401	leak_high	95.16	94.99	95.03	3.55	3.6	3
ViT-B/16	401	leak_low	96.13	95.95	96.00	2.92	3.6	5
ViT-B/16	457	clean_high	95.61	95.44	95.30	3.26	3.6	4
ViT-B/16	457	clean_low	95.80	95.63	95.64	3.13	3.6	0
ViT-B/16	457	leak_high	94.88	94.62	94.43	3.69	3.6	3
ViT-B/16	457	leak_low	95.98	95.82	95.77	2.07	3.6	5
ViT-B/16	503	clean_high	93.21	92.98	93.05	3.40	3.6	4
ViT-B/16	503	clean_low	92.26	91.88	92.19	4.57	3.6	3
ViT-B/16	503	leak_high	96.98	96.90	96.89	1.42	3.5	0
ViT-B/16	503	leak_low	95.47	95.44	95.46	3.08	3.6	5
ViT-B/16	557	clean_high	95.09	94.93	94.97	2.98	3.6	4
ViT-B/16	557	clean_low	96.00	95.84	95.94	2.81	3.6	3
ViT-B/16	557	leak_high	95.64	95.49	95.60	3.09	3.6	0
ViT-B/16	557	leak_low	94.55	94.45	94.47	4.34	3.6	5
ConvNeXt-T	137	clean_high	96.93	96.79	96.96	2.59	2.1	4
ConvNeXt-T	137	clean_low	94.64	94.39	94.54	3.67	2.1	0
ConvNeXt-T	137	leak_high	97.13	96.85	96.91	2.27	2.1	3
ConvNeXt-T	137	leak_low	95.98	95.84	95.94	2.26	2.1	5
ConvNeXt-T	173	clean_high	95.36	95.33	95.26	2.70	2.1	4
ConvNeXt-T	173	clean_low	94.20	94.24	94.02	3.54	2.1	0
ConvNeXt-T	173	leak_high	95.36	95.41	95.34	2.67	2.1	3
ConvNeXt-T	173	leak_low	95.16	95.20	95.41	3.65	2.1	5
ConvNeXt-T	211	clean_high	96.34	96.27	96.32	3.02	2.1	4
ConvNeXt-T	211	clean_low	97.88	97.80	97.79	1.68	2.1	0
ConvNeXt-T	211	leak_high	99.04	99.02	99.01	1.35	2.1	3
ConvNeXt-T	211	leak_low	95.95	95.84	95.90	2.32	2.1	5
ConvNeXt-T	257	clean_high	94.07	93.87	94.34	4.28	2.2	4
ConvNeXt-T	257	clean_low	94.26	94.16	94.18	3.09	2.1	3
ConvNeXt-T	257	leak_high	95.98	95.81	96.22	2.88	2.1	0
ConvNeXt-T	257	leak_low	97.13	97.06	96.91	2.09	2.1	5
ConvNeXt-T	307	clean_high	97.07	96.93	96.89	2.07	2.1	4
ConvNeXt-T	307	clean_low	96.68	96.60	96.69	2.07	2.1	3
ConvNeXt-T	307	leak_high	98.63	98.57	98.57	1.02	2.0	0
ConvNeXt-T	307	leak_low	98.24	98.14	98.09	1.53	2.0	5
ConvNeXt-T	349	clean_high	93.98	93.74	93.86	4.28	2.1	4
ConvNeXt-T	349	clean_low	96.05	95.97	96.07	3.56	2.1	3
ConvNeXt-T	349	leak_high	95.30	95.14	95.21	2.45	2.1	0
ConvNeXt-T	349	leak_low	95.11	94.97	94.93	2.78	2.1	5
ConvNeXt-T	401	clean_high	93.42	93.20	93.20	4.60	2.1	4
ConvNeXt-T	401	clean_low	93.62	93.45	93.48	5.80	2.1	0
ConvNeXt-T	401	leak_high	95.74	95.52	95.61	3.01	2.1	3
ConvNeXt-T	401	leak_low	93.62	93.39	93.28	4.99	2.1	5
ConvNeXt-T	457	clean_high	95.80	95.61	95.62	2.98	2.1	4
ConvNeXt-T	457	clean_low	91.77	91.61	91.28	5.48	2.1	0

Architecture	Seed	Protocol	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	457	leak_high	95.98	95.79	95.68	2.14	2.1	3
ConvNeXt-T	457	leak_low	96.53	96.34	96.42	2.14	2.1	5
ConvNeXt-T	503	clean_high	95.09	94.96	94.96	2.85	2.1	4
ConvNeXt-T	503	clean_low	95.66	95.54	95.64	2.74	2.1	3
ConvNeXt-T	503	leak_high	95.66	95.52	95.53	2.98	2.1	0
ConvNeXt-T	503	leak_low	96.23	96.13	96.25	2.83	2.2	5
ConvNeXt-T	557	clean_high	94.00	93.82	93.66	4.62	2.1	4
ConvNeXt-T	557	clean_low	94.73	94.59	94.53	4.14	2.1	3
ConvNeXt-T	557	leak_high	94.00	93.62	93.45	4.78	2.1	0
ConvNeXt-T	557	leak_low	96.00	95.88	95.90	3.00	2.1	5

Table L.4: Complete 2A gradient-reversal run ledger (280 runs).

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	137	P1	0	97.65	97.66	97.68	1.05	2.7	0
ResNet-50	137	P2	0	95.19	95.24	95.18	2.87	2.6	4
ResNet-50	173	P1	0	97.78	97.80	97.79	1.03	2.6	0
ResNet-50	173	P2	0	95.93	95.97	95.96	2.05	2.6	4
ResNet-50	211	P1	0	97.41	97.42	97.42	1.47	2.6	5
ResNet-50	211	P2	0	96.30	96.32	96.27	2.20	2.6	4
ResNet-50	257	P1	0	97.41	97.42	97.44	1.59	2.6	0
ResNet-50	257	P2	0	96.67	96.68	96.66	2.07	2.6	3
ResNet-50	307	P1	0	98.27	98.28	98.28	1.29	2.6	5
ResNet-50	307	P2	0	97.28	97.28	97.29	1.75	2.6	4
ResNet-50	349	P1	0	96.67	96.69	96.67	2.11	2.6	0
ResNet-50	349	P2	0	96.42	96.45	96.42	2.23	2.6	3
ResNet-50	401	P1	0	98.02	98.03	98.03	1.38	2.6	5
ResNet-50	401	P2	0	97.41	97.42	97.43	1.32	2.6	4
ResNet-50	457	P1	0	98.15	98.14	98.18	1.18	2.6	0
ResNet-50	457	P2	0	94.32	94.31	94.35	3.88	2.6	3
ResNet-50	503	P1	0	97.53	97.54	97.56	1.35	2.6	5
ResNet-50	503	P2	0	96.79	96.79	96.78	2.09	2.6	4
ResNet-50	557	P1	0	97.53	97.53	97.55	1.29	2.6	0
ResNet-50	557	P2	0	98.40	98.40	98.40	1.16	2.6	3
ResNet-50	137	P1	0.01	98.02	98.03	98.05	1.04	2.6	5
ResNet-50	137	P2	0.01	96.54	96.54	96.57	1.88	2.6	3
ResNet-50	173	P1	0.01	96.67	96.67	96.70	2.00	2.6	0
ResNet-50	173	P2	0.01	96.42	96.46	96.44	1.72	2.6	5
ResNet-50	211	P1	0.01	97.53	97.55	97.54	1.36	2.6	5
ResNet-50	211	P2	0.01	96.54	96.57	96.54	2.11	2.6	4
ResNet-50	257	P1	0.01	97.53	97.52	97.56	1.65	2.6	0
ResNet-50	257	P2	0.01	98.02	98.02	98.03	1.21	2.6	3
ResNet-50	307	P1	0.01	96.79	96.81	96.78	1.26	2.6	5
ResNet-50	307	P2	0.01	97.53	97.53	97.53	1.19	2.6	4
ResNet-50	349	P1	0.01	97.65	97.67	97.67	1.17	2.6	0
ResNet-50	349	P2	0.01	97.41	97.42	97.42	1.15	2.6	3
ResNet-50	401	P1	0.01	97.04	97.05	97.07	1.61	2.6	5
ResNet-50	401	P2	0.01	97.28	97.29	97.32	1.80	2.6	4
ResNet-50	457	P1	0.01	98.52	98.52	98.53	1.04	2.6	0
ResNet-50	457	P2	0.01	95.56	95.51	95.59	3.58	2.6	3
ResNet-50	503	P1	0.01	97.28	97.29	97.30	1.62	2.6	5
ResNet-50	503	P2	0.01	96.54	96.56	96.53	2.40	2.6	4
ResNet-50	557	P1	0.01	97.16	97.17	97.16	2.31	2.6	0
ResNet-50	557	P2	0.01	96.79	96.79	96.81	2.13	2.6	3
ResNet-50	137	P1	0.03	96.05	96.05	96.09	2.72	2.7	0
ResNet-50	137	P2	0.03	95.56	95.55	95.57	2.81	2.6	4
ResNet-50	173	P1	0.03	97.04	97.04	97.05	1.39	2.7	3
ResNet-50	173	P2	0.03	97.41	97.43	97.42	2.14	2.7	4
ResNet-50	211	P1	0.03	97.41	97.44	97.41	1.61	2.6	0
ResNet-50	211	P2	0.03	95.56	95.60	95.54	2.77	2.6	3
ResNet-50	257	P1	0.03	97.28	97.29	97.30	1.77	2.6	5
ResNet-50	257	P2	0.03	96.42	96.42	96.41	1.96	2.6	4
ResNet-50	307	P1	0.03	95.68	95.74	95.69	2.32	2.6	0
ResNet-50	307	P2	0.03	97.28	97.28	97.28	1.08	2.6	3
ResNet-50	349	P1	0.03	97.41	97.42	97.41	1.35	2.6	5
ResNet-50	349	P2	0.03	96.79	96.82	96.81	1.79	2.6	4
ResNet-50	401	P1	0.03	95.68	95.73	95.68	2.35	2.6	0
ResNet-50	401	P2	0.03	97.65	97.66	97.65	1.61	2.6	3
ResNet-50	457	P1	0.03	97.53	97.54	97.54	1.76	2.6	5

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	457	P2	0.03	95.43	95.41	95.47	3.32	2.6	4
ResNet-50	503	P1	0.03	98.02	98.05	98.03	1.44	2.6	0
ResNet-50	503	P2	0.03	97.16	97.18	97.16	1.83	2.6	3
ResNet-50	557	P1	0.03	97.04	97.06	97.06	2.21	2.6	5
ResNet-50	557	P2	0.03	97.04	97.04	97.06	1.77	2.6	4
ResNet-50	137	P1	0.1	96.91	96.92	96.94	1.87	2.6	5
ResNet-50	137	P2	0.1	94.81	94.83	94.83	3.44	2.6	3
ResNet-50	173	P1	0.1	96.67	96.68	96.68	1.66	2.6	0
ResNet-50	173	P2	0.1	96.67	96.68	96.68	1.61	2.6	5
ResNet-50	211	P1	0.1	96.91	96.91	96.93	1.30	2.6	5
ResNet-50	211	P2	0.1	96.54	96.59	96.53	2.27	2.6	4
ResNet-50	257	P1	0.1	96.67	96.65	96.69	2.06	2.6	0
ResNet-50	257	P2	0.1	97.04	97.04	97.04	2.03	2.6	3
ResNet-50	307	P1	0.1	97.28	97.28	97.30	1.35	2.6	5
ResNet-50	307	P2	0.1	96.67	96.67	96.65	2.19	2.6	4
ResNet-50	349	P1	0.1	97.04	97.05	97.05	1.46	2.6	0
ResNet-50	349	P2	0.1	97.04	97.07	97.03	1.37	2.6	3
ResNet-50	401	P1	0.1	96.91	96.93	96.94	1.71	2.6	5
ResNet-50	401	P2	0.1	97.16	97.18	97.17	2.14	2.6	4
ResNet-50	457	P1	0.1	97.90	97.91	97.90	1.06	2.6	0
ResNet-50	457	P2	0.1	94.32	94.26	94.36	4.31	2.6	3
ResNet-50	503	P1	0.1	96.67	96.71	96.66	2.24	2.6	5
ResNet-50	503	P2	0.1	96.79	96.81	96.81	1.76	2.6	4
ResNet-50	557	P1	0.1	96.79	96.80	96.80	1.43	2.6	0
ResNet-50	557	P2	0.1	97.04	97.04	97.06	1.73	2.6	3
ResNet-50	137	P1	0.3	97.41	97.40	97.44	1.17	2.6	0
ResNet-50	137	P2	0.3	95.56	95.57	95.57	2.59	2.6	4
ResNet-50	173	P1	0.3	95.68	95.69	95.69	1.80	2.6	3
ResNet-50	173	P2	0.3	97.04	97.04	97.05	0.85	2.7	4
ResNet-50	211	P1	0.3	96.79	96.80	96.82	1.45	2.6	0
ResNet-50	211	P2	0.3	96.42	96.45	96.41	2.13	2.6	3
ResNet-50	257	P1	0.3	97.41	97.40	97.43	1.68	2.6	5
ResNet-50	257	P2	0.3	98.02	98.03	98.04	1.65	2.6	4
ResNet-50	307	P1	0.3	98.02	98.03	98.04	1.22	2.6	0
ResNet-50	307	P2	0.3	96.67	96.67	96.65	1.36	2.6	3
ResNet-50	349	P1	0.3	95.93	95.95	95.92	1.88	2.6	5
ResNet-50	349	P2	0.3	97.65	97.67	97.66	0.96	2.6	4
ResNet-50	401	P1	0.3	96.79	96.80	96.81	1.33	2.6	0
ResNet-50	401	P2	0.3	95.93	95.91	95.97	2.46	2.6	3
ResNet-50	457	P1	0.3	98.02	98.03	98.05	1.42	2.6	5
ResNet-50	457	P2	0.3	95.06	95.04	95.10	4.16	2.6	4
ResNet-50	503	P1	0.3	97.04	97.05	97.04	1.00	2.6	0
ResNet-50	503	P2	0.3	96.05	96.05	96.01	2.57	2.6	3
ResNet-50	557	P1	0.3	97.04	97.06	97.04	1.47	2.6	5
ResNet-50	557	P2	0.3	97.04	97.05	97.05	1.52	2.6	4
ResNet-50	137	P1	1	96.67	96.68	96.70	1.17	2.6	5
ResNet-50	137	P2	1	95.56	95.57	95.58	1.61	2.7	3
ResNet-50	173	P1	1	97.16	97.16	97.19	1.38	2.6	0
ResNet-50	173	P2	1	97.41	97.41	97.44	1.54	2.6	3
ResNet-50	211	P1	1	96.79	96.82	96.80	1.64	2.6	5
ResNet-50	211	P2	1	95.56	95.60	95.55	1.84	2.6	4
ResNet-50	257	P1	1	96.54	96.54	96.56	1.96	2.6	0
ResNet-50	257	P2	1	96.42	96.43	96.43	1.27	2.6	3
ResNet-50	307	P1	1	95.68	95.73	95.69	1.70	2.6	5
ResNet-50	307	P2	1	97.65	97.65	97.65	0.85	2.6	4
ResNet-50	349	P1	1	96.91	96.92	96.91	1.44	2.6	0
ResNet-50	349	P2	1	95.80	95.84	95.80	1.87	2.6	3
ResNet-50	401	P1	1	95.93	95.92	95.95	2.13	2.6	5
ResNet-50	401	P2	1	96.67	96.68	96.66	1.38	2.6	4
ResNet-50	457	P1	1	98.40	98.40	98.41	1.28	2.6	0
ResNet-50	457	P2	1	94.94	94.91	94.97	2.62	2.6	3
ResNet-50	503	P1	1	97.41	97.42	97.42	1.21	2.6	5
ResNet-50	503	P2	1	97.41	97.42	97.42	0.77	2.6	4
ResNet-50	557	P1	1	96.79	96.80	96.82	1.60	2.6	0
ResNet-50	557	P2	1	97.65	97.66	97.67	0.83	2.6	3
ResNet-50	137	P1	3	96.30	96.28	96.34	1.67	2.6	0
ResNet-50	137	P2	3	95.80	95.81	95.82	1.13	2.7	4
ResNet-50	173	P1	3	95.93	95.93	95.91	1.23	2.7	5
ResNet-50	173	P2	3	96.17	96.16	96.20	1.32	2.6	4
ResNet-50	211	P1	3	97.41	97.42	97.43	1.13	2.6	0
ResNet-50	211	P2	3	96.17	96.21	96.15	1.04	2.6	3
ResNet-50	257	P1	3	97.04	97.05	97.05	1.23	2.6	5
ResNet-50	257	P2	3	97.78	97.78	97.79	1.67	2.6	4

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	307	P1	3	97.28	97.29	97.30	1.37	2.6	0
ResNet-50	307	P2	3	96.54	96.54	96.54	2.14	2.6	3
ResNet-50	349	P1	3	95.80	95.80	95.80	2.80	2.6	5
ResNet-50	349	P2	3	94.44	94.42	94.50	1.90	2.6	4
ResNet-50	401	P1	3	95.56	95.59	95.57	1.71	2.6	0
ResNet-50	401	P2	3	96.79	96.80	96.81	0.77	2.6	3
ResNet-50	457	P1	3	97.41	97.40	97.43	1.53	2.6	5
ResNet-50	457	P2	3	93.58	93.52	93.59	3.44	2.6	4
ResNet-50	503	P1	3	96.67	96.67	96.69	1.33	2.6	0
ResNet-50	503	P2	3	95.56	95.58	95.54	1.20	2.6	3
ResNet-50	557	P1	3	97.41	97.43	97.42	1.75	2.6	5
ResNet-50	557	P2	3	96.91	96.91	96.93	1.72	2.6	4
ConvNeXt-T	137	P1	0	98.15	98.16	98.16	1.55	5.2	5
ConvNeXt-T	137	P2	0	96.79	96.79	96.81	2.44	5.2	3
ConvNeXt-T	173	P1	0	97.90	97.90	97.89	1.43	5.1	0
ConvNeXt-T	173	P2	0	96.17	96.19	96.17	2.90	5.2	3
ConvNeXt-T	211	P1	0	97.90	97.90	97.91	1.64	5.2	5
ConvNeXt-T	211	P2	0	96.54	96.56	96.54	2.68	5.2	4
ConvNeXt-T	257	P1	0	97.90	97.90	97.91	1.83	5.2	0
ConvNeXt-T	257	P2	0	97.16	97.16	97.16	2.07	5.2	3
ConvNeXt-T	307	P1	0	98.27	98.28	98.28	1.22	5.2	5
ConvNeXt-T	307	P2	0	96.67	96.68	96.66	2.58	5.2	4
ConvNeXt-T	349	P1	0	97.78	97.77	97.80	1.83	5.2	0
ConvNeXt-T	349	P2	0	93.83	93.89	93.83	5.03	5.2	3
ConvNeXt-T	401	P1	0	97.53	97.55	97.56	1.51	5.2	5
ConvNeXt-T	401	P2	0	98.40	98.40	98.41	1.20	5.2	4
ConvNeXt-T	457	P1	0	98.64	98.65	98.66	1.16	5.2	0
ConvNeXt-T	457	P2	0	95.56	95.55	95.57	3.12	5.2	3
ConvNeXt-T	503	P1	0	97.28	97.30	97.31	1.90	5.2	5
ConvNeXt-T	503	P2	0	97.28	97.28	97.29	1.58	5.2	4
ConvNeXt-T	557	P1	0	96.67	96.70	96.67	2.26	5.2	0
ConvNeXt-T	557	P2	0	98.15	98.16	98.14	1.33	5.2	3
ConvNeXt-T	137	P1	0.01	98.02	98.04	98.04	1.83	5.2	0
ConvNeXt-T	137	P2	0.01	96.79	96.79	96.83	2.16	5.2	4
ConvNeXt-T	173	P1	0.01	97.53	97.54	97.53	1.75	5.2	5
ConvNeXt-T	173	P2	0.01	97.41	97.41	97.42	1.88	5.2	4
ConvNeXt-T	211	P1	0.01	96.91	96.92	96.89	2.16	5.2	0
ConvNeXt-T	211	P2	0.01	96.05	96.07	96.06	3.30	5.2	3
ConvNeXt-T	257	P1	0.01	97.16	97.14	97.20	2.04	5.2	5
ConvNeXt-T	257	P2	0.01	97.53	97.51	97.56	1.75	5.2	4
ConvNeXt-T	307	P1	0.01	97.78	97.77	97.81	1.74	5.1	0
ConvNeXt-T	307	P2	0.01	97.53	97.53	97.53	1.74	5.2	3
ConvNeXt-T	349	P1	0.01	96.42	96.43	96.40	2.58	5.2	5
ConvNeXt-T	349	P2	0.01	97.28	97.30	97.29	1.63	5.2	4
ConvNeXt-T	401	P1	0.01	97.53	97.54	97.56	1.83	5.2	0
ConvNeXt-T	401	P2	0.01	96.67	96.66	96.73	2.77	5.2	3
ConvNeXt-T	457	P1	0.01	98.77	98.77	98.78	1.17	5.2	5
ConvNeXt-T	457	P2	0.01	95.31	95.28	95.35	4.46	5.2	4
ConvNeXt-T	503	P1	0.01	97.65	97.66	97.66	1.85	5.2	0
ConvNeXt-T	503	P2	0.01	97.16	97.17	97.19	2.57	5.2	3
ConvNeXt-T	557	P1	0.01	95.31	95.32	95.35	3.35	5.2	5
ConvNeXt-T	557	P2	0.01	96.42	96.43	96.44	2.67	5.2	4
ConvNeXt-T	137	P1	0.03	95.93	95.99	95.99	3.16	5.2	5
ConvNeXt-T	137	P2	0.03	97.04	97.05	97.05	2.52	5.2	3
ConvNeXt-T	173	P1	0.03	97.65	97.66	97.68	2.19	5.2	0
ConvNeXt-T	173	P2	0.03	97.65	97.66	97.66	1.29	5.2	3
ConvNeXt-T	211	P1	0.03	98.15	98.15	98.15	1.43	5.2	5
ConvNeXt-T	211	P2	0.03	96.42	96.45	96.43	2.82	5.2	4
ConvNeXt-T	257	P1	0.03	97.04	97.05	97.06	2.34	5.2	0
ConvNeXt-T	257	P2	0.03	97.16	97.16	97.17	2.35	5.2	3
ConvNeXt-T	307	P1	0.03	97.28	97.29	97.30	1.92	5.2	5
ConvNeXt-T	307	P2	0.03	96.17	96.17	96.17	2.80	5.2	4
ConvNeXt-T	349	P1	0.03	97.16	97.16	97.17	1.70	5.2	0
ConvNeXt-T	349	P2	0.03	96.42	96.42	96.45	3.01	5.2	3
ConvNeXt-T	401	P1	0.03	97.78	97.78	97.78	1.75	5.2	5
ConvNeXt-T	401	P2	0.03	97.90	97.92	97.88	1.09	5.2	4
ConvNeXt-T	457	P1	0.03	97.16	97.15	97.13	1.89	5.2	0
ConvNeXt-T	457	P2	0.03	93.58	93.53	93.56	5.14	5.2	3
ConvNeXt-T	503	P1	0.03	94.69	94.79	94.69	4.44	5.2	5
ConvNeXt-T	503	P2	0.03	98.15	98.16	98.15	1.43	5.2	4
ConvNeXt-T	557	P1	0.03	96.91	96.92	96.93	2.05	5.2	0
ConvNeXt-T	557	P2	0.03	97.28	97.29	97.30	2.22	5.2	3
ConvNeXt-T	137	P1	0.1	97.28	97.30	97.30	1.50	5.2	0

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	137	P2	0.1	96.42	96.42	96.44	2.67	5.2	4
ConvNeXt-T	173	P1	0.1	97.16	97.17	97.16	2.28	5.2	5
ConvNeXt-T	173	P2	0.1	97.78	97.77	97.76	1.66	5.2	4
ConvNeXt-T	211	P1	0.1	98.40	98.40	98.40	1.15	5.1	0
ConvNeXt-T	211	P2	0.1	96.67	96.69	96.66	2.69	5.2	3
ConvNeXt-T	257	P1	0.1	96.91	96.93	96.91	2.05	5.2	5
ConvNeXt-T	257	P2	0.1	97.16	97.16	97.18	2.16	5.2	4
ConvNeXt-T	307	P1	0.1	96.91	96.92	96.95	2.30	5.2	0
ConvNeXt-T	307	P2	0.1	96.91	96.90	96.92	2.59	5.2	3
ConvNeXt-T	349	P1	0.1	95.80	95.79	95.79	2.88	5.2	5
ConvNeXt-T	349	P2	0.1	96.17	96.22	96.18	2.83	5.2	4
ConvNeXt-T	401	P1	0.1	97.90	97.91	97.93	1.52	5.2	0
ConvNeXt-T	401	P2	0.1	97.28	97.29	97.30	1.37	5.2	3
ConvNeXt-T	457	P1	0.1	98.27	98.27	98.27	1.26	5.2	5
ConvNeXt-T	457	P2	0.1	95.43	95.42	95.41	3.24	5.2	4
ConvNeXt-T	503	P1	0.1	96.79	96.78	96.80	2.37	5.2	0
ConvNeXt-T	503	P2	0.1	97.53	97.55	97.54	1.90	5.2	3
ConvNeXt-T	557	P1	0.1	96.91	96.93	96.93	2.45	5.2	5
ConvNeXt-T	557	P2	0.1	98.15	98.16	98.15	1.06	5.2	4
ConvNeXt-T	137	P1	0.3	96.42	96.38	96.47	2.63	5.2	5
ConvNeXt-T	137	P2	0.3	96.91	96.94	96.91	2.29	5.2	3
ConvNeXt-T	173	P1	0.3	98.15	98.16	98.16	1.20	5.2	0
ConvNeXt-T	173	P2	0.3	97.90	97.90	97.90	1.52	5.2	3
ConvNeXt-T	211	P1	0.3	97.78	97.79	97.78	1.68	5.2	5
ConvNeXt-T	211	P2	0.3	97.04	97.07	97.02	2.67	5.2	4
ConvNeXt-T	257	P1	0.3	97.28	97.28	97.31	1.96	5.2	0
ConvNeXt-T	257	P2	0.3	97.04	97.03	97.06	2.62	5.2	3
ConvNeXt-T	307	P1	0.3	95.06	95.09	95.09	2.90	5.2	5
ConvNeXt-T	307	P2	0.3	96.79	96.79	96.78	2.18	5.2	4
ConvNeXt-T	349	P1	0.3	97.53	97.54	97.55	1.78	5.2	0
ConvNeXt-T	349	P2	0.3	98.02	98.03	98.03	1.64	5.2	3
ConvNeXt-T	401	P1	0.3	98.52	98.53	98.52	1.15	5.2	5
ConvNeXt-T	401	P2	0.3	98.77	98.77	98.77	0.67	5.2	4
ConvNeXt-T	457	P1	0.3	98.15	98.17	98.16	1.51	5.2	0
ConvNeXt-T	457	P2	0.3	94.94	94.92	94.93	3.74	5.2	3
ConvNeXt-T	503	P1	0.3	96.91	96.96	96.94	2.06	5.2	5
ConvNeXt-T	503	P2	0.3	97.28	97.32	97.28	1.71	5.2	4
ConvNeXt-T	557	P1	0.3	97.65	97.66	97.67	1.32	5.2	0
ConvNeXt-T	557	P2	0.3	98.15	98.18	98.15	0.96	5.2	3
ConvNeXt-T	137	P1	1	97.53	97.52	97.57	1.28	5.2	0
ConvNeXt-T	137	P2	1	96.42	96.42	96.44	1.91	5.2	4
ConvNeXt-T	173	P1	1	98.15	98.16	98.15	0.68	5.2	5
ConvNeXt-T	173	P2	1	96.67	96.69	96.67	2.28	5.2	4
ConvNeXt-T	211	P1	1	98.64	98.65	98.65	0.83	5.2	0
ConvNeXt-T	211	P2	1	96.30	96.31	96.31	2.02	5.2	3
ConvNeXt-T	257	P1	1	96.79	96.82	96.79	2.02	5.2	5
ConvNeXt-T	257	P2	1	98.02	98.03	98.02	0.70	5.2	4
ConvNeXt-T	307	P1	1	98.52	98.53	98.54	0.45	5.2	0
ConvNeXt-T	307	P2	1	97.53	97.53	97.53	1.22	5.2	3
ConvNeXt-T	349	P1	1	97.65	97.67	97.64	1.06	5.2	5
ConvNeXt-T	349	P2	1	97.53	97.55	97.53	0.97	5.2	4
ConvNeXt-T	401	P1	1	98.27	98.28	98.28	0.47	5.2	0
ConvNeXt-T	401	P2	1	97.41	97.44	97.38	1.49	5.2	3
ConvNeXt-T	457	P1	1	98.89	98.89	98.91	0.60	5.2	5
ConvNeXt-T	457	P2	1	96.17	96.18	96.19	2.62	5.2	4
ConvNeXt-T	503	P1	1	97.16	97.20	97.17	1.23	5.2	0
ConvNeXt-T	503	P2	1	97.78	97.78	97.79	0.98	5.2	3
ConvNeXt-T	557	P1	1	97.53	97.52	97.56	0.99	5.2	5
ConvNeXt-T	557	P2	1	98.64	98.65	98.65	0.64	5.2	4
ConvNeXt-T	137	P1	3	97.41	97.42	97.44	1.21	5.2	5
ConvNeXt-T	137	P2	3	96.42	96.44	96.42	1.07	5.2	3
ConvNeXt-T	173	P1	3	98.27	98.28	98.27	1.97	5.2	0
ConvNeXt-T	173	P2	3	96.91	96.93	96.92	1.60	5.2	3
ConvNeXt-T	211	P1	3	97.90	97.90	97.92	1.24	5.2	5
ConvNeXt-T	211	P2	3	97.78	97.81	97.78	1.20	5.2	4
ConvNeXt-T	257	P1	3	97.53	97.53	97.55	1.34	5.2	0
ConvNeXt-T	257	P2	3	97.53	97.53	97.53	0.73	5.2	3
ConvNeXt-T	307	P1	3	98.77	98.77	98.78	2.14	5.2	5
ConvNeXt-T	307	P2	3	97.16	97.17	97.19	0.75	5.2	4
ConvNeXt-T	349	P1	3	98.27	98.28	98.28	1.46	5.2	0
ConvNeXt-T	349	P2	3	97.04	97.07	97.04	0.64	5.2	3
ConvNeXt-T	401	P1	3	97.90	97.92	97.92	1.69	5.2	5
ConvNeXt-T	401	P2	3	97.28	97.32	97.28	0.91	5.2	4

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	457	P1	3	98.77	98.78	98.78	1.34	5.2	0
ConvNeXt-T	457	P2	3	96.17	96.16	96.21	2.00	5.2	3
ConvNeXt-T	503	P1	3	97.65	97.69	97.66	1.02	5.2	5
ConvNeXt-T	503	P2	3	98.02	98.02	98.02	1.26	5.2	4
ConvNeXt-T	557	P1	3	98.02	98.03	98.04	1.93	5.2	0
ConvNeXt-T	557	P2	3	98.15	98.15	98.16	1.55	5.2	3

M

Amendment 8 run ledgers

Table M.1: Complete Amendment 8 carrier-factorial run ledger (480 runs).

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	1103	clean_high	colour	93.05	92.62	92.52	4.32	0.9	7
ResNet-50	1103	clean_low	colour	92.34	91.92	91.82	3.36	0.9	5
ResNet-50	1103	leak_high	colour	94.12	93.66	93.41	3.28	1.0	6
ResNet-50	1103	leak_low	colour	95.37	94.95	94.89	2.38	0.9	4
ResNet-50	1151	clean_high	colour	94.11	94.05	93.81	3.19	0.9	4
ResNet-50	1151	clean_low	colour	94.82	94.83	94.82	2.07	0.9	6
ResNet-50	1151	leak_high	colour	94.82	94.77	94.70	1.95	0.9	6
ResNet-50	1151	leak_low	colour	95.54	95.53	95.57	2.17	0.9	5
ResNet-50	1201	clean_high	colour	96.33	96.09	96.00	1.98	0.9	4
ResNet-50	1201	clean_low	colour	95.52	95.17	95.08	2.22	0.9	6
ResNet-50	1201	leak_high	colour	97.35	97.20	97.24	1.11	0.9	7
ResNet-50	1201	leak_low	colour	96.33	96.14	96.10	1.36	0.9	5
ResNet-50	1249	clean_high	colour	94.28	94.29	94.48	3.32	1.0	4
ResNet-50	1249	clean_low	colour	91.85	91.90	92.03	4.43	1.0	5
ResNet-50	1249	leak_high	colour	95.15	95.18	95.25	2.76	1.0	7
ResNet-50	1249	leak_low	colour	97.40	97.43	97.35	2.02	0.9	6
ResNet-50	1301	clean_high	colour	93.37	93.40	94.08	2.74	0.9	4
ResNet-50	1301	clean_low	colour	96.02	96.12	96.22	2.61	0.9	5
ResNet-50	1301	leak_high	colour	96.97	96.99	97.33	2.42	0.9	7
ResNet-50	1301	leak_low	colour	95.83	95.76	96.16	1.89	0.9	6
ResNet-50	1361	clean_high	colour	94.39	94.17	94.20	3.99	0.9	6
ResNet-50	1361	clean_low	colour	94.21	94.02	94.12	3.44	0.9	5
ResNet-50	1361	leak_high	colour	95.84	95.62	95.78	3.57	0.9	7
ResNet-50	1361	leak_low	colour	95.84	95.69	95.75	2.86	0.9	4
ResNet-50	1409	clean_high	colour	94.16	93.70	93.68	3.63	0.9	6
ResNet-50	1409	clean_low	colour	95.53	95.21	95.33	2.32	0.9	5
ResNet-50	1409	leak_high	colour	97.08	96.87	96.92	1.38	0.9	7
ResNet-50	1409	leak_low	colour	97.67	97.46	97.43	2.02	0.9	4
ResNet-50	1451	clean_high	colour	93.47	93.30	93.14	2.72	0.9	6
ResNet-50	1451	clean_low	colour	93.65	93.50	93.43	3.17	0.9	5
ResNet-50	1451	leak_high	colour	95.83	95.72	95.52	2.50	0.9	7
ResNet-50	1451	leak_low	colour	95.10	94.88	94.68	1.79	1.0	4
ResNet-50	1499	clean_high	colour	93.48	93.14	93.18	4.25	0.9	6
ResNet-50	1499	clean_low	colour	94.47	94.17	94.27	4.36	0.9	5
ResNet-50	1499	leak_high	colour	95.65	95.41	95.44	2.46	0.9	7
ResNet-50	1499	leak_low	colour	94.47	94.26	94.43	3.29	0.9	4
ResNet-50	1553	clean_high	colour	91.22	91.03	91.02	5.51	0.9	7
ResNet-50	1553	clean_low	colour	92.61	92.57	92.35	4.07	0.9	5
ResNet-50	1553	leak_high	colour	93.41	93.24	93.34	2.58	0.9	6
ResNet-50	1553	leak_low	colour	94.81	94.67	94.72	2.76	0.9	4
ResNet-50	1103	clean_high	masked	93.40	93.08	92.99	4.98	1.0	7
ResNet-50	1103	clean_low	masked	93.58	93.14	93.00	3.34	1.0	5
ResNet-50	1103	leak_high	masked	94.12	93.84	93.66	2.71	1.0	6
ResNet-50	1103	leak_low	masked	95.72	95.50	95.34	2.14	1.0	4
ResNet-50	1151	clean_high	masked	93.93	93.93	93.69	4.80	1.0	4
ResNet-50	1151	clean_low	masked	93.93	93.91	93.73	3.32	1.0	7
ResNet-50	1151	leak_high	masked	95.18	95.21	95.10	2.50	1.0	6

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	1151	leak_low	masked	96.79	96.79	96.84	1.28	1.0	5
ResNet-50	1201	clean_high	masked	96.33	96.12	96.08	1.54	1.0	4
ResNet-50	1201	clean_low	masked	94.70	94.44	94.37	1.81	1.0	6
ResNet-50	1201	leak_high	masked	97.96	97.82	97.87	1.04	1.0	7
ResNet-50	1201	leak_low	masked	97.96	97.86	97.83	1.56	1.0	5
ResNet-50	1249	clean_high	masked	93.07	93.03	93.36	4.99	1.0	4
ResNet-50	1249	clean_low	masked	93.76	93.78	93.98	4.13	1.0	5
ResNet-50	1249	leak_high	masked	95.15	95.15	95.42	3.24	1.0	7
ResNet-50	1249	leak_low	masked	94.97	95.03	95.02	3.14	1.0	6
ResNet-50	1301	clean_high	masked	93.75	93.70	93.97	3.08	1.0	4
ResNet-50	1301	clean_low	masked	95.08	95.06	95.32	2.48	1.0	5
ResNet-50	1301	leak_high	masked	96.02	96.01	96.34	1.68	1.0	7
ResNet-50	1301	leak_low	masked	95.64	95.66	95.79	2.00	1.0	6
ResNet-50	1361	clean_high	masked	92.59	92.30	92.26	5.37	1.0	6
ResNet-50	1361	clean_low	masked	94.39	94.17	94.28	4.43	1.0	5
ResNet-50	1361	leak_high	masked	95.66	95.49	95.58	2.69	1.0	7
ResNet-50	1361	leak_low	masked	94.21	93.98	93.90	3.93	1.0	4
ResNet-50	1409	clean_high	masked	93.77	93.37	93.53	4.06	1.0	6
ResNet-50	1409	clean_low	masked	96.11	95.82	95.89	2.12	1.0	5
ResNet-50	1409	leak_high	masked	96.50	96.20	96.21	1.85	1.0	7
ResNet-50	1409	leak_low	masked	97.28	97.10	97.16	1.81	1.0	4
ResNet-50	1451	clean_high	masked	93.28	93.13	92.90	3.89	1.0	6
ResNet-50	1451	clean_low	masked	93.28	93.06	93.00	3.54	1.0	5
ResNet-50	1451	leak_high	masked	94.92	94.77	94.65	2.52	1.0	7
ResNet-50	1451	leak_low	masked	96.01	95.93	95.77	0.95	1.0	4
ResNet-50	1499	clean_high	masked	92.69	92.24	92.43	4.04	1.0	6
ResNet-50	1499	clean_low	masked	92.89	92.53	92.63	3.78	1.0	5
ResNet-50	1499	leak_high	masked	95.26	95.11	95.19	2.99	1.0	7
ResNet-50	1499	leak_low	masked	94.47	94.08	94.23	2.83	1.0	4
ResNet-50	1553	clean_high	masked	91.42	91.41	91.34	4.91	1.0	7
ResNet-50	1553	clean_low	masked	91.02	90.87	90.70	4.64	1.0	5
ResNet-50	1553	leak_high	masked	93.81	93.55	93.59	3.65	1.0	6
ResNet-50	1553	leak_low	masked	93.81	93.70	93.46	3.88	1.0	4
ResNet-50	1103	clean_high	standard	94.47	94.06	94.08	3.14	1.0	7
ResNet-50	1103	clean_low	standard	92.69	92.08	91.80	2.96	1.0	5
ResNet-50	1103	leak_high	standard	95.72	95.40	95.32	2.31	1.0	6
ResNet-50	1103	leak_low	standard	95.01	94.64	94.57	2.61	1.0	4
ResNet-50	1151	clean_high	standard	94.46	94.43	94.18	3.08	1.0	4
ResNet-50	1151	clean_low	standard	93.93	93.98	93.72	2.55	1.0	5
ResNet-50	1151	leak_high	standard	96.43	96.43	96.34	1.55	1.0	6
ResNet-50	1151	leak_low	standard	96.07	96.08	95.99	1.83	1.0	7
ResNet-50	1201	clean_high	standard	95.11	94.76	94.77	2.43	1.0	4
ResNet-50	1201	clean_low	standard	95.93	95.73	95.67	1.60	1.0	6
ResNet-50	1201	leak_high	standard	97.15	97.03	96.99	1.44	1.0	7
ResNet-50	1201	leak_low	standard	97.56	97.39	97.41	1.59	1.0	5
ResNet-50	1249	clean_high	standard	94.45	94.51	94.55	3.78	1.0	4
ResNet-50	1249	clean_low	standard	94.63	94.66	94.82	2.75	1.0	5
ResNet-50	1249	leak_high	standard	95.84	95.84	96.01	2.59	1.0	7
ResNet-50	1249	leak_low	standard	96.71	96.71	96.75	1.86	1.0	6
ResNet-50	1301	clean_high	standard	93.75	93.80	94.03	2.90	1.0	4
ResNet-50	1301	clean_low	standard	96.40	96.41	96.77	1.29	1.0	5
ResNet-50	1301	leak_high	standard	96.21	96.25	96.51	1.27	1.0	7
ResNet-50	1301	leak_low	standard	96.40	96.43	96.69	1.96	1.0	6
ResNet-50	1361	clean_high	standard	93.31	93.06	93.06	4.10	1.0	4
ResNet-50	1361	clean_low	standard	93.85	93.52	93.63	4.35	1.0	5
ResNet-50	1361	leak_high	standard	97.29	97.18	97.26	2.21	1.0	7
ResNet-50	1361	leak_low	standard	96.56	96.42	96.54	2.44	1.0	6
ResNet-50	1409	clean_high	standard	96.50	96.21	96.26	2.21	1.0	6
ResNet-50	1409	clean_low	standard	96.89	96.60	96.66	2.49	1.0	5
ResNet-50	1409	leak_high	standard	96.50	96.20	96.25	1.83	1.0	7
ResNet-50	1409	leak_low	standard	97.47	97.31	97.31	1.66	1.0	4
ResNet-50	1451	clean_high	standard	95.10	94.96	94.93	1.89	1.0	6
ResNet-50	1451	clean_low	standard	94.01	93.87	94.09	3.12	1.0	5
ResNet-50	1451	leak_high	standard	97.64	97.55	97.49	1.06	1.0	7
ResNet-50	1451	leak_low	standard	96.73	96.64	96.49	0.94	1.0	4
ResNet-50	1499	clean_high	standard	94.07	93.68	93.94	4.07	1.0	6
ResNet-50	1499	clean_low	standard	94.27	93.99	94.06	3.48	1.0	5
ResNet-50	1499	leak_high	standard	95.26	94.98	94.95	2.93	1.0	7
ResNet-50	1499	leak_low	standard	95.85	95.55	95.57	2.51	1.0	4
ResNet-50	1553	clean_high	standard	88.62	88.16	88.42	7.43	1.0	6
ResNet-50	1553	clean_low	standard	92.22	92.04	91.85	5.06	1.0	5
ResNet-50	1553	leak_high	standard	92.61	92.41	92.47	4.13	1.0	7
ResNet-50	1553	leak_low	standard	95.61	95.56	95.74	3.26	1.0	4

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	1103	clean_high	colour	94.12	93.69	93.58	2.82	1.2	7
DenseNet-121	1103	clean_low	colour	94.12	93.86	94.12	1.82	1.2	5
DenseNet-121	1103	leak_high	colour	95.72	95.40	95.34	1.49	1.2	6
DenseNet-121	1103	leak_low	colour	96.08	95.85	95.62	2.96	1.2	4
DenseNet-121	1151	clean_high	colour	92.86	92.80	92.73	3.15	1.2	4
DenseNet-121	1151	clean_low	colour	93.04	92.95	92.75	2.92	1.2	6
DenseNet-121	1151	leak_high	colour	95.00	95.01	94.93	1.53	1.2	7
DenseNet-121	1151	leak_low	colour	95.54	95.55	95.44	1.51	1.2	5
DenseNet-121	1201	clean_high	colour	96.95	96.76	96.73	1.40	1.2	4
DenseNet-121	1201	clean_low	colour	96.74	96.57	96.50	1.59	1.2	6
DenseNet-121	1201	leak_high	colour	97.15	96.97	97.03	0.83	1.2	7
DenseNet-121	1201	leak_low	colour	96.54	96.35	96.29	1.93	1.2	5
DenseNet-121	1249	clean_high	colour	95.32	95.32	95.46	2.19	1.2	4
DenseNet-121	1249	clean_low	colour	93.59	93.60	93.73	2.92	1.2	5
DenseNet-121	1249	leak_high	colour	96.53	96.58	96.65	2.52	1.2	7
DenseNet-121	1249	leak_low	colour	96.71	96.75	96.81	1.41	1.2	6
DenseNet-121	1301	clean_high	colour	95.27	95.29	95.30	2.15	1.2	4
DenseNet-121	1301	clean_low	colour	95.64	95.78	95.90	2.43	1.2	5
DenseNet-121	1301	leak_high	colour	97.54	97.54	97.79	1.30	1.2	7
DenseNet-121	1301	leak_low	colour	96.78	96.82	97.17	2.08	1.2	6
DenseNet-121	1361	clean_high	colour	94.58	94.30	94.42	2.98	1.2	6
DenseNet-121	1361	clean_low	colour	93.67	93.34	93.37	3.20	1.2	5
DenseNet-121	1361	leak_high	colour	95.30	95.17	95.11	2.48	1.2	7
DenseNet-121	1361	leak_low	colour	96.02	95.88	96.01	2.48	1.2	4
DenseNet-121	1409	clean_high	colour	95.91	95.54	95.56	2.69	1.2	6
DenseNet-121	1409	clean_low	colour	96.89	96.66	96.68	2.24	1.2	5
DenseNet-121	1409	leak_high	colour	97.28	97.06	97.16	1.52	1.2	7
DenseNet-121	1409	leak_low	colour	97.47	97.31	97.37	0.82	1.2	4
DenseNet-121	1451	clean_high	colour	94.37	94.18	93.92	1.89	1.2	6
DenseNet-121	1451	clean_low	colour	95.64	95.51	95.49	2.24	1.2	5
DenseNet-121	1451	leak_high	colour	96.19	96.04	95.85	1.78	1.2	7
DenseNet-121	1451	leak_low	colour	97.10	96.94	96.84	1.63	1.2	4
DenseNet-121	1499	clean_high	colour	93.28	92.92	92.95	3.10	1.2	6
DenseNet-121	1499	clean_low	colour	94.47	94.10	94.01	2.89	1.2	5
DenseNet-121	1499	leak_high	colour	96.64	96.34	96.40	1.29	1.2	7
DenseNet-121	1499	leak_low	colour	96.05	95.80	95.86	1.17	1.2	4
DenseNet-121	1553	clean_high	colour	92.02	91.82	91.79	3.24	1.2	7
DenseNet-121	1553	clean_low	colour	91.62	91.41	91.28	3.37	1.2	5
DenseNet-121	1553	leak_high	colour	94.61	94.43	94.19	2.34	1.2	6
DenseNet-121	1553	leak_low	colour	96.21	96.07	96.32	1.52	1.2	4
DenseNet-121	1103	clean_high	masked	93.58	93.30	93.26	3.54	1.2	7
DenseNet-121	1103	clean_low	masked	93.58	93.26	93.40	2.18	1.2	5
DenseNet-121	1103	leak_high	masked	95.19	94.91	94.96	2.37	1.2	6
DenseNet-121	1103	leak_low	masked	95.19	94.88	94.71	2.46	1.2	4
DenseNet-121	1151	clean_high	masked	93.57	93.51	93.60	3.16	1.2	4
DenseNet-121	1151	clean_low	masked	93.39	93.42	93.31	3.30	1.2	7
DenseNet-121	1151	leak_high	masked	95.36	95.35	95.44	1.62	1.2	6
DenseNet-121	1151	leak_low	masked	95.71	95.69	95.88	1.44	1.2	5
DenseNet-121	1201	clean_high	masked	95.52	95.23	95.33	1.40	1.2	4
DenseNet-121	1201	clean_low	masked	97.15	96.98	96.96	0.64	1.2	6
DenseNet-121	1201	leak_high	masked	97.56	97.43	97.42	2.05	1.2	7
DenseNet-121	1201	leak_low	masked	96.95	96.81	96.80	1.18	1.2	5
DenseNet-121	1249	clean_high	masked	94.45	94.48	94.55	3.64	1.2	4
DenseNet-121	1249	clean_low	masked	92.72	92.74	92.97	3.75	1.2	5
DenseNet-121	1249	leak_high	masked	95.49	95.51	95.77	3.26	1.3	7
DenseNet-121	1249	leak_low	masked	96.01	96.06	96.23	2.47	1.2	6
DenseNet-121	1301	clean_high	masked	94.51	94.51	94.83	1.90	1.2	4
DenseNet-121	1301	clean_low	masked	95.08	95.05	95.53	2.14	1.2	5
DenseNet-121	1301	leak_high	masked	96.59	96.61	96.82	1.29	1.2	7
DenseNet-121	1301	leak_low	masked	96.97	97.01	97.27	2.07	1.2	6
DenseNet-121	1361	clean_high	masked	93.31	93.07	93.11	3.69	1.2	6
DenseNet-121	1361	clean_low	masked	93.49	93.25	93.39	3.43	1.2	5
DenseNet-121	1361	leak_high	masked	95.66	95.46	95.51	2.29	1.2	7
DenseNet-121	1361	leak_low	masked	95.66	95.49	95.57	2.20	1.2	4
DenseNet-121	1409	clean_high	masked	95.72	95.41	95.41	2.12	1.2	6
DenseNet-121	1409	clean_low	masked	96.89	96.71	96.83	1.91	1.2	5
DenseNet-121	1409	leak_high	masked	95.72	95.42	95.57	2.48	1.2	7
DenseNet-121	1409	leak_low	masked	96.89	96.71	96.66	1.33	1.2	4
DenseNet-121	1451	clean_high	masked	94.92	94.75	94.52	2.92	1.3	6
DenseNet-121	1451	clean_low	masked	93.47	93.42	93.39	2.69	1.2	5
DenseNet-121	1451	leak_high	masked	95.64	95.47	95.30	1.90	1.2	7
DenseNet-121	1451	leak_low	masked	97.10	97.01	96.96	2.21	1.2	4
DenseNet-121	1499	clean_high	masked	93.87	93.40	93.59	4.37	1.2	6

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	1499	clean_low	masked	92.69	92.33	92.46	3.37	1.2	5
DenseNet-121	1499	leak_high	masked	96.25	96.01	96.17	2.19	1.2	7
DenseNet-121	1499	leak_low	masked	94.66	94.38	94.48	2.51	1.2	4
DenseNet-121	1553	clean_high	masked	92.61	92.67	92.64	3.52	1.2	7
DenseNet-121	1553	clean_low	masked	91.42	91.27	91.01	4.64	1.3	5
DenseNet-121	1553	leak_high	masked	92.42	92.30	92.04	3.80	1.2	6
DenseNet-121	1553	leak_low	masked	96.81	96.78	96.87	1.60	1.2	4
DenseNet-121	1103	clean_high	standard	95.01	94.78	94.82	1.48	1.2	7
DenseNet-121	1103	clean_low	standard	94.30	93.94	94.02	2.14	1.2	5
DenseNet-121	1103	leak_high	standard	95.54	95.21	95.05	2.44	1.2	6
DenseNet-121	1103	leak_low	standard	96.43	96.17	95.99	1.89	1.2	4
DenseNet-121	1151	clean_high	standard	93.57	93.52	93.46	2.61	1.2	4
DenseNet-121	1151	clean_low	standard	93.93	93.86	93.61	2.68	1.2	7
DenseNet-121	1151	leak_high	standard	94.46	94.40	94.29	3.21	1.2	6
DenseNet-121	1151	leak_low	standard	96.61	96.58	96.62	1.61	1.2	5
DenseNet-121	1201	clean_high	standard	96.74	96.52	96.50	1.13	1.2	4
DenseNet-121	1201	clean_low	standard	97.56	97.46	97.52	1.19	1.2	6
DenseNet-121	1201	leak_high	standard	98.57	98.50	98.46	1.29	1.2	7
DenseNet-121	1201	leak_low	standard	98.17	98.09	98.07	1.23	1.2	5
DenseNet-121	1249	clean_high	standard	94.97	94.97	95.12	2.85	1.2	4
DenseNet-121	1249	clean_low	standard	94.63	94.65	94.80	2.64	1.2	5
DenseNet-121	1249	leak_high	standard	96.53	96.53	96.81	2.58	1.2	7
DenseNet-121	1249	leak_low	standard	96.36	96.40	96.58	1.93	1.2	6
DenseNet-121	1301	clean_high	standard	96.97	96.95	97.11	1.64	1.2	4
DenseNet-121	1301	clean_low	standard	96.02	96.08	96.46	1.57	1.2	5
DenseNet-121	1301	leak_high	standard	97.54	97.52	97.84	1.30	1.2	7
DenseNet-121	1301	leak_low	standard	97.16	97.19	97.62	1.38	1.2	6
DenseNet-121	1361	clean_high	standard	94.76	94.51	94.59	3.01	1.2	4
DenseNet-121	1361	clean_low	standard	94.03	93.75	93.85	3.61	1.2	5
DenseNet-121	1361	leak_high	standard	96.02	95.82	95.94	1.95	1.2	7
DenseNet-121	1361	leak_low	standard	97.47	97.36	97.48	1.53	1.2	6
DenseNet-121	1409	clean_high	standard	96.69	96.51	96.55	1.90	1.2	6
DenseNet-121	1409	clean_low	standard	95.91	95.57	95.65	1.58	1.2	5
DenseNet-121	1409	leak_high	standard	95.91	95.60	95.62	2.01	1.2	7
DenseNet-121	1409	leak_low	standard	97.86	97.72	97.65	1.46	1.2	4
DenseNet-121	1451	clean_high	standard	95.83	95.66	95.65	1.86	1.2	6
DenseNet-121	1451	clean_low	standard	95.83	95.72	95.69	1.64	1.2	5
DenseNet-121	1451	leak_high	standard	97.10	96.97	96.92	1.02	1.2	7
DenseNet-121	1451	leak_low	standard	96.37	96.24	96.03	1.78	1.2	4
DenseNet-121	1499	clean_high	standard	95.26	94.89	95.02	3.30	1.2	6
DenseNet-121	1499	clean_low	standard	94.07	93.62	93.69	2.56	1.2	5
DenseNet-121	1499	leak_high	standard	95.65	95.31	95.41	2.03	1.2	7
DenseNet-121	1499	leak_low	standard	96.44	96.11	96.15	1.81	1.2	4
DenseNet-121	1553	clean_high	standard	91.62	91.37	91.23	4.40	1.2	7
DenseNet-121	1553	clean_low	standard	92.22	92.11	92.08	4.25	1.2	5
DenseNet-121	1553	leak_high	standard	96.01	95.95	95.92	1.93	1.2	6
DenseNet-121	1553	leak_low	standard	97.21	97.09	97.28	1.81	1.2	4
ViT-B/16	1103	clean_high	colour	92.34	91.98	92.25	5.64	3.4	4
ViT-B/16	1103	clean_low	colour	93.58	93.20	93.59	4.10	3.4	5
ViT-B/16	1103	leak_high	colour	94.12	93.50	93.94	4.26	3.4	6
ViT-B/16	1103	leak_low	colour	95.90	95.70	95.65	3.19	3.4	7
ViT-B/16	1151	clean_high	colour	92.50	92.35	92.29	5.01	3.5	4
ViT-B/16	1151	clean_low	colour	93.57	93.36	93.25	4.50	3.4	6
ViT-B/16	1151	leak_high	colour	95.18	95.14	95.21	2.57	3.5	7
ViT-B/16	1151	leak_low	colour	95.36	95.37	95.21	2.57	3.4	5
ViT-B/16	1201	clean_high	colour	96.13	95.91	95.93	2.53	3.4	4
ViT-B/16	1201	clean_low	colour	94.50	94.31	94.38	3.52	3.3	6
ViT-B/16	1201	leak_high	colour	97.96	97.91	98.00	1.56	3.4	7
ViT-B/16	1201	leak_low	colour	97.15	96.95	96.83	2.08	3.4	5
ViT-B/16	1249	clean_high	colour	94.63	94.66	94.55	3.89	3.5	4
ViT-B/16	1249	clean_low	colour	94.80	94.84	94.86	3.31	3.5	5
ViT-B/16	1249	leak_high	colour	92.20	92.12	92.45	2.97	3.5	7
ViT-B/16	1249	leak_low	colour	96.71	96.79	96.73	1.94	3.4	6
ViT-B/16	1301	clean_high	colour	92.99	92.96	93.65	5.14	3.4	4
ViT-B/16	1301	clean_low	colour	92.42	92.23	91.96	4.66	3.4	5
ViT-B/16	1301	leak_high	colour	93.56	93.70	94.31	3.79	3.4	7
ViT-B/16	1301	leak_low	colour	95.27	95.12	95.06	3.44	3.4	6
ViT-B/16	1361	clean_high	colour	93.49	93.16	93.43	4.66	3.4	6
ViT-B/16	1361	clean_low	colour	92.04	91.80	91.59	5.45	3.4	5
ViT-B/16	1361	leak_high	colour	95.66	95.49	95.43	2.41	3.5	7
ViT-B/16	1361	leak_low	colour	91.86	91.50	91.10	5.26	3.4	4
ViT-B/16	1409	clean_high	colour	94.94	94.56	94.64	3.09	3.4	6
ViT-B/16	1409	clean_low	colour	94.55	94.18	94.27	3.98	3.4	5

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ViT-B/16	1409	leak_high	colour	95.91	95.61	95.69	2.10	3.4	7
ViT-B/16	1409	leak_low	colour	93.97	93.65	93.89	3.84	3.4	4
ViT-B/16	1451	clean_high	colour	92.74	92.50	92.29	4.03	3.4	6
ViT-B/16	1451	clean_low	colour	95.10	95.03	94.95	2.59	3.4	5
ViT-B/16	1451	leak_high	colour	95.64	95.49	95.36	2.84	3.5	7
ViT-B/16	1451	leak_low	colour	93.10	92.94	93.34	3.72	3.5	4
ViT-B/16	1499	clean_high	colour	92.69	92.18	92.09	5.37	3.4	6
ViT-B/16	1499	clean_low	colour	92.49	92.09	92.18	5.92	3.4	5
ViT-B/16	1499	leak_high	colour	93.68	93.23	93.28	4.87	3.4	7
ViT-B/16	1499	leak_low	colour	94.66	94.45	94.68	3.45	3.4	4
ViT-B/16	1553	clean_high	colour	92.81	92.74	92.37	5.13	3.4	7
ViT-B/16	1553	clean_low	colour	92.02	91.75	91.59	4.80	3.4	5
ViT-B/16	1553	leak_high	colour	96.61	96.50	96.71	2.28	3.4	6
ViT-B/16	1553	leak_low	colour	96.41	96.41	96.42	2.36	3.5	4
ViT-B/16	1103	clean_high	masked	92.16	91.56	91.09	5.02	3.4	7
ViT-B/16	1103	clean_low	masked	90.02	89.46	89.23	6.42	3.4	5
ViT-B/16	1103	leak_high	masked	95.54	95.20	95.19	3.27	3.4	6
ViT-B/16	1103	leak_low	masked	93.76	93.39	93.48	4.48	3.4	4
ViT-B/16	1151	clean_high	masked	89.11	89.48	88.65	8.19	3.4	4
ViT-B/16	1151	clean_low	masked	93.04	93.06	92.79	4.44	3.4	7
ViT-B/16	1151	leak_high	masked	96.61	96.61	96.65	2.74	3.4	6
ViT-B/16	1151	leak_low	masked	94.46	94.48	94.47	2.89	3.4	5
ViT-B/16	1201	clean_high	masked	94.70	94.38	94.24	2.78	3.4	4
ViT-B/16	1201	clean_low	masked	95.11	94.93	94.86	3.35	3.3	6
ViT-B/16	1201	leak_high	masked	98.78	98.69	98.74	1.27	3.4	7
ViT-B/16	1201	leak_low	masked	94.91	94.73	94.76	2.45	3.4	5
ViT-B/16	1249	clean_high	masked	91.85	91.84	91.90	5.66	3.4	4
ViT-B/16	1249	clean_low	masked	93.76	93.73	93.90	4.62	3.4	5
ViT-B/16	1249	leak_high	masked	92.55	92.49	92.99	5.79	3.5	7
ViT-B/16	1249	leak_low	masked	95.84	95.89	95.89	3.46	3.4	6
ViT-B/16	1301	clean_high	masked	95.64	95.61	95.90	3.31	3.4	4
ViT-B/16	1301	clean_low	masked	93.18	93.15	93.46	4.66	3.4	5
ViT-B/16	1301	leak_high	masked	93.75	93.57	93.84	4.39	3.4	7
ViT-B/16	1301	leak_low	masked	96.02	95.99	96.48	2.76	3.4	6
ViT-B/16	1361	clean_high	masked	93.67	93.42	93.54	4.69	3.4	6
ViT-B/16	1361	clean_low	masked	94.03	93.76	93.94	3.96	3.4	5
ViT-B/16	1361	leak_high	masked	91.14	90.65	90.56	6.56	3.4	7
ViT-B/16	1361	leak_low	masked	92.95	92.73	92.43	5.27	3.4	4
ViT-B/16	1409	clean_high	masked	93.58	93.01	92.92	4.08	3.4	6
ViT-B/16	1409	clean_low	masked	93.00	92.76	93.03	4.74	3.4	5
ViT-B/16	1409	leak_high	masked	96.11	95.91	96.05	1.61	3.4	7
ViT-B/16	1409	leak_low	masked	93.97	93.46	93.39	4.27	3.4	4
ViT-B/16	1451	clean_high	masked	94.92	94.84	94.55	3.52	3.4	6
ViT-B/16	1451	clean_low	masked	92.74	92.49	92.64	3.42	3.4	5
ViT-B/16	1451	leak_high	masked	95.64	95.52	95.33	3.14	3.4	7
ViT-B/16	1451	leak_low	masked	96.55	96.48	96.40	2.22	3.4	4
ViT-B/16	1499	clean_high	masked	92.69	92.28	92.25	4.73	3.3	6
ViT-B/16	1499	clean_low	masked	92.09	91.72	92.10	5.32	3.4	5
ViT-B/16	1499	leak_high	masked	95.65	95.28	95.42	2.54	3.4	7
ViT-B/16	1499	leak_low	masked	95.26	94.94	95.13	3.77	3.4	4
ViT-B/16	1553	clean_high	masked	91.22	91.08	91.17	5.95	3.4	7
ViT-B/16	1553	clean_low	masked	90.42	90.17	90.31	6.12	3.4	5
ViT-B/16	1553	leak_high	masked	93.41	93.32	93.22	3.91	3.3	6
ViT-B/16	1553	leak_low	masked	95.41	95.37	95.40	2.58	3.4	4
ViT-B/16	1103	clean_high	standard	93.05	92.68	92.57	5.52	3.4	7
ViT-B/16	1103	clean_low	standard	93.58	93.02	93.44	4.50	3.4	5
ViT-B/16	1103	leak_high	standard	92.87	91.86	91.49	5.26	3.4	6
ViT-B/16	1103	leak_low	standard	93.58	93.00	93.33	5.48	3.4	4
ViT-B/16	1151	clean_high	standard	94.64	94.51	94.66	3.83	3.4	4
ViT-B/16	1151	clean_low	standard	94.82	94.80	94.68	3.39	3.4	7
ViT-B/16	1151	leak_high	standard	96.25	96.16	96.14	1.88	3.4	6
ViT-B/16	1151	leak_low	standard	95.00	94.92	95.33	3.55	3.4	5
ViT-B/16	1201	clean_high	standard	92.87	92.31	92.56	5.63	3.4	4
ViT-B/16	1201	clean_low	standard	96.13	95.82	95.80	2.58	3.3	6
ViT-B/16	1201	leak_high	standard	96.13	96.03	96.08	1.58	3.4	7
ViT-B/16	1201	leak_low	standard	97.76	97.69	97.69	1.41	3.4	5
ViT-B/16	1249	clean_high	standard	94.28	94.29	94.35	4.39	3.5	4
ViT-B/16	1249	clean_low	standard	95.32	95.34	95.49	3.39	3.4	5
ViT-B/16	1249	leak_high	standard	94.63	94.76	94.68	3.81	3.5	7
ViT-B/16	1249	leak_low	standard	95.15	95.17	94.91	3.54	3.4	6
ViT-B/16	1301	clean_high	standard	95.27	95.30	95.10	3.08	3.4	4
ViT-B/16	1301	clean_low	standard	94.89	94.86	94.94	3.72	3.4	5
ViT-B/16	1301	leak_high	standard	94.32	94.26	94.98	4.43	3.4	7

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ViT-B/16	1301	leak_low	standard	96.59	96.66	96.60	1.93	3.4	6
ViT-B/16	1361	clean_high	standard	95.12	94.86	95.03	4.20	3.4	6
ViT-B/16	1361	clean_low	standard	94.03	93.70	93.88	4.78	3.4	5
ViT-B/16	1361	leak_high	standard	86.44	86.22	85.85	10.16	3.4	7
ViT-B/16	1361	leak_low	standard	96.38	96.22	96.16	2.23	3.4	4
ViT-B/16	1409	clean_high	standard	95.33	95.02	95.09	4.08	3.4	6
ViT-B/16	1409	clean_low	standard	96.69	96.49	96.58	2.79	3.4	5
ViT-B/16	1409	leak_high	standard	87.55	86.87	86.77	7.51	3.4	7
ViT-B/16	1409	leak_low	standard	92.80	91.66	91.76	4.18	3.4	4
ViT-B/16	1451	clean_high	standard	91.29	90.91	91.41	6.29	3.4	6
ViT-B/16	1451	clean_low	standard	92.56	92.22	92.52	4.74	3.4	5
ViT-B/16	1451	leak_high	standard	92.56	92.38	91.88	4.59	3.4	7
ViT-B/16	1451	leak_low	standard	94.37	94.19	93.83	3.26	3.5	4
ViT-B/16	1499	clean_high	standard	90.12	89.75	89.63	6.78	3.3	6
ViT-B/16	1499	clean_low	standard	93.08	92.74	92.94	5.44	3.4	5
ViT-B/16	1499	leak_high	standard	97.23	97.03	97.13	2.61	3.4	7
ViT-B/16	1499	leak_low	standard	92.29	91.67	92.02	6.19	3.4	4
ViT-B/16	1553	clean_high	standard	94.01	93.92	94.05	3.62	3.4	7
ViT-B/16	1553	clean_low	standard	93.81	93.61	93.67	4.38	3.4	5
ViT-B/16	1553	leak_high	standard	96.01	95.85	95.82	2.32	3.3	6
ViT-B/16	1553	leak_low	standard	95.81	95.68	95.53	2.26	3.4	4
ConvNeXt-T	1103	clean_high	colour	94.65	94.42	94.38	3.59	2.0	7
ConvNeXt-T	1103	clean_low	colour	94.30	93.93	93.93	3.21	2.0	5
ConvNeXt-T	1103	leak_high	colour	93.58	93.00	93.42	4.57	2.0	6
ConvNeXt-T	1103	leak_low	colour	96.08	95.91	95.81	3.59	2.0	4
ConvNeXt-T	1151	clean_high	colour	95.18	95.07	94.98	3.07	2.0	4
ConvNeXt-T	1151	clean_low	colour	93.93	93.82	93.60	3.71	2.0	6
ConvNeXt-T	1151	leak_high	colour	93.04	93.02	92.91	5.12	2.0	7
ConvNeXt-T	1151	leak_low	colour	93.93	93.77	93.66	3.27	2.0	5
ConvNeXt-T	1201	clean_high	colour	96.33	96.14	96.10	2.27	2.0	4
ConvNeXt-T	1201	clean_low	colour	96.74	96.61	96.54	2.09	2.0	6
ConvNeXt-T	1201	leak_high	colour	97.76	97.60	97.62	1.42	2.0	7
ConvNeXt-T	1201	leak_low	colour	98.37	98.26	98.30	1.26	2.0	5
ConvNeXt-T	1249	clean_high	colour	94.45	94.47	94.53	3.61	2.0	4
ConvNeXt-T	1249	clean_low	colour	93.41	93.44	93.49	4.68	2.0	5
ConvNeXt-T	1249	leak_high	colour	96.36	96.40	96.46	2.63	2.0	7
ConvNeXt-T	1249	leak_low	colour	94.11	94.03	94.13	3.76	2.0	6
ConvNeXt-T	1301	clean_high	colour	96.40	96.41	96.56	2.92	2.0	4
ConvNeXt-T	1301	clean_low	colour	94.13	93.99	94.38	3.47	2.0	5
ConvNeXt-T	1301	leak_high	colour	96.21	96.16	96.27	2.36	2.0	7
ConvNeXt-T	1301	leak_low	colour	96.78	96.76	96.87	2.25	2.0	6
ConvNeXt-T	1361	clean_high	colour	93.67	93.33	93.34	4.52	2.0	6
ConvNeXt-T	1361	clean_low	colour	93.85	93.65	93.90	4.12	2.0	5
ConvNeXt-T	1361	leak_high	colour	95.48	95.29	95.48	3.26	2.0	7
ConvNeXt-T	1361	leak_low	colour	95.84	95.72	95.65	2.97	2.0	4
ConvNeXt-T	1409	clean_high	colour	95.72	95.30	95.31	2.49	2.0	6
ConvNeXt-T	1409	clean_low	colour	96.50	96.19	96.26	3.00	2.0	5
ConvNeXt-T	1409	leak_high	colour	94.94	94.49	94.60	3.65	2.1	7
ConvNeXt-T	1409	leak_low	colour	95.91	95.74	95.96	2.86	2.0	4
ConvNeXt-T	1451	clean_high	colour	91.65	91.48	91.08	5.98	2.0	6
ConvNeXt-T	1451	clean_low	colour	91.47	91.22	90.99	6.36	2.0	5
ConvNeXt-T	1451	leak_high	colour	96.37	96.27	96.14	3.62	2.0	7
ConvNeXt-T	1451	leak_low	colour	94.37	94.21	94.25	3.46	2.0	4
ConvNeXt-T	1499	clean_high	colour	93.87	93.50	93.55	4.23	2.0	6
ConvNeXt-T	1499	clean_low	colour	92.89	92.49	92.48	5.21	2.0	5
ConvNeXt-T	1499	leak_high	colour	96.84	96.66	96.66	2.65	2.0	7
ConvNeXt-T	1499	leak_low	colour	94.66	94.39	94.67	3.39	2.0	4
ConvNeXt-T	1553	clean_high	colour	90.42	90.18	90.27	6.63	2.0	7
ConvNeXt-T	1553	clean_low	colour	87.23	86.96	86.80	9.65	2.0	5
ConvNeXt-T	1553	leak_high	colour	92.42	92.44	92.33	5.13	2.0	6
ConvNeXt-T	1553	leak_low	colour	97.60	97.57	97.62	0.99	2.0	4
ConvNeXt-T	1103	clean_high	masked	90.20	89.39	89.65	7.48	2.0	7
ConvNeXt-T	1103	clean_low	masked	92.34	91.86	91.72	5.06	2.0	5
ConvNeXt-T	1103	leak_high	masked	95.54	95.28	95.39	2.88	2.0	6
ConvNeXt-T	1103	leak_low	masked	95.90	95.71	95.76	3.19	2.0	4
ConvNeXt-T	1151	clean_high	masked	93.93	93.96	93.72	4.45	2.0	4
ConvNeXt-T	1151	clean_low	masked	93.75	93.79	93.62	4.34	2.0	7
ConvNeXt-T	1151	leak_high	masked	94.82	94.79	94.87	2.51	2.0	6
ConvNeXt-T	1151	leak_low	masked	93.93	93.83	93.70	3.76	2.0	5
ConvNeXt-T	1201	clean_high	masked	96.33	96.10	96.08	2.19	2.0	4
ConvNeXt-T	1201	clean_low	masked	95.52	95.30	95.34	3.60	1.9	6
ConvNeXt-T	1201	leak_high	masked	97.35	97.21	97.31	2.01	2.0	7
ConvNeXt-T	1201	leak_low	masked	97.35	97.25	97.22	1.50	2.0	5

Architecture	Seed	Protocol	condition	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	1249	clean_high	masked	93.76	93.81	93.90	4.57	2.0	4
ConvNeXt-T	1249	clean_low	masked	93.41	93.53	93.38	4.83	2.0	5
ConvNeXt-T	1249	leak_high	masked	95.84	95.87	96.11	3.36	2.0	7
ConvNeXt-T	1249	leak_low	masked	94.97	95.02	95.08	3.96	2.0	6
ConvNeXt-T	1301	clean_high	masked	94.13	94.10	94.70	4.67	2.0	4
ConvNeXt-T	1301	clean_low	masked	94.13	94.13	94.05	3.86	2.0	5
ConvNeXt-T	1301	leak_high	masked	96.59	96.67	96.89	2.67	2.0	7
ConvNeXt-T	1301	leak_low	masked	92.23	92.29	92.29	4.64	2.0	6
ConvNeXt-T	1361	clean_high	masked	93.85	93.64	93.59	4.39	2.0	6
ConvNeXt-T	1361	clean_low	masked	92.95	92.57	92.88	5.45	2.0	5
ConvNeXt-T	1361	leak_high	masked	92.22	91.57	92.21	5.68	2.0	7
ConvNeXt-T	1361	leak_low	masked	95.66	95.50	95.52	2.83	2.0	4
ConvNeXt-T	1409	clean_high	masked	94.75	94.36	94.44	4.18	2.0	6
ConvNeXt-T	1409	clean_low	masked	96.50	96.26	96.31	2.66	2.0	5
ConvNeXt-T	1409	leak_high	masked	94.75	94.47	94.60	3.57	2.0	7
ConvNeXt-T	1409	leak_low	masked	96.89	96.74	96.87	2.53	2.0	4
ConvNeXt-T	1451	clean_high	masked	91.29	91.16	91.54	5.41	2.0	6
ConvNeXt-T	1451	clean_low	masked	93.65	93.54	93.55	4.00	2.0	5
ConvNeXt-T	1451	leak_high	masked	91.83	91.90	91.91	6.38	2.0	7
ConvNeXt-T	1451	leak_low	masked	95.10	94.93	95.02	2.26	2.0	4
ConvNeXt-T	1499	clean_high	masked	93.08	92.71	92.81	4.90	2.0	6
ConvNeXt-T	1499	clean_low	masked	93.87	93.46	93.59	4.74	2.0	5
ConvNeXt-T	1499	leak_high	masked	94.66	94.31	94.58	3.97	2.0	7
ConvNeXt-T	1499	leak_low	masked	96.05	95.69	95.75	2.89	2.0	4
ConvNeXt-T	1553	clean_high	masked	90.22	90.10	90.13	7.40	2.0	7
ConvNeXt-T	1553	clean_low	masked	90.02	90.01	89.52	7.78	2.0	5
ConvNeXt-T	1553	leak_high	masked	94.41	94.37	94.45	3.46	2.0	6
ConvNeXt-T	1553	leak_low	masked	94.81	94.67	94.77	2.76	2.0	4
ConvNeXt-T	1103	clean_high	standard	94.65	94.30	94.38	4.18	2.0	7
ConvNeXt-T	1103	clean_low	standard	93.58	93.37	93.87	5.09	2.0	5
ConvNeXt-T	1103	leak_high	standard	95.54	95.17	95.34	3.16	2.0	6
ConvNeXt-T	1103	leak_low	standard	95.90	95.63	95.66	2.77	2.0	4
ConvNeXt-T	1151	clean_high	standard	94.64	94.64	94.45	4.04	2.0	4
ConvNeXt-T	1151	clean_low	standard	90.18	90.30	89.97	7.78	2.2	7
ConvNeXt-T	1151	leak_high	standard	97.14	97.16	97.25	2.36	2.0	6
ConvNeXt-T	1151	leak_low	standard	95.54	95.49	95.33	2.14	2.2	5
ConvNeXt-T	1201	clean_high	standard	97.15	96.99	96.90	2.20	2.0	4
ConvNeXt-T	1201	clean_low	standard	96.74	96.48	96.43	2.72	2.0	6
ConvNeXt-T	1201	leak_high	standard	97.96	97.83	97.87	1.52	2.0	7
ConvNeXt-T	1201	leak_low	standard	97.35	97.14	97.05	2.28	2.0	5
ConvNeXt-T	1249	clean_high	standard	95.84	95.89	95.91	3.18	2.0	4
ConvNeXt-T	1249	clean_low	standard	94.97	95.00	94.97	3.70	2.0	5
ConvNeXt-T	1249	leak_high	standard	97.05	97.10	97.14	2.26	2.0	7
ConvNeXt-T	1249	leak_low	standard	96.88	96.88	97.03	2.34	2.0	6
ConvNeXt-T	1301	clean_high	standard	93.94	93.83	94.62	4.34	2.0	4
ConvNeXt-T	1301	clean_low	standard	95.83	95.78	96.15	3.01	2.0	5
ConvNeXt-T	1301	leak_high	standard	96.97	96.96	97.36	2.13	2.0	7
ConvNeXt-T	1301	leak_low	standard	96.78	96.76	96.93	2.30	2.0	6
ConvNeXt-T	1361	clean_high	standard	94.58	94.32	94.47	4.09	2.0	4
ConvNeXt-T	1361	clean_low	standard	95.48	95.29	95.30	3.43	2.0	5
ConvNeXt-T	1361	leak_high	standard	95.48	95.30	95.28	3.03	2.0	7
ConvNeXt-T	1361	leak_low	standard	96.38	96.27	96.38	2.27	2.0	6
ConvNeXt-T	1409	clean_high	standard	96.89	96.70	96.81	2.69	2.0	6
ConvNeXt-T	1409	clean_low	standard	94.94	94.58	94.73	3.90	2.0	5
ConvNeXt-T	1409	leak_high	standard	97.67	97.47	97.49	1.86	2.0	7
ConvNeXt-T	1409	leak_low	standard	97.47	97.30	97.39	1.96	2.0	4
ConvNeXt-T	1451	clean_high	standard	94.92	94.80	94.86	3.57	2.0	6
ConvNeXt-T	1451	clean_low	standard	92.20	91.97	91.76	6.05	2.0	5
ConvNeXt-T	1451	leak_high	standard	95.28	95.19	95.22	3.22	2.0	7
ConvNeXt-T	1451	leak_low	standard	97.28	97.16	97.20	2.06	2.0	4
ConvNeXt-T	1499	clean_high	standard	94.66	94.31	94.53	4.48	2.0	6
ConvNeXt-T	1499	clean_low	standard	95.65	95.33	95.35	3.30	2.0	5
ConvNeXt-T	1499	leak_high	standard	96.05	95.74	95.83	2.86	2.0	7
ConvNeXt-T	1499	leak_low	standard	97.04	96.74	96.82	2.46	2.0	4
ConvNeXt-T	1553	clean_high	standard	93.21	93.09	93.21	5.16	2.0	7
ConvNeXt-T	1553	clean_low	standard	90.42	90.47	90.54	7.39	2.0	5
ConvNeXt-T	1553	leak_high	standard	96.41	96.36	96.27	2.76	2.0	6
ConvNeXt-T	1553	leak_low	standard	96.01	95.88	96.04	2.50	2.0	4

Table M.2: Complete Amendment 8 linear-adversary run ledger (300 runs).

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	601	P1	0	97.16	97.18	97.19	1.88	2.6	6
ResNet-50	601	P2	0	97.41	97.42	97.42	1.68	2.6	7
ResNet-50	653	P1	0	97.28	97.31	97.29	2.18	2.6	6
ResNet-50	653	P2	0	96.67	96.69	96.68	2.23	2.6	4
ResNet-50	701	P1	0	97.53	97.54	97.53	1.47	2.6	6
ResNet-50	701	P2	0	96.79	96.81	96.81	2.25	2.6	7
ResNet-50	751	P1	0	97.53	97.56	97.55	1.31	2.6	6
ResNet-50	751	P2	0	95.56	95.55	95.55	3.75	2.6	4
ResNet-50	809	P1	0	97.78	97.77	97.80	1.37	2.6	5
ResNet-50	809	P2	0	97.04	97.05	97.02	1.90	2.6	6
ResNet-50	853	P1	0	97.41	97.41	97.44	2.18	2.6	7
ResNet-50	853	P2	0	96.17	96.17	96.18	2.40	2.6	4
ResNet-50	907	P1	0	98.64	98.65	98.65	0.84	2.6	5
ResNet-50	907	P2	0	97.41	97.41	97.43	1.60	2.6	6
ResNet-50	953	P1	0	97.53	97.55	97.53	1.28	2.6	5
ResNet-50	953	P2	0	95.80	95.87	95.79	2.85	2.6	7
ResNet-50	1009	P1	0	97.16	97.18	97.16	1.72	2.6	4
ResNet-50	1009	P2	0	97.41	97.42	97.41	1.58	2.6	6
ResNet-50	1051	P1	0	98.27	98.29	98.28	0.81	2.6	5
ResNet-50	1051	P2	0	97.41	97.40	97.43	0.99	2.6	4
ResNet-50	601	P1	0.1	97.28	97.31	97.29	1.27	2.6	4
ResNet-50	601	P2	0.1	95.93	95.94	95.96	2.60	2.6	5
ResNet-50	653	P1	0.1	97.53	97.55	97.55	1.11	2.6	5
ResNet-50	653	P2	0.1	96.17	96.16	96.17	2.08	2.6	7
ResNet-50	701	P1	0.1	96.91	96.90	96.93	1.62	2.6	4
ResNet-50	701	P2	0.1	95.56	95.56	95.58	2.70	2.6	5
ResNet-50	751	P1	0.1	97.16	97.18	97.18	1.50	2.6	5
ResNet-50	751	P2	0.1	95.80	95.79	95.79	3.07	2.6	7
ResNet-50	809	P1	0.1	96.67	96.68	96.72	2.09	2.6	7
ResNet-50	809	P2	0.1	96.42	96.45	96.40	2.51	2.6	4
ResNet-50	853	P1	0.1	96.67	96.67	96.68	1.30	2.6	5
ResNet-50	853	P2	0.1	96.30	96.29	96.30	2.59	2.6	6
ResNet-50	907	P1	0.1	97.41	97.42	97.42	1.07	2.6	7
ResNet-50	907	P2	0.1	96.42	96.41	96.39	2.13	2.6	4
ResNet-50	953	P1	0.1	97.16	97.20	97.17	1.30	2.6	4
ResNet-50	953	P2	0.1	96.79	96.82	96.79	2.21	2.6	6
ResNet-50	1009	P1	0.1	97.16	97.18	97.16	1.63	2.6	5
ResNet-50	1009	P2	0.1	97.04	97.05	97.05	1.75	2.6	7
ResNet-50	1051	P1	0.1	97.28	97.29	97.30	1.00	2.6	6
ResNet-50	1051	P2	0.1	96.42	96.43	96.42	1.76	2.6	7
ResNet-50	601	P1	0.3	96.79	96.81	96.81	1.57	2.6	6
ResNet-50	601	P2	0.3	96.67	96.67	96.68	0.96	2.6	7
ResNet-50	653	P1	0.3	97.16	97.17	97.18	1.23	2.6	4
ResNet-50	653	P2	0.3	96.05	96.03	96.05	1.57	2.6	5
ResNet-50	701	P1	0.3	96.91	96.91	96.94	1.17	2.6	6
ResNet-50	701	P2	0.3	95.43	95.48	95.45	1.36	2.6	7
ResNet-50	751	P1	0.3	96.17	96.15	96.22	1.30	2.6	6
ResNet-50	751	P2	0.3	96.17	96.19	96.19	2.31	2.6	4
ResNet-50	809	P1	0.3	96.42	96.43	96.46	1.91	2.6	5
ResNet-50	809	P2	0.3	95.80	95.79	95.79	2.39	2.6	6
ResNet-50	853	P1	0.3	95.06	95.05	95.04	2.75	2.6	7
ResNet-50	853	P2	0.3	96.17	96.14	96.19	1.12	2.6	4
ResNet-50	907	P1	0.3	96.91	96.95	96.90	2.31	2.6	5
ResNet-50	907	P2	0.3	95.80	95.82	95.77	1.89	2.6	6
ResNet-50	953	P1	0.3	97.78	97.79	97.78	1.12	2.6	5
ResNet-50	953	P2	0.3	96.17	96.24	96.17	1.75	2.6	7
ResNet-50	1009	P1	0.3	97.53	97.55	97.53	0.84	2.6	4
ResNet-50	1009	P2	0.3	94.07	94.07	94.05	3.09	2.6	6
ResNet-50	1051	P1	0.3	97.28	97.28	97.31	0.89	2.6	5
ResNet-50	1051	P2	0.3	96.54	96.56	96.56	1.75	2.6	4
ResNet-50	601	P1	1	96.54	96.57	96.58	1.73	2.6	4
ResNet-50	601	P2	1	94.94	94.97	94.96	3.16	2.6	5
ResNet-50	653	P1	1	92.59	92.64	92.62	2.65	2.6	6
ResNet-50	653	P2	1	93.83	93.87	93.80	3.92	2.6	7
ResNet-50	701	P1	1	90.12	89.73	90.05	3.74	2.6	4
ResNet-50	701	P2	1	92.96	92.99	92.98	3.22	2.6	5
ResNet-50	751	P1	1	93.46	93.51	93.52	4.36	2.6	5
ResNet-50	751	P2	1	91.73	91.81	91.71	3.58	2.6	7
ResNet-50	809	P1	1	93.83	93.85	93.86	1.57	2.6	7
ResNet-50	809	P2	1	94.44	94.43	94.41	1.53	2.6	4

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	853	P1	1	94.81	94.79	94.81	3.11	2.6	5
ResNet-50	853	P2	1	93.95	93.88	93.92	2.14	2.6	6
ResNet-50	907	P1	1	94.57	94.56	94.60	3.46	2.6	7
ResNet-50	907	P2	1	91.48	91.53	91.43	1.86	2.6	4
ResNet-50	953	P1	1	95.19	95.17	95.22	1.58	2.6	4
ResNet-50	953	P2	1	95.80	95.84	95.79	2.78	2.6	6
ResNet-50	1009	P1	1	96.42	96.43	96.43	4.57	2.6	5
ResNet-50	1009	P2	1	91.36	91.34	91.34	4.86	2.6	7
ResNet-50	1051	P1	1	94.44	94.48	94.46	2.70	2.6	6
ResNet-50	1051	P2	1	91.60	91.60	91.56	4.29	2.6	7
ResNet-50	601	P1	3	84.20	84.37	84.20	11.37	2.6	6
ResNet-50	601	P2	3	65.68	60.65	65.87	15.25	2.6	7
ResNet-50	653	P1	3	88.89	89.00	88.94	15.06	2.6	4
ResNet-50	653	P2	3	91.85	91.87	91.87	6.43	2.6	5
ResNet-50	701	P1	3	88.27	88.27	88.30	10.16	2.6	6
ResNet-50	701	P2	3	85.19	85.20	85.28	16.72	2.6	7
ResNet-50	751	P1	3	89.38	89.36	89.43	8.57	2.6	6
ResNet-50	751	P2	3	71.36	71.59	71.29	18.14	2.6	4
ResNet-50	809	P1	3	87.41	87.44	87.41	14.36	2.6	5
ResNet-50	809	P2	3	84.81	84.78	84.89	8.87	2.6	6
ResNet-50	853	P1	3	79.88	79.31	79.98	15.51	2.6	7
ResNet-50	853	P2	3	88.77	88.59	88.70	15.66	2.6	4
ResNet-50	907	P1	3	92.10	92.07	92.15	13.84	2.6	5
ResNet-50	907	P2	3	74.94	72.53	75.11	17.50	2.6	6
ResNet-50	953	P1	3	78.02	78.75	78.00	15.11	2.6	5
ResNet-50	953	P2	3	85.19	84.98	85.18	13.01	2.6	7
ResNet-50	1009	P1	3	63.09	59.40	62.60	7.80	2.6	4
ResNet-50	1009	P2	3	80.37	80.19	80.30	23.57	2.6	6
ResNet-50	1051	P1	3	85.43	85.48	85.41	12.46	2.6	5
ResNet-50	1051	P2	3	90.74	90.82	90.75	12.48	2.6	4
DenseNet-121	601	P1	0	97.04	97.05	97.06	1.45	3.3	6
DenseNet-121	601	P2	0	97.16	97.17	97.18	1.39	3.3	7
DenseNet-121	653	P1	0	96.67	96.70	96.67	1.99	3.3	4
DenseNet-121	653	P2	0	97.16	97.18	97.15	1.27	3.3	5
DenseNet-121	701	P1	0	97.90	97.90	97.91	1.39	3.3	6
DenseNet-121	701	P2	0	96.79	96.80	96.83	2.02	3.3	7
DenseNet-121	751	P1	0	98.27	98.29	98.30	0.91	3.3	6
DenseNet-121	751	P2	0	96.79	96.78	96.81	1.87	3.3	4
DenseNet-121	809	P1	0	98.40	98.40	98.41	0.96	3.3	5
DenseNet-121	809	P2	0	97.41	97.39	97.39	1.62	3.3	6
DenseNet-121	853	P1	0	97.41	97.41	97.43	1.44	3.3	7
DenseNet-121	853	P2	0	96.17	96.19	96.17	2.47	3.3	4
DenseNet-121	907	P1	0	97.53	97.54	97.55	1.59	3.3	5
DenseNet-121	907	P2	0	96.91	96.91	96.91	1.25	3.3	6
DenseNet-121	953	P1	0	98.02	98.03	98.04	1.33	3.3	5
DenseNet-121	953	P2	0	97.28	97.31	97.26	1.54	3.3	7
DenseNet-121	1009	P1	0	98.02	98.03	98.03	1.47	3.3	4
DenseNet-121	1009	P2	0	96.79	96.80	96.80	1.75	3.3	6
DenseNet-121	1051	P1	0	97.53	97.55	97.55	1.06	3.3	5
DenseNet-121	1051	P2	0	97.04	97.03	97.06	1.77	3.3	4
DenseNet-121	601	P1	0.1	96.79	96.79	96.81	1.42	3.3	6
DenseNet-121	601	P2	0.1	97.41	97.42	97.42	1.04	3.3	4
DenseNet-121	653	P1	0.1	95.93	95.96	95.95	2.80	3.3	6
DenseNet-121	653	P2	0.1	97.28	97.30	97.29	1.96	3.3	7
DenseNet-121	701	P1	0.1	97.65	97.66	97.67	1.55	3.3	6
DenseNet-121	701	P2	0.1	95.06	95.10	95.08	2.50	3.3	4
DenseNet-121	751	P1	0.1	97.04	97.06	97.04	1.74	3.3	5
DenseNet-121	751	P2	0.1	97.65	97.66	97.67	1.70	3.3	6
DenseNet-121	809	P1	0.1	96.54	96.53	96.57	2.04	3.3	7
DenseNet-121	809	P2	0.1	97.04	97.05	97.02	1.61	3.3	4
DenseNet-121	853	P1	0.1	97.65	97.66	97.68	1.32	3.3	5
DenseNet-121	853	P2	0.1	96.91	96.93	96.91	2.40	3.3	6
DenseNet-121	907	P1	0.1	97.65	97.66	97.65	1.25	3.3	5
DenseNet-121	907	P2	0.1	96.91	96.94	96.90	2.23	3.3	7
DenseNet-121	953	P1	0.1	97.28	97.29	97.31	1.98	3.3	4
DenseNet-121	953	P2	0.1	96.91	96.94	96.90	2.26	3.3	6
DenseNet-121	1009	P1	0.1	97.41	97.41	97.42	1.24	3.3	5
DenseNet-121	1009	P2	0.1	95.93	95.91	95.96	2.82	3.3	4
DenseNet-121	1051	P1	0.1	96.42	96.44	96.43	1.35	3.3	6
DenseNet-121	1051	P2	0.1	97.04	97.03	97.04	0.97	3.3	7
DenseNet-121	601	P1	0.3	93.70	93.70	93.71	2.71	3.3	5
DenseNet-121	601	P2	0.3	96.67	96.67	96.70	1.37	3.3	7
DenseNet-121	653	P1	0.3	97.41	97.42	97.43	0.97	3.3	4

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	653	P2	0.3	95.80	95.82	95.86	1.74	3.3	5
DenseNet-121	701	P1	0.3	97.53	97.53	97.55	2.07	3.3	5
DenseNet-121	701	P2	0.3	94.69	94.72	94.74	2.66	3.3	7
DenseNet-121	751	P1	0.3	96.91	96.95	96.93	1.32	3.3	7
DenseNet-121	751	P2	0.3	97.78	97.78	97.80	1.64	3.3	4
DenseNet-121	809	P1	0.3	96.17	96.19	96.19	1.65	3.3	5
DenseNet-121	809	P2	0.3	97.41	97.41	97.42	1.87	3.3	6
DenseNet-121	853	P1	0.3	97.04	97.04	97.04	1.81	3.3	7
DenseNet-121	853	P2	0.3	96.67	96.65	96.68	1.38	3.3	4
DenseNet-121	907	P1	0.3	96.79	96.79	96.79	1.00	3.3	4
DenseNet-121	907	P2	0.3	95.56	95.58	95.53	2.00	3.3	6
DenseNet-121	953	P1	0.3	97.16	97.17	97.17	0.92	3.3	5
DenseNet-121	953	P2	0.3	96.42	96.44	96.41	1.72	3.3	7
DenseNet-121	1009	P1	0.3	96.67	96.70	96.67	1.45	3.3	6
DenseNet-121	1009	P2	0.3	93.33	93.29	93.36	2.37	3.3	7
DenseNet-121	1051	P1	0.3	96.91	96.92	96.94	0.96	3.3	5
DenseNet-121	1051	P2	0.3	96.17	96.19	96.17	1.06	3.3	4
DenseNet-121	601	P1	1	92.59	92.56	92.58	4.23	3.3	6
DenseNet-121	601	P2	1	90.86	90.96	90.95	5.81	3.3	4
DenseNet-121	653	P1	1	93.21	93.28	93.16	2.32	3.3	6
DenseNet-121	653	P2	1	92.72	92.72	92.72	4.76	3.3	7
DenseNet-121	701	P1	1	86.54	86.52	86.60	5.01	3.3	6
DenseNet-121	701	P2	1	84.69	84.64	84.58	3.33	3.3	4
DenseNet-121	751	P1	1	90.25	90.08	90.26	2.81	3.3	5
DenseNet-121	751	P2	1	88.52	88.40	88.42	3.68	3.3	6
DenseNet-121	809	P1	1	94.57	94.51	94.62	4.61	3.3	7
DenseNet-121	809	P2	1	89.63	89.53	89.54	3.40	3.3	4
DenseNet-121	853	P1	1	92.96	92.89	92.91	4.25	3.3	5
DenseNet-121	853	P2	1	73.58	70.89	73.44	4.99	3.3	6
DenseNet-121	907	P1	1	89.63	89.70	89.50	2.95	3.3	5
DenseNet-121	907	P2	1	91.73	91.76	91.73	4.61	3.3	7
DenseNet-121	953	P1	1	92.59	92.61	92.57	1.10	3.3	4
DenseNet-121	953	P2	1	91.85	91.84	91.91	3.29	3.3	6
DenseNet-121	1009	P1	1	93.09	93.17	93.12	3.88	3.3	5
DenseNet-121	1009	P2	1	88.27	88.23	88.22	2.18	3.3	4
DenseNet-121	1051	P1	1	89.38	89.45	89.37	3.03	3.3	6
DenseNet-121	1051	P2	1	87.78	87.77	87.84	4.71	3.3	7
DenseNet-121	601	P1	3	63.09	62.82	63.10	4.84	3.3	5
DenseNet-121	601	P2	3	69.01	68.02	68.75	17.64	3.3	7
DenseNet-121	653	P1	3	64.44	63.31	64.49	10.23	3.3	4
DenseNet-121	653	P2	3	74.07	74.04	73.74	15.13	3.3	5
DenseNet-121	701	P1	3	63.70	62.00	63.77	9.14	3.3	5
DenseNet-121	701	P2	3	59.26	60.38	59.26	7.80	3.3	7
DenseNet-121	751	P1	3	76.54	75.83	76.58	18.67	3.3	7
DenseNet-121	751	P2	3	68.89	67.64	68.90	11.27	3.3	4
DenseNet-121	809	P1	3	64.57	63.39	64.30	9.81	3.3	5
DenseNet-121	809	P2	3	76.67	76.37	76.48	16.50	3.3	6
DenseNet-121	853	P1	3	59.26	56.41	59.36	6.20	3.3	7
DenseNet-121	853	P2	3	52.35	49.76	52.02	8.03	3.3	4
DenseNet-121	907	P1	3	69.88	69.68	69.68	7.50	3.3	4
DenseNet-121	907	P2	3	81.23	81.01	81.16	15.90	3.3	6
DenseNet-121	953	P1	3	52.96	52.57	53.28	6.27	3.3	5
DenseNet-121	953	P2	3	53.70	53.90	53.64	8.01	3.3	7
DenseNet-121	1009	P1	3	65.68	60.71	65.40	7.08	3.3	6
DenseNet-121	1009	P2	3	64.44	64.16	64.54	16.23	3.3	7
DenseNet-121	1051	P1	3	79.14	78.85	79.00	20.84	3.3	5
DenseNet-121	1051	P2	3	61.48	57.65	61.19	10.02	3.3	4
ConvNeXt-T	601	P1	0	98.40	98.40	98.41	1.60	5.2	4
ConvNeXt-T	601	P2	0	97.16	97.17	97.18	1.93	5.2	5
ConvNeXt-T	653	P1	0	97.65	97.68	97.64	1.82	5.2	6
ConvNeXt-T	653	P2	0	97.53	97.53	97.54	1.75	5.2	7
ConvNeXt-T	701	P1	0	98.52	98.52	98.53	1.12	5.2	4
ConvNeXt-T	701	P2	0	94.94	94.94	94.98	4.14	5.2	5
ConvNeXt-T	751	P1	0	98.40	98.40	98.42	1.28	5.2	5
ConvNeXt-T	751	P2	0	96.67	96.67	96.66	2.46	5.2	7
ConvNeXt-T	809	P1	0	97.16	97.16	97.19	2.57	5.2	7
ConvNeXt-T	809	P2	0	96.42	96.42	96.39	2.49	5.2	4
ConvNeXt-T	853	P1	0	97.53	97.53	97.55	1.67	5.2	5
ConvNeXt-T	853	P2	0	97.16	97.18	97.13	2.17	5.2	6
ConvNeXt-T	907	P1	0	98.02	98.04	98.03	1.56	5.2	7
ConvNeXt-T	907	P2	0	95.93	95.91	95.96	3.26	5.2	4
ConvNeXt-T	953	P1	0	96.42	96.35	96.47	2.74	5.2	4
ConvNeXt-T	953	P2	0	96.42	96.46	96.43	2.76	5.1	6

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	1009	P1	0	97.04	97.08	97.02	1.87	5.2	5
ConvNeXt-T	1009	P2	0	97.04	97.03	97.08	2.44	5.2	7
ConvNeXt-T	1051	P1	0	98.52	98.53	98.53	0.97	5.1	6
ConvNeXt-T	1051	P2	0	96.17	96.19	96.15	2.38	5.2	7
ConvNeXt-T	601	P1	0.1	97.41	97.44	97.42	2.27	5.2	6
ConvNeXt-T	601	P2	0.1	98.15	98.15	98.15	1.15	5.3	7
ConvNeXt-T	653	P1	0.1	98.02	98.04	98.01	1.51	5.2	4
ConvNeXt-T	653	P2	0.1	96.54	96.55	96.53	2.97	5.2	5
ConvNeXt-T	701	P1	0.1	96.67	96.70	96.66	3.02	5.2	6
ConvNeXt-T	701	P2	0.1	95.93	95.93	95.96	2.95	5.2	7
ConvNeXt-T	751	P1	0.1	98.27	98.28	98.29	1.50	5.2	6
ConvNeXt-T	751	P2	0.1	95.93	95.96	95.90	3.19	5.2	4
ConvNeXt-T	809	P1	0.1	97.41	97.41	97.43	1.88	5.2	5
ConvNeXt-T	809	P2	0.1	95.31	95.30	95.26	3.63	5.1	6
ConvNeXt-T	853	P1	0.1	97.41	97.42	97.42	1.86	5.2	7
ConvNeXt-T	853	P2	0.1	96.42	96.40	96.42	2.89	5.2	4
ConvNeXt-T	907	P1	0.1	99.01	99.02	99.01	0.81	5.2	5
ConvNeXt-T	907	P2	0.1	96.54	96.55	96.54	2.88	5.1	6
ConvNeXt-T	953	P1	0.1	96.42	96.38	96.48	2.60	5.2	5
ConvNeXt-T	953	P2	0.1	97.04	97.07	97.04	2.38	5.2	7
ConvNeXt-T	1009	P1	0.1	97.41	97.42	97.44	2.41	5.2	4
ConvNeXt-T	1009	P2	0.1	97.16	97.16	97.17	2.09	5.1	6
ConvNeXt-T	1051	P1	0.1	97.53	97.52	97.57	1.91	5.2	5
ConvNeXt-T	1051	P2	0.1	96.54	96.57	96.52	2.56	5.2	4
ConvNeXt-T	601	P1	0.3	96.91	96.95	96.92	2.43	5.2	4
ConvNeXt-T	601	P2	0.3	96.91	96.93	96.91	1.51	5.2	5
ConvNeXt-T	653	P1	0.3	97.90	97.91	97.90	1.31	5.2	6
ConvNeXt-T	653	P2	0.3	96.54	96.57	96.54	2.31	5.2	7
ConvNeXt-T	701	P1	0.3	97.65	97.66	97.66	1.69	5.2	4
ConvNeXt-T	701	P2	0.3	96.30	96.29	96.35	2.82	5.2	5
ConvNeXt-T	751	P1	0.3	98.64	98.65	98.65	0.61	5.2	5
ConvNeXt-T	751	P2	0.3	91.11	90.63	90.99	6.76	5.2	7
ConvNeXt-T	809	P1	0.3	97.41	97.43	97.43	1.52	5.2	7
ConvNeXt-T	809	P2	0.3	95.93	95.94	95.91	3.09	5.2	4
ConvNeXt-T	853	P1	0.3	97.16	97.15	97.20	1.66	5.2	5
ConvNeXt-T	853	P2	0.3	96.05	96.01	95.98	2.84	5.1	6
ConvNeXt-T	907	P1	0.3	98.27	98.28	98.28	1.17	5.2	7
ConvNeXt-T	907	P2	0.3	96.54	96.55	96.54	1.84	5.2	4
ConvNeXt-T	953	P1	0.3	96.42	96.40	96.46	2.68	5.2	4
ConvNeXt-T	953	P2	0.3	96.54	96.59	96.54	2.67	5.2	6
ConvNeXt-T	1009	P1	0.3	97.90	97.91	97.91	1.79	5.2	5
ConvNeXt-T	1009	P2	0.3	96.54	96.54	96.56	3.03	5.2	7
ConvNeXt-T	1051	P1	0.3	97.53	97.56	97.54	1.65	5.1	6
ConvNeXt-T	1051	P2	0.3	97.78	97.80	97.76	1.37	5.2	7
ConvNeXt-T	601	P1	1	97.16	97.18	97.18	0.86	5.2	6
ConvNeXt-T	601	P2	1	96.17	96.18	96.16	4.22	5.3	7
ConvNeXt-T	653	P1	1	97.16	97.15	97.18	3.60	5.2	4
ConvNeXt-T	653	P2	1	96.67	96.68	96.65	2.62	5.2	5
ConvNeXt-T	701	P1	1	97.28	97.29	97.29	1.58	5.2	6
ConvNeXt-T	701	P2	1	93.21	93.19	93.29	2.63	5.2	7
ConvNeXt-T	751	P1	1	96.79	96.78	96.82	2.46	5.1	6
ConvNeXt-T	751	P2	1	95.19	95.20	95.14	4.02	5.2	4
ConvNeXt-T	809	P1	1	96.17	96.14	96.22	1.65	5.2	5
ConvNeXt-T	809	P2	1	94.44	94.50	94.45	1.87	5.1	6
ConvNeXt-T	853	P1	1	97.90	97.91	97.91	0.90	5.2	7
ConvNeXt-T	853	P2	1	95.56	95.56	95.53	2.03	5.2	4
ConvNeXt-T	907	P1	1	97.41	97.42	97.41	1.05	5.2	5
ConvNeXt-T	907	P2	1	96.54	96.55	96.55	3.32	5.1	6
ConvNeXt-T	953	P1	1	96.05	96.00	96.11	3.54	5.2	5
ConvNeXt-T	953	P2	1	95.31	95.35	95.29	3.00	5.2	7
ConvNeXt-T	1009	P1	1	97.90	97.91	97.92	1.94	5.2	4
ConvNeXt-T	1009	P2	1	95.93	95.98	95.94	4.03	5.2	6
ConvNeXt-T	1051	P1	1	96.17	96.21	96.18	2.23	5.2	5
ConvNeXt-T	1051	P2	1	97.28	97.28	97.27	1.92	5.2	4
ConvNeXt-T	601	P1	3	87.53	87.78	87.58	4.66	5.2	4
ConvNeXt-T	601	P2	3	82.35	83.05	82.19	6.82	5.2	5
ConvNeXt-T	653	P1	3	82.84	82.65	82.79	4.79	5.2	6
ConvNeXt-T	653	P2	3	90.49	90.36	90.58	3.55	5.2	7
ConvNeXt-T	701	P1	3	87.90	88.02	87.97	5.80	5.2	4
ConvNeXt-T	701	P2	3	91.11	91.24	91.06	12.60	5.2	5
ConvNeXt-T	751	P1	3	88.89	89.13	88.94	6.94	5.2	5
ConvNeXt-T	751	P2	3	19.63	6.77	20.12	14.58	5.2	7
ConvNeXt-T	809	P1	3	92.10	92.14	92.18	6.37	5.2	7

Architecture	Seed	Protocol	adversary_lambda	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	809	P2	3	93.33	93.28	93.33	9.11	5.2	4
ConvNeXt-T	853	P1	3	86.17	85.38	86.34	11.14	5.2	5
ConvNeXt-T	853	P2	3	87.28	87.02	87.11	4.75	5.1	6
ConvNeXt-T	907	P1	3	89.63	89.46	89.64	8.63	5.2	7
ConvNeXt-T	907	P2	3	84.44	84.44	84.40	4.86	5.2	4
ConvNeXt-T	953	P1	3	93.58	93.57	93.61	8.92	5.2	4
ConvNeXt-T	953	P2	3	64.94	62.05	64.93	12.03	5.1	6
ConvNeXt-T	1009	P1	3	86.42	86.10	86.52	5.14	5.2	5
ConvNeXt-T	1009	P2	3	85.80	85.78	85.60	6.35	5.2	7
ConvNeXt-T	1051	P1	3	88.52	88.76	88.51	4.31	5.1	6
ConvNeXt-T	1051	P2	3	20.12	6.70	20.00	17.86	5.2	7

Table M.3: Complete Amendment 8 normalisation run ledger (160 runs).

Architecture	Seed	Protocol	cache_name	Acc.	F1	Bal.	ECE	Min	GPU
ResNet-50	601	P1	sipakmed_reinhard100	95.31	95.38	95.29	3.19	2.6	5
ResNet-50	601	P2	sipakmed_reinhard100	94.32	94.33	94.32	3.64	2.6	4
ResNet-50	653	P1	sipakmed_reinhard100	95.68	95.70	95.68	3.46	2.6	5
ResNet-50	653	P2	sipakmed_reinhard100	96.30	96.32	96.29	2.82	2.6	4
ResNet-50	701	P1	sipakmed_reinhard100	95.31	95.31	95.34	2.70	2.6	5
ResNet-50	701	P2	sipakmed_reinhard100	93.46	93.48	93.46	4.74	2.6	4
ResNet-50	751	P1	sipakmed_reinhard100	96.30	96.30	96.34	2.39	2.6	5
ResNet-50	751	P2	sipakmed_reinhard100	94.57	94.55	94.57	3.49	2.6	4
ResNet-50	809	P1	sipakmed_reinhard100	95.68	95.69	95.72	2.27	2.7	5
ResNet-50	809	P2	sipakmed_reinhard100	97.28	97.27	97.31	2.03	2.7	6
ResNet-50	853	P1	sipakmed_reinhard100	96.42	96.44	96.42	1.74	2.7	6
ResNet-50	853	P2	sipakmed_reinhard100	95.31	95.30	95.31	3.15	2.7	7
ResNet-50	907	P1	sipakmed_reinhard100	96.05	96.06	96.05	2.52	2.7	7
ResNet-50	907	P2	sipakmed_reinhard100	92.96	92.94	92.94	4.61	2.7	5
ResNet-50	953	P1	sipakmed_reinhard100	96.17	96.19	96.21	2.49	2.7	5
ResNet-50	953	P2	sipakmed_reinhard100	96.67	96.70	96.66	3.02	2.7	6
ResNet-50	1009	P1	sipakmed_reinhard100	95.80	95.83	95.81	2.00	2.7	6
ResNet-50	1009	P2	sipakmed_reinhard100	95.06	95.07	95.05	2.87	2.7	7
ResNet-50	1051	P1	sipakmed_reinhard100	95.56	95.55	95.58	2.60	2.7	7
ResNet-50	1051	P2	sipakmed_reinhard100	94.20	94.26	94.15	4.07	2.7	5
ResNet-50	601	P1	sipakmed_reinhard50	96.91	96.93	96.92	2.07	2.6	6
ResNet-50	601	P2	sipakmed_reinhard50	96.67	96.69	96.68	1.71	2.6	7
ResNet-50	653	P1	sipakmed_reinhard50	97.41	97.42	97.41	1.49	2.6	6
ResNet-50	653	P2	sipakmed_reinhard50	95.43	95.47	95.42	3.06	2.6	7
ResNet-50	701	P1	sipakmed_reinhard50	97.53	97.52	97.55	1.32	2.6	6
ResNet-50	701	P2	sipakmed_reinhard50	94.69	94.71	94.70	3.19	2.6	7
ResNet-50	751	P1	sipakmed_reinhard50	97.90	97.92	97.92	1.47	2.6	6
ResNet-50	751	P2	sipakmed_reinhard50	96.30	96.29	96.30	2.27	2.6	7
ResNet-50	809	P1	sipakmed_reinhard50	96.42	96.44	96.44	2.22	3.5	4
ResNet-50	809	P2	sipakmed_reinhard50	96.05	96.07	96.04	2.83	2.7	7
ResNet-50	853	P1	sipakmed_reinhard50	97.65	97.66	97.66	1.36	3.6	4
ResNet-50	853	P2	sipakmed_reinhard50	96.67	96.64	96.67	2.04	2.7	5
ResNet-50	907	P1	sipakmed_reinhard50	98.15	98.17	98.16	1.23	3.6	4
ResNet-50	907	P2	sipakmed_reinhard50	96.17	96.19	96.15	2.63	2.7	6
ResNet-50	953	P1	sipakmed_reinhard50	96.79	96.80	96.79	1.83	3.6	4
ResNet-50	953	P2	sipakmed_reinhard50	97.04	97.06	97.04	2.57	2.8	7
ResNet-50	1009	P1	sipakmed_reinhard50	96.67	96.70	96.65	2.25	3.6	4
ResNet-50	1009	P2	sipakmed_reinhard50	96.42	96.40	96.47	2.62	2.7	5
ResNet-50	1051	P1	sipakmed_reinhard50	97.65	97.66	97.66	1.56	3.6	4
ResNet-50	1051	P2	sipakmed_reinhard50	95.80	95.81	95.76	2.38	2.7	6
DenseNet-121	601	P1	sipakmed_reinhard100	94.69	94.71	94.73	2.19	3.3	5
DenseNet-121	601	P2	sipakmed_reinhard100	94.94	94.96	94.94	3.11	3.3	4
DenseNet-121	653	P1	sipakmed_reinhard100	97.16	97.18	97.17	1.29	3.3	5
DenseNet-121	653	P2	sipakmed_reinhard100	95.31	95.37	95.29	2.23	3.3	4
DenseNet-121	701	P1	sipakmed_reinhard100	95.43	95.39	95.47	2.76	3.3	5
DenseNet-121	701	P2	sipakmed_reinhard100	93.83	93.89	93.81	3.80	3.3	4
DenseNet-121	751	P1	sipakmed_reinhard100	96.79	96.80	96.83	2.08	3.3	5
DenseNet-121	751	P2	sipakmed_reinhard100	95.56	95.53	95.55	2.67	4.2	4
DenseNet-121	809	P1	sipakmed_reinhard100	97.28	97.29	97.29	1.76	4.2	4
DenseNet-121	809	P2	sipakmed_reinhard100	96.17	96.15	96.19	2.38	3.3	6
DenseNet-121	853	P1	sipakmed_reinhard100	96.30	96.28	96.32	1.83	4.2	4
DenseNet-121	853	P2	sipakmed_reinhard100	95.43	95.44	95.41	2.13	3.3	7
DenseNet-121	907	P1	sipakmed_reinhard100	96.67	96.69	96.66	1.49	3.3	5
DenseNet-121	907	P2	sipakmed_reinhard100	94.20	94.24	94.17	4.11	4.2	4

Architecture	Seed	Protocol	cache_name	Acc.	F1	Bal.	ECE	Min	GPU
DenseNet-121	953	P1	sipakmed_reinhard100	95.43	95.45	95.47	2.17	3.3	6
DenseNet-121	953	P2	sipakmed_reinhard100	97.04	97.05	97.02	1.92	4.2	4
DenseNet-121	1009	P1	sipakmed_reinhard100	96.05	96.10	96.05	2.35	4.3	4
DenseNet-121	1009	P2	sipakmed_reinhard100	94.94	94.96	94.92	3.36	3.3	7
DenseNet-121	1051	P1	sipakmed_reinhard100	96.17	96.18	96.19	1.86	4.3	4
DenseNet-121	1051	P2	sipakmed_reinhard100	95.56	95.56	95.52	1.84	3.3	5
DenseNet-121	601	P1	sipakmed_reinhard50	97.04	97.05	97.06	1.83	3.3	6
DenseNet-121	601	P2	sipakmed_reinhard50	96.54	96.54	96.56	1.60	3.3	7
DenseNet-121	653	P1	sipakmed_reinhard50	98.27	98.29	98.27	1.06	3.3	6
DenseNet-121	653	P2	sipakmed_reinhard50	96.54	96.59	96.54	2.22	3.3	7
DenseNet-121	701	P1	sipakmed_reinhard50	97.04	97.02	97.06	1.56	3.3	6
DenseNet-121	701	P2	sipakmed_reinhard50	95.43	95.46	95.45	2.90	3.3	7
DenseNet-121	751	P1	sipakmed_reinhard50	97.90	97.92	97.90	1.20	3.3	6
DenseNet-121	751	P2	sipakmed_reinhard50	96.54	96.55	96.57	2.34	3.3	7
DenseNet-121	809	P1	sipakmed_reinhard50	96.91	96.91	96.95	1.87	3.3	7
DenseNet-121	809	P2	sipakmed_reinhard50	96.54	96.56	96.52	2.18	3.3	5
DenseNet-121	853	P1	sipakmed_reinhard50	97.41	97.40	97.44	1.12	3.3	5
DenseNet-121	853	P2	sipakmed_reinhard50	96.42	96.41	96.39	1.96	3.3	6
DenseNet-121	907	P1	sipakmed_reinhard50	97.90	97.91	97.92	1.48	3.3	6
DenseNet-121	907	P2	sipakmed_reinhard50	95.31	95.34	95.28	3.12	3.3	7
DenseNet-121	953	P1	sipakmed_reinhard50	97.16	97.16	97.17	1.39	3.3	7
DenseNet-121	953	P2	sipakmed_reinhard50	96.79	96.84	96.78	2.34	3.3	5
DenseNet-121	1009	P1	sipakmed_reinhard50	97.53	97.55	97.54	1.48	3.3	5
DenseNet-121	1009	P2	sipakmed_reinhard50	95.68	95.71	95.67	2.84	3.3	6
DenseNet-121	1051	P1	sipakmed_reinhard50	97.90	97.91	97.90	1.36	3.3	6
DenseNet-121	1051	P2	sipakmed_reinhard50	97.16	97.16	97.15	0.98	3.3	7
EfficientNet-B0	601	P1	sipakmed_reinhard100	95.93	95.96	95.94	3.67	1.8	5
EfficientNet-B0	601	P2	sipakmed_reinhard100	94.44	94.48	94.43	3.04	1.8	4
EfficientNet-B0	653	P1	sipakmed_reinhard100	96.30	96.33	96.28	2.15	1.8	5
EfficientNet-B0	653	P2	sipakmed_reinhard100	95.93	95.95	95.92	2.51	1.8	4
EfficientNet-B0	701	P1	sipakmed_reinhard100	95.68	95.66	95.70	2.93	1.8	5
EfficientNet-B0	701	P2	sipakmed_reinhard100	95.06	95.10	95.06	3.59	1.8	4
EfficientNet-B0	751	P1	sipakmed_reinhard100	95.93	95.94	95.95	3.01	1.8	5
EfficientNet-B0	751	P2	sipakmed_reinhard100	97.04	97.03	97.05	1.76	1.8	6
EfficientNet-B0	809	P1	sipakmed_reinhard100	96.67	96.66	96.70	2.18	1.8	6
EfficientNet-B0	809	P2	sipakmed_reinhard100	96.05	96.04	96.06	2.35	1.8	7
EfficientNet-B0	853	P1	sipakmed_reinhard100	95.31	95.29	95.33	2.51	1.8	7
EfficientNet-B0	853	P2	sipakmed_reinhard100	96.42	96.41	96.41	2.12	1.8	5
EfficientNet-B0	907	P1	sipakmed_reinhard100	97.04	97.05	97.04	1.92	1.8	5
EfficientNet-B0	907	P2	sipakmed_reinhard100	93.95	94.00	93.93	4.61	1.8	6
EfficientNet-B0	953	P1	sipakmed_reinhard100	96.17	96.16	96.19	2.98	1.8	6
EfficientNet-B0	953	P2	sipakmed_reinhard100	97.28	97.31	97.28	1.69	1.8	7
EfficientNet-B0	1009	P1	sipakmed_reinhard100	96.05	96.09	96.05	3.49	1.8	7
EfficientNet-B0	1009	P2	sipakmed_reinhard100	94.57	94.53	94.57	3.49	1.8	5
EfficientNet-B0	1051	P1	sipakmed_reinhard100	97.04	97.04	97.04	1.29	1.8	5
EfficientNet-B0	1051	P2	sipakmed_reinhard100	95.19	95.18	95.13	3.17	1.8	6
EfficientNet-B0	601	P1	sipakmed_reinhard50	97.16	97.17	97.18	2.23	1.8	6
EfficientNet-B0	601	P2	sipakmed_reinhard50	96.54	96.55	96.54	1.39	1.8	7
EfficientNet-B0	653	P1	sipakmed_reinhard50	98.52	98.53	98.52	0.78	1.8	6
EfficientNet-B0	653	P2	sipakmed_reinhard50	96.67	96.68	96.67	1.74	1.8	7
EfficientNet-B0	701	P1	sipakmed_reinhard50	97.41	97.40	97.41	1.77	1.8	6
EfficientNet-B0	701	P2	sipakmed_reinhard50	95.31	95.34	95.34	3.31	1.8	7
EfficientNet-B0	751	P1	sipakmed_reinhard50	97.78	97.79	97.80	1.37	1.8	6
EfficientNet-B0	751	P2	sipakmed_reinhard50	96.79	96.80	96.79	2.20	1.8	7
EfficientNet-B0	809	P1	sipakmed_reinhard50	97.41	97.41	97.44	1.75	1.8	7
EfficientNet-B0	809	P2	sipakmed_reinhard50	96.05	96.05	96.03	2.68	1.8	5
EfficientNet-B0	853	P1	sipakmed_reinhard50	96.67	96.65	96.69	2.12	1.8	5
EfficientNet-B0	853	P2	sipakmed_reinhard50	96.30	96.28	96.28	2.03	1.8	6
EfficientNet-B0	907	P1	sipakmed_reinhard50	97.53	97.55	97.54	1.35	1.8	6
EfficientNet-B0	907	P2	sipakmed_reinhard50	96.30	96.31	96.28	2.67	1.8	7
EfficientNet-B0	953	P1	sipakmed_reinhard50	96.67	96.68	96.67	2.18	1.8	7
EfficientNet-B0	953	P2	sipakmed_reinhard50	97.41	97.44	97.40	1.38	1.8	5
EfficientNet-B0	1009	P1	sipakmed_reinhard50	96.91	96.95	96.92	1.99	1.8	5
EfficientNet-B0	1009	P2	sipakmed_reinhard50	97.04	97.04	97.04	1.93	1.8	6
EfficientNet-B0	1051	P1	sipakmed_reinhard50	97.65	97.65	97.67	1.50	1.8	6
EfficientNet-B0	1051	P2	sipakmed_reinhard50	97.04	97.05	97.03	1.55	1.8	7
ConvNeXt-T	601	P1	sipakmed_reinhard100	96.05	96.07	96.05	3.46	5.2	5
ConvNeXt-T	601	P2	sipakmed_reinhard100	95.56	95.56	95.58	3.10	5.2	4
ConvNeXt-T	653	P1	sipakmed_reinhard100	97.90	97.92	97.91	1.67	5.2	5
ConvNeXt-T	653	P2	sipakmed_reinhard100	95.06	95.00	95.08	3.33	5.2	4
ConvNeXt-T	701	P1	sipakmed_reinhard100	96.30	96.31	96.31	2.67	5.2	5
ConvNeXt-T	701	P2	sipakmed_reinhard100	93.21	93.22	93.26	5.78	5.2	4
ConvNeXt-T	751	P1	sipakmed_reinhard100	96.42	96.44	96.46	2.76	5.3	5

Architecture	Seed	Protocol	cache_name	Acc.	F1	Bal.	ECE	Min	GPU
ConvNeXt-T	751	P2	sipakmed_reinhard100	96.17	96.16	96.16	2.65	5.9	4
ConvNeXt-T	809	P1	sipakmed_reinhard100	97.41	97.39	97.43	2.02	6.3	4
ConvNeXt-T	809	P2	sipakmed_reinhard100	96.30	96.30	96.27	2.19	5.2	6
ConvNeXt-T	853	P1	sipakmed_reinhard100	96.30	96.29	96.32	2.41	5.3	6
ConvNeXt-T	853	P2	sipakmed_reinhard100	96.54	96.51	96.55	2.70	5.5	7
ConvNeXt-T	907	P1	sipakmed_reinhard100	96.67	96.63	96.71	2.24	5.5	7
ConvNeXt-T	907	P2	sipakmed_reinhard100	94.57	94.65	94.53	3.86	5.4	5
ConvNeXt-T	953	P1	sipakmed_reinhard100	94.69	94.66	94.75	3.77	5.4	5
ConvNeXt-T	953	P2	sipakmed_reinhard100	96.05	96.06	96.08	3.19	5.2	6
ConvNeXt-T	1009	P1	sipakmed_reinhard100	93.58	93.63	93.57	4.53	6.4	4
ConvNeXt-T	1009	P2	sipakmed_reinhard100	94.81	94.82	94.81	3.60	5.5	7
ConvNeXt-T	1051	P1	sipakmed_reinhard100	95.80	95.80	95.85	3.45	5.5	7
ConvNeXt-T	1051	P2	sipakmed_reinhard100	96.42	96.41	96.43	2.74	5.4	5
ConvNeXt-T	601	P1	sipakmed_reinhard50	97.53	97.54	97.54	1.93	5.1	6
ConvNeXt-T	601	P2	sipakmed_reinhard50	97.16	97.17	97.17	2.18	5.2	7
ConvNeXt-T	653	P1	sipakmed_reinhard50	97.78	97.80	97.79	1.70	5.2	6
ConvNeXt-T	653	P2	sipakmed_reinhard50	92.84	92.79	92.78	5.21	5.2	7
ConvNeXt-T	701	P1	sipakmed_reinhard50	96.54	96.55	96.56	2.15	5.1	6
ConvNeXt-T	701	P2	sipakmed_reinhard50	93.70	93.68	93.77	4.95	5.2	7
ConvNeXt-T	751	P1	sipakmed_reinhard50	97.78	97.80	97.80	1.42	5.2	6
ConvNeXt-T	751	P2	sipakmed_reinhard50	95.06	94.96	94.99	3.64	5.4	7
ConvNeXt-T	809	P1	sipakmed_reinhard50	97.16	97.18	97.16	1.98	5.5	7
ConvNeXt-T	809	P2	sipakmed_reinhard50	95.80	95.77	95.76	3.40	5.4	5
ConvNeXt-T	853	P1	sipakmed_reinhard50	97.28	97.28	97.28	1.36	5.4	5
ConvNeXt-T	853	P2	sipakmed_reinhard50	97.04	97.02	97.01	1.67	6.3	4
ConvNeXt-T	907	P1	sipakmed_reinhard50	97.78	97.78	97.80	1.41	5.2	6
ConvNeXt-T	907	P2	sipakmed_reinhard50	94.94	94.94	94.96	3.97	6.3	4
ConvNeXt-T	953	P1	sipakmed_reinhard50	95.68	95.63	95.72	3.09	5.5	7
ConvNeXt-T	953	P2	sipakmed_reinhard50	96.30	96.32	96.32	2.95	6.3	4
ConvNeXt-T	1009	P1	sipakmed_reinhard50	96.42	96.46	96.44	2.40	5.4	5
ConvNeXt-T	1009	P2	sipakmed_reinhard50	95.68	95.66	95.71	3.15	5.3	6
ConvNeXt-T	1051	P1	sipakmed_reinhard50	96.17	96.23	96.17	2.66	5.3	6
ConvNeXt-T	1051	P2	sipakmed_reinhard50	96.91	96.91	96.92	2.42	6.4	4

Table M.4: Complete LEACE ledger (480 models): refitted-head accuracy and recoverability at rank 0 and rank 256 (percent).

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
ResNet-18	137	P1	97.78	97.53	22.8	1.8
ResNet-18	137	P2	96.54	96.42	23.8	2.5
ResNet-18	173	P1	97.90	97.53	25.0	2.0
ResNet-18	173	P2	97.78	97.53	22.9	2.0
ResNet-18	211	P1	97.65	98.02	24.2	2.6
ResNet-18	211	P2	96.91	96.54	22.3	2.8
ResNet-18	257	P1	96.79	97.28	24.8	1.8
ResNet-18	257	P2	98.02	98.40	24.2	2.4
ResNet-18	307	P1	98.52	98.40	22.1	2.5
ResNet-18	307	P2	97.78	97.28	22.9	2.9
ResNet-18	349	P1	97.41	97.28	22.3	1.9
ResNet-18	349	P2	97.16	95.80	22.4	2.5
ResNet-18	401	P1	98.15	97.90	21.9	1.7
ResNet-18	401	P2	98.52	98.77	24.1	1.9
ResNet-18	457	P1	98.40	97.90	25.2	1.8
ResNet-18	457	P2	96.05	94.44	21.8	2.0
ResNet-18	503	P1	98.15	98.27	24.4	2.1
ResNet-18	503	P2	96.91	96.54	23.2	2.4
ResNet-18	557	P1	98.27	97.53	23.3	2.0
ResNet-18	557	P2	97.65	97.78	25.2	2.6
ResNet-18	601	P1	98.02	97.78	23.2	2.0
ResNet-18	601	P2	97.41	96.91	22.9	2.9
ResNet-18	653	P1	97.78	98.02	21.9	2.2
ResNet-18	653	P2	97.28	97.16	24.1	2.2
ResNet-18	701	P1	97.78	97.41	22.8	1.5
ResNet-18	701	P2	96.67	95.80	21.6	1.9
ResNet-18	751	P1	97.78	97.90	21.0	1.9
ResNet-18	751	P2	96.54	96.17	23.8	1.8
ResNet-18	809	P1	97.78	97.65	24.8	1.7
ResNet-18	809	P2	96.91	96.42	23.6	2.9
ResNet-18	853	P1	98.02	97.53	23.2	2.7

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
ResNet-18	853	P2	97.28	97.41	21.9	3.1
ResNet-18	907	P1	97.78	97.90	23.1	1.4
ResNet-18	907	P2	97.78	97.90	22.5	2.6
ResNet-18	953	P1	98.02	97.53	21.5	1.9
ResNet-18	953	P2	97.65	97.90	22.9	0.8
ResNet-18	1009	P1	98.15	98.27	22.4	1.1
ResNet-18	1009	P2	97.28	96.91	25.5	2.2
ResNet-18	1051	P1	97.78	97.65	23.0	2.6
ResNet-18	1051	P2	98.02	97.41	23.5	2.4
ResNet-50	137	P1	98.15	97.90	25.7	10.4
ResNet-50	137	P2	96.30	96.30	26.2	11.2
ResNet-50	173	P1	98.15	98.52	23.6	10.8
ResNet-50	173	P2	97.78	98.27	26.9	10.3
ResNet-50	211	P1	97.78	98.15	24.6	10.6
ResNet-50	211	P2	97.16	97.04	24.5	10.1
ResNet-50	257	P1	98.27	98.02	23.8	9.4
ResNet-50	257	P2	97.04	97.53	25.5	11.2
ResNet-50	307	P1	98.89	98.15	23.0	10.1
ResNet-50	307	P2	97.28	97.41	22.4	9.8
ResNet-50	349	P1	97.41	97.41	25.5	10.0
ResNet-50	349	P2	96.67	96.91	25.1	10.0
ResNet-50	401	P1	98.64	97.90	22.5	9.7
ResNet-50	401	P2	97.41	97.53	24.5	9.1
ResNet-50	457	P1	98.40	98.40	25.2	10.8
ResNet-50	457	P2	95.56	95.06	25.1	10.6
ResNet-50	503	P1	98.15	98.15	24.8	12.3
ResNet-50	503	P2	97.28	97.28	23.2	11.6
ResNet-50	557	P1	97.90	97.65	25.4	11.1
ResNet-50	557	P2	98.27	98.02	25.2	10.1
ResNet-50	601	P1	98.15	98.40	23.8	9.8
ResNet-50	601	P2	98.52	97.65	25.1	9.7
ResNet-50	653	P1	98.15	98.40	26.1	12.1
ResNet-50	653	P2	97.53	97.28	25.9	11.4
ResNet-50	701	P1	98.02	97.53	23.5	10.4
ResNet-50	701	P2	97.04	96.91	25.6	10.7
ResNet-50	751	P1	98.40	98.27	25.5	11.8
ResNet-50	751	P2	96.17	96.05	23.7	9.7
ResNet-50	809	P1	97.90	97.41	25.9	11.4
ResNet-50	809	P2	97.28	97.41	23.9	10.0
ResNet-50	853	P1	97.16	96.79	25.5	11.2
ResNet-50	853	P2	96.91	96.79	26.4	10.4
ResNet-50	907	P1	98.77	98.15	23.0	9.0
ResNet-50	907	P2	97.16	97.78	25.2	8.6
ResNet-50	953	P1	97.78	97.53	24.8	10.6
ResNet-50	953	P2	97.04	96.91	24.5	9.6
ResNet-50	1009	P1	98.40	97.78	23.6	10.6
ResNet-50	1009	P2	97.53	97.16	23.6	9.9
ResNet-50	1051	P1	97.90	97.78	25.7	11.7
ResNet-50	1051	P2	97.28	97.53	24.5	8.6
DenseNet-121	137	P1	98.52	97.65	29.2	3.4
DenseNet-121	137	P2	97.16	97.04	29.1	4.4
DenseNet-121	173	P1	98.15	98.27	30.8	4.5
DenseNet-121	173	P2	98.52	98.52	29.6	4.7
DenseNet-121	211	P1	98.64	98.40	29.7	4.4
DenseNet-121	211	P2	97.41	97.65	28.5	3.4
DenseNet-121	257	P1	97.41	97.28	30.2	3.7
DenseNet-121	257	P2	97.65	96.79	29.1	4.4
DenseNet-121	307	P1	98.77	98.64	29.6	3.2
DenseNet-121	307	P2	98.40	98.27	30.4	5.0
DenseNet-121	349	P1	98.15	97.90	27.9	5.0
DenseNet-121	349	P2	97.65	97.04	27.7	4.0
DenseNet-121	401	P1	98.02	97.90	30.3	4.2
DenseNet-121	401	P2	97.53	97.65	29.8	3.7
DenseNet-121	457	P1	98.27	98.02	27.6	3.9
DenseNet-121	457	P2	96.17	95.80	29.2	4.7
DenseNet-121	503	P1	98.15	97.78	30.8	4.4
DenseNet-121	503	P2	97.90	98.02	27.5	3.4
DenseNet-121	557	P1	98.15	97.65	31.1	2.9
DenseNet-121	557	P2	98.77	98.64	30.3	5.3
DenseNet-121	601	P1	98.64	97.65	28.9	3.1
DenseNet-121	601	P2	98.40	97.53	31.6	4.1
DenseNet-121	653	P1	98.15	98.40	28.7	3.9
DenseNet-121	653	P2	97.53	96.54	28.9	3.8

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
DenseNet-121	701	P1	98.02	97.53	30.2	3.3
DenseNet-121	701	P2	97.16	97.53	29.1	5.2
DenseNet-121	751	P1	98.15	98.27	27.4	4.1
DenseNet-121	751	P2	96.91	96.91	27.9	4.4
DenseNet-121	809	P1	98.27	98.02	29.4	3.9
DenseNet-121	809	P2	97.78	97.53	29.5	5.1
DenseNet-121	853	P1	98.02	97.78	32.5	4.0
DenseNet-121	853	P2	96.54	96.05	30.5	5.7
DenseNet-121	907	P1	98.27	98.15	27.8	2.8
DenseNet-121	907	P2	96.91	96.91	26.3	4.1
DenseNet-121	953	P1	97.90	97.90	30.7	3.3
DenseNet-121	953	P2	97.78	97.41	29.1	4.4
DenseNet-121	1009	P1	98.52	98.02	29.2	2.7
DenseNet-121	1009	P2	97.16	97.28	29.0	4.6
DenseNet-121	1051	P1	98.64	98.52	28.9	3.4
DenseNet-121	1051	P2	97.16	97.65	31.5	3.9
EfficientNet-B0	137	P1	98.15	97.41	21.5	4.7
EfficientNet-B0	137	P2	97.41	97.41	23.5	5.3
EfficientNet-B0	173	P1	98.02	97.78	22.9	6.8
EfficientNet-B0	173	P2	98.15	98.02	23.9	4.1
EfficientNet-B0	211	P1	98.64	98.27	23.3	5.9
EfficientNet-B0	211	P2	97.53	97.16	23.8	7.0
EfficientNet-B0	257	P1	97.90	98.15	24.2	7.0
EfficientNet-B0	257	P2	96.42	97.04	22.2	5.3
EfficientNet-B0	307	P1	98.40	98.15	21.6	4.8
EfficientNet-B0	307	P2	96.79	96.79	21.7	5.8
EfficientNet-B0	349	P1	97.28	97.16	23.6	5.1
EfficientNet-B0	349	P2	97.90	98.15	24.8	5.8
EfficientNet-B0	401	P1	98.27	98.40	23.0	4.4
EfficientNet-B0	401	P2	96.91	96.79	21.3	5.0
EfficientNet-B0	457	P1	97.65	98.15	22.3	6.2
EfficientNet-B0	457	P2	94.57	95.19	22.6	5.1
EfficientNet-B0	503	P1	96.67	96.67	21.1	5.0
EfficientNet-B0	503	P2	97.28	97.78	19.9	5.0
EfficientNet-B0	557	P1	98.02	97.90	22.3	5.1
EfficientNet-B0	557	P2	98.40	98.15	24.3	6.0
EfficientNet-B0	601	P1	98.27	98.52	20.6	5.7
EfficientNet-B0	601	P2	97.65	97.41	23.0	5.2
EfficientNet-B0	653	P1	98.64	98.64	23.0	4.7
EfficientNet-B0	653	P2	97.04	97.53	24.9	6.6
EfficientNet-B0	701	P1	97.53	97.78	21.3	4.8
EfficientNet-B0	701	P2	97.16	97.16	22.2	5.4
EfficientNet-B0	751	P1	98.77	98.27	22.4	6.4
EfficientNet-B0	751	P2	97.41	97.16	21.1	6.8
EfficientNet-B0	809	P1	98.52	98.15	22.6	6.8
EfficientNet-B0	809	P2	96.91	97.16	23.0	5.4
EfficientNet-B0	853	P1	97.65	97.78	23.5	6.2
EfficientNet-B0	853	P2	96.54	96.05	22.8	7.0
EfficientNet-B0	907	P1	98.77	98.40	21.0	4.5
EfficientNet-B0	907	P2	97.53	97.53	21.9	4.6
EfficientNet-B0	953	P1	97.90	98.52	21.0	5.3
EfficientNet-B0	953	P2	98.40	98.02	21.7	3.9
EfficientNet-B0	1009	P1	98.15	98.15	21.3	5.1
EfficientNet-B0	1009	P2	97.53	97.04	22.8	4.8
EfficientNet-B0	1051	P1	97.65	97.65	23.7	5.8
EfficientNet-B0	1051	P2	98.52	98.27	23.8	5.9
MobileNet-V2	137	P1	97.04	97.16	19.0	4.0
MobileNet-V2	137	P2	97.16	96.42	23.3	4.0
MobileNet-V2	173	P1	97.90	97.78	20.0	3.7
MobileNet-V2	173	P2	97.65	98.02	20.3	3.3
MobileNet-V2	211	P1	97.41	97.41	19.7	3.1
MobileNet-V2	211	P2	95.56	95.43	20.8	3.9
MobileNet-V2	257	P1	97.04	96.67	22.3	4.7
MobileNet-V2	257	P2	96.17	96.79	20.6	4.4
MobileNet-V2	307	P1	98.52	98.40	22.2	4.4
MobileNet-V2	307	P2	95.80	96.05	21.3	4.1
MobileNet-V2	349	P1	97.41	96.67	19.2	4.6
MobileNet-V2	349	P2	96.42	96.05	21.1	3.9
MobileNet-V2	401	P1	97.65	97.41	21.5	3.9
MobileNet-V2	401	P2	97.04	96.79	19.7	4.1
MobileNet-V2	457	P1	97.16	97.16	19.2	3.7
MobileNet-V2	457	P2	94.57	94.81	20.0	4.0
MobileNet-V2	503	P1	97.16	97.16	19.9	5.4

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
MobileNet-V2	503	P2	96.79	96.17	20.8	3.9
MobileNet-V2	557	P1	96.54	96.67	20.9	3.8
MobileNet-V2	557	P2	98.02	98.02	20.8	4.8
MobileNet-V2	601	P1	97.78	97.41	19.9	4.6
MobileNet-V2	601	P2	97.28	96.79	20.5	4.1
MobileNet-V2	653	P1	97.78	97.28	21.7	4.7
MobileNet-V2	653	P2	96.05	96.30	22.5	3.3
MobileNet-V2	701	P1	96.91	97.04	19.3	3.7
MobileNet-V2	701	P2	96.05	96.17	20.0	4.7
MobileNet-V2	751	P1	96.79	96.54	21.1	4.6
MobileNet-V2	751	P2	95.68	95.31	21.5	5.2
MobileNet-V2	809	P1	97.04	96.67	21.8	4.8
MobileNet-V2	809	P2	96.91	96.91	21.6	4.2
MobileNet-V2	853	P1	97.53	97.65	21.7	5.9
MobileNet-V2	853	P2	96.05	96.30	20.0	3.8
MobileNet-V2	907	P1	98.15	97.28	19.3	2.9
MobileNet-V2	907	P2	97.41	97.28	19.8	3.7
MobileNet-V2	953	P1	97.41	96.79	21.8	4.7
MobileNet-V2	953	P2	97.90	96.54	20.8	4.7
MobileNet-V2	1009	P1	97.65	97.90	21.8	3.8
MobileNet-V2	1009	P2	96.54	96.17	22.2	5.4
MobileNet-V2	1051	P1	97.28	96.79	19.9	3.3
MobileNet-V2	1051	P2	97.16	97.04	21.5	3.5
ViT-B/16	137	P1	97.53	97.16	30.4	6.8
ViT-B/16	137	P2	97.28	96.30	30.7	5.5
ViT-B/16	173	P1	98.64	98.52	27.4	7.0
ViT-B/16	173	P2	97.53	97.53	26.5	4.5
ViT-B/16	211	P1	98.40	98.52	29.0	7.0
ViT-B/16	211	P2	96.42	96.42	29.6	7.1
ViT-B/16	257	P1	96.91	96.67	29.6	6.6
ViT-B/16	257	P2	97.28	96.79	28.8	6.4
ViT-B/16	307	P1	99.14	99.38	30.2	7.0
ViT-B/16	307	P2	96.17	96.54	30.0	5.9
ViT-B/16	349	P1	97.16	97.16	30.7	5.5
ViT-B/16	349	P2	96.79	96.79	29.6	7.4
ViT-B/16	401	P1	97.78	97.65	28.8	5.5
ViT-B/16	401	P2	97.53	97.16	30.2	6.8
ViT-B/16	457	P1	98.77	98.64	28.7	6.5
ViT-B/16	457	P2	94.69	95.31	29.6	5.1
ViT-B/16	503	P1	96.79	96.91	29.7	6.2
ViT-B/16	503	P2	96.67	97.04	29.5	5.9
ViT-B/16	557	P1	97.53	97.28	27.2	7.2
ViT-B/16	557	P2	97.53	97.04	29.6	8.0
ViT-B/16	601	P1	98.15	98.02	28.5	6.1
ViT-B/16	601	P2	97.90	97.53	30.7	5.0
ViT-B/16	653	P1	98.15	97.90	26.2	4.8
ViT-B/16	653	P2	97.65	97.53	26.8	6.5
ViT-B/16	701	P1	97.41	97.65	30.3	5.3
ViT-B/16	701	P2	96.54	96.54	30.8	7.1
ViT-B/16	751	P1	98.15	98.15	29.1	5.9
ViT-B/16	751	P2	95.93	95.80	28.7	7.0
ViT-B/16	809	P1	97.04	97.16	29.0	5.7
ViT-B/16	809	P2	95.68	95.31	30.2	7.2
ViT-B/16	853	P1	97.28	97.78	28.3	6.1
ViT-B/16	853	P2	96.79	96.67	28.5	6.2
ViT-B/16	907	P1	99.01	98.64	28.3	5.5
ViT-B/16	907	P2	96.42	96.42	29.1	6.1
ViT-B/16	953	P1	96.54	96.79	27.8	7.0
ViT-B/16	953	P2	97.53	97.78	29.1	4.5
ViT-B/16	1009	P1	98.02	97.90	28.4	5.0
ViT-B/16	1009	P2	96.91	96.91	29.1	6.4
ViT-B/16	1051	P1	98.15	98.15	28.3	5.9
ViT-B/16	1051	P2	97.28	97.41	30.9	6.7
ConvNeXt-T	137	P1	98.27	98.27	30.3	5.5
ConvNeXt-T	137	P2	97.28	96.67	32.3	6.7
ConvNeXt-T	173	P1	98.52	98.40	32.0	5.4
ConvNeXt-T	173	P2	98.15	98.27	28.4	4.7
ConvNeXt-T	211	P1	98.27	97.78	28.2	4.8
ConvNeXt-T	211	P2	97.04	97.16	28.9	6.0
ConvNeXt-T	257	P1	98.40	98.27	30.2	5.7
ConvNeXt-T	257	P2	97.16	97.16	28.9	4.2
ConvNeXt-T	307	P1	98.77	98.89	30.3	5.3
ConvNeXt-T	307	P2	96.79	97.41	30.7	6.0

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
ConvNeXt-T	349	P1	97.90	97.90	29.8	5.3
ConvNeXt-T	349	P2	97.04	97.04	27.6	4.8
ConvNeXt-T	401	P1	98.52	98.52	30.0	5.2
ConvNeXt-T	401	P2	98.15	98.15	29.6	6.4
ConvNeXt-T	457	P1	98.64	98.64	29.7	4.8
ConvNeXt-T	457	P2	95.93	95.68	27.2	5.3
ConvNeXt-T	503	P1	97.65	97.78	30.8	6.4
ConvNeXt-T	503	P2	97.90	97.78	29.5	5.4
ConvNeXt-T	557	P1	97.78	97.53	28.1	4.7
ConvNeXt-T	557	P2	99.01	98.89	31.8	5.5
ConvNeXt-T	601	P1	97.90	97.78	29.2	5.5
ConvNeXt-T	601	P2	97.53	97.28	30.8	6.5
ConvNeXt-T	653	P1	98.02	98.40	27.8	4.7
ConvNeXt-T	653	P2	97.53	97.41	31.1	4.8
ConvNeXt-T	701	P1	97.65	97.78	27.0	4.0
ConvNeXt-T	701	P2	96.05	96.05	26.4	4.4
ConvNeXt-T	751	P1	98.64	98.15	28.5	4.8
ConvNeXt-T	751	P2	96.67	96.79	31.4	5.0
ConvNeXt-T	809	P1	97.90	97.65	29.2	6.1
ConvNeXt-T	809	P2	96.91	96.91	30.7	5.0
ConvNeXt-T	853	P1	98.02	97.90	26.8	3.7
ConvNeXt-T	853	P2	96.67	96.79	28.9	4.6
ConvNeXt-T	907	P1	98.64	98.40	29.0	4.5
ConvNeXt-T	907	P2	97.41	97.41	30.5	5.1
ConvNeXt-T	953	P1	97.78	97.65	29.6	5.0
ConvNeXt-T	953	P2	97.78	98.02	30.1	6.0
ConvNeXt-T	1009	P1	98.27	97.90	29.8	5.3
ConvNeXt-T	1009	P2	97.41	97.41	29.0	4.6
ConvNeXt-T	1051	P1	98.40	98.40	30.1	4.8
ConvNeXt-T	1051	P2	97.53	97.41	30.0	4.7
Swin-T	137	P1	97.04	97.41	31.0	6.1
Swin-T	137	P2	96.54	96.67	28.9	6.2
Swin-T	173	P1	98.64	98.52	28.7	6.7
Swin-T	173	P2	97.41	97.65	26.5	7.3
Swin-T	211	P1	98.27	98.27	29.7	7.1
Swin-T	211	P2	96.79	96.54	30.3	5.8
Swin-T	257	P1	97.78	97.90	28.4	6.5
Swin-T	257	P2	97.90	97.41	27.7	7.1
Swin-T	307	P1	98.02	98.02	27.5	6.6
Swin-T	307	P2	97.16	97.04	30.5	7.5
Swin-T	349	P1	98.02	98.02	30.3	6.5
Swin-T	349	P2	96.91	96.67	29.7	6.1
Swin-T	401	P1	97.90	97.78	28.9	5.8
Swin-T	401	P2	97.16	97.53	27.4	6.7
Swin-T	457	P1	98.89	98.52	29.8	7.1
Swin-T	457	P2	95.31	96.17	31.2	7.0
Swin-T	503	P1	98.02	97.78	30.1	7.4
Swin-T	503	P2	98.02	97.78	30.1	7.4
Swin-T	557	P1	98.40	98.27	30.8	7.0
Swin-T	557	P2	98.40	97.78	30.3	8.4
Swin-T	601	P1	97.90	97.78	27.1	8.1
Swin-T	601	P2	98.02	98.40	27.8	6.7
Swin-T	653	P1	98.27	97.90	29.5	7.1
Swin-T	653	P2	97.28	97.28	28.7	5.9
Swin-T	701	P1	97.90	98.15	29.8	5.3
Swin-T	701	P2	96.67	96.54	29.2	7.4
Swin-T	751	P1	98.89	98.89	27.2	6.6
Swin-T	751	P2	96.42	96.79	29.8	7.9
Swin-T	809	P1	97.65	97.53	30.8	5.9
Swin-T	809	P2	95.80	96.42	27.8	5.8
Swin-T	853	P1	98.64	98.52	27.4	6.1
Swin-T	853	P2	97.41	97.04	26.8	5.9
Swin-T	907	P1	98.89	98.77	26.9	5.9
Swin-T	907	P2	96.91	96.91	29.6	6.4
Swin-T	953	P1	97.65	97.78	28.2	6.1
Swin-T	953	P2	98.15	97.90	26.4	5.8
Swin-T	1009	P1	98.77	98.64	30.7	6.4
Swin-T	1009	P2	97.65	96.91	27.6	6.4
Swin-T	1051	P1	97.90	98.27	27.0	4.7
Swin-T	1051	P2	98.15	98.02	29.6	6.6
RegNetY-1.6GF	137	P1	97.04	97.41	21.5	2.1
RegNetY-1.6GF	137	P2	96.17	95.93	22.4	4.4
RegNetY-1.6GF	173	P1	98.27	97.90	21.7	4.1

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
RegNetY-1.6GF	173	P2	98.15	97.78	20.6	2.9
RegNetY-1.6GF	211	P1	98.40	97.65	23.8	3.3
RegNetY-1.6GF	211	P2	96.17	95.80	21.2	3.1
RegNetY-1.6GF	257	P1	97.04	97.41	19.2	2.5
RegNetY-1.6GF	257	P2	96.79	96.91	21.0	4.7
RegNetY-1.6GF	307	P1	98.27	98.15	20.0	4.2
RegNetY-1.6GF	307	P2	96.54	96.30	20.0	3.3
RegNetY-1.6GF	349	P1	97.90	98.02	19.9	2.9
RegNetY-1.6GF	349	P2	97.28	96.54	20.2	2.2
RegNetY-1.6GF	401	P1	98.27	98.02	21.1	3.1
RegNetY-1.6GF	401	P2	97.65	97.41	20.6	2.7
RegNetY-1.6GF	457	P1	98.77	98.15	22.8	4.4
RegNetY-1.6GF	457	P2	95.43	95.06	19.0	3.5
RegNetY-1.6GF	503	P1	97.78	97.28	18.4	3.4
RegNetY-1.6GF	503	P2	97.28	96.67	21.3	3.3
RegNetY-1.6GF	557	P1	97.65	97.28	21.7	2.8
RegNetY-1.6GF	557	P2	97.90	97.65	21.9	3.2
RegNetY-1.6GF	601	P1	97.16	97.04	20.3	3.5
RegNetY-1.6GF	601	P2	98.02	97.53	19.7	2.8
RegNetY-1.6GF	653	P1	97.78	97.28	20.4	3.8
RegNetY-1.6GF	653	P2	97.04	97.41	22.2	3.8
RegNetY-1.6GF	701	P1	97.53	97.16	21.8	3.3
RegNetY-1.6GF	701	P2	95.68	95.43	23.1	4.4
RegNetY-1.6GF	751	P1	98.02	97.90	21.6	3.1
RegNetY-1.6GF	751	P2	96.54	96.30	21.2	2.6
RegNetY-1.6GF	809	P1	97.65	98.15	21.2	3.4
RegNetY-1.6GF	809	P2	97.28	96.54	22.5	3.3
RegNetY-1.6GF	853	P1	98.15	98.27	18.9	2.8
RegNetY-1.6GF	853	P2	96.42	96.54	23.0	3.7
RegNetY-1.6GF	907	P1	98.27	98.15	20.3	4.7
RegNetY-1.6GF	907	P2	96.17	95.93	19.8	3.3
RegNetY-1.6GF	953	P1	97.90	98.02	22.2	3.9
RegNetY-1.6GF	953	P2	98.27	98.27	20.3	3.3
RegNetY-1.6GF	1009	P1	97.90	98.02	23.1	2.4
RegNetY-1.6GF	1009	P2	96.17	95.68	20.6	3.1
RegNetY-1.6GF	1051	P1	97.65	97.16	20.2	2.7
RegNetY-1.6GF	1051	P2	97.16	96.67	21.3	4.0
EfficientNetV2-S	137	P1	96.79	96.79	19.2	1.7
EfficientNetV2-S	137	P2	96.30	95.80	20.3	5.2
EfficientNetV2-S	173	P1	98.02	97.78	18.8	3.5
EfficientNetV2-S	173	P2	98.02	98.15	20.9	5.1
EfficientNetV2-S	211	P1	98.02	98.15	19.1	3.8
EfficientNetV2-S	211	P2	96.79	96.54	19.7	1.8
EfficientNetV2-S	257	P1	96.91	97.04	20.9	3.5
EfficientNetV2-S	257	P2	97.16	97.16	20.8	4.4
EfficientNetV2-S	307	P1	98.40	98.27	20.3	1.2
EfficientNetV2-S	307	P2	96.05	96.67	18.3	4.5
EfficientNetV2-S	349	P1	97.53	97.78	18.4	3.8
EfficientNetV2-S	349	P2	97.41	97.04	21.1	4.1
EfficientNetV2-S	401	P1	97.90	97.90	17.9	5.0
EfficientNetV2-S	401	P2	97.53	97.16	22.1	5.1
EfficientNetV2-S	457	P1	98.15	98.27	21.9	2.6
EfficientNetV2-S	457	P2	95.31	95.93	21.0	4.1
EfficientNetV2-S	503	P1	97.41	97.04	18.8	2.8
EfficientNetV2-S	503	P2	95.93	96.05	21.3	5.4
EfficientNetV2-S	557	P1	97.78	97.90	19.8	4.8
EfficientNetV2-S	557	P2	99.26	98.77	18.5	4.6
EfficientNetV2-S	601	P1	97.28	97.53	19.9	4.1
EfficientNetV2-S	601	P2	97.78	96.91	21.6	5.1
EfficientNetV2-S	653	P1	97.65	97.65	19.7	2.7
EfficientNetV2-S	653	P2	96.17	96.42	21.6	4.7
EfficientNetV2-S	701	P1	97.04	96.79	19.2	1.9
EfficientNetV2-S	701	P2	95.43	95.31	21.0	5.3
EfficientNetV2-S	751	P1	97.90	97.90	21.7	4.7
EfficientNetV2-S	751	P2	96.05	96.17	20.0	4.8
EfficientNetV2-S	809	P1	98.02	97.78	20.3	4.8
EfficientNetV2-S	809	P2	96.91	96.67	22.6	5.1
EfficientNetV2-S	853	P1	97.28	97.04	20.4	1.7
EfficientNetV2-S	853	P2	96.67	95.93	20.9	4.5
EfficientNetV2-S	907	P1	98.52	98.27	17.2	3.9
EfficientNetV2-S	907	P2	96.67	96.30	20.6	5.2
EfficientNetV2-S	953	P1	97.90	97.78	20.2	5.3
EfficientNetV2-S	953	P2	98.52	98.52	19.8	4.8

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
EfficientNetV2-S	1009	P1	97.16	97.16	20.2	3.8
EfficientNetV2-S	1009	P2	96.17	96.30	19.1	2.4
EfficientNetV2-S	1051	P1	98.27	98.27	15.8	1.9
EfficientNetV2-S	1051	P2	97.04	97.28	16.3	1.5
MaxViT-T	137	P1	98.02	97.78	21.2	3.3
MaxViT-T	137	P2	96.42	96.67	24.4	5.4
MaxViT-T	173	P1	98.89	98.27	22.4	3.8
MaxViT-T	173	P2	96.79	96.91	22.3	4.6
MaxViT-T	211	P1	97.90	98.15	22.4	3.3
MaxViT-T	211	P2	96.30	96.54	22.6	2.8
MaxViT-T	257	P1	97.28	97.28	21.7	3.7
MaxViT-T	257	P2	97.53	97.65	21.3	2.9
MaxViT-T	307	P1	98.02	98.02	21.0	3.8
MaxViT-T	307	P2	97.04	96.67	21.1	4.1
MaxViT-T	349	P1	97.90	97.78	23.0	4.1
MaxViT-T	349	P2	97.04	97.28	21.1	5.1
MaxViT-T	401	P1	97.65	97.53	20.3	2.9
MaxViT-T	401	P2	97.78	97.78	22.1	3.8
MaxViT-T	457	P1	98.02	98.02	19.2	3.1
MaxViT-T	457	P2	95.43	95.68	21.3	3.4
MaxViT-T	503	P1	97.04	96.91	21.0	3.8
MaxViT-T	503	P2	97.53	97.65	23.7	4.5
MaxViT-T	557	P1	98.02	98.27	24.1	4.5
MaxViT-T	557	P2	98.52	98.89	25.1	5.4
MaxViT-T	601	P1	98.02	97.78	21.8	4.7
MaxViT-T	601	P2	97.90	97.53	24.8	3.2
MaxViT-T	653	P1	98.52	98.15	19.6	4.1
MaxViT-T	653	P2	97.78	97.90	20.2	2.1
MaxViT-T	701	P1	98.15	98.15	21.3	4.0
MaxViT-T	701	P2	95.93	96.05	22.8	3.5
MaxViT-T	751	P1	97.41	97.53	19.9	3.2
MaxViT-T	751	P2	96.79	96.91	20.8	3.5
MaxViT-T	809	P1	97.16	97.16	22.4	3.7
MaxViT-T	809	P2	96.91	96.54	23.1	4.5
MaxViT-T	853	P1	97.78	97.78	21.9	3.5
MaxViT-T	853	P2	96.67	96.54	25.7	5.0
MaxViT-T	907	P1	98.40	98.27	22.4	3.9
MaxViT-T	907	P2	96.05	96.17	22.6	3.9
MaxViT-T	953	P1	97.41	97.16	21.5	3.5
MaxViT-T	953	P2	98.15	98.02	23.0	4.2
MaxViT-T	1009	P1	98.02	97.90	21.8	4.0
MaxViT-T	1009	P2	97.28	97.16	22.5	4.2
MaxViT-T	1051	P1	97.90	98.15	22.8	4.8
MaxViT-T	1051	P2	98.15	97.90	21.6	3.5
DINOv2 ViT-S/14	137	P1	95.68	93.70	29.6	8.8
DINOv2 ViT-S/14	137	P2	93.21	92.47	32.9	12.5
DINOv2 ViT-S/14	173	P1	95.31	95.80	30.8	10.3
DINOv2 ViT-S/14	173	P2	93.83	91.73	30.9	9.2
DINOv2 ViT-S/14	211	P1	95.93	95.68	31.2	11.0
DINOv2 ViT-S/14	211	P2	93.95	93.33	29.6	10.5
DINOv2 ViT-S/14	257	P1	95.19	93.95	31.1	9.7
DINOv2 ViT-S/14	257	P2	93.46	93.09	30.0	9.9
DINOv2 ViT-S/14	307	P1	97.28	96.17	29.2	9.4
DINOv2 ViT-S/14	307	P2	95.31	93.70	31.0	9.4
DINOv2 ViT-S/14	349	P1	96.54	95.06	32.1	8.5
DINOv2 ViT-S/14	349	P2	93.70	92.84	28.3	7.4
DINOv2 ViT-S/14	401	P1	95.68	93.95	29.7	9.8
DINOv2 ViT-S/14	401	P2	95.43	94.69	28.1	9.8
DINOv2 ViT-S/14	457	P1	96.79	96.91	30.2	9.8
DINOv2 ViT-S/14	457	P2	92.84	92.96	30.4	8.7
DINOv2 ViT-S/14	503	P1	95.19	94.81	28.8	9.7
DINOv2 ViT-S/14	503	P2	95.06	94.57	29.5	10.3
DINOv2 ViT-S/14	557	P1	96.17	96.91	30.9	7.8
DINOv2 ViT-S/14	557	P2	94.94	94.20	33.1	10.0
DINOv2 ViT-S/14	601	P1	96.05	95.19	31.8	9.8
DINOv2 ViT-S/14	601	P2	93.21	90.86	30.2	9.9
DINOv2 ViT-S/14	653	P1	95.06	95.68	30.4	10.1
DINOv2 ViT-S/14	653	P2	94.32	93.09	28.9	11.0
DINOv2 ViT-S/14	701	P1	94.81	95.56	32.0	9.1
DINOv2 ViT-S/14	701	P2	91.48	91.11	29.4	8.7
DINOv2 ViT-S/14	751	P1	97.04	95.93	30.8	8.4
DINOv2 ViT-S/14	751	P2	93.33	93.83	30.7	9.8
DINOv2 ViT-S/14	809	P1	95.93	95.06	30.5	9.4

Architecture	Seed	Prot.	Acc. r=0	Acc. r=256	R r=0	R r=256
DINOv2 ViT-S/14	809	P2	94.20	93.83	30.8	8.3
DINOv2 ViT-S/14	853	P1	94.94	94.07	31.5	9.1
DINOv2 ViT-S/14	853	P2	94.44	93.09	28.9	10.6
DINOv2 ViT-S/14	907	P1	96.67	96.17	31.5	7.7
DINOv2 ViT-S/14	907	P2	92.84	91.73	30.3	11.1
DINOv2 ViT-S/14	953	P1	95.19	94.20	31.7	9.4
DINOv2 ViT-S/14	953	P2	93.95	94.81	29.0	8.6
DINOv2 ViT-S/14	1009	P1	96.79	96.42	31.6	9.8
DINOv2 ViT-S/14	1009	P2	95.43	94.32	32.5	8.8
DINOv2 ViT-S/14	1051	P1	96.05	95.31	30.8	8.3
DINOv2 ViT-S/14	1051	P2	94.20	92.10	31.2	10.5

Table M.5: Complete test-time ablation ledger (160 models; accuracy in percent).

Architecture	Seed	Cell	Orig.	Grey	Stain	Low-pass	Flat	No bg
ResNet-50	137	clean_high	97.32	64.75	90.04	70.11	92.72	88.31
ResNet-50	137	clean_low	96.36	56.70	90.80	51.92	93.87	91.00
ResNet-50	137	leak_high	97.13	63.22	89.85	59.77	93.68	89.46
ResNet-50	137	leak_low	97.89	68.20	90.80	80.65	95.40	82.38
ResNet-50	173	clean_high	95.55	49.32	88.20	39.85	92.26	89.36
ResNet-50	173	clean_low	94.97	70.79	89.17	63.25	91.68	90.52
ResNet-50	173	leak_high	96.71	61.32	88.01	49.13	92.07	90.14
ResNet-50	173	leak_low	96.71	53.77	84.91	74.27	91.49	91.30
ResNet-50	211	clean_high	94.80	57.03	84.01	42.58	92.49	91.71
ResNet-50	211	clean_low	94.22	54.14	85.93	56.26	87.28	89.98
ResNet-50	211	leak_high	96.15	59.73	87.67	40.27	92.29	91.33
ResNet-50	211	leak_low	97.50	64.16	86.71	61.66	94.03	93.45
ResNet-50	257	clean_high	94.07	42.45	88.34	59.46	91.20	91.59
ResNet-50	257	clean_low	95.03	59.85	89.29	68.64	90.44	92.54
ResNet-50	257	leak_high	96.56	52.77	90.06	53.15	94.84	91.20
ResNet-50	257	leak_low	96.75	59.27	90.82	61.57	93.50	91.97
ResNet-50	307	clean_high	94.92	58.59	90.82	59.57	90.04	90.43
ResNet-50	307	clean_low	93.95	65.23	91.60	60.35	88.28	87.89
ResNet-50	307	leak_high	95.51	64.84	91.80	66.60	87.11	90.82
ResNet-50	307	leak_low	97.85	52.34	85.35	54.88	87.50	93.16
ResNet-50	349	clean_high	92.29	61.09	87.97	33.83	88.16	85.90
ResNet-50	349	clean_low	94.74	66.73	93.42	47.18	89.29	88.35
ResNet-50	349	leak_high	94.92	57.52	87.41	51.88	87.78	87.59
ResNet-50	349	leak_low	97.18	65.41	91.92	68.23	93.61	91.92
ResNet-50	401	clean_high	92.07	66.15	86.46	55.32	89.17	85.30
ResNet-50	401	clean_low	93.23	57.83	88.39	41.59	88.20	88.97
ResNet-50	401	leak_high	95.16	69.05	89.17	66.92	89.94	90.14
ResNet-50	401	leak_low	94.78	56.48	88.59	53.19	92.07	89.17
ResNet-50	457	clean_high	96.34	66.18	85.92	45.52	91.96	89.58
ResNet-50	457	clean_low	95.80	63.62	85.37	54.66	87.75	88.85
ResNet-50	457	leak_high	96.89	58.68	85.74	69.47	90.68	89.58
ResNet-50	457	leak_low	96.16	66.54	89.03	52.83	91.59	93.05
ResNet-50	503	clean_high	95.47	55.85	82.83	62.45	92.26	88.49
ResNet-50	503	clean_low	95.85	64.91	84.91	71.89	90.94	91.13
ResNet-50	503	leak_high	96.42	57.17	81.70	51.89	93.77	89.25
ResNet-50	503	leak_low	97.55	54.91	82.26	39.81	93.40	90.75
ResNet-50	557	clean_high	93.64	63.09	86.73	64.55	85.64	88.36
ResNet-50	557	clean_low	94.73	46.91	84.91	40.18	89.27	88.91
ResNet-50	557	leak_high	95.64	69.09	87.82	67.82	85.82	86.18
ResNet-50	557	leak_low	95.64	53.64	86.55	38.18	87.27	89.09
DenseNet-121	137	clean_high	96.93	56.70	92.72	71.26	93.49	70.88
DenseNet-121	137	clean_low	94.25	53.83	86.40	68.39	89.46	77.59
DenseNet-121	137	leak_high	97.51	53.64	90.23	71.26	94.25	76.44
DenseNet-121	137	leak_low	96.36	60.73	87.93	61.30	91.57	75.67
DenseNet-121	173	clean_high	94.58	66.92	91.30	62.48	93.23	81.24
DenseNet-121	173	clean_low	96.32	72.92	89.75	76.02	91.49	77.37
DenseNet-121	173	leak_high	96.52	62.86	92.07	82.40	93.42	85.88
DenseNet-121	173	leak_low	97.68	61.12	91.88	86.07	89.17	81.04
DenseNet-121	211	clean_high	95.95	49.52	89.79	77.46	89.60	79.00
DenseNet-121	211	clean_low	95.18	63.78	87.67	70.33	91.91	91.71
DenseNet-121	211	leak_high	97.69	61.08	89.79	75.53	95.76	84.01
DenseNet-121	211	leak_low	97.69	65.32	85.36	84.59	90.75	92.10
DenseNet-121	257	clean_high	94.84	56.02	89.87	73.04	92.16	90.06
DenseNet-121	257	clean_low	96.75	52.77	91.20	79.35	94.07	87.95

Architecture	Seed	Cell	Orig.	Grey	Stain	Low-pass	Flat	No bg
DenseNet-121	257	leak_high	95.98	47.99	88.15	82.03	92.16	87.19
DenseNet-121	257	leak_low	97.51	58.32	89.87	67.11	93.69	83.56
DenseNet-121	307	clean_high	94.34	57.03	90.43	59.57	90.04	73.83
DenseNet-121	307	clean_low	94.14	66.99	89.84	63.09	91.80	81.84
DenseNet-121	307	leak_high	98.05	73.24	92.97	79.88	93.55	83.98
DenseNet-121	307	leak_low	97.85	62.11	91.80	69.34	93.55	80.66
DenseNet-121	349	clean_high	92.86	46.62	93.42	69.74	88.91	83.83
DenseNet-121	349	clean_low	90.41	44.17	88.91	69.17	81.95	85.15
DenseNet-121	349	leak_high	96.80	61.84	93.42	75.38	86.28	84.96
DenseNet-121	349	leak_low	96.80	63.91	93.05	79.51	90.23	85.71
DenseNet-121	401	clean_high	93.04	62.67	89.56	72.15	92.46	84.91
DenseNet-121	401	clean_low	94.58	67.50	90.14	63.44	91.88	88.01
DenseNet-121	401	leak_high	95.94	74.66	91.88	65.96	93.42	88.01
DenseNet-121	401	leak_low	95.94	58.22	89.56	72.34	93.23	85.49
DenseNet-121	457	clean_high	95.61	56.12	88.48	68.01	91.41	83.91
DenseNet-121	457	clean_low	94.33	49.73	89.03	73.49	91.22	83.91
DenseNet-121	457	leak_high	97.62	58.87	90.31	82.45	93.05	86.11
DenseNet-121	457	leak_low	97.07	54.84	91.22	78.43	91.41	85.01
DenseNet-121	503	clean_high	95.66	70.94	85.85	74.72	94.15	79.62
DenseNet-121	503	clean_low	95.85	60.75	87.17	75.47	92.26	80.38
DenseNet-121	503	leak_high	96.04	61.13	87.74	76.42	94.72	84.34
DenseNet-121	503	leak_low	97.17	71.70	86.42	84.53	93.77	82.83
DenseNet-121	557	clean_high	94.00	56.55	91.27	71.45	88.91	80.36
DenseNet-121	557	clean_low	94.91	49.45	90.00	83.27	83.27	83.82
DenseNet-121	557	leak_high	95.45	62.55	90.55	75.27	90.91	85.09
DenseNet-121	557	leak_low	97.09	61.09	90.36	75.45	91.64	81.45
ViT-B/16	137	clean_high	95.79	54.02	90.04	94.64	92.53	90.61
ViT-B/16	137	clean_low	96.36	50.38	86.97	96.36	89.46	90.61
ViT-B/16	137	leak_high	98.28	54.21	91.38	96.93	93.68	92.72
ViT-B/16	137	leak_low	94.83	40.80	86.78	92.91	89.85	91.57
ViT-B/16	173	clean_high	88.97	70.60	83.95	86.07	84.33	86.85
ViT-B/16	173	clean_low	90.52	68.28	84.53	85.69	79.30	81.43
ViT-B/16	173	leak_high	95.94	59.57	88.59	94.97	90.72	92.65
ViT-B/16	173	leak_low	95.16	59.57	89.56	94.00	86.07	91.10
ViT-B/16	211	clean_high	96.15	61.27	88.63	95.57	87.48	91.52
ViT-B/16	211	clean_low	95.38	57.23	88.05	95.18	86.71	93.45
ViT-B/16	211	leak_high	97.30	61.46	91.52	97.50	89.21	90.37
ViT-B/16	211	leak_low	98.46	61.85	89.79	97.88	91.33	94.61
ViT-B/16	257	clean_high	93.88	51.24	92.16	92.35	91.20	82.79
ViT-B/16	257	clean_low	93.69	70.36	92.35	91.97	90.82	86.81
ViT-B/16	257	leak_high	94.84	56.21	92.93	93.31	91.97	88.72
ViT-B/16	257	leak_low	97.32	55.83	91.78	96.56	90.63	92.93
ViT-B/16	307	clean_high	95.90	56.64	92.58	95.70	88.09	93.95
ViT-B/16	307	clean_low	93.36	52.15	90.62	94.34	88.48	86.72
ViT-B/16	307	leak_high	95.90	53.12	91.80	93.75	90.04	91.80
ViT-B/16	307	leak_low	95.51	55.27	89.65	93.55	86.72	88.87
ViT-B/16	349	clean_high	92.29	49.25	91.35	89.66	90.60	86.84
ViT-B/16	349	clean_low	94.92	57.14	92.29	93.05	92.11	87.59
ViT-B/16	349	leak_high	95.11	60.90	90.04	90.98	88.53	87.03
ViT-B/16	349	leak_low	96.80	62.78	93.23	96.05	94.92	91.17
ViT-B/16	401	clean_high	93.62	64.41	90.91	93.04	91.10	86.65
ViT-B/16	401	clean_low	92.84	47.20	91.49	91.88	92.46	89.94
ViT-B/16	401	leak_high	95.16	54.93	88.39	93.23	89.56	91.30
ViT-B/16	401	leak_low	96.13	53.58	93.23	94.78	90.72	92.26
ViT-B/16	457	clean_high	95.61	52.83	93.42	94.52	89.95	90.86
ViT-B/16	457	clean_low	95.80	58.50	92.69	95.80	91.22	88.67
ViT-B/16	457	leak_high	94.88	51.55	85.01	93.24	89.03	85.37
ViT-B/16	457	leak_low	95.98	52.83	89.95	94.88	91.41	86.47
ViT-B/16	503	clean_high	93.21	63.96	84.15	92.64	83.02	84.53
ViT-B/16	503	clean_low	92.26	64.15	83.40	91.13	89.81	84.72
ViT-B/16	503	leak_high	96.98	58.11	88.11	95.85	90.38	92.45
ViT-B/16	503	leak_low	95.47	56.60	85.47	94.34	92.83	88.49
ViT-B/16	557	clean_high	95.09	65.64	90.00	93.82	83.64	90.55
ViT-B/16	557	clean_low	96.00	53.09	88.36	94.00	88.91	93.64
ViT-B/16	557	leak_high	95.64	52.18	90.00	93.64	85.64	86.73
ViT-B/16	557	leak_low	94.55	49.45	92.18	95.64	84.73	85.82
ConvNeXt-T	137	clean_high	96.93	64.75	93.68	85.82	96.17	89.08
ConvNeXt-T	137	clean_low	94.64	61.49	92.34	81.61	91.19	88.89
ConvNeXt-T	137	leak_high	97.13	71.46	94.64	91.38	94.83	91.19
ConvNeXt-T	137	leak_low	95.98	55.36	92.91	83.91	90.80	91.38
ConvNeXt-T	173	clean_high	95.36	64.41	90.72	79.50	94.58	88.39
ConvNeXt-T	173	clean_low	94.20	66.15	86.65	87.23	92.65	85.49
ConvNeXt-T	173	leak_high	95.36	67.12	89.75	89.17	91.10	92.07

Architecture	Seed	Cell	Orig.	Grey	Stain	Low-pass	Flat	No bg
ConvNeXt-T	173	leak_low	95.16	70.41	88.01	83.75	94.58	87.43
ConvNeXt-T	211	clean_high	96.34	64.93	89.98	83.04	92.68	89.98
ConvNeXt-T	211	clean_low	97.88	69.36	92.68	91.14	93.45	91.71
ConvNeXt-T	211	leak_high	99.04	64.93	93.45	86.32	94.61	92.68
ConvNeXt-T	211	leak_low	95.95	62.43	90.56	89.60	93.83	92.49
ConvNeXt-T	257	clean_high	94.07	69.02	90.63	77.25	92.16	89.48
ConvNeXt-T	257	clean_low	94.26	56.02	92.16	83.56	89.10	89.67
ConvNeXt-T	257	leak_high	95.98	63.10	96.18	79.54	93.50	90.06
ConvNeXt-T	257	leak_low	97.13	65.97	94.46	86.81	93.50	94.07
ConvNeXt-T	307	clean_high	97.07	71.48	95.51	92.38	92.38	93.36
ConvNeXt-T	307	clean_low	96.68	71.88	94.73	86.72	89.84	92.58
ConvNeXt-T	307	leak_high	98.63	69.92	95.90	88.28	92.77	92.19
ConvNeXt-T	307	leak_low	98.24	73.83	95.12	84.77	94.53	92.77
ConvNeXt-T	349	clean_high	93.98	64.66	93.80	71.24	93.42	87.59
ConvNeXt-T	349	clean_low	96.05	63.16	94.17	91.73	93.42	88.72
ConvNeXt-T	349	leak_high	95.30	66.17	94.17	89.66	93.05	83.46
ConvNeXt-T	349	leak_low	95.11	63.91	93.80	88.53	92.11	91.73
ConvNeXt-T	401	clean_high	93.42	66.34	91.30	83.56	89.56	89.36
ConvNeXt-T	401	clean_low	93.62	65.18	93.04	64.60	91.88	89.36
ConvNeXt-T	401	leak_high	95.74	70.60	93.04	70.60	94.58	89.94
ConvNeXt-T	401	leak_low	93.62	50.68	90.52	62.67	91.49	88.20
ConvNeXt-T	457	clean_high	95.80	67.09	94.52	87.39	94.52	90.49
ConvNeXt-T	457	clean_low	91.77	51.01	88.12	81.54	89.40	89.03
ConvNeXt-T	457	leak_high	95.98	65.81	92.69	83.73	94.52	89.21
ConvNeXt-T	457	leak_low	96.53	69.29	94.52	75.87	94.88	88.67
ConvNeXt-T	503	clean_high	95.09	51.32	89.25	84.72	92.64	93.40
ConvNeXt-T	503	clean_low	95.66	67.36	87.36	88.49	93.02	91.70
ConvNeXt-T	503	leak_high	95.66	68.87	87.55	89.81	94.34	92.64
ConvNeXt-T	503	leak_low	96.23	64.15	89.06	88.11	94.34	92.26
ConvNeXt-T	557	clean_high	94.00	62.18	90.36	89.64	91.27	92.00
ConvNeXt-T	557	clean_low	94.73	57.64	92.91	84.00	90.91	90.36
ConvNeXt-T	557	leak_high	94.00	67.09	88.91	74.18	90.55	88.73
ConvNeXt-T	557	leak_low	96.00	67.64	90.91	84.36	90.00	90.73

Table M.6: Every smoke test (Study 2 before registration; Amendment 8 before registration). Only crop-level, leak-cell, or single-protocol runs were smoke-tested, so no registered contrast could be observed.

Run	Kind	Architecture	Protocol	Epochs	Seconds
cric_p1_resnet18	train	resnet18	P1	1.0	8
leak_high_resnet50	train	resnet50	leak_high	1.0	7
p1_convnext_tiny	train	convnext_tiny	P1	1.0	27
p1_densenet121	train	densenet121	P1	1.0	12
p1_dinov2_vits14	train	dinov2_vits14	P1	1.0	13
p1_dinov2_vits14_math	train	dinov2_vits14	P1	1.0	15
p1_efficientnet_b0	train	efficientnet_b0	P1	1.0	7
p1_efficientnet_v2_s	train	efficientnet_v2_s	P1	1.0	13
p1_grl_resnet50	train	resnet50	P1	1.0	8
p1_maxvit_t	train	maxvit_t	P1	1.0	28
p1_mobilenet_v2	train	mobilenet_v2	P1	1.0	6
p1_regnet_y_1_6gf	train	regnet_y_1_6gf	P1	1.0	9
p1_resnet18	train	resnet18	P1	1.0	4
p1_resnet18_repeat	train	resnet18	P1	1.0	4
p1_resnet50	train	resnet50	P1	1.0	10
p1_swin_t	train	swin_t	P1	1.0	17
p1_vit_b_16	train	vit_b_16	P1	1.0	35
p1_vit_b_16_math	train	vit_b_16	P1	1.0	32
p1_vit_b_16_math_repeat	train	vit_b_16	P1	1.0	32
sipakmed_resnet50_leak_high_seed1103_colour	train	resnet50	leak_high	1.0	6
sipakmed_vit_b_16_leak_high_seed1103_masked	train	vit_b_16	leak_high	1.0	15
sipakmed_convnext_tiny_leak_high_seed137	factorial_followup	convnext_tiny	leak_high	nan	15
sipakmed_densenet121_leak_high_seed137	factorial_followup	densenet121	leak_high	nan	8
sipakmed_resnet50_leak_high_seed137	factorial_followup	resnet50	leak_high	nan	8

Run	Kind	Architecture	Protocol	Epochs	Seconds
sipakmed__vit__b__16__leak__high__seed137	factorial_followup	vit__b__16	leak__high	nan	11
sipakmed__resnet18__p1__seed137	leace	resnet18	P1	nan	25
sipakmed__vit__b__16__p1__seed137	leace	vit__b__16	P1	nan	26
sipakmed__convnext__tiny__p1__seed601__linadv1	train	convnext__tiny	P1	2.0	41
sipakmed__densenet121__p1__seed601__linadv3	train	densenet121	P1	1.0	12
sipakmed__efficientnet__b0__p1__seed601__reinhard100	train	efficientnet__b0	P1	1.0	7

N

Variance components of every registered model

Table N.1: Variance components (REML) of every registered mixed model. Correlations of random intercepts and slopes are listed as 'corr.'.

Model	Component	Variance	SD / corr.
H1	split: (Intercept)	0.2271	0.477
	seed: (Intercept)	0.0000	0.000
	architecture: (Intercept)	1.2769	1.130
	architecture: x	0.0104	0.102
	architecture: (Intercept) × x	–	−1.000 (corr.)
	Residual: residual	0.5915	0.769
P2 − P3	split: (Intercept)	0.4033	0.635
	seed: (Intercept)	0.1955	0.442
	architecture: (Intercept)	2.0081	1.417
	architecture: x	0.2253	0.475
	architecture: (Intercept) × x	–	−1.000 (corr.)
	Residual: residual	0.6948	0.834
H2	split: (Intercept)	0.6662	0.816
	seed: (Intercept)	0.0295	0.172
	architecture: (Intercept)	0.2783	0.528
	architecture: x	0.0220	0.148
	architecture: (Intercept) × x	–	−0.639 (corr.)
	Residual: residual	0.7960	0.892
H3	training_set: (Intercept)	0.0000	0.000
	seed: (Intercept)	0.4778	0.691
	architecture: (Intercept)	0.0776	0.279
	architecture: leak	0.0963	0.310
	architecture: (Intercept) × leak	–	0.001 (corr.)
	Residual: residual	1.4223	1.193
H4	training_set: (Intercept)	0.1932	0.440
	seed: (Intercept)	0.7472	0.864
	architecture: (Intercept)	0.1369	0.370
	architecture: leak	0.1412	0.376
	architecture: (Intercept) × leak	–	1.000 (corr.)
	Residual: residual	1.6183	1.272
H5	seed: (Intercept)	1.0619	1.031
	Residual: residual	0.6736	0.821
F2	run: (Intercept)	0.0000	0.000
	training_set: (Intercept)	0.0000	0.000
	seed: (Intercept)	0.7919	0.890
	architecture: (Intercept)	0.7394	0.860
	architecture: leak	0.0004	0.021
	architecture: (Intercept) × leak	–	−1.000 (corr.)
	Residual: residual	4.0889	2.022
F3	training_set: (Intercept)	0.1245	0.353
	seed: (Intercept)	0.9803	0.990
	architecture: (Intercept)	0.2018	0.449
	architecture: leak	0.0042	0.065

Model	Component	Variance	SD / corr.
	architecture: (Intercept) × leak	–	−1.000 (corr.)
	Residual: residual	1.2390	1.113
F4	training_set: (Intercept)	0.2028	0.450
	seed: (Intercept)	0.7043	0.839
	architecture: (Intercept)	0.3280	0.573
	architecture: leak	0.1868	0.432
	architecture: (Intercept) × leak	–	1.000 (corr.)
	Residual: residual	1.7444	1.321
F5	seed: (Intercept)	0.0000	0.000
	architecture: (Intercept)	0.0017	0.042
	Residual: residual	0.3183	0.564
F6	seed: (Intercept)	0.3559	0.597
	Residual: residual	0.9056	0.952
F7	seed: (Intercept)	0.2447	0.495
	Residual: residual	1.4134	1.189
F8	block: (Intercept)	0.4937	0.703
	Residual: residual	0.4997	0.707

O

Complete commit chronology

Table O.1: Complete commit chronology of the project (all branches; IST).

Commit	Date	Message
67e8ded	2026-08-22 04:54	preregister confirmatory analysis
8ebc6d5	2026-08-22 05:27	build leakage verification pipeline
12c9de7	2026-08-22 05:31	add parent identity probe and complete source checksums
4ba6a96	2026-08-22 05:34	make image provenance recovery independent
1dbd6ac	2026-08-22 05:35	separate provenance recovery stages
c551f2d	2026-08-22 05:38	add matched classification harness
14a1569	2026-08-22 05:39	add near duplicate audit and P3 grouping
79b798a	2026-08-22 05:40	amend SIPaKMeD class order before analysis
70b5ffa	2026-08-22 05:44	add bounded GPU probe launcher
4aa3e60	2026-08-22 05:49	fail closed on image provenance confidence
c18d393	2026-08-22 05:54	validate Cx22 against declared source images
f7ca331	2026-08-22 05:54	add bounded matched evaluation launcher
318120f	2026-08-22 05:55	summarize matched runs in one correction family
8f6f76e	2026-08-22 05:57	complete preregistered probe ablations
680fa53	2026-08-22 05:58	record reproducible model run metadata
0c5166e	2026-08-22 06:00	checkpoint exhaustive provenance matching
ea053e9	2026-08-22 06:01	verify structural protocol ordering
65b0230	2026-08-22 06:03	export fixed protocol split assets
3261dc0	2026-08-22 06:06	render fail-closed verification contract
79e5b31	2026-08-22 06:07	require named provenance evidence in report
cded044	2026-08-22 06:09	measure literal cross-split pixel overlap
55708d4	2026-08-22 06:09	keep training commands free of analysis-only imports
6369698	2026-08-22 06:11	generate self-contained thesis figure
da548a4	2026-08-22 06:12	bound clean reproduction matcher to three workers
83d56af	2026-08-22 06:14	verify published class crop counts
03e7442	2026-08-22 06:15	document fail-closed release workflow
be46a53	2026-08-22 06:17	separate canonical and audit provenance maps
c617374	2026-08-22 06:23	bound provenance image cache to class working set
4c6282e	2026-08-22 07:00	use multiscale provenance matching only as fallback
67892b7	2026-08-22 07:19	align provenance validation with registered criteria
268e56a	2026-08-22 07:34	build explicit section 11 acceptance ledger
438fa3f	2026-08-22 07:36	measure duplicate detector precision and recall
62a51ef	2026-08-22 07:39	record virgin-host reproduction evidence
dd0a56e	2026-08-22 07:41	generate five-seed parent probe figure
b90805a	2026-08-22 07:45	reproduce P3 splits and pixel overlap end to end
82e541d	2026-08-22 07:56	add matched protocol results figure
adf5539	2026-08-22 08:04	document full matched classifier matrix
bf8a4ec	2026-08-22 08:06	enforce paper table reporting contract
ab40681	2026-08-22 09:32	separate Figure 2 control labels
a72fb5f	2026-08-22 09:55	record split and image library versions
4da9f62	2026-08-22 10:05	pin CUDA 12.8 training environment
f936802	2026-08-22 10:44	verify classification matrix integrity
157b882	2026-08-22 10:46	report secondary ordering violations
10fb2a3	2026-08-22 10:48	pin reproducible split runtime
970c74e	2026-08-22 10:58	gate acceptance on matrix integrity
8ebb2b9	2026-08-22 11:15	simulate exact P1 allocation rule
525df1d	2026-08-22 12:14	report architecture ranking stability

Commit	Date	Message
1c2da7f	2026-08-22 13:22	record preregistered provenance and five-seed results
2c5d5c3	2026-08-22 13:23	record post-result acceptance evidence
d347b07	2026-08-24 03:15	preregister Cx22 segmentation rescore
a58f72f	2026-08-24 03:24	add resumable Cx22 segmentation extension
635e660	2026-08-24 03:33	decode Cx22 instance labels with bounded memory
2fdb136	2026-08-24 03:46	refresh Cx22 controller heartbeat during epochs
d66dba4	2026-08-24 03:55	add memory-mapped Cx22 training cache
a18ddef	2026-08-24 04:01	preregister mixed-precision Cx22 execution
f10c9c5	2026-08-24 04:02	enable preregistered mixed precision for Cx22
48c9b65	2026-08-28 05:53	stabilize Cx22 MAnet training in FP32
373276a	2026-08-28 06:00	enforce deterministic CUDA workspace for Cx22
a9594b3	2026-09-06 23:32	remove redundant Cx22 instance runtime overhead
01c4c31	2026-09-06 23:38	stage Cx22 checkpoints off network storage
6179cfa	2026-10-18 05:01	add GPU-resident Study 2 training stack
6c12dd2	2026-10-18 05:01	add leakage-by-diversity factorial and transplant designs
d97ea01	2026-10-18 05:01	fetch CRIC slide images with checksums
8bdfd47	2026-10-18 05:01	add Study 2 power simulation and analysis code
5f15787	2026-10-18 05:07	export registered Study 2 splits and run matrix
a62e8e7	2026-10-18 05:07	preregister Study 2
24d372d	2026-10-18 05:12	add structural leakage-risk report and Cx22 literature check
f6f6ab3	2026-10-18 05:21	start single-column Study 2 manuscript and analysis driver
ee057a8	2026-10-18 05:24	measure literal source-pixel overlap for CRIC and Cx22 crop splits
d3dd9f4	2026-10-18 06:18	add Study 2 result figures
66b4091	2026-10-18 06:18	add Study 2 archive-and-vacate script
582327a	2026-10-18 06:19	document Study 2 in the README
7fe2e44	2026-10-18 06:25	dry-run the full Study 2 analysis on synthetic run outputs
ffd9d5a	2026-10-18 06:26	record post hoc pooled reanalyses of Study 1
7b8c96f	2026-10-18 06:29	amend Study 2 failure handling before any 2A run
035ef4d	2026-10-18 06:30	draft outcome-independent manuscript sections
965f20a	2026-10-19 13:42	implement Amendment 8 follow-up experiments (code and designs, not yet registered)
80171fe	2026-10-19 14:02	register Amendment 8: follow-up experiments on how leakage is used
458817d	2026-10-19 14:13	record the registered Study 2 results and draft their manuscript sections
43b6973	2026-10-20 01:25	Amendment 9: fix carrier condition labelling before any F3/F4 estimate
b958b8b	2026-10-20 03:26	record the registered Amendment 8 follow-up results and complete the manuscript
8f4e2f3	2026-10-20 03:27	summarise Study 2 and Amendment 8 results in the README

P

Registration hashes

SHA-256 hashes of every file and cache fixed by the Study 2 registration and by Amendments 7, 8, and 9, as recorded in `reports/study2/registration_hashes.txt`.

```
7b0cafa1d6b305c5c176cbd450bc0e081b515b417c4a901285488ffaaecad79d PREREGISTRATION_STUDY2.md
7822d4a344b06d70629bc7e4419a3a08da9e6c3da44db4f4e163588a0e3f8e8e analysis/study2/protocol_lmm.R
99ac4ebc60c20da9e08004d69d7c5e6a39c3747d6fd984ca7b9a46857a7f4f58 analysis/study2/mechanism_lmm.R
1dd75a0ec895ef93eebe5da8b1948e357ed92b9cb72b033be0af66a595114ada analysis/study2/power_simulation.R
39d2c989c1801cc045f96f17e79509b788d079c744a5e325d2973211172fa6c5 analysis/study2/collect_runs.py
842af403e275229df60241b1f06b9d5ca8af3ef4705e42ef0b42ea50e9e5f722 analysis/study2/prepare_mechanism.py
52ca6a9a01657ce596a13dac28500979dea19a269318b8b665bf909307c0ee8c analysis/study2/secondary.py
c7872775fb85bedc2f2702b3b3a33dcb3d4df93f3692e9baa48f5e5839bcd632 src/cytofair/train_study2.py
aff5ffd89cd66f3c033491d946ace916914c3d177b1e5a10370a5b036378c3ad src/cytofair/gpu_augment.py
39f4485e76bab3a8bdbc9d2234ffd11427c2ad8596430e27e95786385819fce9 src/cytofair/backbones.py
81777104aafabf76ec339ed5a771241d0e0026192d0b798c1828f08bbe3f6e33 src/cytofair/factorial.py
35b6c0b455de6a0d71d8e38309f098763c9ee0ed8ee6f7d4487357baa36993f4 src/cytofair/transplant.py
3f218331e3f78eecc63c105efa831e03138d491ec3f66aed3be02a18319055df src/cytofair/recoverability.py
4e88e1f9ed2b992e51b1f0cb9b55cbd0112438ac67a39510ddea13852a958be4 src/cytofair/study2_design.py
1a2203e7f7b59fae61c4eb9c5181e11ae6a34dd9601045df9e36f5687b2d4d8d src/cytofair/image_cache.py
a952b5c92b945201adbf3102ba16ad4f4c4b1dcb5df6c9c0452ae182800c5101 src/cytofair/fingerprint.py
65a5b4726b730b6919b7c7923f57a9254841033b1fa8d3d44794545c1e46c105 scripts/study2/build_tasks.py
77692934e3cc194e638b6872320f56da46c2baa602d2e91820b1db617e0c8b59 scripts/study2/launch_matrix.py
4cb34f07e9640e632eddaecb128896ae1d9014fa5a82164c7381a6cf381310c1 splits/study2/study2_split_manifest.csv
722d4979be3b63476941deb214d22e0fadbd94669ef2af69a128af3368b18154 splits/study2/study2_transplant_manifest.csv
9e5fac3a0a27b48e4a49184f4d82f8d53c532ff20807911a2853a99682259058 splits/study2/sipakmed_probe_split.csv
d02b7d02519869b417dab81a99d26d2d4a97b0a403439011eb6416e86bca2c37 data_manifests/cric_images_verified.json
06fe16103eae26f970db1c651b8c86660b3eb32c8ff3e56219933e9563e75863 splits/study2/sipakmed_2b/
    background_fraction_seed137.csv
1dd2db6e7d4a06c3995281a7876ce1971ffd45ae455ef3e655e43e2c672f28b3 splits/study2/sipakmed_2b/
    background_fraction_seed173.csv
5063538684a60f94747905d777ee6ba79174c84bd5813da2de237f87cf833cd2 splits/study2/sipakmed_2b/
    background_fraction_seed211.csv
3ef5674738a3901867e2d9744f45ea2d186ac8d6d02674e6f23fbb811f2a6100 splits/study2/sipakmed_2b/
    background_fraction_seed257.csv
914c82d53259068bb7e79e51d7425e2010726c55685221e2e3a820689e62e833 splits/study2/sipakmed_2b/
    background_fraction_seed307.csv
637911ee4155f2593fe472d2be8e7fad606b16525d778bcea5201afcd34b07c0 splits/study2/sipakmed_2b/
    background_fraction_seed349.csv
69a0e0a88ad9de3befa5dfe7ec175612ff8ca5a66087055084a2c351101cb6a3 splits/study2/sipakmed_2b/
    background_fraction_seed401.csv
d87520a7b40aa4a2b43411f80573951ad111faabcfd7f48ab75c2b82ccf258f7 splits/study2/sipakmed_2b/
    background_fraction_seed457.csv
c5e40327ff7a52f86c6a231a73dc52783f31cb93c59245415bfaa7ee37dc3d61 splits/study2/sipakmed_2b/
    background_fraction_seed503.csv
05f936a70d9b5329628cfdfa3ecf9f2f035e0e445d7faaf3d1e5eb95788c37a2 splits/study2/sipakmed_2b/
    background_fraction_seed557.csv
df1d723eba270fdeaac9e42b3663350cdd95503b601d64eb0e31c1e05438ba30 splits/study2/tasks/cric.jsonl
c611865c7919bd329744fa5a03e1e3d27cc70e20f7ac9d768cdfc73d645585fa splits/study2/tasks/factorial.jsonl
ae478ddb914a75c057a07bded9f9edf17ae41ddf216db6a4753d7b1d0e0ada96 splits/study2/tasks/fingerprint.jsonl
c4d96582a084370c448b7b46ec2cbbcde5ae89cd0df8d86476cf5bbbdb412f54 splits/study2/tasks/grl.jsonl
b57b43f34058cad7b6871dfb665e3028adf9609abad3b625004c341170454954 splits/study2/tasks/phase1.jsonl
a52b656719d79879ece36a8cb260789bb2c49ca8567be76799e01c397876ad2a cache:cric:pixels (11534 crops)
b8d905e39e31af99d760039a03c99dc138ef27fa703889729a83a066553c8e1f cache:cric:index
```

```
d5bbaf57f176cf6cdf838a3b4773e2d5fb8c7bf019570344fa6a8b7dcd30c748 cache:sipakmed:pixels (4049 crops)
d29078c8e211819e8fa6ea5f681c5106121bd8f15c0db66769075ea30842d66f cache:sipakmed:index
afe9d620a1de88ab0fdaed3f1abfb5e96d34425819db48be566aa95ea3ea3eca cache:transplant_seed137:pixels (4176 crops)
994672d672c448a1b51f7ae34caca1b21e7d524e06af2b62ec1fcc0e48f0eac6 cache:transplant_seed137:index
87e0e17ab774d40a8f7d4546676f5b281bf4774bf9c8f7a1a62540c25378e35e cache:transplant_seed173:pixels (4136 crops)
ac22a79f280a7f7a3bf735b80d66e9db1d9063d5a4382b4c7327e6b3fbc10c4e cache:transplant_seed173:index
5baadc6106666d174bd7927ad049fcb426ce40bd8cb7a01030ad213141514937 cache:transplant_seed211:pixels (4152 crops)
5f141d0430020d89f35ae080691dc6c5b98a38bc177914f8e9ccb519ff7efe0e cache:transplant_seed211:index
e4ad01639c33080b427f2964922854eebbb44b05c8da85f4ef63bf8351cd6853 cache:transplant_seed257:pixels (4184 crops)
3a07633ca021e8a058a779de4485bdb89ab10c27bc06a37e213457213eaaa1f9 cache:transplant_seed257:index
df36fe07cb47edee3bceeb35943fa1e99bcd2118f7395fa8cf34cca510dd3784 cache:transplant_seed307:pixels (4096 crops)
2f066cbd45475e8eba414c7675d67e607abc92423ba2f23f25452fcacaf2ce16 cache:transplant_seed307:index
eb7ca1021c78dad6e8b20e6728d819359e9498426acd4a2d996a5308620f1e3e cache:transplant_seed349:pixels (4256 crops)
91442b22f10454434cd789b0a8190a01eb59d9321c3f88510327fe0e3ec94ab4 cache:transplant_seed349:index
ce34f911fd43a1c7f5702d9a92acd87ee59444375b47f67291552e64376e071f cache:transplant_seed401:pixels (4136 crops)
4086823828c93e5f7025a57187d11566806bafa2396e16c0284048f5ac00371a cache:transplant_seed401:index
c0fb31f795f94bfd5a5f5f6fe5582d9140a6e0c03d5e5efd025dda89a1164491 cache:transplant_seed457:pixels (4376 crops)
707d08d4f41741aea2bfc8b1a1705f8e52b4b47bb6359eba152296d2a47e0cc6 cache:transplant_seed457:index
9c177a52ce8519b1c1a171c914362db1f91ef5933e8d78df15bef338c998b235 cache:transplant_seed503:pixels (4240 crops)
d6e35027b4894467d4945bee514e6d44689d8ca7ed26c943643604303a2f0f19 cache:transplant_seed503:index
e0b87b9e3d0b0ff04804503b64c4c9deb13987813fe7bcbe02899f8247636189 cache:transplant_seed557:pixels (4400 crops)
065c83b86cbb54d6b05495156cefd3a4e154d75209c19f94922fbb400f0e8383 cache:transplant_seed557:index

# Amendment 7 (2026-10-18): files changed or added after registration, before any 2A run
42bf439fbff44ae9a2d2638ac7fb3f3eb92d47c0fba5acd8d4076272a75e9b86 PREREGISTRATION_STUDY2.md
8d6eb5674bcb6c19b135dc8085e5171dca7a5dfe3f74caee9a306bf1f93f6076 analysis/study2/collect_runs.py
b1cec6d36aaac901dbae55164bcbdccc32835492dab96b9ee89e86d688f6eabf analysis/study2/mechanism_lmm.R
54488ce3f68b0707dbea92da138b3afdffc2d5c5335b0b072f5624e721707d97 analysis/study2/run_all.sh

# Amendment 8 (2026-10-19): follow-up experiments, registered before any follow-up run or analysis
6877eb1ec9598958590e861194098ad0f60a68c64980c88445d118d25d406525 PREREGISTRATION_STUDY2.md
ac286bfed400c57e5f8de377102633132dcbecc6fbe0b2b5fd288da12eb0d68f PREREGISTRATION_STUDY2_AMENDMENT8.md
2d0e79997085d536a02a1d19c31c90103271979d9bf80d7fbfdee4f8ac001639 src/cytofair/carriers.py
689f4ea37617bf13f16d6a067e3ad61bd2cfb58775945e9c2aa550c0a2fc3553 src/cytofair/followup_eval.py
3b181ee8bf9b6824a69a1b1b54ccf7dc3d5424cd232c97dcb242467568a4395b src/cytofair/train_study2.py
e93fe9fa699284d25045abc1d065ef936c37c7fd8e706005a72185c341087847 src/cytofair/gpu_augment.py
1653a502581fa1322d8a412e811ac3d4d9fb8dc7ee9483e717f50bef766acfe8 src/cytofair/backbones.py
e31ef811e9639d4ff78dc6edf4b7dce5c1c713662f1713c36a834f09011588ff src/cytofair/cli.py
4645afa59daa826ec2e14696ae66abc46d52e230d3151e84e2e95f7e43cbf181 src/cytofair/study2_design.py
7afccc8affa20d164c288b851529b05008099a9843280a5fb8b8d8879a0bc900 scripts/study2/build_followup_tasks.py
77692934e3cc194e638b6872320f56da46c2baa602d2e91820b1db617e0c8b59 scripts/study2/launch_matrix.py
6108cbb436b6c14af6925c038bcc7d05ab6725ecb7ed6182f3a3c6c45a199c81 splits/study2/followup/
    followup_split_manifest.csv
560fbb4b81d0faad66433fb6854c2e9f23ecf35e00f353af8cc46de3f22f41c1 splits/study2/followup/
    followup_ablation_manifest.csv
bae4d51de34c37cab5bd1c384b01ce7b3638788089c2c05308163ebd8c4b547c analysis/study2/followup/cric_context.py
8211d214b8476967eedfc59275a8f8731e187707090ded1edd948273060a8179 analysis/study2/followup/followup_lmm.R
8edf79e706a82cb82572422f85ce05d777a9eda6a6ca5482e2bbb40cea9b8103 analysis/study2/followup/prepare_followup.py
f41638914033f1a8895a6ffa89fbf5ba034d8fccc37a8b8aaf8cd6344ebe4094 analysis/study2/followup/run_followup.sh
3f800acc4ac742032d3988df807a0096bfcbe2dc10ac12df3be2e6a4affa79ae analysis/study2/followup/
    simulate_followup_inputs.py
5d53f3715e6432107ab7686d3b2b31c4ee00b3402fcce75fda681af2ae8a8edc splits/study2/followup/ablations/
    ablation_plan_seed137.csv
4bafa40d2251f5837c03cc6e06911f9cd5850ffd501da260bc21b4c0471f46bc splits/study2/followup/ablations/
    ablation_plan_seed173.csv
fbf92dc4c3370a00e8b38c17325c01a7d0b509e9343f70f306a4f1dc94c5cb23 splits/study2/followup/ablations/
    ablation_plan_seed211.csv
29d29e52d9f54a6bb08728a7e577e8674e40cb51a317c4651c82a787f857729d splits/study2/followup/ablations/
    ablation_plan_seed257.csv
a348bee178a67d29497453221f45c7a30f6a1d1a9e31645ac52750ccab8ad9bd splits/study2/followup/ablations/
    ablation_plan_seed307.csv
643b8b42ce1d485d63c511ad77b84dbbe2dc75a228fd7f663d47e021efb0862d splits/study2/followup/ablations/
    ablation_plan_seed349.csv
ed7787d88b9b9e5adf19353027027e1bbed76366eda8da576470cd25bc1b442e splits/study2/followup/ablations/
    ablation_plan_seed401.csv
26cb7c492e9c84050444bf7be8a6cedf9de5d5ef11a39e9fd53d625a329ddab6 splits/study2/followup/ablations/
    ablation_plan_seed457.csv
bd8a91179515b8c30fc437a17ff787faadd1bebd33b27d03687f129b49a1ed8e splits/study2/followup/ablations/
    ablation_plan_seed503.csv
448d5c2bef95c4bc9882234e5516d9098a9f848fbba6112078eb2467541e764e splits/study2/followup/ablations/
    ablation_plan_seed557.csv
9bff878f72a63fc89d1fb374182964174665a0f01c2a2e4b3214f07c94bc45d9 splits/study2/followup/tasks/carrier.jsonl
05a615f1d19c92d4095cd9909b108e4b2dacdb064859c9d3a535e64e7f949d6d splits/study2/followup/tasks/ffact.jsonl
ca8d6db070b0da00457759f905f85f59ccba39383cb05f0f2c04a679d65cb3e2 splits/study2/followup/tasks/leace.jsonl
4f683b6e7a0a90c9fb631df68b2f61777aedf472b6a5ce477a155f6107891642 splits/study2/followup/tasks/linadv.jsonl
```

```
e7d301e0ffa9397cca00af0b60d5675ef7953d6beecfbc76f37aa6d7034a23f3 splits/study2/followup/tasks/normalization.
    jsonl
1117f1219a1c4e7b4f04e3256c1a441ce1d007e1e66b6a2ecfad1de8b9bf862a cache:ablation_seed137:pixels (3132 crops)
879097586ed1b7942e976e5ebccf58fca3a440c8e3ed946b8ee1f8aa88966caa cache:ablation_seed137:index
3d2f43f345a703465888513f1636243d126df06da00e3acd490337d0465ae718 cache:ablation_seed173:pixels (3102 crops)
67e6344f98c824a89729202970ba9f2008a1648144c5e4e26f0315e15e3ab761 cache:ablation_seed173:index
58d57a0b460829781f7510df5048e3604f1542b20e89f4ef57caa99ba61d6a11 cache:ablation_seed211:pixels (3114 crops)
703ac1d02282a9427c49ebc80e9da189547c79bdc4f62ef24b3a25b194d11f37 cache:ablation_seed211:index
96fb0e6729372b4702acb9492b1c35d676c05584423baf00f6449825aef88d4a cache:ablation_seed257:pixels (3138 crops)
483c8612c71cdead087c1387daf655bd3cb1b390b64d60f0928a3e39b5259a2d cache:ablation_seed257:index
850a11d7d3f7ac9ef4507658a087f28ed08f9dfe03062ed0c23ce490fa8a67b9 cache:ablation_seed307:pixels (3072 crops)
3bf6ed06b3c4b1067bd3fe093111a53f847f873e1c60dd275a6b8dd8d565bc8d cache:ablation_seed307:index
7c9f1e2c85cdb4362a6efffbaf36827f774e07ad27edc32a3f19d72673e2e2a0 cache:ablation_seed349:pixels (3192 crops)
6fc1f3245c77071239412cf599d4d658a13512604cb3ef32b5c5b9b7a7f2a1d6 cache:ablation_seed349:index
88e92da6981e7c93ce8befa64d98936ec53768f04e2ba600ae3396558d6a1e5c cache:ablation_seed401:pixels (3102 crops)
c8e9a22fdb2ab9abe33083a6c847c1814b367bd140d67b535be2fd7b8c2865e4 cache:ablation_seed401:index
1c44558cfda145ef285a366d038861d49f1978c0d8d372591ac70de35666718c cache:ablation_seed457:pixels (3282 crops)
11b162ca4f2cc7fd0edd26bc2d987470eaf6485da44137f810ee38dc104e5893 cache:ablation_seed457:index
c316def50ed2e82d750d8b83627602d6bd1d7ad1671c13f923f478fb6a43d57f cache:ablation_seed503:pixels (3180 crops)
2973973f0c623f03da92d70ef37d552843636203655ed4a8ef3df7e9b1b7000d cache:ablation_seed503:index
c868502d5e670ec86040c5aa63d9b9fce3928e65472b2eb1f4928b77b724de26 cache:ablation_seed557:pixels (3300 crops)
f33d0d36634066ef1e7aa6890f500e3a9408928db3e399742ea24724a2e35776 cache:ablation_seed557:index
a0d069a9324563f36fe31ad80003d81ea2efb120e9bcf53ae513efdae9400efe cache:sipakmed_masked:pixels (4049 crops)
d45211370d5e5eee02ab1afc234e2f48572d6aa76e8692e3f3e818dd269d3b99 cache:sipakmed_masked:index
10974f8fe420e4850294f0b959beafc08feb53aa514c272bdc778e4f47366428 cache:sipakmed_reinhard100:pixels (4049
    crops)
017675d683b8f50007c61a0a0305eef7fed8c29ee1e5b0932c68924326c706a8 cache:sipakmed_reinhard100:index
a7265a0f8162cc862c38fab2bedfee783aa19f25260aaf89b5b777eab48b5c05 cache:sipakmed_reinhard50:pixels (4049 crops
    )
e1faf04759b407365319da6852422952818c8d2b72681a25bd09992e0ef34734 cache:sipakmed_reinhard50:index

# Amendment 9 (2026-10-20): preparer bug fix, before any F3/F4 estimate was seen
fd8e2e1f17291592eb3df50cb6ea83fea5910f729852beb15eac6c56f772e22b PREREGISTRATION_STUDY2.md
b58e212b92ce5c2e06b8f85554332bd70f3108aff575bde8d01c8e240fe27f0c PREREGISTRATION_STUDY2_AMENDMENT8.md
63f03ca9082a3a6b59dabb68e8f7550fd28536391aa41f8721ecd9be939a5c79 analysis/study2/followup/prepare_followup.py
94d1fc5cfe1aad4ed0f632d03c6ad6d057b792f31c4f61646eb0ffbd452cec94 tests/test_followup_prepare.py
```

Q

The Study 2 preregistration, verbatim

```
# CytoFair Study 2 preregistration: powered replication and causal decomposition

**Registered:** 2026-10-18 (Asia/Kolkata), in the commit that adds this file.
**Relationship to the original plan:** registered as Amendment 6 of ‘PREREGISTRATION.md‘.
**Status at registration:** no Study 2 model has been trained for any registered cell.
The only Study 2 GPU executions before this commit are one-epoch pipeline smoke tests on
P1 splits (listed in ‘reports/study2/excluded_attempts.md‘); none produced a protocol
contrast, and their metrics were checked only for finiteness.

## 0. Why a second study

Study 1 (frozen 7 September 2026) established structural non-independence, literal pixel
sharing, and a learnable parent fingerprint, but left four limitations that this study
addresses:

1. **Specification.** One question ("does crop-level splitting inflate accuracy?") was tested
   six times, once per architecture, inside a 12-test BH family. Study 2 replaces this with
   one mixed-model test of the protocol effect (H1), powered by simulation.
2. **Availability versus use.** Study 1 showed a fingerprint is available, not that
   classifiers route predictions through it. Study 2 adds a counterfactual background
   transplant (2B, H4) and adversarial fingerprint suppression (2A, H5).
3. **Leakage versus diversity.** At equal crop counts, P1 training sets contain about 914
   distinct parents and P2 training sets about 773, so P1 also has more training
   diversity. Study 2 crosses leakage with diversity at matched crop and parent counts
   over one common test set (2C, H3).
4. **One rescored benchmark.** Study 2 rescores CRIC (H2), whose crop-level split places a
   training parent behind 100% of test cells.

Study 1 results are used only as pilot estimates for the power analysis. No Study 1 run
enters any Study 2 test, and the Study 2 seeds are disjoint from the Study 1 seeds.

While validating the analysis code, the H1 model was fitted to the Study 1 runs
(‘results/study2/power/power_pilot.json‘): P1 - P2 = 1.37 pp, Satterthwaite p = 0.0054.
That reanalysis was chosen after the Study 1 data were seen. It is reported as a post hoc,
exploratory observation and is not a confirmatory result.

## 1. Fixed training recipe

Every run uses the Study 1 classification recipe: ImageNet initialisation, a new linear
head, AdamW (learning rate 1e-4, weight decay 1e-4), batch size 64, 20 epochs, cross
entropy, final-epoch endpoint, no validation split, no early stopping, no checkpoint
selection. Training augmentation is RandomResizedCrop(224, scale 0.8-1.0, ratio 3/4-4/3),
horizontal flip (p = 0.5), vertical flip (p = 0.5), ImageNet normalisation; evaluation uses
Resize(224 x 224) and the same normalisation.

Execution changes relative to Study 1, none of which alters a model, split, seed, or
endpoint:

* The training host limits each user to four CPU cores (cgroup quota 400%). Crops are
  therefore decoded once into checksummed caches on node-local storage, uploaded to the
  GPU, and augmented there (‘cytofair.gpu_augment‘). Crop parameters come from an exact
  port of torchvision’s ‘RandomResizedCrop.get_params‘ (verified identical for 100
  seed/shape pairs, ‘tests/test_gpu_augment.py‘); resizing is antialiased bilinear.
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```
* cuDNN deterministic mode, `torch.use_deterministic_algorithms(True, warn_only=True)`,
  and `CUBLAS_WORKSPACE_CONFIG=:4096:8`.
* Every run writes, beside its JSON: per-crop test logits, a float16 copy of the final
  weights, penultimate features for the balanced 424-parent probe crops, and logits for
  any registered extra evaluation set (Section 7).

At most four GPUs (indices 0, 3, 4, 5 on the training host) run concurrently, one job per
GPU.

## 2. Architectures

Twelve ImageNet-initialised backbones (exact weights in `cytofair.backbones.WEIGHT_NAMES`):
the six Study 1 models (ResNet-18, ResNet-50, DenseNet-121, EfficientNet-B0,
MobileNet-V2, ViT-B/16) and six added for breadth (ConvNeXt-T, Swin-T, RegNetY-1.6GF,
EfficientNetV2-S, MaxViT-T, and self-supervised DINOv2 ViT-S/14 at 224 px).

## 3. Seeds and stopping rule

`STUDY2_SEEDS` = 137, 173, 211, 257, 307, 349, 401, 457, 503, 557, 601, 653, 701, 751,
809, 853, 907, 953, 1009, 1051. The first five were named as robustness seeds in the
original preregistration and were never run. Phase 1 uses all twenty; the mechanism arms
and CRIC use the first ten (`MECHANISM_SEEDS`).

No seed, architecture, or cell is added, removed, or replaced after any result is seen.
A failed run (infrastructure failure or non-finite loss) is retried once with the identical
seed and split; a cell that fails twice is reported as missing, and every test that needs it
fails closed rather than substituting a run.

## 4. Splits

All splits were generated before registration and are committed under `splits/study2/`
with SHA-256 hashes in `study2_split_manifest.csv`. Trainers refuse a split whose hash
differs. P1, P1-prime, P2, and P3 follow the original definitions. For all twenty SIPaKMeD
seeds, P2 has exactly 810 test crops, so P1-prime is byte-identical to P1 and is not trained
separately (P1-prime - P2 equals P1 - P2 by construction). For CRIC, P1-prime differs from
P1 by 2-11 training crops and is trained.

## 5. Phase 1: powered SIPaKMeD replication (H1, the primary endpoint)

**Design.** 12 architectures x 20 seeds x {P1, P2, P3} = 720 runs.

**Primary endpoint.** Test accuracy in percentage points.

**Confirmatory model (H1).** On P1 and P2 runs:

    accuracy ~ x + (1 + x | architecture) + (1 | seed) + (1 | split)

with `x = +0.5` (P1) or `-0.5` (P2), so the fixed effect is P1 - P2; `split` is the
seed-by-protocol partition shared by all architectures (omitting it would treat twelve
evaluations of one test set as independent). REML, `bobyqa`, lme4 with lmerTest
Satterthwaite degrees of freedom, two-sided alpha = 0.05.
**Decision rule:** H1 is supported if p < 0.05 and the estimate is positive.

**Convergence.** A singular (boundary) fit is accepted. Only on convergence failure, the
model is simplified in this order: uncorrelated architecture intercept and slope; then
dropping the seed intercept. The architecture slope and the split effect are never dropped.

**Assumptions and fallback.** Conditional residual skewness, excess kurtosis, Shapiro-Wilk,
architecture-slope BLUPs, and residual SD by protocol are reported. If no model converges,
or residual |skewness| > 1, or excess kurtosis > 3, the primary result is the two-way
cluster bootstrap (architectures and seeds resampled independently, 10,000 replicates,
percentile CI and two-sided bootstrap p).

**Sensitivity (reported, not decisive).** Kenward-Roger degrees of freedom; binomial GLMM
on correct counts with an observation-level random effect; seed-level exact sign-flip
test; the cluster bootstrap.

**Power.** `analysis/study2/power_simulation.R` simulated the confirmatory model 1,000
times per design from the Study 1 pilot (effect 1.37 pp; architecture intercept SD 0.44,
slope SD 0.36; split SD 0.31 for P1 and 0.59 for P2; residual SD 0.69; seed SD 0;
intercept-slope correlation set to 0 because six architectures put it on its -1
boundary). The registered rule was: choose the smallest seed count in {10, 15, 20, 25, 30},
with twelve architectures, whose power for a 0.8-pp effect is at least 0.90 under pilot
variances and at least 0.75 with every variance inflated by 1.5. Twenty seeds give 0.977
and 0.782 (fifteen give 0.949 and 0.698). 0.8 pp exceeds the 0.62-pp spread between
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published SIPaKMeD scores that motivates the question.

**Secondary analyses (non-confirmatory, no multiplicity claim).** P2 - P3 accuracy (same
model); P1 - P2 for macro-F1 and balanced accuracy; per-architecture paired differences
(descriptive t intervals); Kendall's tau between P1 and P2 architecture rankings with a
seed bootstrap; the P3 <= P2 <= P1 ordering per architecture; pooled confusion matrices,
NLL, and ECE; and two crop-level analyses from the saved predictions:
**M1** within-crop P1 - P2 accuracy for crops that appear in both P1 and P2 test sets
(crop-level bootstrap), and **M2** within-P1 accuracy as a function of the number of
training siblings and literal source-pixel overlap (logistic GLM with crop-clustered
standard errors, adjusting for class and the crop's P2 accuracy).

**Gate A.** If H1 is supported, the inflation effect is reported as certified. If not, the
paper is framed as a careful null replication. The mechanism arms are run regardless
(they are registered now), but their interpretation follows Gate A.

## 6. Arm 2C: leakage x diversity at matched size (H3)

**Design** (`cytofair.factorial`, `FACTORIAL_CONFIG`). For each of the ten mechanism seeds
and each class: 30% of parents with at least two crops are evaluation parents E; each E
parent contributes floor(k/2) test crops (the common test set, about 527 crops) and its
remaining crops are donor siblings. Six parents per class with at least three crops are
reserved as unseen background donors U (never trained on or tested). The rest form the
clean pool B. Four training sets each contain exactly 220 crops per class:

* `clean_high`: every B parent;
* `leak_high`: the E parents' donor siblings replace the last |E| parents of one fixed B
  permutation;
* `clean_low` and `leak_low`: the same construction on the first 70% of that permutation.

Within a diversity level, leak and clean have identical crop counts and identical numbers
of distinct parents (720 versus 504 in total; effective parent numbers 633 and 632 versus
432 and 427), and differ only in whether the test parents' siblings are present. All test
crops have a training sibling in the leak cells (mean 2.0 high, 3.0 low) and none in the
clean cells. Low-diversity parent sets are nested in high-diversity sets. Configurations
with 240 crops per class were infeasible for seed 307 and were rejected before any training.

**Models.** ResNet-50, DenseNet-121, ConvNeXt-T, ViT-B/16; 4 x 10 x 4 = 160 runs.

**H3.** `accuracy ~ leak * div + (1 + leak | architecture) + (1 | seed) + (1 | training_set)`
with leak and div coded +-0.5. H3 is the leak main effect (leakage at matched diversity),
predicted positive. Simplification order on convergence failure: uncorrelated slope, then
intercepts only.

**Secondary.** The diversity effect is estimated only within clean cells
(`clean_high - clean_low`), because the leak dose (siblings per test crop) is higher in the
low-diversity leak cell; the leak x diversity interaction is exploratory for the same
reason. Embedding diversity of every training set is reported in the space of a retrained
RGB parent-probe encoder (`cytofair.fingerprint`). A descriptive decomposition of the
Phase 1 P1 - P2 gap combines the H3 estimate (scaled by the 91.7% leak rate of natural P1)
with the clean diversity effect scaled log-linearly from the 720/504 contrast to the natural
914/773 contrast; it is labelled model-based and approximate.

## 7. Arm 2B: counterfactual background transplant (H4)

Every 2C model is also evaluated on composites of its seed's common test set
(`cytofair.transplant`, plans in `splits/study2/sipakmed_2b/`). The cell foreground is kept
(the Study 1 background-only mask: Otsu on saturation and luminance, 5 x 5 closing) and
feathered (Gaussian sigma 2) onto a donor background inpainted with Telea (radius 5):

* `original`; `self` (own inpainted background: seam and inpainting artifacts only);
* `sibling` (background of a donor sibling of the same parent: own acquisition context);
* `unseen_same_class` (background from a U parent of the same class);
* `unseen_other_class` (one composite per other class, from U parents).

**H4 (context-route difference-in-differences).** For each run,
`route = acc(sibling) - acc(unseen_same_class)` measures how much the recipient's own
acquisition context contributes beyond class-typical context. Leak models have seen that
context with the correct label; clean models have not. H4 is the leak coefficient in
`route ~ leak + div + (1 + leak | architecture) + (1 | seed) + (1 | training_set)`,
predicted positive (fallback: intercepts only).

**Secondary.** The ratio of H4 to the leak effect on original images (the share of the
leakage inflation attributable to the context route); the donor-class probability shift for
other-class backgrounds relative to same-class backgrounds (whether backgrounds carry class
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information that classifiers use, independent of leakage); ‘self‘ versus ‘original‘
(artifact size); and H4 restricted to recipients whose background covers at least 30% of
the crop.

## 8. Arm 2A: adversarial fingerprint suppression (H5)

A class-conditional parent head (512-unit MLP over the training parents, logits of
other-class parents masked) sits behind a gradient-reversal layer whose coefficient follows
the DANN ramp ‘lambda * (2 / (1 + exp(-10 p)) - 1)‘. The class head is unchanged. Because
the adversary only distinguishes parents within a class, suppressing it never requires
removing class information.

**Design.** ResNet-50 and ConvNeXt-T; lambda in {0, 0.01, 0.03, 0.1, 0.3, 1, 3}; natural P1
and P2 splits of the ten mechanism seeds; 280 runs. Lambda = 0 trains the same head with no
reversed gradient and is the matched baseline. The full sweep, including any degenerate
region, is reported.

**Recoverability R.** Within-class parent identification accuracy of a multinomial logistic
regression (C = 1, standardised features) on the frozen penultimate features of the
balanced 424-parent probe split (two fitting and two evaluation crops per parent; seed 17),
reported with its class-only floor (the accuracy of a representation that knows only the
class). Overall 424-way and nearest-centroid accuracies are secondary.

**Non-degenerate regime.** For each architecture, the lambdas whose mean P2 accuracy is
within 1.0 pp of its lambda = 0 mean. **Manipulation check:** at the largest such lambda,
mean R must fall by at least 25% of its above-floor value at lambda = 0; an architecture that
fails is excluded from H5, and if both fail H5 is reported as untestable, not refuted.

**H5.** Within the regime, per architecture-seed-lambda pair,
‘gap = acc(P1) - acc(P2)‘ and ‘R_pair‘ = mean R of the two runs;
‘gap ~ architecture + R_pair + (1 | seed)‘. H5 is the R_pair slope, predicted positive (the
inflation shrinks as the fingerprint is suppressed).

## 9. CRIC rescore (H2)

CRIC cells are 100 x 100 windows centred on the published nucleus coordinates (shifted
inward at image borders) from the 400 Figshare slide images (MD5-verified,
‘data_manifests/cric_images_verified.json‘). Six Bethesda classes. Models: ResNet-18,
ResNet-50, DenseNet-121, EfficientNet-B0, ConvNeXt-T, ViT-B/16; ten mechanism seeds;
P1, P1-prime, P2; 180 runs. **H2** is the H1 model applied to CRIC accuracy (P1 - P2),
predicted positive. Secondary: P1-prime - P2, balanced accuracy and macro-F1 (the classes
are imbalanced), and the CRIC versus SIPaKMeD effect sizes against their structural
leakage, which is exploratory.

## 10. Multiplicity

H1 is the single primary endpoint, tested alone at alpha = 0.05. H2, H3, H4, and H5 form
one Holm family (family-wise alpha = 0.05); each is supported only if its Holm-adjusted p
is below 0.05 and its estimate has the predicted sign. Everything else is secondary or
exploratory and carries no confirmatory claim.

## 11. Exclusions and deviations

Runs are excluded only for logged infrastructure failure, non-finite loss, or a failed
integrity check (wrong split hash, cache hash, or missing artifact); never for magnitude or
direction. Every attempt is logged. Any deviation from this document is recorded as a dated
amendment before the affected analysis is run; a deviation discovered afterwards is
reported as such.

## 12. Gate B and venue framing

Gate B is interpretive, not statistical: the causal evidence is "clean" if H3 and H4 are
supported and H5 is either supported or untestable for a stated reason, "murky" otherwise.
It determines framing only. All registered results are reported whatever the gates show.

## 13. Registered artifacts

The analysis code (‘analysis/study2/*.R‘, ‘analysis/study2/*.py‘), split manifest, task
builder, and caches are fixed by this commit; their SHA-256 hashes are listed in
‘reports/study2/registration_hashes.txt‘.

## Amendment 7 -- failure handling (2026-10-18, Asia/Kolkata)

Registered while Phase 1 was running and before any 2A run had started; no Study 2
result had been analysed or inspected. Two cases were not handled by Section 3:
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1. **Infrastructure failures.** A run that fails because of the host rather than the run
   itself (out-of-memory caused by another user's process on the same GPU, loss of the
   host, session, or controller, or a filesystem error, each identifiable from the
   attempt log) does not count toward the one-retry limit. It is rerun with the identical
   seed, split, and code, and every attempt stays in the log.
2. **Divergence in the 2A sweep.** Runs are deterministic, so a run whose training loss
   becomes non-finite diverges again on retry. In the gradient-reversal arm only, such a
   run is recorded as *diverged* rather than missing: a lambda with any diverged run of an
   architecture lies outside that architecture's non-degenerate regime, the diverged runs
   are excluded from H5, and the sweep table reports them with the epoch at which the loss
   became non-finite. A diverged run in any other arm still makes the tests that need it
   fail closed, as registered.

The collector (`analysis/study2/collect_runs.py --diverged-logs`) and `mechanism_lmm.R`
implement point 2; their new hashes are appended to `reports/study2/registration_hashes.txt`.

## Amendment 8 -- follow-up experiments (2026-10-19, Asia/Kolkata)

Registered after the Study 2 analysis was complete, before any follow-up model was trained
or any follow-up analysis was run on existing outputs. It adds eight new hypotheses (F1-F8)
on how leakage is used -- CRIC slide-context pull, test-time carrier ablations, carrier
removal during training, rank-limited LEACE erasure, a stain-normalisation dose, a
probe-matched linear adversary, and a pooled dose-response -- in their own Holm family. It
does not change H1-H5 or their verdicts. Full text: `PREREGISTRATION_STUDY2_AMENDMENT8.md`.

## Amendment 9 -- preparer bug fix (2026-10-20, Asia/Kolkata)

A labelling bug in `analysis/study2/followup/prepare_followup.py` (standard-recipe carrier runs
labelled `colour`) was fixed before any F3 or F4 estimate was seen. Details in
`PREREGISTRATION_STUDY2_AMENDMENT8.md`.
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Amendments 8 and 9, verbatim

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# Study 2, Amendment 8 -- Follow-up experiments on how leakage is used

Registered 2026-10-19 (Asia/Kolkata) as an amendment to ‘PREREGISTRATION_STUDY2.md‘.

## 0. Status, motivation, and what was known

This amendment was written **after** the registered Study 2 analysis was complete. Its
verdicts stand and are not revisited here: H1 (SIPaKMeD P1 - P2) and the Holm-family
hypotheses H2 (CRIC) and H3 (leakage at matched diversity) were supported; H4 (context route
through backgrounds) and H5 (gap tracks gradient-reversal suppression) were not.

The eight hypotheses below (F1-F8) are new questions, motivated by two observations from
Study 2: (i) leak-trained 2C models kept most of their advantage on backgrounds transplanted
from unseen fields, which points to a carrier inside the cell region rather than the
background; and (ii) for ConvNeXt-T the gradient-reversal adversary was evaded (independent
probe recoverability rose), so H5 had only one usable architecture.

**Known when writing:** every registered Study 2 output, including transplant condition
means by cell and the gradient-reversal sweep means. **Not known:** any per-cell CRIC result
broken down by slide composition; any evaluation of ablated images; any evaluation of erased
features; any Grad-CAM map; any model trained with masked backgrounds, colour augmentation,
stain normalisation, or the linear adversary. One-epoch smoke tests used only crop-level or
leak-cell runs, so no follow-up contrast could be observed before registration.

Results for F1-F8 are reported as registered follow-ups to Study 2, separately from H1-H5,
whatever they show.

## 1. Common elements

The Study 2 recipe (section 1 of the parent document) is unchanged: ImageNet initialisation,
AdamW 1e-4 / weight decay 1e-4, batch 64, 20 epochs, final-epoch endpoint, random resized crop
(0.8-1.0) and flips on the GPU, deterministic kernels. Only the factor each arm manipulates
differs. Every new training run saves per-crop logits, final weights, and probe-split
features. Failure handling follows Amendment 7: infrastructure failures are rerun; a
linear-adversary run whose attempts both end in a non-finite loss is recorded as *diverged*,
its lambda leaves that architecture’s regime, and it is excluded from F7 and F8; any other
failure fails the arm closed.

**Recoverability R** (F5-F8) is within-class parent identification accuracy on the balanced
424-parent probe split (seed 17; two fitting and two evaluation crops per parent): multinomial
logistic regression on standardised features with sklearn’s objective (C = 1), solved by
L-BFGS on the GPU (‘cytofair.followup_eval.fit_logistic‘), restricted at prediction to parents
of the crop’s class. The same function scores every arm. Its agreement with the registered
H5 sklearn probe on the 280 Study 2 gradient-reversal runs is reported (secondary).

## 2. F1 -- CRIC slide-context pull (existing predictions; no training)

For every CRIC cell, the **context class** is the most frequent class among the *other*
cells of its slide image; cells whose slide has no other cell, or a tied majority, are
excluded. A cell is **discordant** if its own class differs from its context class. For each
of the 120 registered CRIC P1 and P2 runs (6 architectures x 10 seeds), the softmax
probability of the context class is computed for every test cell. Per cell and protocol,
probabilities are averaged over all runs in which the cell is a test cell.
```

```
**F1** is the mean, over discordant cells tested under both protocols, of
‘100 * (p_context(P1) - p_context(P2))‘, predicted positive: when a cell’s slide is in
training, predictions are pulled toward the slide’s majority label. Inference: percentile
cluster bootstrap over slides (10,000 replicates, seed 20261019), two-sided
‘p = 2 * min(P(boot <= 0), P(boot >= 0))‘.
Secondary: the within-cell P1 - P2 accuracy difference in concordant and discordant cells
and their contrast (predicted concordant > discordant); the rate of predicting the context
class; per-architecture F1.

## 3. F2 -- test-time carrier ablations (existing 2C checkpoints; no training)

The 160 registered 2C models are re-evaluated on their seed’s common test set under six
conditions, each rendered at native resolution (‘cytofair.carriers‘): ‘original‘; ‘grey‘
(luminance only); ‘stain_transfer‘ (Reinhard transfer of the crop’s LAB mean and standard
deviation to those of the 2B unseen same-class donor crop); ‘low_pass‘ (Gaussian blur, sigma =
2 px at the 224-px input); ‘illumination_flat‘ (subtract a sigma = 16 px Gaussian, add back the
crop mean); ‘background_grey‘ (pixels outside the Study 1 foreground mask set to the ImageNet
mean). Ablation plans are ‘splits/study2/followup/ablations/ablation_plan_seed*.csv‘.

For each ablation ‘a‘, with run-level accuracy in pp and ‘ablated = +-1/2‘:
‘acc ~ leak * ablated + div + (1 + leak | architecture) + (1 | seed) + (1 | training set) + (1 | run)‘
(fallback: random intercepts only). The reduction of the leak effect is ‘-coef(leak:ablated)‘.
**F2** is that reduction for ‘stain_transfer‘, predicted positive (the leak advantage depends
on the crop’s own colour statistics). Secondary: the reductions for the other four ablations;
the leak effect within each condition; agreement of re-evaluated originals with the recorded
predictions (fp16 weights).

**Grad-CAM (descriptive only).** For ResNet-50, DenseNet-121, and ConvNeXt-T 2C models, the
Grad-CAM map of the true-class logit at the last convolutional stage is upsampled to the crop
and the fraction of positive attribution outside the foreground mask is averaged per run. The
leak effect on that fraction is estimated with the H4 intercept model and reported with its
CI and no test. A sanity check compares each map with the map obtained after replacing the
classification head by a fixed random head (median Spearman correlation).

## 4. F3 and F4 -- carrier removal during training (new runs)

Ten seeds never used for any model, each checked feasible for the registered 2C
configuration: 1103, 1151, 1201, 1249, 1301, 1361, 1409, 1451, 1499, 1553. Their 2C designs
and cell splits are ‘splits/study2/followup/sipakmed_2c/‘ (manifest
‘splits/study2/followup/followup_split_manifest.csv‘). ResNet-50, DenseNet-121, ConvNeXt-T,
and ViT-B/16 are trained on all four cells under three conditions: ‘standard‘ (Study 2
recipe), ‘masked‘ (every training and test crop has its background set to the ImageNet mean;
cache ‘sipakmed_masked‘), and ‘colour‘ (training batches additionally receive HED-strong
optical-density jitter, alpha in [0.8, 1.2] and beta in [-0.2, 0.2] per stain channel,
followed by brightness and contrast factors in [0.8, 1.2]; test crops unmodified). 480 runs.

**F3**: within ‘masked‘, the H3 model
‘acc ~ leak * div + (1 + leak | architecture) + (1 | seed) + (1 | training set)‘ (H3 fallback
ladder); the leak coefficient, predicted positive (leakage survives without any background).
**F4**: within ‘standard‘ and ‘colour‘, with ‘treated = +-1/2‘,
‘acc ~ leak * treated + div + (1 + leak | architecture) + (1 | seed) + (1 | training set)‘
(fallback: random intercepts); ‘-coef(leak:treated)‘, predicted positive (colour augmentation
shrinks the leak effect). Secondary: the H3 model on ‘standard‘ (an independent replication
of H3 on fresh seeds); the same contrast for ‘masked‘ versus ‘standard‘ (background share of
the leak effect); cell means.

## 5. F5 -- LEACE erasure of within-class parent identity (existing Phase 1 checkpoints)

For every registered Phase 1 P1 and P2 model (12 architectures x 20 seeds; 480 models),
penultimate features of all SIPaKMeD crops are extracted (eval transform). On the model’s own
training crops, with ‘W = Sigma^{-1/2}‘ (ridge 1e-6 x mean eigenvalue) and the within-class
between-parent scatter ‘S = sum_j (n_j / n)(mu_j - mu_class(j))(mu_j - mu_class(j))^T‘, the
rank-r eraser removes the top-r eigendirections ‘U_r‘ of ‘W S W‘:
‘x’ = x - (x - mu) W U_r U_r^T W^{-1}‘ (rank-limited LEACE; Belrose et al., 2023). Ranks
r in {0, 4, 16, 64, 256}. At each rank the linear class head is refitted on the erased
training features (the logistic objective above) and test accuracy is measured; R is
measured on the erased probe features.

Per architecture, the **regime** is the set of ranks whose mean P2 accuracy is within 1 pp of
rank 0 (refitted heads); r* is its largest rank. **Manipulation check**: mean R_pair at r* is at
least 25% of the above-floor rank-0 value below it (as for H5); architectures that fail, or
whose regime is {0}, are excluded. For each included architecture-seed,
‘reduction = gap(0) - gap(r*)‘ with ‘gap = acc(P1) - acc(P2)‘. **F5**: intercept of
‘reduction ~ 1 + (1 | architecture) + (1 | seed)‘ (fallback ‘(1 | architecture)‘), predicted
```

```
positive (erasing linearly decodable within-class field identity closes part of the gap).
If no architecture is included, F5 is untestable, not refuted.

## 6. F6 -- stain-normalisation dose (new runs)

Reinhard normalisation of every crop toward one reference (the mean over all SIPaKMeD crops of
per-crop LAB means and of per-crop LAB standard deviations), blended in LAB space at strength
s in {0.5, 1.0} and applied to training, test, and probe crops alike (caches
‘sipakmed_reinhard50‘, ‘sipakmed_reinhard100‘). ResNet-50, ConvNeXt-T, DenseNet-121, and
EfficientNet-B0 on natural P1 and P2 for the ten Study 2 seeds no mechanism arm used (601,
653, 701, 751, 809, 853, 907, 953, 1009, 1051): 160 runs. Strength 0 is the corresponding
registered Phase 1 run (identical deterministic recipe, split, and seed). **F6**: in
‘gap ~ architecture + s + (1 | seed)‘ over the three strengths, ‘-coef(s)‘, predicted positive.
The regime and manipulation check of section 5 decide which strengths enter F8.

## 7. F7 -- probe-matched linear adversary (new runs)

The 2A design with a different adversary: per-feature standardisation (BatchNorm without
affine parameters) followed by one linear layer over the training parents, class-conditional
masking, gradient reversal with the DANN ramp; the adversary has its own AdamW optimizer
(learning rate 1e-3, no weight decay) and is re-initialised, with fresh optimizer state, every
5 epochs from a forked random stream. ResNet-50, ConvNeXt-T, and DenseNet-121; lambda in
{0, 0.1, 0.3, 1, 3}; natural P1 and P2; seeds as in F6; 300 runs. Regime, manipulation check,
and model exactly as H5: ‘gap ~ architecture + R_pair + (1 | seed)‘ over included
architectures and in-regime lambdas. **F7** is the R_pair slope, predicted positive.

## 8. F8 -- pooled dose-response

All included architecture x level cells of F5 (ranks), F6 (strengths), and F7 (lambdas),
after their regimes and manipulation checks:
‘gap ~ cell + R_pair + (1 | manipulation:seed)‘ with one fixed intercept per
manipulation x architecture cell. **F8** is the R_pair slope, predicted positive (across
independent ways of removing the fingerprint, the inflation tracks how much of it remains).

## 9. Multiplicity and interpretation

F1-F8 form one Holm family (family-wise alpha = 0.05), separate from H1-H5. Each is supported
only if its Holm-adjusted p < 0.05 and its estimate is positive; estimates are oriented so the
predicted direction is positive. Everything else here is secondary or descriptive.

Pre-specified reading: F2 and F3 supported together locate the leaked signal in the cell’s own
appearance rather than its background; F4 supported identifies colour augmentation as a
partial mitigation; F5, F7, or F8 supported means the inflation tracks recoverable field
identity (the shortcut account); F5 not supported while R falls means the inflation is not
carried by linearly decodable within-class field identity in pooled features.

## 10. Analysis order and artifacts

Each arm is analysed once all of its registered tasks are complete (‘--allow-incomplete‘
outputs are progress reports only); the Holm family is computed once all eight are available.
The code (‘src/cytofair/carriers.py‘, ‘src/cytofair/followup_eval.py‘, the trainer options,
‘scripts/study2/build_followup_tasks.py‘, ‘analysis/study2/followup/*‘), split and ablation
manifests, and task files are fixed by this commit; their SHA-256 hashes are listed under
"Amendment 8" in ‘reports/study2/registration_hashes.txt‘, together with the hashes of the
derived caches. A synthetic dry run of ‘followup_lmm.R‘
(‘analysis/study2/followup/simulate_followup_inputs.py‘) recovered every planted effect before
registration.

## Amendment 9 -- preparer bug fix (2026-10-20, Asia/Kolkata)

When the registered models were first run on the completed carrier arm, ‘followup_lmm.R‘
stopped at F4 because ‘prepare_followup.py‘ had labelled all 160 standard-recipe runs as
‘colour‘: the label was derived from a DataFrame row in which a missing colour augmentation
(‘None‘ in the run record) had become a truthy NaN. The masked runs were labelled correctly.
The script halted before writing any output, so no F3 or F4 estimate was seen. The fix
derives each run’s condition from its own record (‘carrier_condition‘: background-masked
cache -> ‘masked‘; ‘color_augment == "hed_strong_bc"‘ -> ‘colour‘; otherwise ‘standard‘), which
is the registered definition; no model, rule, or data changes. A unit test
(‘tests/test_followup_prepare.py‘) covers it. The synthetic dry run had not exercised the
preparer, which is why the error was not caught before registration.

Known when this amendment was written: the arm-level results of F1, F2 (with the Grad-CAM
description), and F5, analysed with the registered code once their arms were complete, as
section 10 allows. Not known: any F3, F4, F6, F7, or F8 estimate.
```

S

Phase 0 checklist, Cx22 literature check, and excluded attempts

S.1 Phase 0 checklist

```
# Phase 0: arXiv v1 readiness

Status of each pre-posting item from the scaling roadmap, as checked on 2026-10-18.
"Verified" means checked directly; "Authors" means only the authors can do it.

| Item | Status | Evidence or action |
|---|---|---|
| Public repository installs and tests pass | Verified | Fresh ‘git clone‘ of ‘github.com/optimisticexquisite
    /cytofair‘ at ‘2c5d5c3‘: ‘uv sync --extra dev‘ succeeds, 45/45 tests pass, ‘cytofair --help‘ runs. |
| Repository ships provenance maps and split manifests | Verified | ‘provenance/*.parquet‘ (SIPaKMeD Branch A
    /B, P3 map, Cx22, CRIC, Herlev) and ‘splits/generated/‘ (20 SIPaKMeD split CSVs + manifest) are tracked.
    |
| Repository contains the work the paper reports | **Gap** | The public ‘main‘ stops at the 22 August results.
     The 11 Cx22-extension commits and all Study 2 work exist only locally (branch ‘study2‘). Push before
    posting so the paper’s repository link resolves to the reported code. |
| Software licence | **Authors** | The repository has no ‘LICENSE‘ file, so the toolkit is legally all-rights-
    reserved and cannot be reused. Choose a code licence (MIT or Apache-2.0 are common) and state that
    derived artifacts follow each source’s terms (below). |
| Redistribution terms for derived artifacts | **Authors** (partly verified) | No source images are
    redistributed. Derived parent IDs, coordinates, hashes, and split assignments: CRIC and the Herlev
    repackaging are CC BY 4.0 (attribution suffices); SIPaKMeD is "public for experimental use, cite Plissiti
     et al." (derived IDs very likely fine; confirm); Cx22 is non-commercial research and education only;
    CCEDD forbids redistribution of the images (the released maps contain only source file names and
    coordinates; confirm this is acceptable to the CCEDD authors). |
| CRIC images for Study 2 | Verified | All 400 slide images are CC BY 4.0 on Figshare (collection 4960286)
    and were fetched with Figshare MD5 checks; ‘data_manifests/cric_images_verified.json‘ records MD5 and SHA
    -256 per file. Images are not committed. |
| All co-authors approve the exact posted version | **Authors** | Required before posting. The current draft
    still has e-mail and repository-URL placeholders. |
| PDF text extraction | Fixed in sources | The project record’s extracted text shows broken small caps ("
    CytoF air"), ligature glyphs, and split equations. The Study 2 manuscript preamble sets T1 font encoding
    and Unicode glyph mapping; the same two lines are suggested for the ISBI draft (see ‘paper/v2/README.md‘)
    . |
| "Replication in preparation" sentence | Suggested text | "A preregistered, adequately powered replication
    and a causal decomposition of the effect are in progress." (Study 2 is registered in this repository as ‘
    PREREGISTRATION_STUDY2.md‘.) |
| Anonymity policy of the target venue | **Authors** | Posting a preprint is usually allowed; promoting it
    during double-blind review may not be. Check the chosen venue’s policy before announcing the preprint. |

## Note on public timestamps

The Study 2 preregistration is committed locally before any Study 2 result exists, matching
how Study 1 was registered (its plan commit predated its first result commit, and both were
pushed later). A local commit is weak evidence of timing. For an independent timestamp,
push the registration commit, or deposit ‘PREREGISTRATION_STUDY2.md‘ on OSF, as early as
possible; the commit hash and file hashes in ‘reports/study2/registration_hashes.txt‘ let a
```

```
reader confirm that the deposited text is the one that governed the analysis.
```

S.2 Cx22 literature check

```
# Do published Cx22 studies pool the release subsets?

**Question (roadmap, Part 5).** Cx22's official Multi-Train / Multi-Test partition is
parent-disjoint, but the separately released Pair subset shares 233 CCEDD source images
with Multi-Train and 55 with Multi-Test. Pooling the 1,320 released ROIs and re-splitting
them at the ROI level therefore reintroduces source-image leakage. Is that risk realised in
published work, or only theoretical?

**Method (2026-10-18).** Europe PMC search `"Cx22" AND (cervical OR cytology)` (43 hits),
then the open-access full text (Europe PMC `fullTextXML`) of every hit that uses Cx22
images, searched for sentences describing training, testing, splitting, or folds. Hits
where "Cx22" denotes something else (a connexin, a compound) were discarded. Papers without
open full text were not assessed.

**Stated protocols (verbatim excerpts).** The table records what each paper states; it
does not infer anything the text does not say.

| Paper | Use of Cx22 | Stated protocol |
|---|---|---|
| BMC Medical Imaging 2023, deep ensemble segmentation (PMC10514950) | train and test | "The Cx22 dataset
    contained a training dataset and a testing dataset with 400 and 100 samples, respectively." Official
    Multi split. |
| Scientific Reports 2025, Mask2Former with denoising (PMC12095581) | train and test | "The Cx22 dataset
    contains 400 images in the training dataset and 100 ones in the testing dataset." Official Multi split. |
| Cell Reports Medicine 2026, UniCAS foundation model (PMC12866161) | segmentation benchmark | "Cx22-multi:
    This public dataset consists of 500 cytology images, divided into 400 for training and 100 for testing."
    Official Multi split. |
| Cancer Cell International 2026, joint U-Net and CNN ensemble (PMC13041069) | train and validate | Describes
     Cx22 as "1,320 images" and states "a stratified five-fold cross-validation protocol, in which the data
    were divided into five mutually exclusive folds", with splitting "at the image level, ensuring that all
    cell instances derived from a single image remained within the same subset". It does not state that Pair
    was excluded or that CCEDD source images were kept within one fold. The same paper treats SIPaKMeD crops
    as "independent, pre-cropped single-cell samples" and separates them "at the image level". |
| Scientific Data 2024 (PMC11227563); Cureus 2025 (PMC11885184) | cited only | No Cx22 training or evaluation.
     |
| Frontiers in Oncology 2025 (PMC12929100) | unclear | Describes "Cx22 clinical data" modelled with an MLP;
    the text does not match the image dataset and was not assessed further. |

**Answer.** Of four open-access studies that train or evaluate on Cx22 images, three use
the official 400/100 Multi-Train/Multi-Test split, which is parent-disjoint. One describes
image-level five-fold cross-validation over the 1,320-image collection without stating that
the Pair subset or the CCEDD source images were kept apart, which is the downstream reuse
pattern the Cx22 stress test models. The risk is therefore realised in at least one recent
publication, while most users follow the safe official split, consistent with Cx22 serving
as an existence proof that a hierarchy-aware release is used in practice. The same study's
description of SIPaKMeD crops as independent samples is the misconception this project
addresses.

**Limits.** Open-access full texts only; closed-access Cx22 users (for example a 2026
Scientific Reports background-gating network, a 2024 Computers in Biology and Medicine
unsupervised instance-segmentation method, and a 2025 Medical & Biological Engineering &
Computing method) were not assessed. Whether a study pooled subsets can be stated only from
what the paper reports.

**Reporting choice.** In keeping with the project's rule that no individual paper is named
as deficient, the manuscript reports these counts in aggregate. This file keeps the
verbatim evidence so the count can be checked.
```

S.3 Excluded attempts (Study 2 smoke tests)

```
# Study 2 excluded attempts

No attempt below is eligible for any Study 2 analysis. Each was an execution check run
```

```
before registration; none trained a P2, P3, clean-cell, or CRIC P2 model, so none could
reveal a protocol contrast. Their metrics were checked only for finiteness and completeness
of artifacts, never compared across protocols.

| Attempt | Architecture | Split | Epochs | Purpose |
|---|---|---|---|---|
| `p1_<architecture>` (12 runs) | all twelve registered backbones | SIPaKMeD P1, seed 137 | 1 | pipeline,
    artifacts, throughput |
| `p1_resnet18_repeat` | ResNet-18 | SIPaKMeD P1, seed 137 | 1 | bitwise determinism (logits and features
    identical to `p1_resnet18`) |
| `leak_high_resnet50` | ResNet-50 | 2C `leak_high`, seed 137 | 1 | transplant extra-evaluation artifact |
| `p1_grl_resnet50` | ResNet-50 | SIPaKMeD P1, seed 137, lambda = 1 | 1 | adversary bookkeeping and DANN ramp
    |
| `cric_p1_resnet18` | ResNet-18 | CRIC P1, seed 137 | 1 | CRIC window cache and six-class labels |
| `p1_vit_b_16_math`, `p1_vit_b_16_math_repeat`, `p1_dinov2_vits14_math` | ViT-B/16, DINOv2 ViT-S/14 |
    SIPaKMeD P1, seed 137 | 1 | deterministic (math) attention kernels: speed and repeatability |

The first ViT-B/16 smoke run used fused memory-efficient attention, whose backward pass
PyTorch reports as non-deterministic. Registration therefore fixes the deterministic math
attention kernel for every run (Section 1 of `PREREGISTRATION_STUDY2.md`).

Outputs are kept on the training host under `smoke/` and are not copied into `results/`.

Before registration, the 2C factorial configuration with 240 crops per class was rejected
because it was infeasible for registered seed 307 (and for 10 of 100 unrelated seeds); no
model was trained on it.
```

T

Reproduction cookbook

T.1 Environment

Python 3.13 with the locked dependencies (`uv sync -extra dev`); PyTorch 2.9.0 with CUDA 12.8 on the GPU host; R 4.4 with lme4 2.0 and lmerTest 3.2 for the mixed models. Software versions recorded by every run are in Table 27.2.

T.2 Study 2

```
uv run cytofair study2-design                          # regenerate every registered split (hash-identical)
scripts/study2/node_prepare.sh ROOT PYTHON CRIC_DIR    # crop caches and transplant caches on the GPU host
python scripts/study2/build_tasks.py --root ROOT --repository REPO --python PYTHON --seeds 20
python scripts/study2/launch_matrix.py splits/study2/tasks/*.jsonl \
    --gpus 0,3,4,5 --logs ROOT/runs/logs --status ROOT/runs/status.json
analysis/study2/run_all.sh ROOT PYTHON RSCRIPT         # registered analysis; fails closed
python scripts/study2/paper_numbers.py                 # manuscript numbers and tables
python scripts/study2/figures_results.py               # Study 2 figures
```

T.3 Amendment 8

```
uv run cytofair followup-design                        # fresh-seed 2C designs and ablation plans
uv run cytofair build-carrier-caches --cache-dir ROOT/cache --cache-name sipakmed --output ROOT/cache
python scripts/study2/build_followup_tasks.py --root ROOT --repository REPO \
    --study2-runs ROOT/runs --python PYTHON
python scripts/study2/launch_matrix.py splits/study2/followup/tasks/*.jsonl \
    --gpus 4,5,6,7 --logs ROOT/runs/followup/logs --status ROOT/runs/followup/status.json
analysis/study2/followup/run_followup.sh ROOT PYTHON RSCRIPT
```

```
python analysis/study2/followup/exploratory_overlap.py --study2-runs ROOT/runs \
    --followup-runs ROOT/runs/followup --output results/study2/followup/exploratory/overlap.json
python scripts/study2/figures_followup.py            # follow-up figures
```

T.4 This record

```
python record/scripts/split_study1.py                # Study 1 chapters from monograph/main.tex
python record/scripts/export_run_records.py RECORD_DATA results/study2/run_records
python record/scripts/make_record_tables.py          # tables and record macros
python record/scripts/make_record_figures.py         # record figures
record/build.sh                                      # full record and standalone part PDFs
```

U

Where everything is

Table U.1: Key files and directories.

Path	Content
`PREREGISTRATION.md`	Study 1 plan and Amendments 1–6
`PREREGISTRATION_STUDY2.md`	Study 2 plan, Amendments 7–9 pointers
`PREREGISTRATION_STUDY2_AMENDMENT8.md`	Amendments 8 and 9 in full
`provenance/`	crop-to-parent maps for the four datasets
`splits/study2/`	registered Study 2 splits, designs, plans, tasks, manifests
`splits/study2/followup/`	Amendment 8 designs, ablation plans, tasks, manifests
`src/cytofair/train_study2.py`	the Study 2 trainer
`src/cytofair/gpu_augment.py`	GPU-resident augmentation
`src/cytofair/image_cache.py`	packed, hashed crop caches
`src/cytofair/factorial.py`, `transplant.py`	2C designs and 2B composites
`src/cytofair/carriers.py`, `followup_eval.py`	Amendment 8 transforms, LEACE, Grad-CAM, device probe
`src/cytofair/risk.py`	structural closed forms (`risk-report`)
`scripts/study2/`	task builders, scheduler, figures, numbers, archiving
`analysis/study2/`	registered Study 2 analysis (R and Python)
`analysis/study2/followup/`	registered Amendment 8 analysis, exploratory overlap
`results/study2/analysis/`	Study 2 registered outputs
`results/study2/followup/`	Amendment 8 registered and exploratory outputs
`results/study2/run_records/`	exported records of every run
`reports/study2/`	registration hashes, Phase 0, literature check, excluded attempts
`paper/v2/`	the single-column manuscript
`monograph/`	the Study 1 record source
`record/`	this record
`~/cytofair_study2_archive` (GPU host)	121 GB archive of every run artifact

V

Glossary

Parent
The acquisition unit a crop was cut from: a SIPaKMeD cluster field, a Cx22 (CCEDD) source image, or a CRIC slide image.

P1, P1′, P2, P3
Crop-level, size-matched crop-level, parent-disjoint, and component-disjoint splits.

Straddle
A parent with crops on both sides of a split.

Leak and clean cells
2C training sets that do or do not contain the test fields' sibling crops, at matched size and diversity.

Recoverability R
Within-class field-identification accuracy of a linear probe on frozen features.

Regime
The manipulation levels whose mean P2 accuracy stays within 1 pp of baseline.

Manipulation check
At least a quarter of the above-floor recoverability removed at the strongest in-regime level.

Holm family
A set of tests whose p-values are adjusted together by Holm's step-down procedure.

LEACE
Least-squares concept erasure: a linear projection after which no linear function of the features covaries with the concept.

GRL
Gradient-reversal layer: identity in the forward pass, negated and scaled gradient in the backward pass.

HED jitter
Random scaling and shifting of stain channels in optical-density space.

Reinhard transfer
Matching an image's LAB mean and standard deviation to target values.

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